

COUNTEREXAMPLES TO GRAD'S CONJECTURE

JAVIER GÓMEZ-SERRANO, LUKAS LIEHR, AND MITCHELL A. TAYLOR

ABSTRACT. For each sufficiently large N , we construct smooth solutions of the magnetohydrostatic equations on embedded solid tori whose regular pressure levels are nested tori, whose magnetic field vanishes exactly on a round magnetic axis, and whose group of Euclidean symmetries, even when isometries reversing the sign of the field are admitted, is exactly the cyclic group C_N of rotations through multiples of $2\pi/N$. For each such N , these equilibria form a smooth one-parameter family which is locally nontrivial in the moduli space of embedded configurations and have none of the plane-reflection, axial, or helical symmetries conjectured by Grad. A Lean 4 certification of the above results is also provided.

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1. INTRODUCTION

The objective of this article is to construct exotic smooth solutions of the magnetohydrostatic equations

$$(1.1) \quad B \times \operatorname{curl} B + \nabla P = 0, \quad \operatorname{div} B = 0 \quad \text{in } \Omega, \quad B \cdot n = 0 \quad \text{on } \partial\Omega,$$

on solid tori $\Omega \subset \mathbb{R}^3$, with a prescribed symmetry group. Here, B is the magnetic field, P is the pressure, n is the outward unit normal, and by a *solid torus* we mean a compact domain diffeomorphic to $\overline{\mathbb{D}^2} \times \mathbb{T}$. The system (1.1) describes the static equilibria of an ideal, perfectly conducting plasma. Indeed, the first equation states that the Lorentz force balances the pressure gradient, the second states that the field is *solenoidal*, and the third states that the boundary $\partial\Omega$ is a *magnetic surface*. Since the first equation carries no external force, we call (1.1) the *unforced magnetohydrostatic (MHS) equations*.

Taking the inner product of the first equation in (1.1) with B gives $B \cdot \nabla P = 0$, so field lines lie on the level sets of the pressure. In plasma confinement, one is interested in equilibria whose regular pressure levels are nested tori filling Ω and collapsing in the interior onto a closed curve, the *magnetic axis*, where $\nabla P = 0$. On each torus, the first equation imposes a solvability condition along the field lines [22, 34, 43], and this condition obstructs the construction of nonsymmetric equilibria with nonconstant pressure. Grad [20, 21] argued that, outside the classical symmetric configurations, no smooth equilibrium with nested toroidal pressure surfaces should belong to a family depending smoothly on a parameter. A precise formulation of this conjecture was given by Constantin, Drivas and Ginsberg [9], and it states that a smooth unforced MHS equilibrium on a domain of toroidal or cylindrical topology, whose regular pressure surfaces foliate the domain, whose field does not vanish away from the magnetic axis, and which is not isolated, has plane-reflection, axial, or helical symmetry. Here, *non-isolation* means that in some suitable topology there are nearby equilibria other than trivial rescalings and translations of the given one.

In this article, we construct equilibria for which none of the three alternatives holds. More precisely, for every sufficiently large integer N , we produce a smooth one-parameter family of solutions of (1.1) on embedded solid tori. The pressure levels of these solutions are nested tori, their field vanishes exactly on a round magnetic axis, and their full group of Euclidean symmetries, even allowing isometries that reverse the sign of B , is the cyclic group C_N of rotations through multiples of $2\pi/N$ about the axis of the torus.

1.1. Embedded configurations and their moduli topology. We compare equilibria on different domains by pulling them back to a common reference domain. Let

$$Q = \overline{\mathbb{D}^2} \times \mathbb{T}, \quad \mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z}),$$

and, for an integer $k \geq 3$, represent an equilibrium by the triple

$$(X, b, p) = (X, B \circ X, P \circ X) \in \operatorname{Emb}^k(Q, \mathbb{R}^3) \times C^{k-1}(Q, \mathbb{R}^3) \times C^k(Q),$$

where X is an embedding of Q onto $\Omega = X(Q)$. The above indices are matched so that $\operatorname{curl} B$ is continuously differentiable, and the product carries its usual topology. Two triples describe the same equilibrium if they

differ by a reparametrization of the reference domain,

$$(1.2) \quad (X, b, p) \mapsto (X \circ \Phi, b \circ \Phi, p \circ \Phi), \quad \Phi \in \text{Diff}^k(Q),$$

or by a spatial similarity, a rescaling of the magnetic amplitude, or a shift of the pressure, all of which preserve (1.1) and can be represented by

$$(1.3) \quad (X, b, p) \mapsto (sOX \circ \Phi + c, \kappa Ob \circ \Phi, \kappa^2 p \circ \Phi + c_0),$$

with $s > 0$, $O \in O(3)$, $c \in \mathbb{R}^3$, $\kappa \in \mathbb{R} \setminus \{0\}$, and $c_0 \in \mathbb{R}$. We write $\mathcal{M}_{\text{emb}}^k(Q)$ for the quotient with the natural quotient topology, and call it the *moduli space of embedded configurations*. For smooth configurations, we use the usual C^∞ topology and quotient by smooth reparametrizations and the same physical action. The embedded domain is part of the configuration.

For a Euclidean isometry $g(x) = Ox + c$, we set $(g_*B)(x) = OB(g^{-1}x)$ and define the *extended stabilizer* by

$$(1.4) \quad \text{Stab}_\pm(\Omega, B, P) = \{g : g\Omega = \Omega, P \circ g^{-1} = P, g_*B = \pm B\}.$$

Thus, this stabilizer includes isometries that reverse the field. We will prove that distinct members of each family we construct are inequivalent even under the larger group (1.3).

1.2. Historical comments. The existence of three-dimensional equilibria with nested pressure surfaces is the central mathematical question of stellarator theory. We refer to Freidberg [18] and Helander [25] for an account of the underlying physics. Grad's argument [20] rests on the analysis of the real characteristics of the MHS system along field lines. Near a torus with irrational rotational transform, the solvability condition of [22, 34, 43] must hold on a dense set of rational tori, and this appears to force the pressure to flatten. One way around this obstruction is to admit current sheets. Bruno and Laurence [2] proved the existence of equilibria with piecewise constant pressure in tori that are small perturbations of an axisymmetric one. In their setting, the pressure jumps across finitely many interfaces carrying current sheets, and the rotational transform on each interface must be sufficiently irrational (see also Kaiser and Salat [32]). This picture underlies the equilibrium program of Hudson, Dewar, and their colleagues [28, 29, 30, 33, 39]. Notably, Enciso, Luque and Peralta-Salas [15] recently constructed piecewise smooth equilibria with arbitrarily many current sheets in toroidal domains that need not be close to axisymmetric and may be knotted. In [2, 15], the pressure is constant between the interfaces and the solutions are not C^1 .

Smooth equilibria without continuous symmetries are known to exist in a few special situations. Lortz [38] constructed toroidal equilibria with nonconstant pressure whose domains are symmetric with respect to a plane but otherwise have no symmetry. In this construction, all field lines are closed and the rotational transform is zero. The planar symmetry makes it consistent with Grad's conjecture. Salat and Kaiser [48] produced exact equilibria in an infinite cylinder with a straight axis but no cylindrical, helical, or mirror symmetry. Their field lines are closed curves in the planes orthogonal to the axis, and their pressure surfaces have elliptical cross-sections whose eccentricity and orientation vary freely along the axis. Sato [49] obtained equilibria in bounded domains without continuous Euclidean symmetry, and Sato and Yamada [50] constructed nested invariant tori without such symmetry for anisotropic magnetohydrodynamics. Weitzner obtained formal expansions of nonsymmetric MHS equilibria to all orders in doubly periodic domains [52, 53] and about a circular magnetic axis [53]. Weitzner and Sengupta [54] treated nearly parallel plasma flows in a doubly periodic domain. For related work on MHS equilibria and their symmetry properties, we also refer to [3, 6, 19, 35].

On the numerical side, three-dimensional MHS equilibria have been computed for decades, and the difficulty behind Grad’s conjecture is visible in the design of the codes. VMEC [27] and, more recently, DESC [13] assume nested flux surfaces from the outset and minimize the energy or the force residual within this class. Hudson and Kraus [30, 33] have argued that nested surfaces with a smooth pressure profile produce infinite currents at the rational surfaces, so that the pressure gradient must vanish on an interval around each of them and the resulting profile is fractal. PIES [47], HINT [24] and SIESTA [26] drop the assumption of nested surfaces and resolve islands and stochastic regions, while SPEC [28] implements the stepped-pressure picture described above. Loizu, Hudson, Bhattacharjee and their collaborators [36, 37] computed the singular currents that form at rational surfaces in the presence of a pressure gradient, and Zhou, Huang, Qin and Bhattacharjee [56] followed the formation of current singularities in topologically constrained plasmas. More recently, physics-informed neural networks [46], which minimize the residual of the equations over a mesh-free parametrization of the unknowns, have been applied to the Grad–Shafranov equation [31, 55] and to fully three-dimensional stellarator equilibria with nested surfaces [51], where they reach lower force residuals than the conventional solvers.

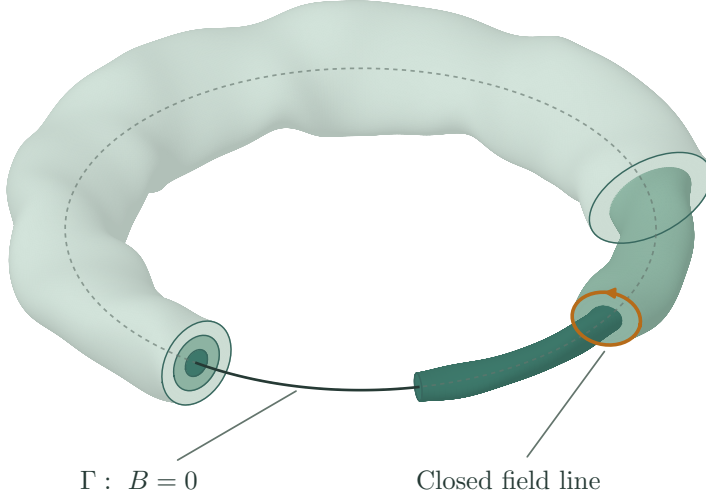
The formulation of Grad’s conjecture stated above is due to Constantin, Drivas and Ginsberg [9]. In their paper, they also proved rigidity and flexibility theorems for steady Euler, Boussinesq, and MHS flows. Their flexibility results deform a given equilibrium to nearby domains, which is the point of view we take in Section 1.1. In [10], the same authors constructed smooth nonsymmetric equilibria with nested flux surfaces at the price of a small forcing term, and in [11] they studied the free boundary version of the problem. Cardona, Duignan and Perrella showed in [5] that on a Riemannian manifold the absence of Killing symmetries is generic among adapted metrics. On the other hand, in [45] it is proved that an orientation-reversing isometry of an equilibrium with toroidally nested pressure surfaces forces all field lines to be periodic, and in [7, 17, 44] it is shown that Beltrami fields with nonconstant proportionality factor are rare. Interestingly, Enciso, Kepplinger and Peralta-Salas [14] found steady Euler flows on certain closed three-manifolds that are isolated in C^1 , suggesting that the non-isolation hypothesis in the conjecture is not vacuous. For structure theorems and steady Euler flows of low regularity, we refer to [1, 16].

The closest result to ours is the construction by Drivas, Elgindi and Ginsberg [12] of smooth steady Euler flows in a periodic cylinder, fibered by invariant cylinders that are level sets of the Bernoulli function. Their flows have an m -fold rotational symmetry and a plane-reflection symmetry but no continuous Euclidean symmetry. The solutions that they construct come in infinite-dimensional families, the perturbation parameter is $1/m$, and the iteration scheme is motivated by the approach of Lortz. They ask whether all Euclidean symmetries, including the reflection, can be broken, and they propose to refine Grad’s conjecture by requiring ergodic orbits on almost every invariant torus.

In the present article, we construct new MHS solutions that exhibit the following properties.

- (i) a smooth pressure with nested embedded tori in a solid torus and a magnetic axis on which the field vanishes;
- (ii) a prescribed symmetry group, excluding reflections as well as continuous symmetries, even when isometries reversing the field are admitted;
- (iii) for each of the prescribed symmetry groups, a locally nontrivial smooth family of solutions obtained by a Nash–Moser argument.

(a) Nested pressure surfaces



(b) Cross-section

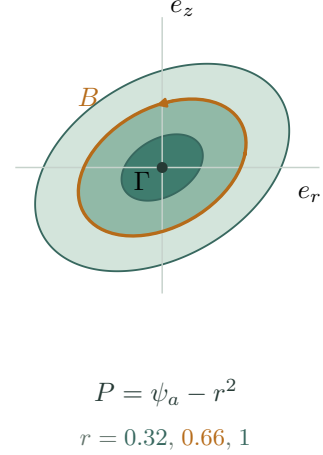


FIGURE 1. Nested pressure surfaces for the model $X^0(y, \phi) = R_\phi(NLe_x + \iota M_\lambda(N\phi)y)$. (a) A cutaway showing three pressure surfaces, a closed field line, and the magnetic axis Γ . The axis is dashed where hidden. (b) A cross-section through the highlighted field line. We use $N = 8$, $L = 0.65$, $\rho_* = 0.40$, $\delta_* = 0.425$, $\alpha_0 = \pi/6$, and $\lambda = 0.375$. The cross-sections have axis ratio approximately 1.53 : 1. The eccentricity is enlarged for visibility and lies outside the theorem's small-eccentricity regime. The nonlinear correction in (1.11) is omitted.

LLM usage disclosure: The work on this project began in July 2026. In July, the authors set up the problem and provided a detailed roadmap to construct the solutions. After this, they used a combination of the LLMs GPT-5.6 Sol, Claude Fable 5, and Claude Opus 5 to fill in the technical details, assist with calculations, and provide mathematical feedback on the parts of the blueprint that were not detailed enough. The Lean code was produced by the same models with continuous guidance from the authors. During the latter stages of the Lean verification, the authors switched to the newer models GPT-6 Astra and Claude Fable 5.1. These models were also used for final proofreading but did not contribute to any of the mathematical content of the article. All mathematical statements and proofs have been checked by the authors, who take full responsibility for their correctness.

We also remark that our examples share some resemblances to the recent and spectacular counterexamples [4, 8] to Schiffer's conjecture. Our approach is closer in spirit to that of [4], but we also ran an independent computer-assisted approach. This latter attempt was abandoned once it became clear that the approach we present here would close, but it would nonetheless be interesting to systematically pursue such methods as they should have immense value.

1.3. The main result. Let e_x, e_y, e_z be the standard basis of $\mathbb{R}^3$ and R_ϕ the rotation through ϕ about the e_z -axis. Throughout, N is the number of periods of the field. Note that the pressure is determined by (1.1) up to an additive constant, which we fix by prescribing its value ψ_a on the magnetic axis.

Figure 1 illustrates the geometry and terminology. The C_N symmetry consists of rotations about the e_z -axis. The magnetic axis is the circle Γ where the nested tori collapse and the magnetic field vanishes. A *cell* is one of the N repeated pieces of the torus. Consecutive cells are related by a rotation through $2\pi/N$. The magnetic axis has radius NL , so its arc within each cell has length $2\pi L$. At leading order, λ controls how the elliptical cross-sections turn along a cell.

Our main result may be stated as follows.

Theorem 1.1 (Equilibria with exact cyclic symmetry). *Fix $L > 0$. There exist a compact interval $I \subset (0, \frac{1}{2})$ with nonempty interior and an integer $N_0 < \infty$ such that the following holds. For every integer $N \geq N_0$ and every $\lambda \in I$, there is a smooth embedding*

$$X_{N,\lambda} : Q \longrightarrow \mathbb{R}^3$$

and smooth fields $B_{N,\lambda}, P_{N,\lambda}$ on $\Omega_{N,\lambda} = X_{N,\lambda}(Q)$ satisfying (1.1), with the following properties.

(i) (Nested pressure surfaces). *The unique critical set of $P_{N,\lambda}$ is the round circle*

$$(1.5) \quad \Gamma_N = \{NL(\cos \phi, \sin \phi, 0) : \phi \in \mathbb{T}\},$$

every regular pressure level is an embedded torus, and these levels foliate $\Omega_{N,\lambda} \setminus \Gamma_N$.

(ii) (Magnetic axis). *The field vanishes exactly on the axis, that is,*

$$(1.6) \quad B_{N,\lambda}(x) = 0 \iff x \in \Gamma_N.$$

(iii) (Exact stabilizer). *The extended stabilizer is the cyclic group of rotations,*

$$(1.7) \quad \text{Stab}_\pm(\Omega_{N,\lambda}, B_{N,\lambda}, P_{N,\lambda}) = \{R_{2\pi j/N} : j = 0, \dots, N-1\} \cong C_N.$$

In particular, the equilibrium has neither a continuous Euclidean symmetry nor an orientation-reversing Euclidean symmetry.

(iv) (Non-isolation). *For each fixed $N \geq N_0$, the representatives may be chosen to depend smoothly on λ , and the map*

$$(1.8) \quad \lambda \longmapsto [X_{N,\lambda}, B_{N,\lambda} \circ X_{N,\lambda}, P_{N,\lambda} \circ X_{N,\lambda}] \in \mathcal{M}_{\text{emb}}^k(Q)$$

is continuous and injective for every finite $k \geq 3$, as well as in the C^∞ topology. Consequently, every member with $\lambda \in \text{int } I$ is non-isolated in the moduli space of embedded configurations.

The rotations in (1.7) exhaust the symmetry group, including isometries sending B to $-B$. This group has none of the plane-reflection, axial, or helical symmetries posited in [9], and thus Theorem 1.1 refutes Grad's conjecture.

As illustrated in Figures 1 and 3, our construction depends on the three constants

$$0 < \rho_* < \frac{1}{4}, \quad \delta_* \neq 0, \quad \alpha_0 \notin \frac{\pi}{2}\mathbb{Z},$$

which we will later fix. These constants allow us to make the equilibria completely explicit to leading order in $1/N$. Writing $\zeta = N\phi$ for the coordinate along a cell, we set

$$(1.9) \quad \alpha_\lambda(\zeta) = \alpha_0 + \delta_*(\cos \zeta + \lambda \sin 2\zeta),$$

$$(1.10) \quad M_\lambda(\zeta) = R_{\alpha_\lambda(\zeta)} \begin{pmatrix} \sqrt{1+\rho_*} & 0 \\ 0 & \sqrt{1-\rho_*} \end{pmatrix} R_{-\alpha_\lambda(\zeta)},$$

where R_α is the planar rotation through α , so that $M_\lambda(\zeta)$ is a symmetric matrix with eigenvalues $\sqrt{1 \pm \rho_*}$ whose principal axis turns by $\alpha_\lambda(\zeta)$ around the circle. We set $\iota(a, b) = ae_x + be_z$ and $v_{0,\lambda}(y, \zeta) = \iota M_\lambda(\zeta)y$. When the axis is straightened, that is, in the limit $N \rightarrow \infty$, the map $(y, \zeta) \mapsto L\zeta e_y + v_{0,\lambda}(y, \zeta)$, with $\zeta \in \mathbb{R}$, parametrizes an exact equilibrium. Its pressure surfaces have elliptical cross-sections whose orientation varies along the axis, and its nonconstant field lines are the images under this map of the circles $|y| = r > 0$ at fixed ζ . The parameter ρ_* governs the eccentricity of the ellipse, α_0 represents the mean tilt of the major axis of the ellipse measured from the plane of the torus, and δ_* controls the amplitude of the twisting. We refer the reader to the associated GitHub repository for an interactive plot that may be used to visualize the geometry. Up to normalization, these maps describe a subfamily of solutions constructed by Salat and Kaiser [48]. [Theorem 1.1](#) says that these solutions can be bent into a closed torus of major radius NL while keeping the pressure surfaces nested and the field smooth. Indeed, for every small curvature parameter ε , the proof constructs a 2π -periodic map $v_{\varepsilon,\lambda}(y, \zeta)$ close to $v_{0,\lambda}$. Setting $\varepsilon = 1/N$, the equilibrium is

$$(1.11) \quad X_{N,\lambda}(y, \phi) = R_\phi(NLe_x + v_{\varepsilon,\lambda}(y, N\phi)),$$

$$(1.12) \quad P_{N,\lambda} \circ X_{N,\lambda} = \psi_a - |y|^2, \quad B_{N,\lambda} \circ X_{N,\lambda} = \partial_\theta X_{N,\lambda},$$

with $\partial_\theta = -y_2 \partial_{y_1} + y_1 \partial_{y_2}$. In the normalized chart, we write the perturbed map as its linear Taylor term in y at the axis $y = 0$ plus a remainder,

$$(1.13) \quad v_{\varepsilon,\lambda}(y, \zeta) = P_{M_\lambda, \tau_{\varepsilon,\lambda}}(\zeta)y + \tilde{v}_{\varepsilon,\lambda}(y, \zeta), \quad j_y^1 \tilde{v}_{\varepsilon,\lambda}|_{y=0} = 0.$$

Here,

$$P_{M,\tau} = a(\tau)\iota M + e_y \otimes \tau, \quad a(\tau) = \sqrt{1 - |\tau|^2/2},$$

and the unknown function $\tau = \tau_{\varepsilon,\lambda} : \mathbb{T} \rightarrow \mathbb{R}^2$ determines the component of this linear term along the straightened axis, namely $(\tau(\zeta) \cdot y)e_y$. The condition $j_y^1 \tilde{v}_{\varepsilon,\lambda}|_{y=0} = 0$ means that $\tilde{v}_{\varepsilon,\lambda}$ and its first derivatives in y vanish at $y = 0$. We will also prove that, uniformly for $\lambda \in I$,

$$(1.14) \quad \|\tau_{\varepsilon,\lambda}\|_{C^2} + \|\tilde{v}_{\varepsilon,\lambda}\|_{C^2} \leq C|\varepsilon|.$$

The term $\cos \zeta$ in (1.9) fixes the period of the tilt at $2\pi/N$ in the variable ϕ . The term $\lambda \sin 2\zeta$, which is odd in ζ , distinguishes the equilibrium from its mirror images and from nearby members of the family. The vanishing condition in (1.13) makes the Hessian of the pressure on the magnetic axis independent of $\tilde{v}$. Restricted to the normal plane of the axis, its inverse can be normalized to remove the dependence on τ . This normalized inverse Hessian determines $\alpha_\lambda(N\phi)$ modulo π , which gives the exact symmetry group.

1.4. Strategy of the proof. The proof has three parts. First, we reduce (1.1) to a problem on one period, posed on a fixed domain with the curvature as a parameter. Second, we invert the linearization of this problem, with estimates that are uniform on an open set of states. Third, we apply the Nash-Moser theorem to solve the nonlinear equations, close the cell into a torus, and compute the symmetry group. Most of the paper is devoted to the second part of the argument.

Choosing coordinates. We look for an immersion $X = X(\psi, \theta, \zeta)$ with $B \circ X = X_\theta$ and $P \circ X = \psi$, so that pressure surfaces and field lines are built into the coordinates. This makes (1.1) equivalent to three scalar equations for X (see [Section 2](#)). Two of these equations are in conservation form, and the third says that the Jacobian of X is independent of θ . To isolate one period of the field, we write

$$X(\psi, \theta, \zeta) = R_{\varepsilon\zeta} \left(\frac{L}{\varepsilon} e_x + v(\psi, \theta, \zeta) \right),$$

where ε is a curvature parameter. Although L/ε diverges as $\varepsilon \rightarrow 0$, the equations for v remain well defined at $\varepsilon = 0$, since the curvature enters only through $D_\varepsilon v = v_\zeta + \varepsilon e_z \times v + L e_y$, which replaces v_ζ . The unknown v lives on the fixed domain $Q = \overline{\mathbb{D}^2} \times \mathbb{T}$, and we solve for all small ε before setting $\varepsilon = 1/N$ and joining N cells into a torus. This is why [Theorem 1.1](#) holds for every large N .

At $\varepsilon = 0$, the equations are solved exactly by every $v = \iota M(\zeta)y$ with M a smooth periodic 2×2 matrix satisfying $\det M > 0$ and $\text{tr}(M^T M) = 2$ (see [Lemma 2.2](#)). We perturb this infinite-dimensional family rather than a single state, and we choose the subfamily (1.10) so that the symmetry group can be read off at the end. The degeneracies of the problem, namely reparametrizations of the cell and motion along the axis, both lie in the kernel of the linearization. We remove them with the normalized chart (1.13), two gauge conditions, and a condition on the outer boundary (see [Section 3](#)).

Cell frequency. The variable $\zeta \in \mathbb{T}$ parametrizes the period cell, and we expand every unknown in a Fourier series in it,

$$u(y, \zeta) = \sum_{n \in \mathbb{Z}} \hat{u}_n(y) e^{in\zeta}.$$

We call the index n the *cell frequency*. Once $\varepsilon = 1/N$, the *cell variable* is $\zeta = N\phi$, so $e^{in\zeta}$ is the toroidal mode $e^{inN\phi}$. Thus, cell frequency n corresponds to toroidal mode number nN , and the modes with $n = 0$ are the axisymmetric ones. We write $\langle n \rangle = (1 + n^2)^{1/2}$ and $\Lambda = \langle D_\zeta \rangle$. The operator Λ multiplies the n -th Fourier coefficient by $\langle n \rangle$. Since the coefficients of the linearized operator depend on ζ through M , different cell frequencies are coupled. Nevertheless, the whole linear analysis is organized by the size of $\langle n \rangle$. A cutoff 2^J separates the *high frequencies* $|n| \geq 2^J$, which we treat by energy estimates and a Neumann series, from the finitely many *low frequencies*, which we treat by comparison with an explicit reference.

The cell frequency also determines the scale at which we split the disk. We write $y = re^{i\theta}$. In polar coordinates, the equations have singular coefficients at the axis $r = 0$. The transition from the behavior near the axis to a radial evolution occurs when $r\langle n \rangle \sim 1$. Accordingly, at cell frequency n we split the disk into a *cap* of radius comparable to $\langle n \rangle^{-1}$ around the axis, where the equations are of Bessel type, and the complementary *annulus*, where they are a first-order evolution in r with $\langle n \rangle$ as the large parameter. The cap shrinks as the frequency grows, and the analysis on it is a rescaled version of a single fixed problem.

The linearized problem. Three difficulties arise in constructing a uniform inverse. First, the cap and the annulus require different methods, and we must match the solutions at their interface with estimates uniform in the cell frequency n . Second, the system on the annulus has three parts that require different estimates. At a constant ellipse, the *center part* consists of two coupled oscillators whose speeds depend on the ellipse parameter ρ and coincide at the circle $\rho = 0$. We estimate them together so that the bounds remain uniform as the ellipse approaches a circle. The *affine part* comes from the solutions at zero curvature that are linear in the disk variable y , of the form $v(y, \zeta) = \iota M(\zeta)y$. Its equations allow linear growth in r . The remaining part is elliptic and satisfies a coercive estimate. Finally, the source must satisfy compatibility conditions, and converting it from Cartesian variables to the components used in these estimates requires differentiation. This conversion costs a fixed number ℓ of derivatives, which must be accounted for in the inverse estimate.

The uniform inverse. The construction is summarized in [Figure 2](#). To control the growth of the center and affine solutions on the annulus, we introduce an exponential weight in r whose strength grows with the frequency. We work in fixed spaces for the unknowns and the sources, equipped with the analytic weight

$$W_\gamma(r, n) = \exp\left\{\sigma_0 \langle n \rangle - \gamma(\sqrt{1 + r^2 \langle n \rangle^2} - 1)\right\}.$$

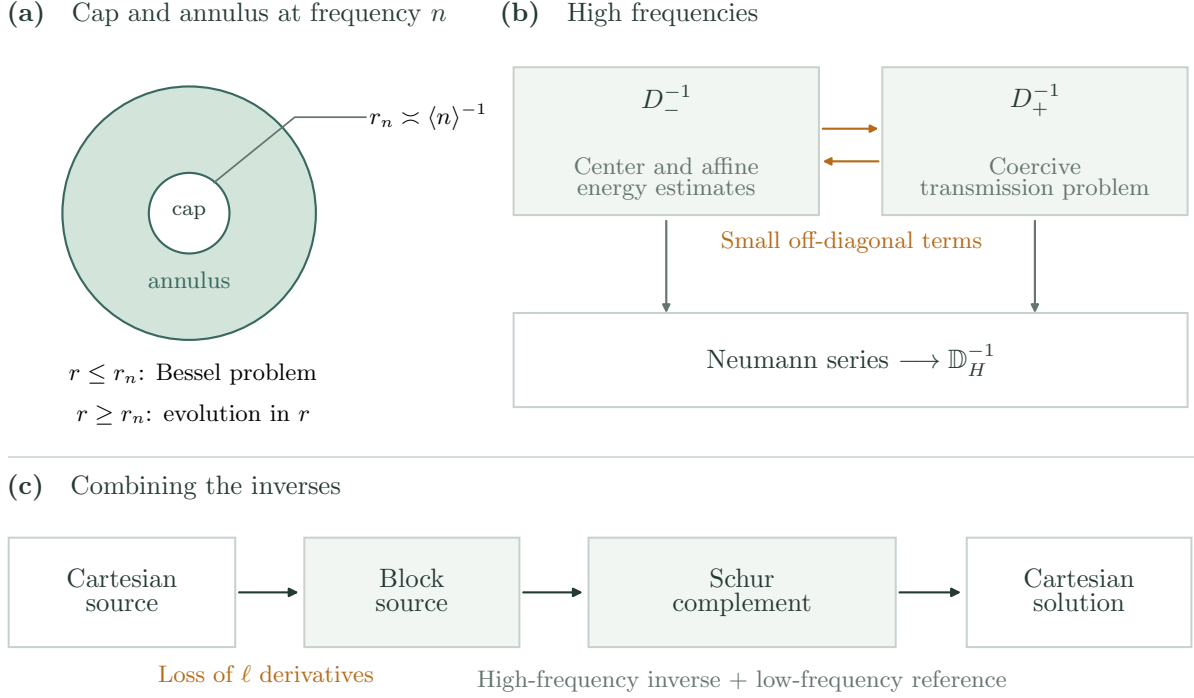


FIGURE 2. Construction of the uniform inverse. (a) We match the cap and annulus at $r_n \asymp \langle n \rangle^{-1}$. (b) A Neumann series combines the diagonal inverses and the small off-diagonal terms to give the inverse at high frequencies. (c) A Schur complement joins this inverse to the low-frequency reference. We convert the Cartesian source to block variables once, before applying the inverse, to avoid incurring this loss of derivatives multiple times.

Here, $0 < \gamma < \min\{\sigma_0, 1\}$. On the axis, W_γ is the weight $e^{\sigma_0 \langle n \rangle}$ of analyticity in ζ on a strip of width σ_0 . Outside the cap, the exponent decreases like $\gamma r \langle n \rangle$, so the analytic width shrinks linearly in r and, after applying the weight, the evolution in r acquires a damping of size $\gamma \langle n \rangle$. Near the reference state, this damping controls the growth on the annulus. On the cap, the ratio $W_\gamma(r, n)/W_\gamma(0, n)$ is bounded above and below by positive constants independent of n , so the weight does not alter the cap estimates except by fixed factors.

We rewrite the unknowns and sources in *block variables*, grouping the center and affine components together and the elliptic components separately. Each component has a weight chosen for its equation. In these variables, the high-frequency part of the linearized operator is a 2×2 matrix

$$\mathbb{D}_H = \begin{pmatrix} D_- & E_{-+} \\ E_{+-} & D_+ \end{pmatrix}.$$

The block D_- contains the center and affine equations. To construct its inverse, we match the cap and annulus solutions (see [Sections 8.2](#) and [8.3](#)). On the cap, we start from the explicit inverse at the circle, obtained from Bessel equations for the center components and integral formulas for the affine components, with prescribed data on the axis. The cap solution supplies the initial data for the radial equations on the annulus. There, we estimate all four center modes together using a positive quadratic energy, so the bounds remain uniform even when the two oscillator speeds coincide. A separate, modified energy controls the affine

components (see [Section 8.3](#)). To combine these estimates, we use a change of variables that removes the leading coupling between the center and affine components (see [\(8.95\)](#)). The block D_+ contains the elliptic part. We construct its inverse by solving on the cap and annulus and matching the values and conormal derivatives at their interface (see [Section 9](#)). We correct these inverses for the full operator and estimate the coupling between its diagonal blocks. The highest-order terms in the off-diagonal blocks cancel, and the remaining blocks vanish at the circular reference state and are small nearby (see [Section 10](#)). A Neumann series then combines the diagonal inverses to give the inverse of $\mathbb{D}_H$. We handle the low frequencies by comparison with the inverse at a constant ellipse (see [Section 11](#)). A Schur complement combines this with the high-frequency inverse (see [Section 12](#)). Importantly, we only convert the Cartesian source to block variables once, before applying the block inverse, to avoid incurring this loss of derivatives multiple times. The resulting two-sided inverse satisfies tame estimates on an open set of states (see [Theorem 13.2](#)). The constants are fixed in [Proposition 13.1](#), with the frequency cutoff chosen before the ellipse parameter $\rho_* > 0$.

The Nash-Moser scheme and the resulting geometry. We apply Hamilton's Nash-Moser theorem [\[23, 40, 41, 42\]](#) with ε and λ as parameters. The inverse estimates established above allow us to solve the nonlinear equations for $|\varepsilon| < \varepsilon_H$ and λ in a compact interval, obtaining solutions that depend smoothly on both parameters (see [Section 14](#)). The correction vanishes at $\varepsilon = 0$, since the unperturbed family already solves the equations. Smooth dependence on ε therefore gives [\(1.14\)](#), uniformly for λ in the compact interval. At $\varepsilon = 1/N$, we bound the Jacobian of $X_{N,\lambda}$ away from zero by a perturbation estimate and obtain $|B| \geq c|y|$. Together with $B = 0$ on the axis, this shows that the field vanishes exactly there. We remark, however, that global injectivity does not follow from the Jacobian bound, since the fast variable $N\phi$ could produce distant self-intersections. To establish global injectivity, we use coordinates around the axis and allow the angular coordinate to range over $\mathbb{R}$. In these coordinates, the map is close to the identity together with its first derivatives, which rules out self-intersections. The major radius NL compensates for the factor N from differentiation of the fast variable (see [Section 15.2](#)). To determine the symmetry group, we use the normalized chart to compute

$$\mathcal{S}_\phi := \frac{(-H_\phi)^{-1}}{\text{tr}((-H_\phi)^{-1})} = \frac{1}{2} M_\lambda(N\phi) M_\lambda(N\phi)^T,$$

where H_ϕ is the Hessian of the pressure restricted to the normal plane of the axis, written in the radial and vertical basis $(R_\phi e_x, e_z)$. The tensor $\mathcal{S}_\phi$ is independent of τ and $\tilde{v}$. An isometry preserving the equilibrium preserves the axis. It therefore acts on the axis by $\phi \mapsto \sigma\phi + c$ with $\sigma = \pm 1$ and preserves or flips e_z . Preservation of this tensor imposes a relation on the angle α_λ . Averaging this relation excludes a flip of e_z because $2\alpha_0 \notin \pi\mathbb{Z}$. The first harmonic $\cos\zeta$ then forces $c \in (2\pi/N)\mathbb{Z}$, and the odd second harmonic $\sin 2\zeta$ excludes $\sigma = -1$. This proves [\(1.7\)](#). The same tensor shows that distinct values of λ give inequivalent equilibria.

1.5. Outline of the paper. The paper follows the three parts of the strategy described above.

The first part, [Sections 2 to 5](#), sets up the problem on one cell. In [Section 2](#), we write [\(1.1\)](#) in flux coordinates, pass to one rotating cell with the curvature ε as a parameter, integrate the equations once to obtain a first-order system, and exhibit the family of exact solutions at $\varepsilon = 0$. In [Section 3](#), we introduce the normalized chart, specify the target space of H_ε through its compatibility conditions, and choose the scale of spaces used in our estimates. In [Section 4](#), we construct extension operators, projections, and smoothing operators, and identify the varying function spaces with one fixed space. In [Section 5](#), we define the nonlinear map on

the resulting fixed spaces, verify that the family at zero curvature consists of exact zeros, and compute the linearization $L_{\mathbf{b}}$ at a general state $\mathbf{b}$.

The second part, [Sections 6 to 13](#), inverts $L_{\mathbf{b}}$ and is the bulk of the paper. In [Section 6](#), we introduce the block variables, the transfer maps between them and the Cartesian variables, and the augmented block operator $\mathbb{D}_{\mathbf{b}}$. This is also where the conversion of Cartesian sources is performed and its loss of derivatives is recorded. In [Section 7](#), we construct the inverse at a fixed ellipse and analyze the circular case. [Sections 8 and 9](#) construct inverses at high frequencies for the center and affine equations and for the elliptic boundary problem, respectively. In [Section 10](#), we estimate the off-diagonal blocks and correct these inverses to obtain inverses of the diagonal blocks of the full operator. In [Section 11](#), we invert the low-frequency part at the constant ellipse, and in [Section 12](#) we assemble the inverse of $\mathbb{D}_{\mathbf{b}}$ by a Neumann series and a Schur complement, estimate its derivatives with respect to the state, and return to real Cartesian variables. In [Section 13](#), we fix the constants in their required order and prove [Theorem 13.2](#), the uniform inverse theorem for $L_{\mathbf{b}}$.

The third part, [Sections 14 and 15](#), produces the desired equilibria. In [Section 14](#), we verify the hypotheses of the Nash-Moser theorem and obtain a smooth branch of solutions in the curvature and the parameter λ . In [Section 15](#), we set $\varepsilon = 1/N$, close the cell into a torus, bound the Jacobian away from zero, prove global injectivity, reconstruct the magnetic field and pressure, verify the MHS equations, and show that the regular pressure levels are nested tori. We then determine the symmetry group, show that distinct values of λ give inequivalent equilibria, and complete the proof of [Theorem 1.1](#). An explanation of the Lean formalization of this theorem is given in [Section 16](#).

The article is designed to be modular. With the nonlinear setup and parameters fixed, [Theorem 13.2](#) supplies the inverse and tame estimates used in the Nash-Moser argument of [Section 14](#). A reader interested in the linear analysis may take the block operator of [Section 6](#) as the starting point. [Sections 7 and 11](#) contain the explicit computations at the constant ellipse that give the inverse formulas and estimates used later.

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2. COORDINATES AND EQUATIONS

In this section, we derive the equations on one period cell and describe the explicit family of solutions at zero curvature.

Throughout, e_x, e_y, e_z denote the standard basis vectors of $\mathbb{R}^3$. We write R_φ for rotation about e_z through the angle φ , and R_α for rotation in $\mathbb{R}^2$ through the angle α . We set

$$Aq = e_z \times q, \quad q \in \mathbb{R}^3,$$

so that $\partial_\varphi R_\varphi = AR_\varphi$. All angular variables have period 2π .

2.1. Flux coordinates. We first work away from the magnetic axis. Let $X = X(\psi, \theta, \zeta)$ be a smooth immersion, and locally prescribe

$$B \circ X = X_\theta, \quad P \circ X = \psi, \quad J = \det(X_\psi, X_\theta, X_\zeta).$$

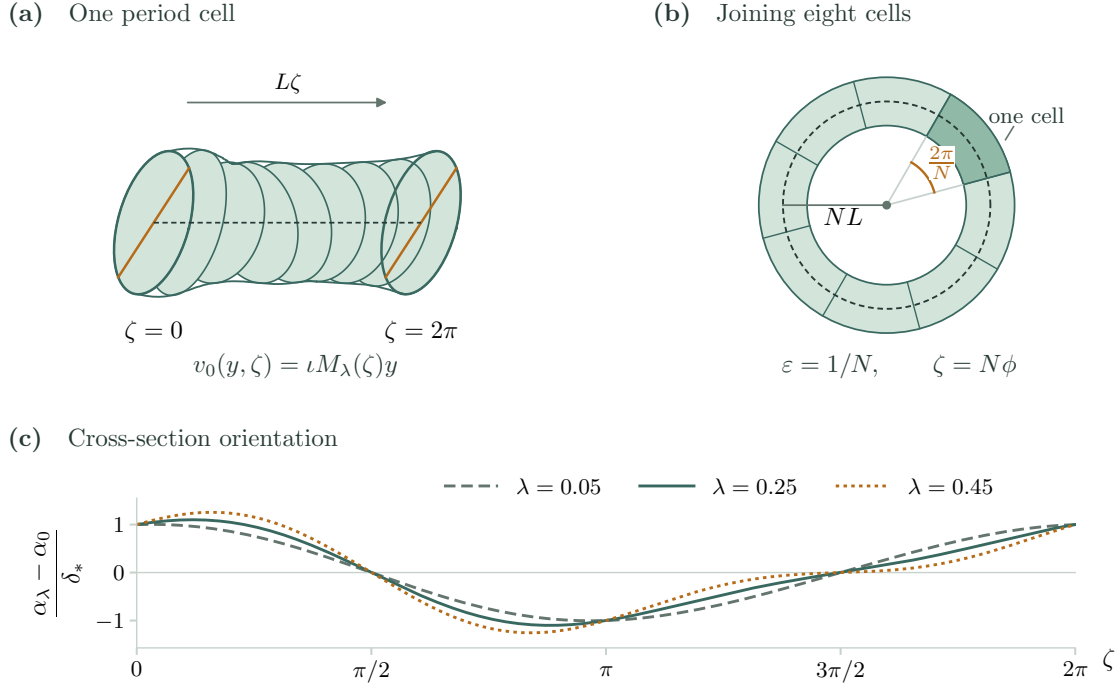


FIGURE 3. From one period cell to a torus. (a) A straight cell at zero curvature, with matching end cross-sections and the parameters of Figure 1. (b) A schematic view of eight cells joined around a circle of radius $NL = 5.2$. Each cell spans an angle $2\pi/N$, and the radial lines mark where the cells meet. (c) The cross-section angle over one period, plotted as $(\alpha_\lambda - \alpha_0)/\delta_* = \cos \zeta + \lambda \sin 2\zeta$ for $\lambda = 0.05, 0.25$, and 0.45 .

In these *flux coordinates*, ψ labels pressure surfaces and θ parametrizes field lines. Since $J \neq 0$, the vectors $X_\psi, X_\theta, X_\zeta$ form a basis of $\mathbb{R}^3$, so the *force imbalance* $\mathcal{Q} = B \times \text{curl } B + \nabla P$ vanishes if and only if its inner product with each basis vector vanishes. Combining the identity

$$B \times \text{curl } B = \nabla \frac{|B|^2}{2} - (B \cdot \nabla)B$$

with the chain rule $((B \cdot \nabla)f) \circ X = \partial_\theta(f \circ X)$, we obtain

$$(2.1) \quad (\mathcal{Q} \circ X) \cdot X_\theta = 0,$$

$$(2.2) \quad (\mathcal{Q} \circ X) \cdot X_\psi = -F_1,$$

$$(2.3) \quad (\mathcal{Q} \circ X) \cdot X_\zeta = F_2,$$

where

$$(2.4) \quad F_1(X) = X_\psi \cdot X_{\theta\theta} - X_\theta \cdot X_{\psi\theta} - 1,$$

$$(2.5) \quad F_2(X) = X_\theta \cdot X_{\theta\zeta} - X_\zeta \cdot X_{\theta\theta}.$$

Moreover,

$$(2.6) \quad (\text{div } B) \circ X = J^{-1} \partial_\theta J,$$

so the magnetohydrostatic equations in these coordinates can be written as

$$(2.7) \quad F_1 = F_2 = F_3 = 0, \quad F_3 := \partial_\theta J.$$

The boundary condition $B \cdot n = 0$ holds automatically when the physical boundary is a level set of ψ , since X_θ is tangent to that level set. To obtain a first-order system, we write the two force equations in conservation form,

$$(2.8) \quad F_1 = \partial_\theta(X_\psi \cdot X_\theta) - \partial_\psi |X_\theta|^2 - 1,$$

$$(2.9) \quad F_2 = \partial_\zeta |X_\theta|^2 - \partial_\theta(X_\theta \cdot X_\zeta).$$

We now bend one period cell around a circle. We fix $L > 0$, with $2\pi L$ the length of the reference axis within one cell. For $\varepsilon \neq 0$, we take v to be 2π -periodic in ζ and set

$$(2.10) \quad X_{\varepsilon,v}(\psi, \theta, \zeta) = R_{\varepsilon\zeta} \left(\frac{L}{\varepsilon} e_x + v(\psi, \theta, \zeta) \right),$$

where ζ is allowed to range over $\mathbb{R}$. Increasing ζ by 2π rotates the image through $2\pi\varepsilon$. We also set

$$(2.11) \quad D_\varepsilon v = v_\zeta + \varepsilon A v + L e_y.$$

Since $AR_\varphi = R_\varphi A$ and $Ae_x = e_y$, we obtain from (2.10) the identities

$$X_\psi = R_{\varepsilon\zeta} v_\psi, \quad X_\theta = R_{\varepsilon\zeta} v_\theta, \quad X_\zeta = R_{\varepsilon\zeta} D_\varepsilon v.$$

Thus, (2.7) becomes $F_{1,\varepsilon} = F_{2,\varepsilon} = F_{3,\varepsilon} = 0$, where

$$(2.12) \quad F_{1,\varepsilon}(v) = v_\psi \cdot v_{\theta\theta} - v_\theta \cdot v_{\psi\theta} - 1,$$

$$(2.13) \quad F_{2,\varepsilon}(v) = v_\theta \cdot v_{\theta\zeta} - D_\varepsilon v \cdot v_{\theta\theta},$$

$$(2.14) \quad F_{3,\varepsilon}(v) = \partial_\theta \det(v_\psi, v_\theta, D_\varepsilon v),$$

on the cell, away from the magnetic axis. The equations remain well defined at $\varepsilon = 0$, although (2.10) does not. At $\varepsilon = 0$, they are the equations for the map $X_{0,v} = L\zeta e_y + v$. For $\varepsilon = 1/N$ with $N \in \mathbb{N}$, the map is periodic after N cells. Global injectivity of the constructed solutions is proved in Section 15.2. Notably, we will solve the system for all small ε and only restrict to the discrete values $\varepsilon = 1/N$ at the very end of the argument.

2.2. A first-order system. We use Cartesian coordinates $y = (y_1, y_2)$ on the disk and set

$$y = r(\cos \theta, \sin \theta), \quad \psi = \psi_a - r^2, \quad 0 \leq r \leq 1,$$

where ψ_a is the pressure on the axis. The unknown v is now viewed as a map on $Q = \overline{\mathbb{D}^2} \times \mathbb{T}$. For a scalar or vector function f , we define its *angular average* by

$$(\Pi f)(y, \zeta) = \frac{1}{2\pi} \int_0^{2\pi} f(R_s y, \zeta) ds.$$

Thus, Π averages at fixed r and ζ . We set

$$E = |v_\theta|^2, \quad g_{\psi\theta} = v_\psi \cdot v_\theta, \quad g_{\theta\zeta}^\varepsilon = v_\theta \cdot D_\varepsilon v, \quad J_\varepsilon = \det(v_\psi, v_\theta, D_\varepsilon v),$$

and introduce an auxiliary scalar w to integrate the force equations in θ . Note that its additive freedom is fixed by $\Pi w = 0$.

For fixed ε , we define

$$(2.15) \quad G_0(v, w) = w_\theta - (E - r^2),$$

$$(2.16) \quad G_1(v, w) = (I - \Pi)(g_{\psi\theta} - w_\psi),$$

$$(2.17) \quad G_2(v, w) = (I - \Pi)(g_{\theta\zeta}^\varepsilon - w_\zeta),$$

$$(2.18) \quad G_3(v, w) = (I - \Pi)J_\varepsilon.$$

The equations $G_0 = G_1 = G_2 = G_3 = 0$ involve only first derivatives of (v, w) . Note that only G_3 contains terms of degree three. The following lemma shows that introducing w does not change the solutions.

Lemma 2.1. *Let $\varepsilon \in \mathbb{R}$ and let $v \in C^\infty(Q; \mathbb{R}^3)$. The following are equivalent.*

- (i) $F_{1,\varepsilon}(v) = F_{2,\varepsilon}(v) = F_{3,\varepsilon}(v) = 0$ on $0 < r \leq 1$.
- (ii) *There is a unique $w \in C^\infty(Q)$ with $\Pi w = 0$ such that $G_0(v, w) = G_1(v, w) = G_2(v, w) = G_3(v, w) = 0$ on $0 < r \leq 1$.*

Proof. Suppose first that $G_0 = G_1 = G_2 = G_3 = 0$. The equation $G_0 = 0$ gives $w_\theta = E - r^2$, from which we obtain

$$\begin{aligned} \partial_\theta(g_{\psi\theta} - w_\psi) &= F_{1,\varepsilon}, \\ \partial_\theta(g_{\theta\zeta}^\varepsilon - w_\zeta) &= -F_{2,\varepsilon}. \end{aligned}$$

Here, we used that $\partial_\psi r^2 = -1$ and the identity

$$v_\theta \cdot \partial_\theta(D_\varepsilon v) = v_\theta \cdot (v_{\theta\zeta} + \varepsilon A v_\theta) = v_\theta \cdot v_{\theta\zeta},$$

which is a consequence of the skew-symmetry of A . Hence, $G_1 = G_2 = 0$ gives the two force equations, and $G_3 = 0$ gives $F_{3,\varepsilon} = 0$.

Conversely, suppose that $F_{1,\varepsilon} = F_{2,\varepsilon} = F_{3,\varepsilon} = 0$. The first conservation identity becomes $F_{1,\varepsilon} = \partial_\theta g_{\psi\theta} - \partial_\psi E - 1$, so averaging in θ gives $\partial_\psi \Pi E = -1$. Since v is smooth in y , we have $v_\theta = 0$ and hence $E = 0$ on the axis. Integrating from $\psi = \psi_a$, we obtain $\Pi E = \psi_a - \psi = r^2$. We therefore set

$$w(y, \zeta) = \frac{1}{2\pi} \int_0^{2\pi} s(E - r^2)(R_s y, \zeta) ds.$$

This function is smooth on Q . Since $\Pi(E - r^2) = 0$, integration by parts gives $w_\theta = E - r^2$, and averaging the formula gives $\Pi w = 0$. Any two such functions differ by a function independent of θ , so the mean zero condition gives uniqueness. The preceding derivative identities and $F_{3,\varepsilon} = 0$ show that $g_{\psi\theta} - w_\psi$, $g_{\theta\zeta}^\varepsilon - w_\zeta$, and J_ε are independent of θ . Applying $I - \Pi$ therefore gives $G_1 = G_2 = G_3 = 0$. $\square$

The equations $G_1 = 0$ and $G_3 = 0$ still contain the derivative ∂_ψ , which is singular at the axis. To obtain equations that are smooth in y , we set

$$(2.19) \quad \mathcal{D} = y_1 \partial_{y_1} + y_2 \partial_{y_2}, \quad \mathcal{R} = -y_2 \partial_{y_1} + y_1 \partial_{y_2},$$

so that when $r > 0$ we have

$$\partial_\theta = \mathcal{R}, \quad \partial_\psi = -\frac{1}{2r^2} \mathcal{D}.$$

We multiply the two singular equations by the appropriate factors of r^2 and define

$$(2.20) \quad H_{0,\varepsilon}(v, w) = \mathcal{R}w - (|\mathcal{R}v|^2 - r^2),$$

$$(2.21) \quad H_{1,\varepsilon}(v, w) = (I - \Pi)(\mathcal{D}w - \mathcal{D}v \cdot \mathcal{R}v),$$

$$(2.22) \quad H_{2,\varepsilon}(v, w) = (I - \Pi)(\mathcal{R}v \cdot D_\varepsilon v - w_\zeta),$$

$$(2.23) \quad H_{3,\varepsilon}(v, w) = (I - \Pi) \det(\mathcal{D}v, \mathcal{R}v, D_\varepsilon v).$$

We unify the above by defining $H_\varepsilon = (H_{0,\varepsilon}, H_{1,\varepsilon}, H_{2,\varepsilon}, H_{3,\varepsilon})$. On $r > 0$, these equations satisfy

$$(2.24) \quad H_{0,\varepsilon} = G_0, \quad H_{1,\varepsilon} = 2r^2 G_1, \quad H_{2,\varepsilon} = G_2, \quad H_{3,\varepsilon} = -2r^2 G_3.$$

The components G_1 and G_3 need not extend smoothly to the axis, whereas every component of H_ε does. Thus, $H_\varepsilon = 0$ is a system on the whole disk with the same solutions as the original equations on $r > 0$. When the reconstructed map is smooth and has nonzero Jacobian in Cartesian coordinates, the physical MHS equations also hold on the axis by continuity.

2.3. A family of solutions at zero curvature. At zero curvature, the equations admit an explicit family of solutions that are linear in y , including the straight-axis family of Salat and Kaiser [48]. We recall the inclusion $\iota : \mathbb{R}^2 \rightarrow \mathbb{R}^3$,

$$\iota(q_1, q_2) = q_1 e_x + q_2 e_z,$$

whose ordering fixes the sign of the Jacobian. For a smooth periodic matrix $M : \mathbb{T} \rightarrow GL^+(2, \mathbb{R})$, we set

$$(2.25) \quad v_M(y, \zeta) = \iota M(\zeta) y.$$

The following trace condition makes v_M an exact solution.

Lemma 2.2. *Suppose that*

$$(2.26) \quad \text{tr}(M^T M) = 2$$

for every ζ . Then $F_{1,0}(v_M) = F_{2,0}(v_M) = F_{3,0}(v_M) = 0$ on $r > 0$. Let $w_M \in C^\infty(Q)$ be the unique function with $\Pi w_M = 0$ satisfying

$$(2.27) \quad (w_M)_\theta = |(v_M)_\theta|^2 - r^2.$$

Then $H_0(v_M, w_M) = 0$ on the whole disk.

Proof. With $e_r = (\cos \theta, \sin \theta)$, $e_\theta = (-\sin \theta, \cos \theta)$, $rr_\psi = -1/2$, and $M' = \partial_\zeta M$, we have

$$\begin{aligned} (v_M)_\psi &= r_\psi \iota M e_r, & (v_M)_\theta &= r \iota M e_\theta, & (v_M)_{\theta\theta} &= -r \iota M e_r, \\ (v_M)_\zeta &= r \iota M' e_r, & (v_M)_{\theta\zeta} &= r \iota M' e_\theta. \end{aligned}$$

Inserting these at $\varepsilon = 0$, where $D_0 v_M = r \iota M' e_r + L e_y$, we obtain

$$(2.28) \quad F_{1,0}(v_M) = \frac{1}{2} (|M e_r|^2 + |M e_\theta|^2) - 1 = \frac{1}{2} \text{tr}(M^T M) - 1,$$

$$(2.29) \quad F_{2,0}(v_M) = r^2 (M e_\theta \cdot M' e_\theta + M' e_r \cdot M e_r) = \frac{r^2}{2} \partial_\zeta \text{tr}(M^T M),$$

$$(2.30) \quad J_0(v_M) = \frac{L}{2} \det M(\zeta),$$

by orthonormality of e_r, e_θ and by the identity $\det(\iota a, \iota b, e_y) = -\det[a \ b]$. The first two lines vanish by (2.26), the last is independent of θ , and the assertion for w_M now follows from Lemma 2.1. $\square$

The trace condition alone makes the equations vanish. To reconstruct smooth physical fields, we also need M to be invertible. Our assumption $M(\zeta) \in GL^+(2, \mathbb{R})$, together with compactness of $\mathbb{T}$, gives

$$(2.31) \quad \inf_{\zeta \in \mathbb{T}} \det M(\zeta) > 0.$$

For $\zeta \in \mathbb{R}$, the map $X_{0,v_M} = L\zeta e_y + \iota M(\zeta)y$ is injective, since its e_y -component determines ζ and $M(\zeta)$ is invertible. Its Jacobian in Cartesian coordinates is $-L \det M$, so it defines smooth physical fields through the axis.

We use the following family in the construction. For $0 < \rho < 1$, we set

$$(2.32) \quad \alpha_\lambda(\zeta) = \alpha_0 + \delta(\cos \zeta + \lambda \sin 2\zeta),$$

$$(2.33) \quad M_\lambda(\zeta) = R_{\alpha_\lambda(\zeta)} \begin{pmatrix} \sqrt{1+\rho} & 0 \\ 0 & \sqrt{1-\rho} \end{pmatrix} R_{-\alpha_\lambda(\zeta)},$$

so that

$$(2.34) \quad \text{tr}(M_\lambda^T M_\lambda) = 2, \quad \det M_\lambda = \sqrt{1-\rho^2}$$

and both (2.26) and (2.31) hold identically. We require $\delta \neq 0$, $0 < \lambda < 1/2$, and $\alpha_0 \notin (\pi/2)\mathbb{Z}$. The orientation α_0 is fixed at the beginning of the argument, and the small parameter $\delta \neq 0$ is chosen only after the linear analysis. As in Section 1.3, the term $\cos \zeta$ fixes the period of the tilt at 2π in ζ , or $2\pi/N$ in ϕ , when $\varepsilon = 1/N$. The term $\lambda \sin 2\zeta$ excludes the reversal $\zeta \mapsto -\zeta$. Both are used to determine the symmetry group.

We finally compute the error caused by bending the cell. For nonzero curvature, v_M still satisfies the two force equations. Indeed, $\varepsilon A v_M = \varepsilon r(M e_r)_1 e_y$ is orthogonal to the range of ι , which contains $(v_M)_{\theta\theta}$, so that $F_{1,\varepsilon}(v_M) = F_{2,\varepsilon}(v_M) = 0$ while the Jacobian becomes

$$(2.35) \quad J_\varepsilon(v_M) = \frac{1}{2} \det M(\zeta) (L + \varepsilon r(M(\zeta) e_r)_1),$$

$$(2.36) \quad F_{3,\varepsilon}(v_M) = \frac{\varepsilon}{2} \det M(\zeta) \partial_\theta (r(M(\zeta) e_r)_1).$$

Thus, only the divergence equation fails. Since $r(M(\zeta) e_r)_1 = (M(\zeta) y)_1$, its residual is smooth through the axis and is $O(|\varepsilon|)$ in every fixed $C^k(Q)$ norm. This is the error that the nonlinear argument must correct.

3. FUNCTION SPACES AND CONSTRAINTS

We now choose coordinates for the unknowns, specify the constraints on these coordinates and on the right-hand sides, and define the norms used in the analysis. The constraints remove the freedom described in Section 1.4, while the norms control both smoothness and analyticity in the periodic variable.

3.1. The normalized chart. We let $e_T = e_y$ be tangent to the limiting straight axis and let $\iota : \mathbb{R}^2 \rightarrow e_T^\perp$ be the inclusion into the (e_x, e_z) -plane. The unperturbed solution at zero curvature is the pair given by

$$v_M(y, \zeta) = \iota M(\zeta) y, \quad \text{tr}(M^T M) = 2,$$

together with the auxiliary scalar w_M of (2.27). To allow the linear Taylor term of v to vary along the axis while preserving its trace normalization, we introduce an unknown function $\tau : \mathbb{T} \rightarrow \mathbb{R}^2$. For $|\tau(\zeta)|^2 < 2$, we define

$$(3.1) \quad a(\tau) = \sqrt{1 - \frac{|\tau|^2}{2}}, \quad P_{M,\tau} = a(\tau) \iota M + e_T \otimes \tau.$$

Since $\iota^T e_T = 0$ gives $P_{M,\tau}^T P_{M,\tau} = a(\tau)^2 M^T M + \tau \tau^T$, we obtain

$$(3.2) \quad \text{tr}(P_{M,\tau}^T P_{M,\tau}) = 2.$$

We use the triple $(\tau, \tilde{v}, s)$ as coordinates for the unknown pair (v, w) , writing

$$(3.3) \quad v = P_{M,\tau}(\zeta)y + \tilde{v}(y, \zeta), \quad w = w_M + s, \quad j_y^1 \tilde{v}|_{y=0} = 0.$$

Here, $j_y^1 u|_{y=0}$ denotes the value and first derivatives of u in y on the axis, which we call its *first jet*. These are the coefficients of its Taylor expansion through first order. Thus, $P_{M,\tau}(\zeta)y$ is the linear term in the Taylor expansion of v at $y = 0$, and $\tilde{v}$ and its first derivatives in y vanish there. At $\tau = 0$, the derivative of $P_{M,\tau}$ in τ is $e_T \otimes \dot{\tau}$, so a variation of τ changes this linear term only along the axis direction e_T . Its component in the normal plane is unchanged to first order in τ .

With Π denoting the angular average in y and $\mathcal{R} = -y_2 \partial_{y_1} + y_1 \partial_{y_2}$ as in (2.19), we fix the remaining freedom to change coordinates by imposing the two *gauge conditions*

$$(3.4) \quad \Pi(\tilde{v} \cdot \mathcal{R} v_M) = 0, \quad \Pi(\tilde{v} \cdot D_0 v_M) = 0,$$

where we use D_0 in place of D_ε so that the constraints do not depend on the curvature. We also impose the following condition on the outer boundary:

$$(3.5) \quad \Pi_{|m| \geq 3} [e_r \cdot M(\zeta)^{-1} \iota^T \tilde{v}(1, \theta, \zeta)] = 0.$$

Here, P_m projects onto the angular Fourier mode $e^{im\theta}$, and $\Pi_{|m| \geq 3} = \sum_{|m| \geq 3} P_m$. Thus, the boundary condition removes the modes $|m| \geq 3$ of the displayed radial component. Finally, we require $\Pi s = 0$, so that $w = w_M + s$ retains the mean zero normalization of Lemma 2.1.

For fixed M , we let $\mathcal{X}_M^\infty$ be the linear space of smooth triples $x = (\tau, \tilde{v}, s)$ satisfying these constraints. With

$$(3.6) \quad B_M u := \Pi_{|m| \geq 3} [e_r \cdot M(\zeta)^{-1} \iota^T u|_{r=1}],$$

we put

$$(3.7) \quad \mathcal{C}_M(\tau, \tilde{v}, s) = (j_y^1 \tilde{v}|_{y=0}, \Pi s, \Pi(\tilde{v} \cdot \mathcal{R} v_M), \Pi(\tilde{v} \cdot D_0 v_M), B_M \tilde{v}),$$

so that $\mathcal{X}_M^\infty = \ker \mathcal{C}_M$, while the tangential coefficient τ is unrestricted in this linear space. The nonlinear chart (3.3) is defined on the open subset where $|\tau(\zeta)|^2 < 2$ for every ζ . Thus, the conditions on the axis, the two gauges, the angular mean of s , and the outer boundary are all included in the domain.

3.2. The target space of H_ε . The components of $H_\varepsilon(v, w)$ satisfy identities on the axis and have prescribed angular means. We build these conditions into the target space. We call a quadruple $f = (f_0, f_1, f_2, f_3)$ a *residual* and introduce its *quotient variables*, initially for $y \neq 0$, by

$$(3.8) \quad g_+ = \frac{f_1 + i f_0}{\bar{z}}, \quad g_- = \frac{f_1 - i f_0}{z}, \quad g_3 = \frac{f_3}{z \bar{z}}, \quad z = y_1 + i y_2.$$

We call f *compatible* if these quotients are smooth in Cartesian coordinates and

$$(3.9) \quad \Pi f_1 = \Pi f_2 = \Pi f_3 = 0, \quad \frac{1}{4} \Delta_y \Pi f_0(0, \zeta) = 0.$$

We write $\mathcal{Y}_{\text{comp}}^\infty$ for the space of compatible residuals. We take $Z = (g_+, g_-, g_3, h)$, with $h = f_2$, as the range coordinate and reconstruct

$$(3.10) \quad J_Y Z = \left(\frac{\bar{z} g_+ - z g_-}{2i}, \frac{\bar{z} g_+ + z g_-}{2}, h, z \bar{z} g_3 \right).$$

Thus, passing from Z to f only multiplies by smooth factors. Passing from f to Z , however, requires the divisibility conditions in (3.8). We write

$$(3.11) \quad \mathcal{Q}_Y(f_0, f_1, f_2, f_3) = (g_+, g_-, g_3, f_2)$$

for the inverse of J_Y on such residuals. We measure regularity in these quotient coordinates. In particular, the norm of f is the product norm of $\mathcal{Q}_Y f$, rather than the product norm of its four original components. In Section 4.3, we construct a bounded projection that imposes (3.9) in these coordinates.

We now check that H_ε takes values in this target space when (v, w) is written in the chart (3.3). The components $H_{1,\varepsilon}$, $H_{2,\varepsilon}$, and $H_{3,\varepsilon}$ carry the factor $I - \Pi$, so their angular means vanish. For the zeroth entry, we have $\Pi f_0 = r^2 - \Pi|\mathcal{R}v|^2$. Since $\tilde{v}$ vanishes to second order on the axis, so does $\mathcal{R}\tilde{v}$, and only the linear Taylor term $P_{M,\tau}y$ contributes to the coefficient of r^2 in $\Pi|\mathcal{R}v|^2$. That coefficient is $\frac{1}{2} \text{tr}(P_{M,\tau}^T P_{M,\tau}) = 1$ by (3.2), so $\Pi f_0 = O(|y|^3)$. Its quadratic Taylor coefficient therefore vanishes, as required by the last condition in (3.9).

It remains to see that the quotients (3.8) of $H_\varepsilon(v, w)$ are smooth across the axis. With

$$\mathbf{m}_1(v, w) := \Pi(\mathcal{D}w - \mathcal{D}v \cdot \mathcal{R}v),$$

and with the curvature subscript suppressed, we have

$$(3.12) \quad H_1 + iH_0 = (\mathcal{D} + i\mathcal{R})w - (\mathcal{D} + i\mathcal{R})v \cdot \mathcal{R}v + ir^2 - \mathbf{m}_1(v, w),$$

$$(3.13) \quad H_1 - iH_0 = (\mathcal{D} - i\mathcal{R})w - (\mathcal{D} - i\mathcal{R})v \cdot \mathcal{R}v - ir^2 - \mathbf{m}_1(v, w),$$

$$(3.14) \quad \det(\mathcal{D}v, \mathcal{R}v, D_\varepsilon v) = -2iz\bar{z} \det(\partial_z v, \partial_{\bar{z}} v, D_\varepsilon v).$$

Here, $\partial_z = \frac{1}{2}(\partial_{y_1} - i\partial_{y_2})$ and $\partial_{\bar{z}} = \frac{1}{2}(\partial_{y_1} + i\partial_{y_2})$, so $\mathcal{D} + i\mathcal{R} = 2\bar{z}\partial_{\bar{z}}$ and $\mathcal{D} - i\mathcal{R} = 2z\partial_z$. Since $\mathbf{m}_1$ is an angular average of a smooth function, it is smooth in (r^2, ζ) . It vanishes on the axis and is therefore divisible by $r^2 = z\bar{z}$. The first two identities then give the factors $\bar{z}$ and z in $H_1 + iH_0$ and $H_1 - iH_0$. The last identity gives a factor r^2 in the determinant. Applying $I - \Pi$ preserves this factor because r^2 is independent of θ . Thus, these identities define the quotients (3.8) of $H_\varepsilon(v, w)$ on the axis by smooth expressions, so $H_\varepsilon(v, w)$ is compatible whenever (v, w) is written in the chart (3.3) and is smooth in y .

3.3. A fixed scale of spaces. We use a weight that controls analyticity in ζ and supplies radial damping in the linear estimates. Let $\Lambda = \langle D_\zeta \rangle$, whose symbol is $\langle n \rangle = (1 + n^2)^{1/2}$, and fix $\sigma_0 > 0$ and $0 < \gamma < \min\{\sigma_0, 1\}$. For $n \in \mathbb{Z}$, we set

$$(3.15) \quad \Phi_\gamma(r, \langle n \rangle) = \sigma_0 \langle n \rangle - \gamma(\sqrt{1 + r^2 \langle n \rangle^2} - 1), \quad W_\gamma = e^{\Phi_\gamma(r, \Lambda)}.$$

We require $\mathcal{R}v_M$, $D_0 v_M$, and M^{-1} to be analytic in ζ on a strip wider than σ_0 . Note that our unperturbed solutions satisfy this condition. We define the norms in the interior by applying the relevant Sobolev norms to $W_\gamma u$. Thus, we set

$$(3.16) \quad \|u\|_{\mathcal{A}_\gamma^s} = \|W_\gamma u\|_{H^s(Q)}.$$

Here, W_γ acts on the n -th Fourier coefficient by multiplication by $W_\gamma(r, n) = e^{\Phi_\gamma(r, \langle n \rangle)}$. We weight functions on the axis, including τ , by $e^{\sigma_0 \Lambda}$, and outer boundary traces by $e^{\Phi_\gamma(1, \Lambda)}$, which is equivalent to $e^{(\sigma_0 - \gamma)\Lambda}$. For

the target space, we apply these norms to the quotient coordinates Z . The three properties of the weight used below are

$$(3.17) \quad e^{(\sigma_0 - \gamma r)\langle n \rangle} \leq W_\gamma(r, n) \leq e^\gamma e^{(\sigma_0 - \gamma r)\langle n \rangle},$$

$$(3.18) \quad W_\gamma(r, n_1 + n_2) \leq W_\gamma(r, n_1) W_\gamma(r, n_2),$$

$$(3.19) \quad \Phi_\gamma(0, \langle n \rangle) = \sigma_0 \langle n \rangle, \quad \nabla_y \Phi_\gamma(0, \langle n \rangle) = 0.$$

We also use the following product estimate: For a fixed base index $s_b > 3/2$ and $s \geq s_b$, we have

$$(3.20) \quad \|uv\|_{\mathcal{A}_\gamma^s} \leq C_s (\|u\|_{\mathcal{A}_\gamma^s} \|v\|_{\mathcal{A}_\gamma^{s_b}} + \|u\|_{\mathcal{A}_\gamma^{s_b}} \|v\|_{\mathcal{A}_\gamma^s}).$$

Note that the weight is smooth in $|y|^2$, so multiplication by it preserves vanishing Cartesian jets on the axis. Its negative logarithmic radial derivative,

$$(3.21) \quad d_\gamma(r, \langle n \rangle) = -\partial_r \Phi_\gamma = \frac{\gamma r \langle n \rangle^2}{\sqrt{1 + r^2 \langle n \rangle^2}},$$

is the damping introduced when we apply the weight to an outward radial equation. For $r \langle n \rangle \geq 1$, it is comparable to $\gamma \langle n \rangle$. We use this damping to control growth on the annulus without changing the chosen spaces during the iteration.

4. RELATIONS BETWEEN DIFFERENT SPACES

We now construct projections that impose the constraints of [Section 3](#), identify the domains for different M , and build smoothing operators that preserve the constraints. These constructions allow us to apply the Nash-Moser theorem on fixed spaces.

Throughout this section, we fix a compact family of periodic matrix functions

$$(4.1) \quad M \in C^\infty(\mathbb{T}; GL(2, \mathbb{R})), \quad \text{tr}(M^T M) = 2, \quad \|M\| + \|M^{-1}\| \leq C_M,$$

and every estimate below is locally uniform in this family. For the weighted estimates, we also require the matrix functions to satisfy the analyticity bounds on the strip of [Section 3.3](#). We first work in the standard Sobolev spaces on $Q = \overline{\mathbb{D}^2} \times \mathbb{T}$, and we then conjugate by the weight [\(3.15\)](#). We also fix the projections

$$(4.2) \quad H_J := \mathbf{1}_{\{|n| \geq 2^J\}}(D_\zeta), \quad P_J := I - H_J = \mathbf{1}_{\{|n| < 2^J\}}(D_\zeta),$$

onto the high and low cell frequencies, where the integer J sets the cutoff at 2^J . Superscripts indicate the fixed coordinate space on which a cutoff acts. We write P_J^X, H_J^X for the domain and P_J^Y, H_J^Y for the range.

For the unconstrained unknowns, we use the product space

$$\mathbb{E}^s = H^s(\mathbb{T}; \mathbb{R}^2) \times H^s(Q; \mathbb{R}^3) \times H^s(Q)$$

for $(\tau, \tilde{v}, s)$. In the range, we set

$$\mathbb{B}^s = J_Y((H^s(Q; \mathbb{C}))^4), \quad \|f\|_{\mathbb{B}^s} = \|\mathcal{Q}_Y f\|_{(H^s)^4}.$$

Thus, $\mathcal{X}_M^s$ and $\mathcal{Y}_{\text{comp}}^s$ are the subspaces of $\mathbb{E}^s$ and $\mathbb{B}^s$ satisfying the respective constraints. We use the same notation for their complexifications and for the weighted spaces, specifying which norms are in use. The superscript ∞ denotes the intersection over all regularity indices. Throughout the projection constructions, we take $s > 2$, so that the first derivatives have well-defined traces on the axis.

We recall the terminology of Hamilton [23] that we use below. A *graded Fréchet space* is a Fréchet space whose topology is defined by an increasing sequence of norms $\|\cdot\|_s$, such as $\bigcap_s \mathcal{X}_M^s$ with the norms of Section 3.3. We prove that our spaces are *tame* in Hamilton's sense by realizing them as ranges of bounded projections on standard spaces of smooth functions. We also construct *smoothing operators* S_ν , $\nu \geq 1$, satisfying, whenever $b \geq a$ are above the base index,

$$(4.3) \quad \|S_\nu q\|_b \leq C_{a,b} \nu^{(b-a)_+} \|q\|_a,$$

$$(4.4) \quad \|(I - S_\nu)q\|_a \leq C_{a,b} \nu^{a-b} \|q\|_b.$$

Thus, $S_\nu q$ is as regular as we wish, at the price of a power of ν , and it approximates q with an error that is small in a weaker norm when q is regular. A map F between graded spaces is called *tame* if, on a neighborhood of each point of its domain, it satisfies an estimate of the form

$$\|F(x)\|_s \leq C_s (1 + \|x\|_{s+r})$$

for all s above a fixed base index, with a loss of derivatives r independent of s . A smooth map is called *smooth tame* if every derivative is tame, viewed as a map of the base point and the direction variables together. The main result of this section may be stated as follows.

Proposition 4.1. *There is an index s_* such that the following hold for every $s \geq s_*$, with constants locally uniform in the family (4.1).*

(i) *There are bounded projections*

$$P_{X,M} : \mathbb{E}^s \longrightarrow \mathcal{X}_M^s, \quad P_Y : \mathbb{B}^s \longrightarrow \mathcal{Y}_{\text{comp}}^s,$$

and the derivatives of $P_{X,M}$ in M of every fixed order are bounded at the same regularity level. The constants may depend on the regularity index and the number of parameter derivatives.

(ii) *For every M in the family and a fixed reference matrix function M_* in it, there is a linear isomorphism*

$$T_M : \mathcal{X}_{M_*}^\infty \xrightarrow{\sim} \mathcal{X}_M^\infty,$$

which is bounded with bounded inverse at every regularity level, depends analytically on M with every derivative tame, and commutes with complex conjugation.

(iii) *The constrained scales $\mathcal{X}_M^s$ and $\mathcal{Y}_{\text{comp}}^s$ admit smoothing operators $S_{\nu,M}^X$ and S_ν^Y satisfying (4.3)–(4.4). Their intersections over all regularity indices are tame Fréchet spaces.*

The same conclusions hold on the weighted analytic scales of Section 3.3. In particular, neither the projections nor the smoothing operators lose derivatives.

We prove Proposition 4.1 in the next five subsections. We begin with the extension operators from which the projections are built.

4.1. Extension operators on the axis and on the boundary. The constraints (3.7) prescribe the first jet on the axis and a trace on the outer boundary. To project onto them, we need operators which produce from data on the axis, on the boundary, or on an interface, a function on Q having these data as its trace, with the gain of derivatives given by the trace theorem. We call such operators *extension operators*. We

choose a radial function $\phi \in C_c^\infty(\mathbb{D}^2)$, supported in $|y| < 1/4$ and equal to one near zero, and we put $\lambda_n = \langle n \rangle$. For $a(\zeta) = \sum_n a_n e^{in\zeta}$, we define

$$(4.5) \quad E_0 a = \sum_n a_n e^{in\zeta} \phi(\lambda_n y), \quad E_j a = \sum_n a_n e^{in\zeta} y_j \phi(\lambda_n y), \quad j = 1, 2.$$

The function $E_0 a$ has value a and vanishing first derivatives on the axis, and $E_j a$ has value zero and derivative a in the direction y_j there. Note that the support of the term at cell frequency n has radius comparable to λ_n^{-1} . Rescaling in y , we therefore obtain

$$(4.6) \quad \|E_0 a\|_{H^s(Q)} \leq C_s \|a\|_{H^{s-1}(\mathbb{T})}, \quad \|E_j a\|_{H^s(Q)} \leq C_s \|a\|_{H^{s-2}(\mathbb{T})}.$$

Thus, if $J_A u = (u, \partial_{y_1} u, \partial_{y_2} u)|_{y=0}$ is the first jet on the axis and E_A is the componentwise combination of (4.5), then $J_A E_A = I$, and

$$(4.7) \quad Q_A = I - E_A J_A$$

is a bounded projection onto the fields whose first jet vanishes on the axis, which is the first of the constraints in (3.7). Note that the support must shrink with the frequency. Indeed, multiplying the axis trace by a fixed cutoff would lose the one or two derivatives gained in (4.6).

At the outer boundary, we use the inward normal coordinate $t = 1 - r$. We choose a cutoff χ_∂ supported in a thin neighborhood of the boundary, called a *collar*, and equal to one near $t = 0$. With $\lambda_{mn} = \langle m, n \rangle = (1 + m^2 + n^2)^{1/2}$, we define the localized Poisson extension

$$(4.8) \quad E_\partial b = \chi_\partial(t) \sum_{m,n} b_{mn} e^{i(m\theta + n\zeta)} e^{-\lambda_{mn} t}.$$

It has trace b on the outer boundary and satisfies

$$(4.9) \quad \|E_\partial b\|_{H^s(Q)} \leq C_s \|b\|_{H^{s-1/2}(\partial\mathbb{D}^2 \times \mathbb{T})}.$$

Indeed, integration in t contributes the factor λ_{mn}^{-1} that accounts for the half derivative.

The same estimates hold with the analytic weight. On the support of E_0 and E_j , we have

$$W_\gamma(r, n) e^{-\sigma_0 \langle n \rangle} = \exp\{-\gamma(\sqrt{1 + r^2 \langle n \rangle^2} - 1)\},$$

and this ratio is bounded, together with all of its derivatives in the rescaled variable $\lambda_n y$. In the collar of the outer boundary, we have instead

$$(4.10) \quad e^{-\lambda_{mn} t} e^{\Phi_\gamma(r, \langle n \rangle) - \Phi_\gamma(1, \langle n \rangle)} \leq e^{-(1-\gamma)\lambda_{mn} t}.$$

The choice $\gamma < 1$ leaves positive exponential decay, so the same integration gives the weighted boundary estimate.

In the linear analysis, a field on Q will be represented by its restrictions to a neighborhood of the axis and to the complementary region, and we will need to prescribe the value and the normal derivative on the interface between the two regions at the same time. For the local construction, we fix a smooth interface Σ between the cap and the annulus. In the later application, the cap radius depends on the cell frequency, and the cap is rescaled before these extensions are used. On either side of Σ , we choose an inward collar coordinate $t \geq 0$, and we set $\Lambda_\Sigma = \langle D_\theta, D_\zeta \rangle$. With a cutoff χ_Σ equal to one near $t = 0$, we define

$$(4.11) \quad \mathcal{E}_{\Sigma, D}^0 d = \chi_\Sigma(t) e^{-t\Lambda_\Sigma} d, \quad \mathcal{E}_{\Sigma, N}^0 n = \chi_\Sigma(t) t e^{-t\Lambda_\Sigma} n.$$

The first function has value d and inward normal derivative $-\Lambda_\Sigma d$ on Σ . The second has value zero and inward normal derivative n . We will correct the first extension to make its conormal derivative vanish.

For an operator with principal part $-\partial_i(a^{ij}\partial_j)$, the outward *conormal derivative* is $\nu_i a^{ij}\partial_j u$, where ν is the outward unit normal and repeated indices are summed. We call the value and conormal derivative on Σ the *Cauchy data*. We consider a second-order operator on Q whose coefficients depend on a parameter $\mathbf{b}$, and we write γ_D for the trace on Σ , $\gamma_{N,\mathbf{b}}^{\text{out}}$ for the outward conormal derivative of this operator on Σ , and $e_{\Sigma,\mathbf{b}}$ for the coefficient of ∂_t in $\gamma_{N,\mathbf{b}}^{\text{out}}$. We assume that $e_{\Sigma,\mathbf{b}}$ is pointwise invertible, with an inverse bounded uniformly in $\mathbf{b}$. The extension estimates below depend on this bound and on the coefficient norms. We set

$$(4.12) \quad C_{\Sigma,N,\mathbf{b}}n := \mathcal{E}_{\Sigma,N}^0(e_{\Sigma,\mathbf{b}}^{-1}n),$$

$$K_{\Sigma,\mathbf{b}}d := \gamma_{N,\mathbf{b}}^{\text{out}}\mathcal{E}_{\Sigma,D}^0d,$$

$$(4.13) \quad C_{\Sigma,D,\mathbf{b}}d := \mathcal{E}_{\Sigma,D}^0d - C_{\Sigma,N,\mathbf{b}}K_{\Sigma,\mathbf{b}}d.$$

Since $\mathcal{E}_{\Sigma,N}^0n$ vanishes at $t = 0$, its tangential derivatives and the zeroth-order terms vanish there as well, so the Cauchy data of the two extensions are exactly the prescribed ones,

$$(4.14) \quad \begin{pmatrix} \gamma_D \\ \gamma_{N,\mathbf{b}}^{\text{out}} \end{pmatrix} (C_{\Sigma,D,\mathbf{b}}, C_{\Sigma,N,\mathbf{b}}) = \begin{pmatrix} I & 0 \\ 0 & I \end{pmatrix}.$$

Integrating each Fourier mode in t , we obtain, for every fixed s ,

$$(4.15) \quad \|C_{\Sigma,D,\mathbf{b}}d\|_{H^s} + \|C_{\Sigma,N,\mathbf{b}}n\|_{H^s} \leq C_s(\|d\|_{H^{s-1/2}(\Sigma)} + \|n\|_{H^{s-3/2}(\Sigma)}).$$

The same construction applies on the other side of Σ , with its own outward orientation. The constants may depend on the interface and the chosen collar. Conjugating by the weight and using (4.10), we obtain the weighted form of (4.15), in the norms in which the value on Σ carries the weight $\Lambda^{1/2}$ and the conormal derivative carries the weight $\Lambda^{-1/2}$. When we differentiate (4.12)–(4.13) in $\mathbf{b}$, the derivatives fall only on the coefficients of the conormal derivative and on their pointwise inverse. If these coefficients depend analytically on $\mathbf{b}$, the resulting estimates are therefore tame, with at most one factor measured in a higher norm.

4.2. The projection onto the domain. We now construct the projection $P_{X,M}$ by imposing the constraints (3.7) one at a time, in an order in which each correction preserves the constraints already imposed. We write

$$e_r = (\cos \theta, \sin \theta), \quad e_\theta = (-\sin \theta, \cos \theta)$$

in the disk plane, and we define

$$(4.16) \quad \mathcal{P}_T q = e_\theta \Pi(q \cdot e_\theta),$$

which is the angular average of the component of q along e_θ , times e_θ . Although e_θ is not smooth at the origin, $\mathcal{P}_T$ is bounded on every Sobolev space in the Cartesian variables. Indeed, averaging the action $q(y) \mapsto R_\varphi q(R_{-\varphi}y)$ of the rotations over φ gives a smooth field of the form $\Pi(q \cdot e_r)e_r + \Pi(q \cdot e_\theta)e_\theta$, and $\mathcal{P}_T q$ is the part of this field that is odd under reflection. This construction preserves the homogeneous degree in y and the vanishing of the first jet.

For the unperturbed solution at zero curvature, we have

$$\mathcal{R}v_M = r\iota M e_\theta, \quad D_0 v_M = L e_T + r\iota M' e_r,$$

and we define the corrections for the two gauge conditions in (3.4) by

$$(4.17) \quad C_{p,M}u = \iota M \mathcal{P}_T(M^T \iota^T u), \quad C_{t,M}u = \frac{e_T}{L} \Pi(u \cdot D_0 v_M).$$

Since $\Pi(e_\theta^T M^T M e_\theta) = \frac{1}{2} \text{tr}(M^T M) = 1$, the functional $u \mapsto \Pi(u \cdot \mathcal{R} v_M)$ of the first gauge condition takes the same value at $C_{p,M}u$ as at u , so $I - C_{p,M}$ is a projection onto the fields satisfying the first gauge condition. Since $e_T \cdot D_0 v_M = L$, the same holds for $I - C_{t,M}$ and the second gauge condition. Both corrections preserve the vanishing of the first jet.

For the constraint on the outer boundary, we write the condition on the high angular modes and its right inverse as

$$(4.18) \quad B_M u = \Pi_{|m| \geq 3} [e_r \cdot M^{-1} \iota^T u|_{r=1}], \quad C_{\partial,M} b = \iota M e_r E_{\partial} b,$$

where E_{∂} is the extension (4.8). Note that $B_M C_{\partial,M} = I$, since $M^{-1} \iota^T \iota M e_r = e_r$ and $E_{\partial} b$ has trace b . Note also that the boundary correction $C_{\partial,M} b$ preserves the two gauge conditions. Indeed, the only angular factors that can contribute to the corresponding means are $e_r^T C e_r$ and $e_r^T C e_\theta$ for a matrix C depending on ζ alone, whose Fourier support in θ lies in $\{0, \pm 2\}$, whereas the datum in (4.18) has $|m| \geq 3$. Conversely, the gauge corrections leave the boundary condition unchanged, since $e_r \cdot M^{-1} \iota^T \iota M e_\theta = 0$ and $\iota^T e_T = 0$.

We may therefore put

$$(4.19) \quad P_{X,M}^u = (I - C_{\partial,M} B_M)(I - C_{t,M})(I - C_{p,M}) Q_A,$$

and we let $P_{X,M}$ act as the identity on τ , as $P_{X,M}^u$ on $\tilde{v}$, and as $I - \Pi$ on s . The rightmost factor acts first. Thus, we first remove the value and first derivatives on the axis, then impose the two gauges, and finally correct the outer boundary trace. Each factor is a projection, and this order preserves the constraints already imposed. Indeed, the second gauge correction adds a multiple of e_T , which does not affect the first gauge condition, and the boundary correction preserves both gauge conditions and the jet on the axis, since $E_{\partial} b$ vanishes outside a collar of the boundary. Thus, $P_{X,M}$ maps $\mathbb{E}^s$ into $\mathcal{X}_M^s$ and is the identity on $\mathcal{X}_M^s$, so it is a projection onto $\mathcal{X}_M^s$. Its bounds follow from (4.6) and (4.9), and its derivatives in M fall on M , M^{-1} , and M' . At each fixed derivative order, the same estimates apply, with constants depending on the corresponding coefficient norms.

4.3. The projection onto the range. We write the quotient coordinates of (3.8) as $Z = (g_+, g_-, g_3, h)$, with the reconstruction J_Y of (3.10), and we impose the conditions (3.9) one at a time. We remove the means of h and g_3 by applying $I - \Pi$ to these two components, and we denote the resulting projections by Q_2 and Q_3 . To remove the mean of f_1 , we let P_m be the projection onto the angular mode m , we put $(\mathcal{K}q)(r, \theta, \zeta) = q(r, -\theta, \zeta)$, and we define

$$(4.20) \quad p = \frac{1}{2}(P_1 g_+ + \mathcal{K} P_{-1} g_-), \quad K_1 Z = (p, \mathcal{K} p, 0, 0).$$

Since $f_1 + i f_0 = \bar{z} g_+$ and $f_1 - i f_0 = z g_-$, the means of f_0 and f_1 come from the mode $m = 1$ of g_+ and from the mode $m = -1$ of g_- , and

$$\bar{z} p = z \mathcal{K} p = \frac{1}{2}(\bar{z} P_1 g_+ + z P_{-1} g_-) = \Pi f_1.$$

Hence, $J_Y K_1 Z$ has vanishing f_0 -component and f_1 -component exactly Πf_1 . Thus, $I - K_1$ removes the mean of f_1 and leaves f_0 unchanged.

It remains to impose the last condition in (3.9), which is given by the functional

$$(4.21) \quad \mathfrak{a}(Z) = \frac{1}{4} \Delta_y \Pi \left(\frac{\bar{z}g_+ - zg_-}{2i} \right) (0, \zeta).$$

This functional depends on the first derivatives of the quotient variables on the axis, so we use the axis extensions and (4.6). Writing $\Phi_{\text{ax}} = E_0$ for the extension in (4.5), we set

$$(4.22) \quad E_{\text{aff}}a = (iz\Phi_{\text{ax}}a, -i\bar{z}\Phi_{\text{ax}}a, 0, 0).$$

Then, $\mathfrak{a}E_{\text{aff}} = I$, since $J_Y E_{\text{aff}}a$ has $f_0 = r^2\Phi_{\text{ax}}a$ and $f_1 = 0$, and $K_1 E_{\text{aff}} = 0$, since $\mathcal{K}\bar{z} = z$ and the radial cutoff makes $\Phi_{\text{ax}}a$ independent of θ . We also have the estimate

$$\|E_{\text{aff}}a\|_{(H^s)^4} \leq C_s \|a\|_{H^{s-2}(\mathbb{T})}.$$

Here, the second gained derivative comes from the factor z , which is of size λ_n^{-1} on the support of Φ_{ax} . We thus obtain the projection onto the range coordinates and the projection onto the compatible range,

$$(4.23) \quad P_Z = (I - E_{\text{aff}}\mathfrak{a})(I - K_1)Q_3Q_2, \quad P_Y = J_Y P_Z J_Y^{-1}.$$

Each correction preserves the conditions already imposed. In particular, $E_{\text{aff}}a$ has $g_3 = h = 0$ and $K_1 E_{\text{aff}} = 0$. Complex conjugation of the residual acts in quotient coordinates by the rule

$$\mathfrak{c}(g_+, g_-, g_3, h) = (\overline{g_-}, \overline{g_+}, \overline{g_3}, \overline{h}).$$

Each factor of P_Z respects this rule. Since $J_Y \mathfrak{c}Z = \overline{J_Y Z}$, the projection P_Y therefore preserves real residuals.

4.4. Identification of the domains. The domain $\mathcal{X}_M^\infty$ depends on M through the gauge conditions and through the condition on the outer boundary. We now identify it with the domain at the reference matrix function M_* by an explicit isomorphism. We let A_M be the multiplication by the matrix

$$(4.24) \quad A_M := \iota M M_*^{-1} \iota^T + e_T \otimes e_T,$$

which depends only on ζ , is invertible with inverse $A_M^{-1} = \iota M_* M^{-1} \iota^T + e_T \otimes e_T$, and is the identity at $M = M_*$. Since $M^{-1} \iota^T A_M u = M_*^{-1} \iota^T u$, we have $B_M A_M = B_{M_*}$, so A_M maps the fields satisfying the boundary condition for M_* onto those satisfying it for M . It also preserves the vanishing of the first jet on the axis, but it does not preserve the gauge conditions. We therefore correct the gauges afterwards and define

$$(4.25) \quad T_M(\tau, \tilde{v}, s) := (\tau, (I - C_{t,M})(I - C_{p,M})A_M \tilde{v}, s).$$

The properties of this map are summarized in the following lemma.

Lemma 4.2. *For every M in the family (4.1), the map (4.25) is a linear isomorphism*

$$(4.26) \quad T_M : \mathcal{X}_{M_*}^\infty \longrightarrow \mathcal{X}_M^\infty,$$

whose inverse is

$$T_M^{-1}(\tau, u, s) = (\tau, (I - C_{t,M_*})(I - C_{p,M_*})A_M^{-1}u, s).$$

Both maps are bounded at every regularity level of the Sobolev and analytic scales, they depend analytically on M with every derivative tame, and they commute with complex conjugation.

Proof. Let $x = (\tau, \tilde{v}, s) \in \mathcal{X}_{M_*}^\infty$. The jet condition on the axis and the condition $\Pi s = 0$ are preserved by every factor of (4.25), the boundary condition for M holds after A_M and is preserved by the gauge corrections, and the two gauge conditions for M hold after the corrections. Thus, T_M maps $\mathcal{X}_{M_*}^\infty$ into $\mathcal{X}_M^\infty$, and the same argument with the roles of M and M_* exchanged shows that the map T_M^{-1} in the statement, which we denote by S_M for the moment, maps $\mathcal{X}_M^\infty$ into $\mathcal{X}_{M_*}^\infty$. To see that $S_M T_M = I$, we note that the gauge corrections at M add to $A_M \tilde{v}$ a multiple of $\iota M e_\theta$ and a multiple of e_T , with coefficients that depend on (r, ζ) only, so that

$$A_M^{-1}(I - C_{t,M})(I - C_{p,M})A_M \tilde{v} = \tilde{v} + \iota M_* e_\theta \varphi + e_T \psi$$

for some functions φ and ψ of (r, ζ) . Since $\tilde{v}$ satisfies the gauge conditions for M_* , and since $\Pi(e_\theta^T M_*^T M_* e_\theta) = 1$ and $e_T \cdot D_0 v_{M_*} = L$, the correction $I - C_{p,M_*}$ removes the second term and leaves the other two unchanged, and $I - C_{t,M_*}$ then removes the third. Hence, $S_M T_M = I$, and $T_M S_M = I$ follows in the same way. The factors are matrix multiplications and angular averages. Their coefficients depend analytically on M through M , M^{-1} , and M' , with M_* fixed. The multiplication estimates therefore give boundedness at each regularity level and tame bounds for every parameter derivative. On the weighted scales, we use the analyticity assumption on the coefficients and (3.20). All coefficients and averages are real, so both maps commute with complex conjugation. $\square$

The map T_M identifies the fixed function space $\mathcal{X}_{M_*}^\infty$ with $\mathcal{X}_M^\infty$ for each M . Thus, we can work on one fixed space even though the constraints depend on M . Unlike the projections, T_M uses no trace or extension operator.

4.5. Smoothing operators. We first smooth in the unconstrained spaces and then apply the projections to restore the constraints. We let $\tilde{Q}$ be the double of Q , a compact manifold without boundary obtained by gluing two copies of Q along the outer boundary. We let E^0 be an extension operator from Q to $\tilde{Q}$ which is bounded on every Sobolev space, we let R^0 be the restriction to Q , and we take the nonnegative Laplace operator $\Delta_{\tilde{Q}}$. We set $S_\nu^0 = \chi(\nu^{-2} \Delta_{\tilde{Q}})$, where χ is smooth, compactly supported, and equal to one near zero. This operator removes frequencies above a constant multiple of ν . For the weighted components defined on Q , we set

$$(4.27) \quad \mathcal{S}_\nu^\gamma = W_\gamma^{-1} R^0 S_\nu^0 E^0 W_\gamma.$$

Since the weight is applied before the extension and removed after the restriction, the estimates (4.3)–(4.4) for $\mathcal{S}_\nu^\gamma$ on the weighted scale follow from the spectral estimates for S_ν^0 on the unweighted one. For the component τ , which is defined on $\mathbb{T}$, we use the Fourier cutoff $\chi(\nu^{-2} \Lambda^2)$. It commutes with the axis weight $e^{\sigma_0 \Lambda}$. We use $\mathcal{S}_\nu^\gamma$ for the resulting componentwise smoothing on the product space and define

$$(4.28) \quad S_{\nu,M}^X = P_{X,M} \mathcal{S}_\nu^\gamma, \quad S_\nu^Y = J_Y P_Z \mathcal{S}_\nu^{\gamma,Z} J_Y^{-1},$$

where $\mathcal{S}_\nu^{\gamma,Z}$ denotes the same smoothing operator applied to each of the four coordinates in Z . By applying the projection after the smoothing, we preserve the constraints, and since the projections are bounded at the same index, no derivatives are lost. If $Pq = q$, then $q - P \mathcal{S}_\nu^\gamma q = P(I - \mathcal{S}_\nu^\gamma)q$, which reduces (4.4) on the constrained space to its unconstrained version, and (4.3) follows from the boundedness of P at the index b . Smoothing operators on $\mathcal{X}_M^\infty$ can also be obtained from S_{ν,M_*}^X by conjugation with T_M .

Finally, the standard spaces of smooth functions on $\mathbb{T}$ and on Q are tame. For Q , extension to $\tilde{Q}$ and restriction give a tame projection from a space on a closed manifold. The analytic weights identify the unconstrained weighted spaces with these standard spaces, and J_Y identifies the residual space with its

quotient coordinates. The bounded projections constructed above therefore make the constrained spaces tame as well, by [23, Part II, Lemma 1.3.3]. This proves the remaining assertion of [Proposition 4.1](#).

5. THE NONLINEAR MAP AND ITS LINEARIZATION

We now express the equilibrium equations as a nonlinear map between the fixed spaces of [Sections 3](#) and [4](#). We establish its smooth tame dependence on the unknowns and parameters, then compute the linearization whose inverse is constructed in [Sections 6–13](#).

5.1. The map and its zeros at zero curvature. We collect the parameters in $q = (\rho, \varepsilon, p)$, where ρ is the ellipse parameter, ε is the curvature, and $p = (\alpha_0, \delta, \lambda)$ contains the remaining parameters of [\(2.33\)](#). We write $M_q = M_{\rho, p}$ for the matrix in that equation, which is independent of ε , and $a_{\rho, p}^0 = (v_{M_q}, w_{M_q})$ for the corresponding solution at zero curvature. The chart of [Section 3.1](#) sends $x = (\tau, \tilde{v}, s) \in \mathcal{X}_{M_q}^\infty$, with $|\tau(\zeta)|^2 < 2$, to the pair

$$(5.1) \quad \mathcal{S}_q x = (v, w), \quad v = P_{M_q, \tau} y + \tilde{v}, \quad w = w_{M_q} + s.$$

Here, $x = 0$ represents the unperturbed solution. The component τ varies the linear Taylor term, while $\tilde{v}$ and s are the remaining corrections. We collect the four equations [\(2.20\)–\(2.23\)](#) in the residual

$$(5.2) \quad \mathbf{H}_\varepsilon(v, w) = \begin{pmatrix} \mathcal{R}w - (|\mathcal{R}v|^2 - r^2) \\ (I - \Pi)(\mathcal{D}w - \mathcal{D}v \cdot \mathcal{R}v) \\ (I - \Pi)(\mathcal{R}v \cdot D_\varepsilon v - w_\zeta) \\ (I - \Pi)\det(\mathcal{D}v, \mathcal{R}v, D_\varepsilon v) \end{pmatrix}, \quad D_\varepsilon v = v_\zeta + \varepsilon A v + L e_y.$$

By [Section 3.2](#), this residual is compatible for every smooth pair in the chart. Its quotient coordinates are therefore well-defined and are fixed by the projection P_Z of [\(4.23\)](#), so

$$(5.3) \quad P_Z \mathcal{Q}_Y \mathbf{H}_\varepsilon(\mathcal{S}_q x) = \mathcal{Q}_Y \mathbf{H}_\varepsilon(\mathcal{S}_q x).$$

Reconstructing the residual gives the nonlinear map

$$(5.4) \quad \mathcal{F}(q, x) := J_Y P_Z \mathcal{Q}_Y \mathbf{H}_\varepsilon(\mathcal{S}_q x) = \mathbf{H}_\varepsilon(\mathcal{S}_q x) \in \mathcal{Y}_{\text{comp}}^\infty.$$

We retain the quotient maps in this formula to make the target norm explicit. The domain $\mathcal{X}_{M_q}^\infty$ still depends on q . We fix a reference parameter q_* and set $M_* = M_{q_*}$. Using $T_q := T_{M_q}$ from [\(4.25\)](#) to write $x = T_q \hat{x}$, we obtain a map on the fixed domain $\mathcal{X}_{M_*}^\infty$,

$$(5.5) \quad \hat{\mathcal{F}}(q, \hat{x}) = \mathcal{F}(q, T_q \hat{x}).$$

The range maps $\mathcal{Q}_Y$, P_Z , and J_Y are fixed. Thus, differentiating $\hat{\mathcal{F}}$ only differentiates the residual, the chart, and the domain identification. A zero of $\hat{\mathcal{F}}$ is precisely a solution of the four equilibrium equations in the chart, with all domain constraints imposed.

We choose the base index s_* so that [Proposition 4.1](#) applies and $s_* \geq s_b + 3$, where s_b is the index in the product estimate [\(3.20\)](#). We work on $\mathcal{U}_0 = \mathcal{B}_{\text{par}} \times \mathcal{B}_{\text{st}}$, where $\mathcal{B}_{\text{par}}$ is a ball centered at q_* , and $\mathcal{B}_{\text{st}}$ is a ball about zero in $\mathcal{X}_{M_*}^{s_*}$ intersected with $\mathcal{X}_{M_*}^\infty$. We choose these balls small enough that $|\tau(\zeta)|^2$ stays uniformly below 2 for $x = T_q \hat{x}$. The following proposition gives the properties needed for the Nash-Moser theorem. We use the meanings of tame and smooth tame given in [Section 4](#).

Proposition 5.1. *After decreasing $\mathcal{B}_{\text{par}}$ and $\mathcal{B}_{\text{st}}$, the following hold.*

- (i) The domain $\mathcal{X}_{M_q}^\infty$ and the compatible range $\mathcal{Y}_{\text{comp}}^\infty$, graded by the norms of [Section 3.3](#), are tame Fréchet spaces. They admit smoothing operators that preserve the constraints, with bounds locally uniform in q .
- (ii) The identifications T_q depend analytically on q , and the charts $\mathcal{S}_q$ depend analytically on (q, x) . Every derivative is tame. These maps, as well as $\mathcal{Q}_Y$ and J_Y , preserve real data. In quotient coordinates, complex conjugation is given by the rule $\mathfrak{c}$ of [Section 4.3](#).
- (iii) The map $\widehat{\mathcal{F}}$ in (5.5) is real and smooth tame. Its four original residual components are polynomials in (v, w) and their first derivatives, composed with the analytic chart and domain identification.
- (iv) Every unperturbed solution at zero curvature is an exact zero, in the sense that

$$(5.6) \quad \widehat{\mathcal{F}}(\rho, 0, p, 0) = 0.$$

Proof. Part (i) and the assertions about T_q in (ii) follow from [Proposition 4.1](#). The chart depends on q through M_q and w_{M_q} . Its nonlinear dependence on x comes only from $a(\tau) = \sqrt{1 - |\tau|^2/2}$, while its dependence on $\tilde{v}$ and s is linear. On the chosen ball, the power series for $a(\tau)$ and its derivatives converge. The analytic composition and product estimates give tame bounds for every derivative of the chart. All its coefficients are real. The reality assertions for $\mathcal{Q}_Y$ and J_Y follow from the conjugation rule in [Section 4.3](#). This proves (ii).

For (iii), we must estimate the quotient coordinates, since they define the target norm. With $\mathbf{m}_1$ as in [Section 3.2](#), we set $\mu = \mathbf{m}_1(v, w)/r^2$. The identities (3.12)–(3.14) give

$$\begin{aligned} g_+ &= 2\partial_{\bar{z}}w - 2\partial_{\bar{z}}v \cdot \mathcal{R}v + iz - z\mu, \\ g_- &= 2\partial_zw - 2\partial_zv \cdot \mathcal{R}v - i\bar{z} - \bar{z}\mu, \\ g_3 &= -2i(I - \Pi) \det(\partial_zv, \partial_{\bar{z}}v, D_\varepsilon v), \quad h = H_{2,\varepsilon}(v, w). \end{aligned}$$

Thus, the only division still to estimate is that defining μ . For any smooth function F that is radial in y and vanishes on the axis, integration of the radial Laplacian gives

$$\frac{F(y, \zeta)}{|y|^2} = \int_0^1 t \log(1/t) (\Delta_y F)(ty, \zeta) dt, \quad \left\| \frac{F}{r^2} \right\|_{\mathcal{A}_\gamma^k} \leq C_k \|F\|_{\mathcal{A}_\gamma^{k+2}}.$$

Here, k is any nonnegative integer, and the integral defines the value on the axis. To obtain the weighted estimate, we work at each cell frequency n . By (3.15), the ratio $W_\gamma(r, n)/W_\gamma(tr, n)$ is at most one, and its derivatives of order j in y are bounded by $C_j \langle n \rangle^j$, uniformly for $0 < t \leq 1$. Dilation by t costs at most t^{-1} in $L^2(\mathbb{D}^2)$, and $\int_0^1 \log(1/t) dt = 1$. Differentiating the integral and summing in n therefore bounds it by $C_k \|\Delta_y F\|_{\mathcal{A}_\gamma^k}$, which is at most $C_k \|F\|_{\mathcal{A}_\gamma^{k+2}}$.

We apply this estimate to the radial function $\mathbf{m}_1$, which vanishes on the axis. Before division, $\mathbf{m}_1$ and all the other terms are polynomials in (v, w) and their first derivatives. The product estimate (3.20), the division estimate, and (ii) consequently give, for every integer $s \geq s_*$,

$$\|\widehat{\mathcal{F}}(q, \widehat{x})\|_{\mathcal{Y}_{\text{comp}}^s} \leq C_s (1 + \|\widehat{x}\|_{\mathcal{X}_{M_*}^{s+3}}),$$

locally uniformly in q . The loss of three derivatives allows one for the original residual and two for the division. Differentiating the same formulas with respect to q and $\widehat{x}$ gives the corresponding tame bounds at every order. Indeed, division is a fixed linear map on radial functions vanishing on the axis, and the remaining operations are products, first derivatives, and the analytic maps in (ii). This proves (iii).

Finally, $T_q 0 = 0$ and $P_{M_q, 0} = \iota M_q$, so $\mathcal{S}_q 0 = (v_{M_q}, w_{M_q})$. This pair is a zero of $\mathbf{H}_0$ by [Lemma 2.2](#), which proves (iv). $\square$

5.2. The linearization. Holding q fixed, we differentiate with respect to $\hat{x}$ and define the *linearization*

$$(5.7) \quad L_{q,\hat{x}} = D_{\hat{x}}\hat{\mathcal{F}}(q,\hat{x}) : \mathcal{X}_{M_*}^\infty \longrightarrow \mathcal{Y}_{\text{comp}}^\infty.$$

We set $x = T_q\hat{x}$ and $(v, w) = \mathcal{S}_q x$. A direction $\hat{h}$ in the fixed domain corresponds to $h = T_q\hat{h} = (\dot{\tau}, \dot{\tilde{v}}, \dot{s})$ in $\mathcal{X}_{M_q}^\infty$, and hence to the physical variation

$$(5.8) \quad \begin{aligned} (\dot{v}, \dot{w}) &= D_x \mathcal{S}_q(x)h, \quad \dot{w} = \dot{s}, \\ \dot{v} &= \left[-\frac{\tau \cdot \dot{\tau}}{2a(\tau)} \iota M_q + e_T \otimes \dot{\tau} \right] y + \dot{\tilde{v}}, \\ \dot{D}_\varepsilon \dot{v} &:= \dot{v}_\zeta + \varepsilon A \dot{v}. \end{aligned}$$

The coefficient of $\iota M_q y$ comes from differentiating $a(\tau)$, which gives $a'(\tau)\dot{\tau} = -\tau \cdot \dot{\tau} / (2a(\tau))$. In the last line, $\dot{D}_\varepsilon \dot{v}$ is the variation of $D_\varepsilon v$. The constant term Le_y disappears upon differentiation. Applying the chain rule to the four entries of (5.2), we obtain

$$(5.9) \quad L_{q,\hat{x}}\hat{h} = J_Y P_Z \mathcal{Q}_Y(\dot{H}_0, \dot{H}_1, \dot{H}_2, \dot{H}_3),$$

where

$$(5.10) \quad \dot{H}_0 = \mathcal{R}\dot{w} - 2\mathcal{R}v \cdot \mathcal{R}\dot{v},$$

$$(5.11) \quad \dot{H}_1 = (I - \Pi)(\mathcal{D}\dot{w} - \mathcal{D}\dot{v} \cdot \mathcal{R}v - \mathcal{D}v \cdot \mathcal{R}\dot{v}),$$

$$(5.12) \quad \dot{H}_2 = (I - \Pi)(\mathcal{R}\dot{v} \cdot D_\varepsilon v + \mathcal{R}v \cdot \dot{D}_\varepsilon \dot{v} - \dot{w}_\zeta),$$

$$(5.13) \quad \begin{aligned} \dot{H}_3 &= (I - \Pi)[\det(\mathcal{D}\dot{v}, \mathcal{R}v, D_\varepsilon v) + \det(\mathcal{D}v, \mathcal{R}\dot{v}, D_\varepsilon v) \\ &\quad + \det(\mathcal{D}v, \mathcal{R}v, \dot{D}_\varepsilon \dot{v})]. \end{aligned}$$

Since the domain constraints are linear for fixed q , the direction h satisfies $\mathcal{C}_{M_q}h = 0$. Differentiating (5.3) also shows that the four linearized residuals are compatible. Thus, P_Z fixes their quotient coordinates, and J_Y reconstructs the same residuals.

For the following sections, we record the physical state as $a = \mathcal{S}_q x = (v, w)$ and collect it with the parameters in $\mathbf{b} = (\rho, a, \varepsilon, p)$. On the fixed domain, we write $L_{\mathbf{b}} = L_{q,\hat{x}}$. In the coordinates $h = T_q\hat{h}$, the same linearization is represented by $D_x \mathcal{F}(q, x) = L_{\mathbf{b}} T_q^{-1}$. We call this the *Cartesian operator*, since its inputs are variations of $(\tau, \tilde{v}, s)$ and its outputs are the four original residual components. We now construct its inverse, with estimates uniform on an open set of states.

6. DECOMPOSING THE LINEARIZED EQUATIONS

We decompose the Cartesian linearization (5.7) into a minus block, containing the center and affine components, and a positive block, governed by an elliptic boundary value problem. The center and affine equations are solved outward from the axis. The resulting *block variables* separate the highest-order terms, allowing us to use a norm suited to each component. We construct *transfer maps* between the Cartesian and block variables, and allow independent interface and constraint data when solving the blocks. At the end of the section, we show how the block inverse gives a solution of the original Cartesian problem.

Throughout the linear analysis, a *state* is a quadruple

$$\mathbf{b} = (\rho, a, \varepsilon, p), \quad a = \mathcal{S}_q(T_q\hat{x}), \quad q = (\rho, \varepsilon, p),$$

where ρ is the ellipse parameter, $a = (v, w)$ is the pair of unknowns obtained from the chart (5.1), ε is the curvature parameter, and p collects the parameters $(\alpha_0, \delta, \lambda)$ of (2.33). Here, $T_q = T_{M_q}$ is the domain identification (4.26), so $\hat{x}$ belongs to the fixed domain and $T_q \hat{x}$ to the domain for M_q . We write $L_{\mathbf{b}}$ for the Cartesian linearization (5.7). The fixed constrained domain $\mathcal{X}_{M_*}^s$ and compatible source space $\mathcal{Y}_{\text{comp}}^s$, equipped with the analytic norms of (3.16), will be denoted by $\mathcal{X}_\gamma^s$ and $\mathcal{Y}_\gamma^s$. The source norm is applied to the quotient coordinates (3.8). The block spaces depend on $\mathbf{b}$, so we will identify them with fixed reference spaces before differentiating with respect to the state.

We fix $\alpha_0 \notin (\pi/2)\mathbb{Z}$ and $\lambda_* \in (0, 1/2)$, and in the chart $(\alpha_0, \delta, \lambda)$ we set $p_e = (\alpha_0, 0, \lambda_*)$. Since $\delta = 0$, the matrix M_{ρ, p_e} is constant in ζ , and the corresponding unperturbed solution has the same elliptical cross-section with ellipse parameter ρ at every ζ . We define the *constant reference state* by

$$\mathbf{b}_e = (\rho_*, a_e, 0, p_e), \quad a_e = a_{\rho_*, p_e}^0.$$

Thus, $a_{\rho, p_e}^0 = (v_{M_{\rho, p_e}}, w_{M_{\rho, p_e}})$, with v_M and w_M given by (2.25) and (2.27). We choose a small threshold $\rho_0 < 1/4$ below and fix $0 < \rho_* < \rho_0$ at the end of the linear analysis. The inverse is built around $\mathbf{b}_e$. For a state $\mathbf{b} = (\rho, a, \varepsilon, p)$, we also write

$$\mathbf{b}_\rho := (\rho, a_{\rho, p_e}^0, 0, p_e)$$

for the constant ellipse with the same value of ρ as the state $\mathbf{b}$. Thus, $\mathbf{b}_e = \mathbf{b}_{\rho_*}$. We measure the full perturbation from $\mathbf{b}_e$ and use $\mathbf{b}_\rho$ for comparisons at a fixed ellipse parameter. We use the base regularity index s_* of Proposition 4.1. For a tangent direction $h = (\dot{\rho}, \dot{a}, \dot{\varepsilon}, \dot{p})$, we set

$$\begin{aligned} \|h\|_{\text{tan}, s} &:= \|\dot{a}\|_{\mathcal{A}_\gamma^s} + |\dot{\rho}| + |\dot{\varepsilon}| + |\dot{p}|, \\ \|\mathbf{b} - \mathbf{b}_e\|_{\mathcal{B}^s} &:= \|a - a_e\|_{\mathcal{A}_\gamma^s} + |\rho - \rho_*| + |\varepsilon| + |p - p_e|. \end{aligned}$$

For an m -th derivative, we collect the directions as $\mathbf{h} = (h_1, \dots, h_m)$. For a nonnegative integer k_m , fixed in each estimate, we set

$$\begin{aligned} \mathcal{H}_{0,m}(\mathbf{h}) &:= \prod_{i=1}^m \|h_i\|_{\text{tan}, s_* + k_m}, \\ \mathcal{H}_{s,m}(\mathbf{h}) &:= \sum_{i=1}^m \|h_i\|_{\text{tan}, s + k_m} \prod_{j \neq i}^m \|h_j\|_{\text{tan}, s_* + k_m}, \end{aligned} \tag{6.1}$$

with $\mathcal{H}_{0,0} = 1$ and $\mathcal{H}_{s,0} = 0$. The first quantity in the above uses lower regularity norms for all directions. In each term of the second quantity, one direction is measured in a higher regularity norm and the others in lower regularity norms. These quantities will be used in the tame estimates for derivatives of order m .

Finally, we decompose the high cell frequencies $|n| \geq 2^J$ into the sets $2^j \leq |n| < 2^{j+1}$, which we call *dyadic shells*, and we let

$$Q_j = \mathbf{1}_{\{2^j \leq |n| < 2^{j+1}\}}(D_\zeta), \quad j \geq J, \quad Q_j Q_k = \delta_{jk} Q_j, \quad \sum_{j \geq J} Q_j = H_J, \tag{6.2}$$

be the corresponding Fourier projections. Note that these projections are orthogonal and bounded on every Sobolev space and on every analytic space (3.16).

6.1. The block norms. The variables are grouped into two blocks. The *minus block* contains the center and affine components and corresponds to D_- in Section 1.4. The *positive block* contains the component controlled by a positive quadratic energy and corresponds to D_+ . Their inverses are constructed in Sections 8 and 9, respectively.

The block spaces contain scalar functions on the cap and annulus, together with data on the interface, axis, and outer boundary. They also contain components recording the prescribed angular averages, the scale of the linear Taylor term, and the right-hand sides of the two gauge conditions (3.4). These conditions fix the remaining freedom to change coordinates. The mean, scale, and gauge data form a finite list of functions that may depend on r and ζ . The original domain constraints are collected in (3.7). In the block problem, we allow their right-hand sides to be prescribed independently.

An element of a block space is a tuple of the scalar functions and data described above. We call each entry of this tuple a *component* and index it by α . The domain norms measure the unknowns, while the range norms measure the sources and prescribed data. To specify the weights in cell frequency, we set

$$(6.3) \quad \lambda_n = \langle n \rangle = (1 + n^2)^{1/2}, \quad \mu_n = (\delta_A + \lambda_n^{-1})^{1/2}, \quad \delta_A > 0,$$

where δ_A is a small positive parameter chosen in the energy estimate for the affine equations in Section 8. The Fourier coefficient of the component α at cell frequency n carries a weight $d_\alpha^X(n)$ in the domain and $d_\alpha^Y(n)$ in the range. These weights are listed in Table 1, with the common factor λ_n^s omitted. They account for the different orders of the equations and traces. The radial and angular norms will be specified with the equations in Sections 8 and 9.

Component	$d_\alpha^X(n)$	$d_\alpha^Y(n)$
center component on the annulus	1	1
affine component on the annulus, upper entry	μ_n	μ_n
affine component on the annulus, lower entry	$\lambda_n^{1/2}$	$\lambda_n^{1/2}$
positive component on the annulus	λ_n	λ_n^{-1}
cap component of the minus block	1	1
cap component of the positive block	1	1
value on an interface or on the boundary	$\lambda_n^{1/2}$	$\lambda_n^{1/2}$
conormal derivative on an interface or on the boundary	$\lambda_n^{-1/2}$	$\lambda_n^{-1/2}$
value on the axis	λ_n^{-1}	λ_n^{-1}
first derivatives on the axis, or affine mean	λ_n^{-2}	λ_n^{-2}
mean, scale, and gauge conditions	1	1

TABLE 1. The weights of the block components at cell frequency n , on the domain side and on the range side.

The annular center component contains the two oscillator variables and their radial derivatives described in Section 1.4. Writing the oscillator variables as ξ_1 and ξ_2 , the four coordinates are

$$(\xi_1, \xi_2, \partial_r \xi_1, \partial_r \xi_2).$$

The affine component consists of two pairs (x_1, y_1) and (x_2, y_2) , called *affine chains*. We call x_j the upper entry and y_j the lower entry. At the reference state, the homogeneous part of highest order in the cell variable

is

$$\partial_r x_{j,n} = \lambda_n y_{j,n}, \quad \partial_r y_{j,n} = 0, \quad j = 1, 2,$$

where $x_{j,n}$ and $y_{j,n}$ are the Fourier coefficients at cell frequency n . Thus, in this system, the lower entry is constant in r and drives the radial evolution of the upper entry. The full equations also contain lower-order terms and sources. These different roles account for the two affine weights in the table.

The positive component is the part of the solution to the linearized problem governed by the elliptic block D_+ . We measure this component in its *energy norm*, with H^{s+1} regularity, and its source in the corresponding dual scale, with H^{s-1} regularity. The factors λ_n and λ_n^{-1} record this difference. On a cap rescaled to a fixed disk, the weights in the table are one, while the disk norms retain their regularity shifts. For the minus block, the solution of the equation on this disk is measured in H^{s+2} and its right-hand side in H^s . For the positive block, the corresponding spaces are H^{s+1} and H^{s-1} .

For any two cell frequencies n and m , each weight in Table 1 satisfies

$$(6.4) \quad \|d_\alpha(n)d_\alpha(m)^{-1}\| \leq C_\alpha \langle n - m \rangle^{q_\alpha}, \quad q_\alpha \leq 2.$$

Here, d_α denotes either d_α^X or d_α^Y . The constants depend on the component and may depend on δ_A , but are independent of any upper Fourier cutoff and of the number N of periods of the field. This inequality compares the input and output weights when an operator changes the cell frequency. We denote the products of the component spaces by $\mathbb{X}_\sharp^s$ and $\mathbb{Y}_\sharp^s$ and call them the *block spaces*, abbreviated to $X_\sharp^s$ and $Y_\sharp^s$ when the side is clear.

At radius r , we use the phase

$$(6.5) \quad \Phi_r(n) = \sigma_0 \lambda_n - \gamma(\sqrt{1 + r^2 \lambda_n^2} - 1),$$

which is the phase (3.15) written as a function of the cell frequency at a fixed radius. Thus, for $u = \sum_n u_n(r, \theta) e^{in\zeta}$,

$$(W_\gamma u)_n(r, \theta) = e^{\Phi_r(n)} u_n(r, \theta).$$

We take Φ_0 on the axis and the phase at the corresponding radius for each trace. By (3.18), the exponential weight $e^{\Phi_r(n)}$ is submultiplicative in n at a fixed radius. Extension operators turn boundary or interface data into functions defined at nearby radii. Since the analytic weight depends on the radius, estimating the extended function requires comparing its weight with that of the original data. The Poisson decay in (4.10) compensates for any increase in the weight, so the same extension estimates hold in the analytic norms.

6.2. Kernel bounds for operators between components. Multiplication by a coefficient depending on ζ mixes cell frequencies, so we estimate operators through their Fourier kernels. Here, α labels a component of the unknown and β a component of the source. For example, suppose a scalar center variable u_c appears as $c(\zeta)u_c$ in the positive-block equation. Then α labels this center variable and β the positive source component, and the operator between them is multiplication by c .

In this subsection, m and n both denote cell frequencies. We write $A_{\beta\alpha}(n, m) : E_{\alpha,m} \rightarrow F_{\beta,n}$ for the map from input component α at frequency m to source component β at frequency n . In the example above, this map is multiplication by c_{n-m} , the $(n-m)$ -th Fourier coefficient of c . The spaces $E_{\alpha,m}$ and $F_{\beta,n}$ carry the remaining radial, angular, or trace norms. Incorporating the component weights gives the *weighted kernel*

$$(6.6) \quad K_A^{\beta\alpha}(n, m) = e^{\Phi_\beta(n)} d_\beta^Y(n) A_{\beta\alpha}(n, m) (d_\alpha^X(m))^{-1} e^{-\Phi_\alpha(m)}.$$

Here, Φ_α and Φ_β are the phases of (6.5) for the input and output components, evaluated at the corresponding radius for a trace. All kernel norms below are operator norms from $E_{\alpha,m}$ to $F_{\beta,n}$. For an integer $q \geq 0$, we set

$$(6.7) \quad |A|_{\mathfrak{S}^0[q]} := \max_{\alpha,\beta} \max \left\{ \sup_m \sum_n \langle n-m \rangle^q \|K_A^{\beta\alpha}(n,m)\|, \right. \\ \left. \sup_n \sum_m \langle n-m \rangle^q \|K_A^{\beta\alpha}(n,m)\| \right\},$$

which we call the *moment* of order q of A . At $q = 0$, Schur's test gives, for each pair of components α, β ,

$$\left(\sum_n \left\| \sum_m K_A^{\beta\alpha}(n,m) u_m \right\|_{F_{\beta,n}}^2 \right)^{1/2} \leq |A|_{\mathfrak{S}^0[0]} \left(\sum_m \|u_m\|_{E_{\alpha,m}}^2 \right)^{1/2}.$$

Here, $u_m \in E_{\alpha,m}$, and the sums run over cell frequencies. Since K_A includes the weights, this bounds the weighted output norm of $A_{\beta\alpha}$ by $|A|_{\mathfrak{S}^0[0]}$ times the weighted input norm. Higher moments measure decay away from the diagonal and control additional Sobolev weights. The following lemma combines bounds for individual operators into bounds for their sums and compositions.

Lemma 6.1. *For each $q \geq 0$, finite sums and compositions preserve finiteness of the moment (6.7), provided that the intermediate spaces match. Bounds for all moments give bounds at every Sobolev index at which the component kernels are bounded.*

For the tame estimates, fix ρ , ε , and p , and write $a = (v, w)$. Let $A(a) : X_{\mathfrak{h}}^s \rightarrow Y_{\mathfrak{h}}^s$ be a finite sum of compositions with matching intermediate spaces. For each factor $T(a)$, let E^s and F^s denote its input and output component spaces with the weights of Table 1. Suppose $T(a) : E^s \rightarrow F^s$ satisfies, for $s \geq s_*$,

$$\|T(a)u\|_{F^s} \leq C_{s,T} [M_{T,0}(a)\|u\|_{E^s} + M_{T,s}(a)\|u\|_{E^{s_*}}].$$

Here, $M_{T,0}(a)$ is polynomial in norms of a at fixed regularity indices. Each term of $M_{T,s}(a)$ is a product of such norms with at most one factor measured at a higher index $s + k_T$, where k_T is independent of s . Assume the same form of estimate for each fixed-order derivative of T with respect to $\mathbf{b}$, with at most one coefficient or tangent direction measured in a higher regularity norm. Then

$$(6.8) \quad \|A(a)u\|_{Y_{\mathfrak{h}}^s} \leq C_s [\mathcal{L}_0(a)\|u\|_{X_{\mathfrak{h}}^s} + \mathcal{L}_s(a)\|u\|_{X_{\mathfrak{h}}^{s_*}}],$$

where $\mathcal{L}_0$ and $\mathcal{L}_s$ have the same form as $M_{T,0}$ and $M_{T,s}$, respectively. Their regularity shifts may depend on the factors and the number of compositions, but not on s . If $A(a_\circ) = 0$, then there is an integer $k \geq 0$, independent of s , such that, for every a with $\|a - a_\circ\|_{s_*+k}$ sufficiently small, and with the other components of the state held fixed, we have

$$(6.9) \quad \|A(a)u\|_{Y_{\mathfrak{h}}^s} \leq C_s [\|a - a_\circ\|_{s_*+k}\|u\|_{X_{\mathfrak{h}}^s} + \|a - a_\circ\|_{s+k}\|u\|_{X_{\mathfrak{h}}^{s_*}}].$$

Here, $\|\cdot\|_s$ denotes the norm of $\mathcal{A}_\gamma^s$. Derivatives with respect to $\mathbf{b}$ satisfy the corresponding tame estimates. For each fixed number of derivatives, the regularity shift is finite and independent of s .

Proof. Schur's test gives the bound without the common cell-frequency weight λ_n^s . Summing over the finitely many components gives the bound for the full operator. At index s , the additional factor λ_n^s/λ_m^s is bounded by $C_s \langle n-m \rangle^{|s|}$, so a moment of order at least $|s|$ controls the cell Sobolev weights. For a composition, the intermediate weights cancel, giving

$$K_{BA}(n, m) = \sum_k K_B(n, k) K_A(k, m).$$

The inequality $\langle n - m \rangle^q \leq C_q \langle n - k \rangle^q \langle k - m \rangle^q$ then gives the required row and column sums.

Applying the assumed tame bounds for the individual factors in succession and interpolating the intermediate regularity norms proves (6.8), with at most one higher regularity norm in each term. If $A(a_\circ) = 0$, the mean value formula along the segment from $a_\circ$ to a gives (6.9). The product rule expresses each derivative with respect to the state as a finite sum of compositions of derivatives of the factors. Their assumed bounds give the corresponding tame estimate. $\square$

For multiplication by an analytic coefficient a , the Fourier kernel is a_{n-m} . Submultiplicativity of the exponential weight at a fixed radius and (6.4) therefore bound its moments by the corresponding coefficient norms. We also use smooth Fourier cutoffs $\tilde{Q}_j = \tilde{q}_j(D_\zeta)$ at cell-frequency scale 2^j , distinct from the sharp projections Q_j in (6.2). Their commutators with multiplication satisfy

$$(6.10) \quad \begin{aligned} ([\tilde{Q}_j, a]u)_n &= \sum_m (\tilde{q}_j(n) - \tilde{q}_j(m)) a_{n-m} u_m, \\ |\tilde{q}_j(n) - \tilde{q}_j(m)| &\leq C \min\{1, 2^{-j}|n - m|\}. \end{aligned}$$

Thus, the first moment of the multiplication operator bounds the zeroth moment of the commutator, with a gain of 2^{-j} .

For the axis extensions E_0, E_1, E_2 of (4.5) and the boundary extension of (4.8), we use their explicit Fourier formulas. Their supports and the Poisson decay (4.10) give the weighted bounds (4.6) and (4.9).

An operator norm bound alone does not imply finite moments. If a factor G is known only through its operator norm, we estimate compositions containing it by

$$(6.11) \quad \|A_2 G A_1\| \leq \|A_2\| \|G\| \|A_1\|.$$

6.3. The principal symbol and its two blocks. We now choose coordinates in which the highest-order terms do not couple the two blocks. On the annulus, we begin with

$$(6.12) \quad p = \mathcal{D}v, \quad q_1 = \mathcal{R}v, \quad T = L^{-1}D_\varepsilon v, \quad \mathbb{B} = (p, q_1, T), \quad U = \mathbb{B}c.$$

Here, U is the variation of v , and $\mathcal{D}$ and $\mathcal{R}$ are the operators $\mathcal{D} = r\partial_r$ and $\mathcal{R} = \partial_\theta$ from (2.19). The determinant satisfies

$$\det \mathbb{B} = \frac{r^2}{L} \det(v_{y_1}, v_{y_2}, D_\varepsilon v), \quad |\det \mathbb{B}| \geq c_{\det} r^2$$

on a sufficiently small neighborhood of the constant reference state, for some $c_{\det} > 0$. Indeed, at that state, $\det \mathbb{B} = -r^2 \det M_{\rho, p_e}$, and (2.34) bounds $\det M_{\rho, p_e}$ away from zero. Thus, on the annulus, $\mathbb{B}$ is a *frame*, that is, a basis of $\mathbb{R}^3$ at each point, and c in (6.12) is the coordinate vector of U in this basis. In this calculation, p denotes the frame vector $\mathcal{D}v$ rather than the parameter triple in $\mathbf{b}$. We also set

$$(6.13) \quad A = p \cdot U, \quad B = q_1 \cdot U, \quad C = T \cdot U, \quad X = S - B.$$

Here, S is the variation of w , as in (5.8). The inner products (A, B, C) differ from the frame coordinates c by the relation $(A, B, C)^T = \mathbb{B}^* \mathbb{B}c$, while $X = S - B$ simplifies the first three linearized equations. Since the constant term Le_y in $D_\varepsilon v$ has zero variation, the derivative acting on variations and frame columns is

$$(6.14) \quad \dot{D}_\varepsilon U := U_\zeta + \varepsilon e_z \times U,$$

and the derivatives of the frame are recorded by

$$(6.15) \quad \Gamma_{\mathcal{D}} = \mathbb{B}^{-1}(\mathcal{D}\mathbb{B}), \quad \Gamma_{\mathcal{R}} = \mathbb{B}^{-1}(\mathcal{R}\mathbb{B}), \quad \Gamma_\varepsilon = \mathbb{B}^{-1}(\dot{D}_\varepsilon \mathbb{B}).$$

Writing the linearized equations (5.10)–(5.13) in this frame and expanding the derivatives, we obtain

$$\begin{aligned}
 N_0 &= \mathcal{R}S - 2\mathcal{R}(q_1 \cdot U) + 2(\mathcal{R}q_1) \cdot U, \\
 N_1 &= (I - \Pi)[\mathcal{D}S - \mathcal{D}(q_1 \cdot U) - \mathcal{R}(p \cdot U) + (\mathcal{D}q_1 + \mathcal{R}p) \cdot U], \\
 N_2 &= (I - \Pi)[L\mathcal{R}(T \cdot U) + \partial_\zeta(q_1 \cdot U - S) - (L\mathcal{R}T + \dot{D}_\varepsilon q_1) \cdot U], \\
 N_3 &= (I - \Pi)[(\det \mathbb{B})\hat{N}_3], \\
 \hat{N}_3 &= Le_1^*(\mathcal{D}c + \Gamma_{\mathcal{D}}c) + Le_2^*(\mathcal{R}c + \Gamma_{\mathcal{R}}c) + e_3^*(c_\zeta + \Gamma_\varepsilon c).
 \end{aligned}
 \tag{6.16}$$

Here, $N_0, \dots, N_3$ are the expressions $\dot{H}_0, \dots, \dot{H}_3$ in the frame, and e_j^* selects the j -th entry of the three-vector to its right. The operator Π is angular averaging,

$$(\Pi u)(r, \zeta) = \frac{1}{2\pi} \int_0^{2\pi} u(r, \theta, \zeta) \, d\theta.$$

For example, $\mathcal{D}U = \mathbb{B}(\mathcal{D}c + \Gamma_{\mathcal{D}}c)$, so $\Gamma_{\mathcal{D}}c$ accounts for differentiation of the frame. In the fourth equation, $I - \Pi$ acts on $(\det \mathbb{B})\hat{N}_3$ before division by $\det \mathbb{B}$, since multiplication by this determinant need not commute with angular averaging. The angular mean equations are retained separately and will be included among the auxiliary equations.

To find the *principal symbol*, we evaluate the coefficients at a fixed point and retain the terms of highest order after rescaling by the cell frequency. This evaluation is called *freezing the coefficients*. We replace derivatives by their frequency variables, except that we initially retain the angular operator $\mathcal{R}$.

We begin with a state $v = Py$ linear in y , where P is the 3×2 matrix of (3.1). Then, $p = Py$ and $q_1 = PJ_2 y$, with J_2 the rotation of the plane by a quarter turn. Taking a constant vector $T \in \mathbb{R}^3$ to complete the frame and writing $E_\theta = (e_r, e_\theta)$, we set

$$\begin{aligned}
 g &= T^*T, & t &= P^*T, & G_T &= P^* \left(I - \frac{T \otimes T}{g} \right) P, \\
 \vartheta &= E_\theta^* t, & K &= (E_\theta^* G_T E_\theta)^{-1}, & h &= \frac{K\vartheta}{g}, & \kappa &= \frac{1}{g} + \frac{\vartheta^* K \vartheta}{g^2}.
 \end{aligned}
 \tag{6.17}$$

Here, G_T is the *Gram matrix*, or matrix of pairwise inner products, of the columns of P projected onto $T^\perp$, and e_r, e_θ are the radial and angular unit vectors. The determinant bound makes G_T invertible, so these coefficients are smooth. Writing $A = ra_1$, $B = ra_2$, and $\mathbf{u} = (a_1, a_2, C, X)^T$, we rescale the radial variable on a dyadic shell by the cell frequency. For $\varsigma = \text{sgn } n \in \{+1, -1\}$, the principal symbol is affine in the rescaled radial frequency d ,

$$Q_{P,T}^\varsigma(d) = L_0^\varsigma + dL_1, \tag{6.18}$$

where

$$L_0^\varsigma = \begin{pmatrix} -2I & -\mathcal{R} & 0 & 0 \\ -\mathcal{R} & 2I & 0 & 0 \\ 0 & 0 & \mathcal{R} & -i\varsigma I \\ -i\varsigma h_1 & -i\varsigma h_2 & i\varsigma \kappa I & 0 \end{pmatrix}, \quad L_1 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & I \\ 0 & 0 & 0 & 0 \\ K_{11} & K_{12} & -h_1 & 0 \end{pmatrix}.
 \tag{6.19}$$

To obtain these matrices from (6.16), we work on nonzero angular modes, where $I - \Pi$ is the identity, substitute $S = X + B$, divide the third and fourth rows by L , and divide the fourth row by $\det \mathbb{B}$. Rescaling

gives the first three rows, and (6.17) gives the last. The term $\mathcal{R}X$ in the first equation is one order lower after rescaling and belongs to the remainder. The same computation at a general state gives

$$(6.20) \quad Q_{\mathbf{b}}(d, \omega, \nu) = L_{0,\mathbf{b}}(\omega, \nu) + dL_{1,\mathbf{b}}(\omega, \nu),$$

where ω and ν are the angular and cell frequencies. The remainder includes commutators and terms in which derivatives fall on the coefficients, except for those already retained in the normalized symbol, such as the entries $\pm 2I$ in (6.19). This algebraic computation does not use an inverse estimate for the affine equations.

At fixed (ω, ν) , we replace the radial frequency by a complex variable and write $Q_{\mathbf{b}}(\eta) = L_{0,\mathbf{b}} + \eta L_{1,\mathbf{b}}$. The values of η for which this matrix is singular are its *characteristic roots*. Since $L_{1,\mathbf{b}}$ is singular, we use distinct projections on the unknowns and on the equations. To select the center and affine components, we choose a fixed union of contours Γ_- , invariant under complex conjugation, enclosing their roots and excluding the positive-block roots. The corresponding *spectral projections* are

$$(6.21) \quad \begin{aligned} P_{-,\mathbf{b}}^D &= \frac{1}{2\pi i} \int_{\Gamma_-} Q_{\mathbf{b}}(\eta)^{-1} L_{1,\mathbf{b}} \, d\eta, \\ P_{-,\mathbf{b}}^R &= \frac{1}{2\pi i} \int_{\Gamma_-} L_{1,\mathbf{b}} Q_{\mathbf{b}}(\eta)^{-1} \, d\eta, \\ P_{+,\mathbf{b}}^{D,R} &= I - P_{-,\mathbf{b}}^{D,R}. \end{aligned}$$

The contours are fixed throughout the neighborhood of states and normalized frequency range under consideration. Uniform bounds for the *resolvent* $Q_{\mathbf{b}}(\eta)^{-1}$ on these contours give bounded projections depending analytically on the state. The superscripts D and R distinguish the domain and range projections. Their factors occur in different orders so that

$$(6.22) \quad L_{1,\mathbf{b}} P_{-,\mathbf{b}}^D = P_{-,\mathbf{b}}^R L_{1,\mathbf{b}}.$$

Since $L_{0,\mathbf{b}} = Q_{\mathbf{b}}(\eta) - \eta L_{1,\mathbf{b}}$ and the contour integral of a constant vanishes, both $L_{0,\mathbf{b}} P_{-,\mathbf{b}}^D$ and $P_{-,\mathbf{b}}^R L_{0,\mathbf{b}}$ are equal to

$$(6.23) \quad -\frac{1}{2\pi i} \int_{\Gamma_-} \eta L_{1,\mathbf{b}} Q_{\mathbf{b}}(\eta)^{-1} L_{1,\mathbf{b}} \, d\eta.$$

The complementary projections satisfy the same identities, and therefore

$$(6.24) \quad L_{j,\mathbf{b}} P_{\pm,\mathbf{b}}^D = P_{\pm,\mathbf{b}}^R L_{j,\mathbf{b}} \quad (j = 0, 1), \quad Q_{\mathbf{b}} P_{\pm,\mathbf{b}}^D = P_{\pm,\mathbf{b}}^R Q_{\mathbf{b}}.$$

These identities apply to the complete normalized symbol $Q_{\mathbf{b}}$, with the terms of each equation kept together. The mean, scale, gauge, and chart equations form a separate *triangular system*. This means that, after applying the inverses of the diagonal operators, the equations for the remaining unknowns $\eta_1, \dots, \eta_{N_{\text{aux}}}$ have the form

$$\eta_j + \sum_{k < j} T_{jk} \eta_k = g_j, \quad j = 1, \dots, N_{\text{aux}}.$$

Here, N_{aux} is the number of equations, g_j are the prescribed data, and T_{jk} are the operators coupling the unknowns. The unknowns and data may be functions of r and ζ . We solve first for $\eta_1 = g_1$, then for η_2 , and continue in this order.

For $\alpha, \beta \in \{-, +\}$, let $r_{\alpha,\mathbf{b}}$ reconstruct the normalized column $\mathbf{u} = (a_1, a_2, C, X)^T$ of (6.18) from the coordinates of block α . Let $e_{\beta,\mathbf{b}}$ extract the scalar source coordinates of block β after applying the range

projection $P_{\beta, \mathbf{b}}^R$. The highest-order operator from block α to block β is then

$$(6.25) \quad \ell_{\beta\alpha, \mathbf{b}}^{\text{pr}} = e_{\beta, \mathbf{b}} P_{\beta, \mathbf{b}}^R Q_{\mathbf{b}} P_{\alpha, \mathbf{b}}^D r_{\alpha, \mathbf{b}}.$$

For $\beta \neq \alpha$, (6.24) gives

$$(6.26) \quad P_{\beta, \mathbf{b}}^R Q_{\mathbf{b}} P_{\alpha, \mathbf{b}}^D = 0, \quad \ell_{\beta\alpha, \mathbf{b}}^{\text{pr}} = 0.$$

Thus, the highest-order part of the linearization is block diagonal. The cancellation follows before applying $e_{\beta, \mathbf{b}}$ or $r_{\alpha, \mathbf{b}}$, so it is independent of these coordinate choices.

We estimate the projections $P_{\pm, \mathbf{b}}^{D, R}$ in (6.21) by the weighted kernels (6.6). Applying the analytic weight to a radial equation replaces ∂_r by $\partial_r + d_\gamma(r, \Lambda)$, with d_γ as in (3.21). The commutator $[d_\gamma(r, \Lambda), P_{\pm, \mathbf{b}}^{D, R}]$ is estimated using the difference of the damping symbol at the input and output cell frequencies, as in (6.10). The coefficient estimates follow from (3.20). For a state variation $\dot{\mathbf{b}}$, differentiating $Q_{\mathbf{b}} Q_{\mathbf{b}}^{-1} = I$ gives

$$D_{\mathbf{b}}(Q_{\mathbf{b}}^{-1})[\dot{\mathbf{b}}] = -Q_{\mathbf{b}}^{-1}(D_{\mathbf{b}}Q_{\mathbf{b}}[\dot{\mathbf{b}}])Q_{\mathbf{b}}^{-1}.$$

We apply this identity under the contour integrals in (6.21), also differentiating $L_{1, \mathbf{b}}$, and use Lemma 6.1 to combine the bounds for the individual factors.

6.4. The source variables. For a compatible Cartesian source $f = (f_0, f_1, f_2, f_3)$, we use the quotient coordinates $Z = (g_+, g_-, g_3, h)$ of Section 3.2, with $h = f_2$. The reconstruction map is

$$(6.27) \quad J_Y Z = \left(\frac{\bar{z}g_+ - zg_-}{2i}, \frac{\bar{z}g_+ + zg_-}{2}, h, z\bar{z}g_3 \right),$$

as in (3.10). It only multiplies by smooth factors, and its inverse on compatible residuals is the quotient extraction $\mathcal{Q}_Y$ of (3.11). In particular, $J_Y \mathcal{Q}_Y f = f$ for compatible f .

To obtain the scalar equation used for the positive block, we differentiate and combine the equations (6.16). Applying the same operations to their right-hand sides, written as $f = J_Y Z$ in (6.27), gives its source $\Psi_{\mathbf{b}} Z$. We retain both Z and $\Psi_{\mathbf{b}} Z$ in the block coordinates, so that the original source can be recovered without inverting a differential operator.

The enlarged block problem also includes independent cap, trace, mean, scale, gauge, and chart data, collected in r_{aug} . These include data for the angular modes $m = \pm 2$, called the *exceptional modes*, where the angular multipliers used in the elimination vanish. Here and below, m denotes an angular mode and n a cell frequency. We define

$$(6.28) \quad \mathbf{G}_{Y, \mathbf{b}}(Z, r_{\text{aug}}) = (Z, \Psi_{\mathbf{b}} Z, r_{\text{aug}}),$$

and its inverse on its image is

$$(6.29) \quad (Z, \Psi_{\mathbf{b}} Z, r_{\text{aug}}) \mapsto (Z, r_{\text{aug}}).$$

The image of $\mathbf{G}_{Y, \mathbf{b}}$ is the *graph* of $Z \mapsto \Psi_{\mathbf{b}} Z$, together with the independent data r_{aug} . Thus, $\Psi_{\mathbf{b}} Z$ is determined by Z , and the inverse simply discards this derived entry. In the triangular elimination leading to the scalar equations, one Cartesian source component is replaced by $\Psi_{\mathbf{b}} Z$, with a nonzero coefficient on that component.

The derivative orders in $\Psi_{\mathbf{b}} Z$, before applying the angular multipliers, are listed in Table 2.

	g_+	g_-	g_3	h
derivatives (D_y, D_ζ) at zero curvature	(1, 1)	(1, 1)	(0, 1)	(2, 0)
total order at a general state	≤ 2	≤ 2	≤ 1	≤ 2

TABLE 2. Derivative orders in the scalar source $\Psi_{\mathbf{b}}Z$, before the angular multipliers are applied.

At the unperturbed solution at zero curvature, the terms are, up to fixed nonzero factors, $\partial_\zeta \partial_z g_+$, $\partial_\zeta \partial_{\bar{z}} g_-$, $\partial_\zeta g_3$, and $\Delta_y h$. This gives the first row of the table. The total orders in the second row follow from (6.16). Each entry is a composition of at most two first-order operators, and differentiation of a coefficient leaves one fewer derivative on the source.

Since $\mathcal{R} = \partial_\theta$, the angular multipliers act by

$$(\mathcal{R} - 2i)e^{im\theta} = i(m - 2)e^{im\theta}, \quad (\mathcal{R} + 2i)e^{im\theta} = i(m + 2)e^{im\theta}.$$

They can therefore be inverted on the odd modes and on the even modes with $|m| \geq 4$. At $m = 2$ and $m = -2$, respectively, one multiplier vanishes. We retain the corresponding equation and its source as part of the affine component, instead of dividing by that multiplier. The detailed reconstruction appears in Section 7.

We retain the equations for the angular mean and zero cell frequency, together with the remaining mean, scale, gauge, chart, and compatibility conditions, and solve them successively in the triangular system. Their right-hand sides, together with the coordinates in (6.28), give the source variables used by the block inverses.

6.5. The order of the reduced scalar equation. We eliminate A , B , and X from (6.16) to obtain the scalar equation for $C = T \cdot U$ introduced in (6.13). Its positive spectral part supplies the equation for the positive block. Although the calculation appears to produce third derivatives, an angular factor cancels and leaves an equation of order two. We first verify this at zero curvature, using

$$D = r\partial_r, \quad R = \partial_\theta, \quad E = D^2 + R^2.$$

On a nonzero angular mode, R is invertible. Keeping the terms of (6.16) in which a derivative falls on an unknown gives

$$(6.30) \quad \begin{aligned} n_0 &= RX - RB, & n_1 &= DX - RA, \\ n_2 &= LRC - \partial_\zeta X, & n_3 &= LDA + LRB + r^2 \partial_\zeta C. \end{aligned}$$

We write $n = (n_0, n_1, n_2, n_3)$ for these four expressions. Up to a nonzero normalization, the highest-order part of the source conversion is the combination

$$(6.31) \quad \Phi^{\text{pr}} n = (LR)^{-1} \left\{ En_2 + \partial_\zeta \left(Dn_1 + Rn_0 + \frac{R}{L} n_3 \right) \right\}.$$

Since D , R , and ∂_ζ commute, substitution gives

$$(6.32) \quad Dn_1 + Rn_0 + \frac{R}{L} n_3 = EX + \frac{r^2}{L} R \partial_\zeta C, \quad \Phi^{\text{pr}} n = \left(E + \frac{r^2}{L^2} \partial_\zeta^2 \right) C.$$

Thus, the A , B , and X terms cancel, and the third-order numerator in (6.31) contains the angular factor R that appears in the denominator. Canceling this factor gives the second-order equation for C .

For a general frame, we first absorb the nonzero factors of L into the third and fourth equations. Thus, ν below is the frequency corresponding to $L^{-1}\partial_\zeta$. We set

$$(6.33) \quad G_{\mathbf{b}} = \mathbb{B}_{\mathbf{b}}^* \mathbb{B}_{\mathbf{b}}, \quad \xi = (d, \omega, \nu), \quad (\alpha, \beta, \chi) = \xi^* G_{\mathbf{b}}^{-1},$$

so that the principal symbol at the state $\mathbf{b}$ has the rows

$$n_0 = \omega(X - B), \quad n_1 = dX - \omega A, \quad n_2 = \omega C - \nu X, \quad n_3 = \alpha A + \beta B + \chi C,$$

where d and ω are the radial and angular frequencies. For $\omega \neq 0$, we set

$$(6.34) \quad s_{G_{\mathbf{b}}}(\xi) = \left(\frac{\nu\beta}{\omega}, \frac{\nu\alpha}{\omega}, \frac{\alpha d + \beta\omega}{\omega}, \nu \right).$$

We let $\ell_{1, G_{\mathbf{b}}}(\xi)$ be the matrix of $n_0, \dots, n_3$ in the unknowns (A, B, C, X) , and let e_C^* select the C entry. Multiplying the four equations by the entries of $s_{G_{\mathbf{b}}}(\xi)$ and adding, we obtain

$$(6.35) \quad s_{G_{\mathbf{b}}}(\xi) \ell_{1, G_{\mathbf{b}}}(\xi) = (\xi^* G_{\mathbf{b}}^{-1} \xi) e_C^*.$$

Indeed, the coefficients of A , B , and X cancel, while the coefficient of C is $\alpha d + \beta\omega + \chi\nu = \xi^* G_{\mathbf{b}}^{-1} \xi$. Multiplying by ω gives a polynomial identity. The angular mean is kept in the triangular system, and the exceptional modes are treated by the affine equations. On the compatible Cartesian spaces on the cap, that is, on sources whose quotients (3.8) are smooth, (6.35) extends across the axis. When the coefficients vary, a derivative falling on the frame, on the inverse of $G_{\mathbf{b}}$, or on a projection, leaves one fewer derivative on the unknown. Thus, the reduced scalar operator is still of order two, and after canceling its highest-order off-diagonal part, the remaining terms have order at most one.

6.6. From Cartesian to block variables. We now combine the chart derivative, the annular frame map, and the spectral projections to express a Cartesian variation in block variables. We give the inverse of each step so that the variation can be reconstructed. The restrictions to the cap and annulus, and the interface data, are treated in the next subsection.

We write $\mathbf{b} = (\rho, a, \varepsilon, p)$, with $a = (v, w) = \mathcal{S}_q(x)$, $x = T_q \hat{x}$, and $q = (\rho, \varepsilon, p)$. For a direction $\hat{h}$ in the fixed domain $\mathcal{X}_\gamma^s$, the chart derivative gives the physical variation (U, S) ,

$$(6.36) \quad \Theta_{\text{sl}, \mathbf{b}} := D_x \mathcal{S}_q(x) T_q, \quad \Theta_{\text{sl}, \mathbf{b}} \hat{h} = (U, S),$$

as in (5.8). Its inverse first extracts the tangential coefficient $\dot{\tau}$ of the linear Taylor term and then sets

$$(6.37) \quad \dot{\tilde{v}} = U - \left[-\frac{\tau \cdot \dot{\tau}}{2a(\tau)} \iota M_q + e_T \otimes \dot{\tau} \right] y, \quad \hat{h} = T_q^{-1}(\dot{\tau}, \dot{\tilde{v}}, S).$$

Here, $a(\tau) = \sqrt{1 - |\tau|^2/2}$ is the scalar factor in (3.1), whereas $a = (v, w)$ denotes the state. The tangential first derivatives on the axis determine $\dot{\tau}_j = e_T \cdot \partial_{y_j} U(0, \zeta)$ for $j = 1, 2$. Subtracting their linear Taylor contribution leaves $\dot{\tilde{v}}$ with vanishing value and first derivatives on the axis.

On the annulus, we express this variation using the frame (6.12) and its Gram matrix,

$$(6.38) \quad p = \mathcal{D}v, \quad q_1 = \mathcal{R}v, \quad T = L^{-1} D_\varepsilon v, \quad \mathbb{B} = (p, q_1, T), \quad G_{\mathbf{b}} = \mathbb{B}^* \mathbb{B},$$

and we define the pointwise map and its inverse by

$$(6.39) \quad \Theta_{\text{fr}, \mathbf{b}}(U, S) = (A, B, C, X) = (p \cdot U, q_1 \cdot U, T \cdot U, S - q_1 \cdot U),$$

$$(6.40) \quad \Theta_{\text{fr}, \mathbf{b}}^{-1}(A, B, C, X) = (\mathbb{B}c, X + B), \quad c = G_{\mathbf{b}}^{-1}(A, B, C)^T.$$

On the cap, we keep the Cartesian components (U, S) , so the frame map acts as the identity there. On the annulus, the bound $|\det \mathbb{B}| \geq c_{\det} r^2$ following (6.12) makes (6.39)–(6.40) mutually inverse analytic maps of order zero.

Finally, the projections (6.21) split the unknowns and sources into the two blocks,

$$(6.41) \quad \begin{aligned} \Theta_{\text{sp}, \mathbf{b}}^X z &= (P_{-, \mathbf{b}}^D z, P_{+, \mathbf{b}}^D z), & (\Theta_{\text{sp}, \mathbf{b}}^X)^{-1}(z_-, z_+) &= z_- + z_+, \\ \Theta_{\text{sp}, \mathbf{b}}^Y f &= (P_{-, \mathbf{b}}^R f, P_{+, \mathbf{b}}^R f), & (\Theta_{\text{sp}, \mathbf{b}}^Y)^{-1}(f_-, f_+) &= f_- + f_+. \end{aligned}$$

The inverses are defined on the images of the corresponding projections. We then write the minus component in center and affine coordinates and the positive component in scalar coordinates, retaining the exceptional angular modes $m = \pm 2$ separately. The source graph map (6.28) is applied after this splitting. For the mean, scale, gauge, chart, and zero cell frequency equations, we apply the inverse of each diagonal operator, giving the triangular form described after (6.24).

6.7. Cutting and gluing at the interfaces. When solving the cap and annulus equations separately, we allow their solutions to have different values and conormal derivatives on the common interface. These values and derivatives are the *Cauchy data*. Their difference, with the signs specified below, is the *mismatch*, which we keep as an independent unknown. Subtracting an extension of the mismatch from the annular field gives a matching pair. Retaining the mismatch as a separate coordinate makes this change invertible.

For the value, conormal, and outer-boundary coordinates, we use $\Lambda = \langle D_\zeta \rangle$, so that $(\Lambda^t u)_n = \lambda_n^t u_n$ with λ_n as in (6.3). The change of trace variables is

$$(6.42) \quad \begin{aligned} \mathcal{T}_\Lambda(z_D, z_N, z_B) &= (\Lambda^{1/2} z_D, \Lambda^{-1/2} z_N, \Lambda^{-1/2} z_B), \\ \mathcal{T}_\Lambda^{-1}(d, n, b) &= (\Lambda^{-1/2} d, \Lambda^{1/2} n, \Lambda^{1/2} b). \end{aligned}$$

Here, z_D and z_N are the interface value and conormal data, and z_B is the datum for the first-order outer boundary condition. The factors $\Lambda^{1/2}$ and $\Lambda^{-1/2}$ account for the one-derivative difference in (4.15). We call the resulting norms the *balanced trace norms*.

At the interface Σ between a cap and an annulus, we write u_c and u_a for the two restrictions, and we define their *value and outward conormal mismatches* by

$$(6.43) \quad d_{\mathbf{b}}(u_c, u_a) = \gamma_D^a u_a - \gamma_D^c u_c, \quad n_{\mathbf{b}}(u_c, u_a) = \gamma_{N, \mathbf{b}}^{a, \text{out}} u_a + \gamma_{N, \mathbf{b}}^{c, \text{out}} u_c.$$

Here, γ_D is the trace, $\gamma_{N, \mathbf{b}}^{\text{out}}$ is the outward conormal derivative at $\mathbf{b}$, and the superscripts a and c indicate the annular and cap sides. The value mismatch is independent of the state and will be denoted by d . To compare the conormal data with those at the constant ellipse $\mathbf{b}_\rho$, we also define the *reference conormal mismatch*

$$(6.44) \quad h_\rho(u_c, u_a) := \gamma_{N, \mathbf{b}_\rho}^{a, \text{out}} u_a + \gamma_{N, \mathbf{b}_\rho}^{c, \text{out}} u_c.$$

Using the annular outward normal ν as a common direction, we set $q_\Sigma = \partial_\nu u_a - \partial_\nu u_c$. The opposite outward normals explain the sum in (6.43). Since the coefficients agree on Σ , both conormal mismatches can be expressed in terms of q_Σ and d ,

$$(6.45) \quad n_{\mathbf{b}} = e_{\Sigma, \mathbf{b}} q_\Sigma + T_{\Sigma, \mathbf{b}} d, \quad h_\rho = e_{\Sigma, \rho} q_\Sigma + T_{\Sigma, \rho} d,$$

where $e_{\Sigma, \mathbf{b}}$ and $e_{\Sigma, \rho}$ are the invertible coefficients of the common normal derivative, and $T_{\Sigma, \mathbf{b}}$ and $T_{\Sigma, \rho}$ are tangential operators of order at most one. Eliminating q_{Σ} gives

$$(6.46) \quad \begin{pmatrix} d \\ n_{\mathbf{b}} \end{pmatrix} = \mathcal{T}_{\Sigma, \mathbf{b}, \rho}^{CN} \begin{pmatrix} d \\ h_{\rho} \end{pmatrix}, \quad \mathcal{T}_{\Sigma, \mathbf{b}, \rho}^{CN} = \begin{pmatrix} I & 0 \\ K_{\Sigma, \mathbf{b}, \rho} & E_{\Sigma, \mathbf{b}, \rho} \end{pmatrix},$$

$$E_{\Sigma, \mathbf{b}, \rho} := e_{\Sigma, \mathbf{b}} e_{\Sigma, \rho}^{-1}, \quad K_{\Sigma, \mathbf{b}, \rho} := T_{\Sigma, \mathbf{b}} - E_{\Sigma, \mathbf{b}, \rho} T_{\Sigma, \rho},$$

$$(6.47) \quad (\mathcal{T}_{\Sigma, \mathbf{b}, \rho}^{CN})^{-1} = \begin{pmatrix} I & 0 \\ -E_{\Sigma, \mathbf{b}, \rho}^{-1} K_{\Sigma, \mathbf{b}, \rho} & E_{\Sigma, \mathbf{b}, \rho}^{-1} \end{pmatrix}.$$

The map and its inverse are uniformly bounded on the balanced trace spaces. Indeed, the one-derivative difference between the value and conormal norms absorbs the order of $K_{\Sigma, \mathbf{b}, \rho}$. In particular,

$$(6.48) \quad d = 0, \ n_{\mathbf{b}} = 0 \iff d = 0, \ h_{\rho} = 0,$$

so using the conormal derivative at $\mathbf{b}$ or at $\mathbf{b}_{\rho}$ gives the same matching condition. For the minus block, the cap uses Cartesian coordinates and the annulus uses the frame and spectral coordinates of (6.39) and (6.41). We denote the induced change of Cauchy coordinates by $T_{-, \mathbf{b}}$. If $\Gamma_{c, -, \mathbf{b}} u$ and $\Gamma_{a, -, \mathbf{b}} u$ represent the Cauchy data of the same field u in these two coordinate systems, then

$$T_{-, \mathbf{b}} \Gamma_{c, -, \mathbf{b}} u = \Gamma_{a, -, \mathbf{b}} u.$$

This map has order zero in the balanced trace norms. We apply it to the cap data before subtracting them from the annular data. For the positive block, we use the value and conormal mismatches directly.

We now subtract their annular extensions, (4.12)–(4.13), by setting

$$(6.49) \quad u_a^0 = u_a - C_{\Sigma, D, \mathbf{b}}^a d_{\mathbf{b}}(u_c, u_a) - C_{\Sigma, N, \mathbf{b}}^a n_{\mathbf{b}}(u_c, u_a).$$

By (4.14), the pair (u_c, u_a^0) has matching values and opposite outward conormal derivatives. It therefore agrees to first order across Σ . Keeping the original mismatch with this pair gives the mutually inverse maps

$$(6.50) \quad \mathfrak{B}_{\Sigma, \mathbf{b}}(u_c, u_a) = (u_c, u_a^0, d_{\mathbf{b}}(u_c, u_a), n_{\mathbf{b}}(u_c, u_a)),$$

$$(6.51) \quad \mathfrak{B}_{\Sigma, \mathbf{b}}^{-1}(u_c, u_a^0, d, n) = (u_c, u_a^0 + C_{\Sigma, D, \mathbf{b}}^a d + C_{\Sigma, N, \mathbf{b}}^a n).$$

On the axis, $J_A u = (u, \partial_{y_1} u, \partial_{y_2} u)|_{y=0}$ records the value and first derivatives, and E_A is their extension from (4.5). The identity $J_A E_A = I$ gives

$$u \mapsto ((I - E_A J_A)u, J_A u), \quad (u_0, z) \mapsto u_0 + E_A z, \quad J_A u_0 = 0.$$

For the outer boundary condition, the corresponding maps are

$$(6.52) \quad u \mapsto ((I - C_{\partial, M} B_M)u, B_M u), \quad (u_0, b) \mapsto u_0 + C_{\partial, M} b, \quad B_M u_0 = 0,$$

where B_M is the boundary operator (3.6) and $C_{\partial, M}$ its extension (4.18). The identity $B_M C_{\partial, M} = I$ ensures that the remainder satisfies the boundary condition and that the two maps are inverse.

The shell decomposition. On each dyadic shell, we use the extensions (4.12)–(4.13) at the constant ellipse $\mathbf{b}_{\rho}$ with the same ellipse parameter as the state, and we write

$$(6.53) \quad C_{\Sigma, N, \rho} := C_{\Sigma, N, \mathbf{b}_{\rho}}, \quad C_{\Sigma, D, \rho} := C_{\Sigma, D, \mathbf{b}_{\rho}}.$$

The reference coefficients are independent of ζ , so these extensions commute with Q_j . At a general state, coefficients may mix cell frequencies. We therefore reconstruct the full field before converting the reference conormal data to those at the state.

For the shell Q_j of (6.2), we use

$$r_j = 2TL2^{-j}, \quad 0 \leq r \leq r_j \text{ on the cap}, \quad t_j = 2^j r/L \in [0, 2T].$$

Here, $T > 0$ is the fixed matching scale, and J is large enough that $r_J < 1$. We restrict $Q_j u$ to the cap and its complementary annulus $r_j \leq r \leq 1$, rescale the cap by $t_j = 2^j r/L$, and apply (6.50) at $\mathbf{b}_\rho$. Taking the square sum over $j \geq J$ gives the spaces

$$(6.54) \quad \mathcal{X}_{\text{br},H}^s = \bigoplus_{j \geq J}^{\ell_s^2} (X_{c,j}^s \oplus X_{a,j}^s), \quad \mathcal{X}_{\text{aux},H}^s = \bigoplus_{j \geq J}^{\ell_s^2} Z_j^s.$$

Here, $X_{c,j}^s$ and $X_{a,j}^s$ are the cap and annulus spaces on shell j , while Z_j^s contains the interface pair (d, h_ρ) . For the minus block, this pair is replaced by the equivalent Cauchy datum for the outward evolution on the annulus, as described in Section 8. The notation ℓ_s^2 means that, for a sequence (u_j) of shell components, its squared norm is $\sum_{j \geq J} \|u_j\|_s^2$. The component norm $\|\cdot\|_s$ already includes the cell-frequency weights of Table 1 and the common factor λ_n^s .

On each rescaled cap at the state $\mathbf{b}$, let $\mathcal{L}_{c,-}$ and $\mathcal{L}_{c,+}$ denote the differential operators for the minus and positive equations in the rescaled coordinates. A *graph norm* measures both a solution u and its source $\mathcal{L}_{c,\pm} u$. The two norms have the form

$$\begin{aligned} \|u\|_{c,-,s} &= \|u\|_{H^{s+2}} + \|\mathcal{L}_{c,-} u\|_{H^s}, \\ \|u\|_{c,+,s} &= \|u\|_{H^{s+1}} + \|\mathcal{L}_{c,+} u\|_{H^{s-1}}. \end{aligned}$$

Here, the Sobolev norms are taken on the fixed rescaled cap, with the analytic weight (6.5). The values and conormal derivatives of the positive component on the interface have regularities $H^{s+1/2}$ and $H^{s-1/2}$, respectively, on both sides. We return each cap to the original coordinates and join it to its own complementary annulus before summing over the shells. Variable coefficients then act on the reconstructed field. This avoids comparing norms on disks rescaled by different shell frequencies.

For a field u and source F , we write $\mathcal{A}_{a,j}^X u$ and $\mathcal{A}_{a,j}^Y F$ for the restrictions of $Q_j u$ and $Q_j F$ to the annulus of shell j . Together with the rescaled cap restrictions, these give the domain and source maps for the shell decomposition,

$$(6.55) \quad \begin{aligned} \mathfrak{B}_{\text{sh},J,\rho}^X &:= \bigoplus_{j \geq J}^{\ell_s^2} \mathfrak{B}_{\Sigma_j, \mathbf{b}_\rho}(\mathcal{A}_{X,j}, \mathcal{A}_{a,j}^X), \\ \mathfrak{B}_{\text{sh},J,\rho}^{Y,\text{raw}} &:= \bigoplus_{j \geq J}^{\ell_s^2} (\mathcal{A}_{Y,j}, \mathcal{A}_{a,j}^Y, 0_{Z_j}). \end{aligned}$$

Here, $\mathcal{A}_{X,j} u = \mathcal{R}_j^X Q_j u$ and $\mathcal{A}_{Y,j} F = \mathcal{R}_j^Y Q_j F$. The maps $\mathcal{R}_j^X$ and $\mathcal{R}_j^Y$ restrict an unknown and a source to $r \leq r_j$ and pull them back to the fixed cap by $y = L2^{-j} \tilde{y}$, with $|\tilde{y}| \leq 2T$. The zero entry 0_{Z_j} records that an original Cartesian interior source prescribes no independent interface data. The reference extensions preserve each shell, and we denote the reconstruction maps by

$$(6.56) \quad \begin{aligned} \mathfrak{R}_{\text{sh},J,\rho}^X &:= (\mathfrak{B}_{\text{sh},J,\rho}^X)^{-1}, \\ \mathfrak{R}_{\text{sh},J,\rho}^{Y,\text{raw}} &:= (\mathfrak{B}_{\text{sh},J,\rho}^{Y,\text{raw}})^{-1} \quad \text{on the image of the Cartesian interior sources.} \end{aligned}$$

On the domain, we apply $\mathfrak{B}_{\Sigma_j, \mathbf{b}_\rho}^{-1}$ and join each cap to its annulus before summing over the shells. On the range, we join the interior entries and discard the zero auxiliary entry. The local inverse identities and $\sum_{j \geq J} Q_j = H_J$ show that these maps invert the restrictions of Cartesian fields.

The remaining constraints. We add the low-frequency components $|n| < 2^J$ and the chart coordinates to (6.54). Allowing arbitrary entries gives the product $\mathcal{X}_{\gamma, \text{amb}, \mathbf{b}}^s$, before matching and the other constraints are imposed. We let $\mathcal{C}_{X, \mathbf{b}}$ record the reference mismatches (d, h_ρ) on each shell, the low-frequency interface data, and the constraints in (3.7). Its target $\mathcal{Z}_{\text{aux}, \mathbf{b}}^s$ contains all these constraint values. At fixed ρ , the shell extensions are fixed, and the remaining state dependence comes from the frame and chart maps. Since the conormal derivatives at $\mathbf{b}_\rho$ and $\mathbf{b}$ impose the same matching condition by (6.48), we identify the Cartesian domain with

$$(6.57) \quad \mathcal{X}_\gamma^s \simeq \ker(\mathcal{C}_{X, \mathbf{b}} : \mathcal{X}_{\gamma, \text{amb}, \mathbf{b}}^s \longrightarrow \mathcal{Z}_{\text{aux}, \mathbf{b}}^s).$$

Subtracting the extensions in (6.49), (4.5), and (6.52), and reconstructing the field, gives a bounded map $\mathcal{R}_{X, \mathbf{b}}^{\text{br}} : \mathcal{X}_{\gamma, \text{amb}, \mathbf{b}}^s \rightarrow \mathcal{X}_\gamma^s$. It removes the mismatch and constraint data recorded by $\mathcal{C}_{X, \mathbf{b}}$ before reconstructing a constrained Cartesian field. Conversely, $\mathcal{I}_{X, \mathbf{b}}^{\text{br}}$ restricts a constrained field to the caps and annuli with zero constraint data. The matching identities and $\sum_{j \geq J} Q_j = H_J$ give

$$(6.58) \quad \mathcal{R}_{X, \mathbf{b}}^{\text{br}} \mathcal{I}_{X, \mathbf{b}}^{\text{br}} = I, \quad \mathcal{C}_{X, \mathbf{b}} \mathcal{I}_{X, \mathbf{b}}^{\text{br}} = 0,$$

$$(6.59) \quad \mathcal{I}_{X, \mathbf{b}}^{\text{br}} \mathcal{R}_{X, \mathbf{b}}^{\text{br}} U = U \quad \text{if } \mathcal{C}_{X, \mathbf{b}} U = 0.$$

Including the constraint data, we obtain the isomorphism

$$(6.60) \quad \mathcal{B}_{X, \mathbf{b}} U := (\mathcal{R}_{X, \mathbf{b}}^{\text{br}} U, \mathcal{C}_{X, \mathbf{b}} U) : \mathcal{X}_{\text{br}, \mathbf{b}}^s \xrightarrow{\sim} \mathcal{X}_{\text{sp}, \mathbf{b}}^s \oplus \mathcal{Z}_{\text{aux}, \mathbf{b}}^s,$$

whose inverse is

$$(6.61) \quad \mathcal{B}_{X, \mathbf{b}}^{-1}(u, z) = \mathcal{I}_{X, \mathbf{b}}^{\text{br}} u + \mathcal{E}_{X, \mathbf{b}}^{\text{br}} z.$$

Here, $\mathcal{X}_{\text{br}, \mathbf{b}}^s$ denotes the unconstrained product $\mathcal{X}_{\gamma, \text{amb}, \mathbf{b}}^s$, and $\mathcal{X}_{\text{sp}, \mathbf{b}}^s$ its constrained subspace, identified with $\mathcal{X}_\gamma^s$. The map $\mathcal{E}_{X, \mathbf{b}}^{\text{br}}$ extends the constraint data using the reference shell extensions and the axis and boundary extensions of Section 4. We normalize it by $\mathcal{R}_{X, \mathbf{b}}^{\text{br}} \mathcal{E}_{X, \mathbf{b}}^{\text{br}} = 0$ and $\mathcal{C}_{X, \mathbf{b}} \mathcal{E}_{X, \mathbf{b}}^{\text{br}} = I$, which gives

$$(6.62) \quad \mathcal{B}_{X, \mathbf{b}}^{-1} \mathcal{B}_{X, \mathbf{b}} = I, \quad \mathcal{B}_{X, \mathbf{b}} \mathcal{B}_{X, \mathbf{b}}^{-1} = I.$$

6.8. The augmented block operator. We now assemble the transfer maps and define the operator $\mathbb{D}_{\mathbf{b}}$ to be inverted in the following sections. Besides the interior equations, it records the interface, boundary, axis, and remaining constraints. We call these additional equations the *auxiliary equations* and the full operator the *augmented block operator*. Their right-hand sides vanish for the original Cartesian problem.

Fixed block spaces. Let $\Theta_{\text{bulk}, \mathbf{b}}^{Y, \text{raw}}$ collect the interior source restrictions and cap rescalings in (6.55), together with the low-frequency source coordinates. It does not include the auxiliary data. Before applying the trace weights and identifying the spaces with fixed reference spaces, we denote the domain product by $\mathcal{X}_{\text{ret}}^s$ and the range product by $\mathcal{Y}_{\text{aug}}^s$. The former keeps the cap and annulus fields separate, while the latter contains their sources and the auxiliary data. Their low- and high-frequency parts are

$$(6.63) \quad \begin{aligned} \mathcal{X}_{\text{ret}, L}^s &= P_J^X \mathcal{X}_{\text{ret}}^s, & \mathcal{X}_{\text{ret}, H}^s &= H_J^X \mathcal{X}_{\text{ret}}^s, \\ \mathcal{Y}_{\text{aug}, L}^s &= P_J^Y \mathcal{Y}_{\text{aug}}^s, & \mathcal{Y}_{\text{aug}, H}^s &= H_J^Y \mathcal{Y}_{\text{aug}}^s, \end{aligned}$$

using the projections of (4.2). The two diagonal blocks act on the high-frequency products. The projections are applied in these coordinates without commuting them through the state coefficients.

Axis values and first derivatives carry, respectively, the factors Λ^{-1} and Λ^{-2} of Table 1. We write $\Theta_{\text{tr}}^{X,Y}$ for the direct sum of (6.42), these axis factors, and the identity on the interior components and the remaining constraint coordinates.

On the range, the extraction of the quotients is the bounded isomorphism

$$(6.64) \quad \Theta_Q = \mathcal{Q}_Y : \mathcal{Y}_{\text{comp}}^s \longrightarrow \text{Ran } P_Z, \quad \Theta_Q^{-1} = J_Y,$$

from (3.11) and (3.10). Here, P_Z is the projection (4.23) imposing compatibility in quotient coordinates. We apply Θ_Q separately to the two outputs of (6.41), followed by the source graph map of (6.28),

$$(6.65) \quad \Theta_{\text{src},\mathbf{b}}(Z, r_{\text{aug}}) = (Z, \Psi_{\mathbf{b}}Z, r_{\text{aug}}), \quad \Theta_{\text{src},\mathbf{b}}^{-1}(Z, \Psi_{\mathbf{b}}Z, r_{\text{aug}}) = (Z, r_{\text{aug}}).$$

Only this step differentiates the prescribed source. Its inverse discards the derived entry $\Psi_{\mathbf{b}}Z$. We record the loss of derivatives by

$$(6.66) \quad \ell_Y := \max_{\alpha} \text{ord}_{\mathcal{Y}_{\gamma} \rightarrow \mathbb{Y}_{\mathfrak{q}}} \Psi_{\alpha,\mathbf{b}} < \infty,$$

where $\Psi_{\alpha,\mathbf{b}}$ gives the scalar source of component α , and ord denotes the loss between the indicated scales. Thus, ℓ_Y is the number of additional derivatives required of the Cartesian source to control every converted component. In the component norms, this means

$$\|\Psi_{\alpha,\mathbf{b}} \mathcal{Q}_Y f\|_{Y_{\alpha}^s} \leq C_s(\mathbf{b}) \|f\|_{\mathcal{Y}_{\gamma}^{s+\ell_Y}}.$$

Here, Y_{α}^s is the source space for component α , including its weight from Table 1. The orders in Table 2 give a finite bound ℓ_Y , independent of s and uniform on a sufficiently small ball of states. We also write $\ell := \ell_Y$. The conversion is applied once, before the block inverse, to avoid incurring this loss of derivatives multiple times. In particular, every term in the Neumann series acts on sources already in block coordinates.

To express the state-dependent spaces in fixed coordinates, we use, for each domain, range, cap, or trace projection $P_{\mathbf{b}}$ with reference projection P_* , the map

$$(6.67) \quad U_{P,\mathbf{b}} = P_{\mathbf{b}}P_* + (I - P_{\mathbf{b}})(I - P_*).$$

It equals the identity at $P_{\mathbf{b}} = P_*$ and is invertible when the projections are sufficiently close. Since $P_{\mathbf{b}}U_{P,\mathbf{b}} = U_{P,\mathbf{b}}P_*$, its inverse maps the range at $\mathbf{b}$ to the fixed reference range. We denote the direct sums of these inverses by $\mathcal{K}_{X,\mathbf{b}}$ and $\mathcal{K}_{Y,\mathbf{b}}$, using the weighted ℓ^2 sums of (6.54) on the high-frequency components. All subsequent direct sums over shells are understood in this sense. The fixed contours and the uniform cap rescaling bounds give uniform bounds for these maps.

Let $\mathbb{Y}_{\text{raw},\mathfrak{q}}^s$ be the block source space obtained from the compatible Cartesian sources, including the derived entry $\Psi_{\mathbf{b}}Z$. The subscript raw indicates that the independent auxiliary data have not yet been added. Inclusion into the full space $\mathbb{Y}_{\mathfrak{q}}^s$ with zero auxiliary entries and extraction of the interior source give the maps

$$(6.68) \quad \iota_{Y,\mathbf{b}} : \mathbb{Y}_{\text{raw},\mathfrak{q}}^s \longrightarrow \mathbb{Y}_{\mathfrak{q}}^s, \quad p_{Y,\mathbf{b}} : \mathbb{Y}_{\mathfrak{q}}^s \longrightarrow \mathbb{Y}_{\text{raw},\mathfrak{q}}^s, \quad p_{Y,\mathbf{b}}\iota_{Y,\mathbf{b}} = I.$$

In coordinates (F, z) , where F is an interior source and z is the auxiliary data, these maps act by

$$\iota_{Y,\mathbf{b}}F = (F, 0), \quad p_{Y,\mathbf{b}}(F, z) = F, \quad \text{aux}_{Y,\mathbf{b}}(F, z) = z.$$

Thus, the reverse composition sends (F, z) to $(F, 0)$.

The transfer maps. Composing the chart and frame maps, the spectral projections, the restrictions, the trace weights, and the identifications with the reference spaces gives the domain transfer

$$(6.69) \quad J_{X,\mathbf{b}} := \mathcal{K}_{X,\mathbf{b}} \Theta_{\text{tr}}^X \mathcal{J}_{X,\mathbf{b}}^{\text{br}} \Theta_{\text{sp},\mathbf{b}}^X \Theta_{\text{fr},\mathbf{b}} \Theta_{\text{sl},\mathbf{b}} : \mathcal{X}_\gamma^s \longrightarrow \mathbb{X}_{\mathfrak{q}}^s.$$

Reading the composition from right to left, we apply the chart derivative (6.36), the frame map (6.39), the projections (6.41), and the restriction map characterized by (6.58). The trace weights and the inverse identifications (6.67) then put these components in the fixed block spaces. Each factor acts as the identity on the components it does not affect. To return to the constrained Cartesian domain, we subtract the extensions of the constraint values and reverse the coordinate changes,

$$(6.70) \quad R_{X,\mathbf{b}} := \Theta_{\text{sl},\mathbf{b}}^{-1} \Theta_{\text{fr},\mathbf{b}}^{-1} (\Theta_{\text{sp},\mathbf{b}}^X)^{-1} \mathcal{D}_{X,\mathbf{b}}^{\text{br}} (\Theta_{\text{tr}}^X)^{-1} \mathcal{K}_{X,\mathbf{b}}^{-1} : \mathbb{X}_{\mathfrak{q}}^s \longrightarrow \mathcal{X}_\gamma^s.$$

Writing the constraint map $\mathcal{C}_{X,\mathbf{b}}$ in the same fixed coordinates gives

$$(6.71) \quad C_{X,\mathbf{b}} : \mathbb{X}_{\mathfrak{q}}^s \longrightarrow \mathcal{Z}_{\text{aux},\mathfrak{q}}^s.$$

From (6.58)–(6.61), we then obtain the identities

$$(6.72) \quad R_{X,\mathbf{b}} J_{X,\mathbf{b}} = I, \quad C_{X,\mathbf{b}} J_{X,\mathbf{b}} = 0, \quad J_{X,\mathbf{b}} R_{X,\mathbf{b}} U = U \quad \text{if } C_{X,\mathbf{b}} U = 0.$$

We write the extension $\mathcal{E}_{X,\mathbf{b}}^{\text{br}}$ of (6.61) in the same fixed coordinates and denote it by $E_{X,\mathbf{b}} : \mathcal{Z}_{\text{aux},\mathfrak{q}}^s \rightarrow \mathbb{X}_{\mathfrak{q}}^s$. The normalization of this extension gives

$$(6.73) \quad R_{X,\mathbf{b}} E_{X,\mathbf{b}} = 0, \quad C_{X,\mathbf{b}} E_{X,\mathbf{b}} = I, \quad U = J_{X,\mathbf{b}} R_{X,\mathbf{b}} U + E_{X,\mathbf{b}} C_{X,\mathbf{b}} U.$$

Thus, a block field is determined by its constrained Cartesian part and its constraint data. Equivalently, $(R_{X,\mathbf{b}}, C_{X,\mathbf{b}})$ identifies $\mathbb{X}_{\mathfrak{q}}^s$ with $\mathcal{X}_\gamma^s \oplus \mathcal{Z}_{\text{aux},\mathfrak{q}}^s$, while $J_{X,\mathbf{b}}$ identifies the Cartesian domain with $\ker C_{X,\mathbf{b}}$.

For the source transfer, we split the compatible source into blocks, extract its quotient coordinates, and restrict to the cap and annulus. The map $\iota_{\text{aux}}^{\text{pre}}$ acts at this stage by $F \mapsto (F, 0)$, before the trace weights and fixed-space identifications are applied. Applying the source graph map, trace weights, and identifications with the reference spaces then gives

$$(6.74) \quad \iota_{Y,\mathbf{b}} J_{Y,\mathbf{b}}^{\text{raw}} := \mathcal{K}_{Y,\mathbf{b}} \Theta_{\text{tr}}^Y \Theta_{\text{src},\mathbf{b}} \iota_{\text{aux}}^{\text{pre}} \Theta_{\text{bulk},\mathbf{b}}^{Y,\text{raw}} \Theta_Q \Theta_{\text{sp},\mathbf{b}}^Y : \mathcal{Y}_\gamma^{s+\ell_Y} \longrightarrow \mathbb{Y}_{\mathfrak{q}}^s.$$

Reversing these steps, we discard the derived entry $\Psi_{\mathbf{b}} Z$, recombine the interior Fourier components, and recover the Cartesian source from its quotient coordinates. This gives a left inverse

$$(6.75) \quad R_{Y,\mathbf{b}}^{\text{raw}} : \mathbb{Y}_{\text{raw},\mathfrak{q}}^s \longrightarrow \mathcal{Y}_\gamma^s.$$

On smooth Cartesian source coordinates, these maps are mutually inverse. At finite Sobolev regularity, the forward conversion loses ℓ_Y derivatives, so the two compositions are the inclusions

$$(6.76) \quad R_{Y,\mathbf{b}}^{\text{raw}} J_{Y,\mathbf{b}}^{\text{raw}} = \iota_{s+\ell_Y,s}^{\mathcal{Y}},$$

$$(6.77) \quad J_{Y,\mathbf{b}}^{\text{raw}} R_{Y,\mathbf{b}}^{\text{raw}} = \iota_{s+\ell_Y,s}^{\mathbb{Y}_{\text{raw}}} \quad \text{on the Cartesian source image,}$$

where the right-hand sides include the same source from index $s + \ell_Y$ into index s . In particular, $\iota_{Y,\mathbf{b}} J_{Y,\mathbf{b}}^{\text{raw}}$ is injective on the compatible scale.

The augmented operator. The domain coordinates record the reference conormal mismatch, whereas the interface equations use the conormal derivative at $\mathbf{b}$. We therefore apply $(\mathcal{T}_{\mathbf{b},\rho}^{CN})^{-1}$ from (6.47) to the interface equations. Here, $\mathcal{T}_{\mathbf{b},\rho}^{CN}$ denotes the collection of the maps $\mathcal{T}_{\Sigma,\mathbf{b},\rho}^{CN}$ over all interfaces. This conversion acts after

$\mathfrak{R}_{\text{sh},J,\rho}^X$ reconstructs the full field, so its variable coefficients may couple different shells. It and its inverse are analytic and bounded without loss on the balanced trace product, and equal the identity at $\mathbf{b} = \mathbf{b}_\rho$.

We let $\mathbb{N}_{\mathbf{b}}^{\text{cur}} : \mathbb{X}_{\mathfrak{d}}^s \rightarrow \mathbb{Y}_{\mathfrak{d}}^s$ be the operator obtained by applying the interior equations at the state $\mathbf{b}$ separately on the cap and on the annulus and recording the interface pair $(d, n_{\mathbf{b}})$ at the state. We let $\mathcal{R}_{CN,\mathbf{b},\rho}$ act as the identity on the other data and as $(\mathcal{T}_{\mathbf{b},\rho}^{CN})^{-1}$ on this pair, and we set

$$(6.78) \quad \mathbb{N}_{\mathbf{b}} := \mathcal{R}_{CN,\mathbf{b},\rho} \mathbb{N}_{\mathbf{b}}^{\text{cur}},$$

so that its auxiliary pair is the reference pair

$$(6.79) \quad (\mathcal{T}_{\mathbf{b},\rho}^{CN})^{-1} \begin{pmatrix} d \\ n_{\mathbf{b}} \end{pmatrix} = \begin{pmatrix} d \\ h_\rho \end{pmatrix}.$$

The other auxiliary entries record the mismatch between the cap and annular Cauchy data for the minus block, together with the outer boundary, axis, mean, scale, gauge, and chart conditions. The separate block inverses are first constructed with the conormal data at $\mathbf{b}$, and the conversion is then included in their perturbation estimates. In particular, an inverse $G_{\mathbf{b}}^{\text{cur}}$ of $\mathbb{N}_{\mathbf{b}}^{\text{cur}}$ gives the inverse

$$(6.80) \quad G_{\mathbf{b}}^{\mathbb{N}} = G_{\mathbf{b}}^{\text{cur}} \mathcal{R}_{CN,\mathbf{b},\rho}^{-1}$$

of $\mathbb{N}_{\mathbf{b}}$, with the converse obtained by composing with $\mathcal{R}_{CN,\mathbf{b},\rho}$.

There is one further adjustment. The extension of nonzero constraint data contributes to the interior equations, and we subtract this contribution so that the interior output depends only on the constrained field. Let $\mathbb{P}_{\text{bulk},\mathbf{b}} U$ be the output $\mathbb{N}_{\mathbf{b}} U$ with its auxiliary entries set to zero, namely

$$\mathbb{P}_{\text{bulk},\mathbf{b}} = \iota_{Y,\mathbf{b}} p_{Y,\mathbf{b}} \mathbb{N}_{\mathbf{b}}.$$

We also let $j_{Z,\mathbf{b}} : \mathcal{Z}_{\text{aux},\mathfrak{d}}^s \rightarrow \mathbb{Y}_{\mathfrak{d}}^s$ insert auxiliary data with zero interior source. Thus,

$$(6.81) \quad \text{aux}_{Y,\mathbf{b}} j_{Z,\mathbf{b}} = I, \quad p_{Y,\mathbf{b}} j_{Z,\mathbf{b}} = 0.$$

Without using any inverse of $L_{\mathbf{b}}$, we define

$$(6.82) \quad \mathbb{B}_{\mathbf{b}} := \mathbb{P}_{\text{bulk},\mathbf{b}} \mathbb{E}_{X,\mathbf{b}} : \mathcal{Z}_{\text{aux},\mathfrak{d}}^s \longrightarrow \ker \text{aux}_{Y,\mathbf{b}},$$

so that $\mathbb{B}_{\mathbf{b}} z$ is the interior source produced by the extension. Applying $\mathbb{N}_{\mathbf{b}}$ to the decomposition (6.73) gives

$$(6.83) \quad \mathbb{N}_{\mathbf{b}} U = \iota_{Y,\mathbf{b}} \mathbf{J}_{Y,\mathbf{b}}^{\text{raw}} L_{\mathbf{b}} R_{X,\mathbf{b}} U + \mathbb{B}_{\mathbf{b}} C_{X,\mathbf{b}} U + j_{Z,\mathbf{b}} C_{X,\mathbf{b}} U.$$

We remove the second term by a change of range variables that leaves the auxiliary data unchanged. For a range element $F + j_{Z,\mathbf{b}} z$, with $F \in \ker \text{aux}_{Y,\mathbf{b}}$, we set

$$(6.84) \quad \begin{aligned} \mathcal{S}_{Y,\mathbf{b}}(F + j_{Z,\mathbf{b}} z) &= F - \mathbb{B}_{\mathbf{b}} z + j_{Z,\mathbf{b}} z, \\ \mathcal{S}_{Y,\mathbf{b}}^{-1}(F + j_{Z,\mathbf{b}} z) &= F + \mathbb{B}_{\mathbf{b}} z + j_{Z,\mathbf{b}} z. \end{aligned}$$

Both maps are analytic and bounded without loss of derivatives. We can now define the augmented block operator by

$$(6.85) \quad \mathbb{D}_{\mathbf{b}} := \mathcal{S}_{Y,\mathbf{b}} \mathbb{N}_{\mathbf{b}} : \mathbb{X}_{\mathfrak{d}}^s \longrightarrow \mathbb{Y}_{\mathfrak{d}}^s,$$

$$(6.86) \quad \mathbb{D}_{\mathbf{b}} U = \iota_{Y,\mathbf{b}} \mathbf{J}_{Y,\mathbf{b}}^{\text{raw}} L_{\mathbf{b}} R_{X,\mathbf{b}} U + j_{Z,\mathbf{b}} C_{X,\mathbf{b}} U.$$

The auxiliary output records the constraints, while on the constrained domain the interior output agrees with the Cartesian equations. For $U \in \mathbb{X}_{\mathfrak{b}}^\infty$ and $u \in \mathcal{X}_\gamma^\infty$, the preceding formula gives

$$(6.87) \quad \text{aux}_{Y,\mathfrak{b}} \mathbb{D}_{\mathfrak{b}} U = C_{X,\mathfrak{b}} U,$$

$$(6.88) \quad \mathbb{D}_{\mathfrak{b}} J_{X,\mathfrak{b}} u = \iota_{Y,\mathfrak{b}} J_{Y,\mathfrak{b}}^{\text{raw}} L_{\mathfrak{b}} u, \quad u \in \mathcal{X}_\gamma^\infty.$$

By boundedness, these identities extend to the closures of the smooth elements in the indicated norms. If $\mathbb{D}_{\mathfrak{b}} U = \iota_{Y,\mathfrak{b}} J_{Y,\mathfrak{b}}^{\text{raw}} f$, the first identity gives $C_{X,\mathfrak{b}} U = 0$, so $U = J_{X,\mathfrak{b}} u$ with $u = R_{X,\mathfrak{b}} U$. The second identity and injectivity of the source transfer then give $L_{\mathfrak{b}} u = f$. Thus, solving the augmented problem with zero auxiliary data recovers a solution of the Cartesian problem.

The domains, targets, and roles of the transfer maps are listed in Table 3.

Map	Domain	Target	Role
$J_{X,\mathfrak{b}}$	$\mathcal{X}_\gamma^s$	$\mathbb{X}_{\mathfrak{b}}^s$	Expresses a Cartesian unknown in block variables.
$R_{X,\mathfrak{b}}$	$\mathbb{X}_{\mathfrak{b}}^s$	$\mathcal{X}_\gamma^s$	Returns its constrained Cartesian component.
$C_{X,\mathfrak{b}}$	$\mathbb{X}_{\mathfrak{b}}^s$	$\mathcal{Z}_{\text{aux},\mathfrak{b}}^s$	Records the constraint data.
$E_{X,\mathfrak{b}}$	$\mathcal{Z}_{\text{aux},\mathfrak{b}}^s$	$\mathbb{X}_{\mathfrak{b}}^s$	Extends the constraint data.
$J_{Y,\mathfrak{b}}^{\text{raw}}$	$\mathcal{Y}_\gamma^{s+\ell_Y}$	$\mathbb{Y}_{\text{raw},\mathfrak{b}}^s$	Converts the Cartesian source.
$R_{Y,\mathfrak{b}}^{\text{raw}}$	$\mathbb{Y}_{\text{raw},\mathfrak{b}}^s$	$\mathcal{Y}_\gamma^s$	Reconstructs the Cartesian source.
$\iota_{Y,\mathfrak{b}}$	$\mathbb{Y}_{\text{raw},\mathfrak{b}}^s$	$\mathbb{Y}_{\mathfrak{b}}^s$	Adds zero auxiliary data.
$p_{Y,\mathfrak{b}}$	$\mathbb{Y}_{\mathfrak{b}}^s$	$\mathbb{Y}_{\text{raw},\mathfrak{b}}^s$	Extracts the interior source coordinates.
$\text{aux}_{Y,\mathfrak{b}}$	$\mathbb{Y}_{\mathfrak{b}}^s$	$\mathcal{Z}_{\text{aux},\mathfrak{b}}^s$	Extracts the auxiliary data.
$JZ,\mathfrak{b}$	$\mathcal{Z}_{\text{aux},\mathfrak{b}}^s$	$\mathbb{Y}_{\mathfrak{b}}^s$	Inserts auxiliary data with zero interior source.
$\mathcal{R}_{CN,\mathfrak{b},\rho}$	$\mathbb{Y}_{\mathfrak{b}}^s$	$\mathbb{Y}_{\mathfrak{b}}^s$	Converts Cauchy data to reference coordinates.
$\mathcal{S}_{Y,\mathfrak{b}}$	$\mathbb{Y}_{\mathfrak{b}}^s$	$\mathbb{Y}_{\mathfrak{b}}^s$	Subtracts the extension's interior contribution.

TABLE 3. The transfer maps. The last two maps are invertible on the indicated spaces.

The separate high-frequency block operators. We construct inverses for the minus and positive blocks separately. For $\alpha \in \{-, +\}$, let $\iota_\alpha^{X,Y}$ include block α in the full domain or range, and let $\pi_\alpha^{X,Y}$ extract that block. We first take the diagonal blocks of the two factors separately, $N_{\alpha\alpha} := \pi_\alpha^Y \mathbb{N}_{\mathfrak{b}} \iota_\alpha^X$ and $\mathcal{S}_{\alpha\alpha} := \pi_\alpha^Y \mathcal{S}_{Y,\mathfrak{b}} \iota_\alpha^Y$. If $G_{\alpha\alpha}^{\mathbb{N}}$ is the inverse of $N_{\alpha\alpha}$, we obtain mutually inverse operators on the spaces of block α by setting

$$(6.89) \quad \widehat{D}_{\alpha,\mathfrak{b}} := \mathcal{S}_{\alpha\alpha} N_{\alpha\alpha}, \quad \widehat{G}_{\alpha,\mathfrak{b}} := G_{\alpha\alpha}^{\mathbb{N}} \mathcal{S}_{\alpha\alpha}^{-1}.$$

The diagonal block of the full product also contains a term passing through the complementary range,

$$\pi_\alpha^Y \mathbb{D}_{\mathfrak{b}} \iota_\alpha^X = \widehat{D}_{\alpha,\mathfrak{b}} + \pi_\alpha^Y \mathcal{S}_{Y,\mathfrak{b}} (I - \iota_\alpha^Y \pi_\alpha^Y) \mathbb{N}_{\mathfrak{b}} \iota_\alpha^X.$$

The additional term passes through the other high-frequency block or the low-frequency part. Thus, the separately constructed $\widehat{D}_{\alpha,\mathfrak{b}}$ need not equal the diagonal block of $\mathbb{D}_{\mathfrak{b}}$. We keep the auxiliary equations in this

comparison and estimate the additional term in [Section 10](#). On the constrained Cartesian domain, we have

$$(6.90) \quad p_{Y,\mathbf{b}} \mathbb{D}_{\mathbf{b}} J_{X,\mathbf{b}} = J_{Y,\mathbf{b}}^{\text{raw}} L_{\mathbf{b}}.$$

The mapping properties of the transfer maps and the estimates for their derivatives with respect to $\mathbf{b}$ are summarized in the following proposition.

Proposition 6.2 (Transfer maps between Cartesian and block variables). *On a sufficiently small ball of states in the analytic norm, the source loss ℓ_Y in (6.66) is finite and independent of the Sobolev index. The transfer maps satisfy*

$$(6.91) \quad J_{X,\mathbf{b}}, R_{X,\mathbf{b}}, C_{X,\mathbf{b}}, E_{X,\mathbf{b}}, \mathcal{R}_{CN,\mathbf{b},\rho}^{\pm 1}, \mathcal{S}_{Y,\mathbf{b}}^{\pm 1} \text{ are bounded between the spaces listed in Table 3,}$$

$$(6.92) \quad J_{Y,\mathbf{b}}^{\text{raw}} : \mathcal{Y}_{\gamma}^{s+\ell_Y} \longrightarrow \mathbb{Y}_{\text{raw},\mathfrak{q}}^s, \quad R_{Y,\mathbf{b}}^{\text{raw}} : \mathbb{Y}_{\text{raw},\mathfrak{q}}^s \longrightarrow \mathcal{Y}_{\gamma}^s.$$

The maps in (6.68) lose no derivatives. All the maps above preserve real data, with the conjugation rule of [Section 4.3](#) in quotient coordinates. They depend analytically on $\mathbf{b}$ with the domain and target regularities listed in [Table 3](#) and (6.92). Every derivative of fixed order with respect to $\mathbf{b}$ satisfies a tame estimate with at most one factor measured in a higher regularity norm and with regularity shifts independent of s . Once the inverse $G_{\mathbf{b}}$ of $\mathbb{D}_{\mathbf{b}}$ has been constructed, the corresponding Cartesian inverse is

$$(6.93) \quad R_{X,\mathbf{b}} G_{\mathbf{b}} \iota_{Y,\mathbf{b}} J_{Y,\mathbf{b}}^{\text{raw}}.$$

The source conversion contributes the loss ℓ_Y once, including after differentiation with respect to $\mathbf{b}$, in addition to any loss in the estimates for $G_{\mathbf{b}}$.

Proof. Inverse and constraint identities. The extension identity (4.14), together with $J_A E_A = I$ from (4.5) and $B_M C_{\partial,M} = I$ from (4.18), shows that subtracting the extensions imposes the constraints and adding them back reconstructs the field. On the shells, this gives (6.58)–(6.62). Composing with the chart and frame maps, the spectral projections, and the identifications with the reference spaces gives (6.72) and (6.73). The reference extensions commute with Q_j , and each cap is joined to its annulus before summation. Thus, the coefficients at the state act on the reconstructed field in the original coordinates.

Discarding the derived entry $\Psi_{\mathbf{b}} Z$ and reconstructing the Cartesian source from its quotient coordinates gives (6.76)–(6.77). Adding zero auxiliary entries and then extracting the interior source gives (6.68). Substitution in (5.10)–(5.13) yields the interior part of (6.83), while (6.47) converts the interface data to the reference mismatch without changing the matching condition. Applying $\mathcal{S}_{Y,\mathbf{b}}$ cancels the interior contribution of the extension and gives (6.86). The identities (6.81) and (6.72) then imply (6.87) and (6.88).

Bounds for the transfer maps. We use the weights of [Table 1](#). The chart, frame, and quotient maps are bounded at the same regularity index on their respective spaces. The spectral projections are controlled by the uniform resolvent bounds on the fixed contours, and (6.67) is invertible with bounded inverse because the state projections are close to their reference projections.

The trace and extension bounds follow from (4.6), (4.9), and (4.15). Orthogonality of the shells gives the bounds for their infinite sums. For the conormal conversion, the one-derivative difference between the value and conormal norms absorbs $K_{\Sigma,\mathbf{b},\rho}$, so (6.46)–(6.47) are bounded without loss, as are their state derivatives. The map (6.82), which gives the interior source produced by an extension, is bounded on the same scales.

Hence, subtracting this source as in (6.84) is also bounded. Only $\Theta_{\text{src}, \mathbf{b}}$ differentiates the prescribed source, with the loss ℓ_Y of (6.66).

For derivatives with respect to $\mathbf{b}$, differentiating $G_{\mathbf{b}} G_{\mathbf{b}}^{-1} = I$, with $G_{\mathbf{b}}$ as in (6.38), gives

$$D_{\mathbf{b}}(G_{\mathbf{b}}^{-1})[h] = -G_{\mathbf{b}}^{-1}(D_{\mathbf{b}}G_{\mathbf{b}}[h])G_{\mathbf{b}}^{-1}.$$

We combine this identity with the product estimate (3.20), differentiation of the contour formulas (6.21), and the Neumann series for the inverse of (6.67). Differentiation changes the coefficients of the source map without increasing its differential order. Since each term contains exactly one source conversion factor, the loss ℓ_Y still occurs only once. Finally, each factor respects complex conjugation in its coordinates, so the transfer maps preserve real data. $\square$

We close this section with the decomposition of $\mathbb{D}_{\mathbf{b}}$ into low and high cell frequencies, which organizes the rest of the linear analysis. In the notation of (4.2), we set

$$P = P_{J, \mathfrak{h}} := \mathbf{1}_{\{|n| < 2^J\}}(D_{\zeta}), \quad H = H_{J, \mathfrak{h}} := \mathbf{1}_{\{|n| \geq 2^J\}}(D_{\zeta}),$$

with superscripts X and Y distinguishing the domain and range. Although P selects finitely many cell frequencies, its range remains infinite-dimensional because the radial and angular variables are unrestricted. With the low-frequency variables first, the decomposition is

$$(6.94) \quad \mathbb{D}_{\mathbf{b}} = \begin{pmatrix} A_{\mathbf{b}} & B_{\mathbf{b}} \\ C_{\mathbf{b}} & D_{H, \mathbf{b}} \end{pmatrix}, \quad \begin{aligned} A_{\mathbf{b}} &= P^Y \mathbb{D}_{\mathbf{b}} P^X, & B_{\mathbf{b}} &= P^Y \mathbb{D}_{\mathbf{b}} H^X, \\ C_{\mathbf{b}} &= H^Y \mathbb{D}_{\mathbf{b}} P^X, & D_{H, \mathbf{b}} &= H^Y \mathbb{D}_{\mathbf{b}} H^X. \end{aligned}$$

7. THE INVERSE AT A FIXED ELLIPSE

We now invert the Cartesian linearization (5.7) at a constant reference ellipse, with a finite loss of derivatives of the source. Since its coefficients are independent of ζ , we can solve separately at each cell frequency. We also show that the estimates for the even and affine components remain uniform as the ellipse approaches a circle.

7.1. The reference ellipse and the inverse estimate. We fix a constant orientation α_0 and set

$$(7.1) \quad M_e = R_{\alpha_0} \begin{pmatrix} \sqrt{1 + \rho_*} & 0 \\ 0 & \sqrt{1 - \rho_*} \end{pmatrix} R_{-\alpha_0}, \quad 0 < \rho_* < \rho_0 < \frac{1}{4},$$

which is the matrix (2.33) with $\delta = 0$ and $\rho = \rho_*$. Here, $\text{tr}(M_e^T M_e) = 2$, as in (3.2), so M_e belongs to the family (4.1), and the corresponding unperturbed solution maps every cross-section of the cell to the same ellipse with parameter ρ_* . We let

$$L_e : \mathcal{X}_{M_e}^{\infty} \longrightarrow \mathcal{Y}_{\text{comp}}^{\infty}$$

denote the linearization (5.7) in the normalized chart. Its domain satisfies the axis, mean, scale, gauge, and outer boundary conditions of Section 3.1, and its range is the compatible space $\mathcal{Y}_{\text{comp}}^{\infty}$ of Section 3.2. We impose these conditions before separating the angular modes $e^{im\theta}$ by the *parity* of m , so that the domain and range have matching odd and even components. In the estimate below, we use the analytic scale of Section 3.3 and suppress the weight subscript.

Proposition 7.1 (The reference inverse at a constant ellipse). *If ρ_0 is sufficiently small, then for every fixed $0 < \rho_* < \rho_0$, the map L_e is an invertible linear map. Its inverse V_e satisfies, for every $s \geq s_*$,*

$$(7.2) \quad \|V_e f\|_{\mathcal{X}^s} \leq C_{e,s} \|f\|_{Y^{s+\ell_e}}$$

for one finite ℓ_e , which is independent of the cell frequency and of every angular or cell Fourier cutoff. A constant change of orientation conjugates this inverse by a unitary map.

The proof occupies the next seven subsections. In scalar coordinates, the problem splits into odd and even angular modes. For the odd block, we solve on the cap and annulus, match at the interface, and establish the analytic estimates. The even block also contains the affine modes $m = \pm 2$, which require a separate calculation. We then combine the two inverses and return to Cartesian variables.

7.2. Scalar variables and even and odd angular modes. A constant rotation is unitary on every Cartesian scale, so we may take $\alpha_0 = 0$, and we write $\rho = \rho_*$. For the tangent vector U and the scalar S , we introduce the scalar coordinates

$$A = y \cdot U_\perp, \quad B = \mathcal{R}y \cdot U_\perp, \quad C = U \cdot e_T.$$

Here, S is the variation of w , $U_\perp = M_e^T \iota^T U$, and $\mathcal{R}y = (-y_2, y_1)$. Thus, $A = (\mathcal{D}v_{M_e}) \cdot U$ and $B = (\mathcal{R}v_{M_e}) \cdot U$ are the combinations appearing in (5.10)–(5.13), while C is the tangential component of U . We retain the prescribed angular averages and the scale coordinate associated with (3.2). We set $Z = A + iB$, $W = A - iB$, and $z = y_1 + iy_2$. With C^∞ denoting smooth functions of y , the divisibility conditions

$$Z \in \bar{z}C^\infty, \quad W \in zC^\infty, \quad f_1 + if_0 \in \bar{z}C^\infty, \quad f_1 - if_0 \in zC^\infty, \quad f_3 \in z\bar{z}C^\infty$$

are part of the definitions of the domain and of the compatible range in Section 3, so we may use them before taking any quotient. The maps to and from the scalar coordinates are bounded, with a finite shift in the Sobolev index, which we call ℓ_{alg} in the proof of Proposition 7.1.

At a nonzero cell frequency n , we set

$$t = \frac{|n|r}{L}, \quad D = t\partial_t, \quad \mathcal{R} = \partial_\theta, \quad \sigma = \text{sgn } n.$$

We also rescale the tangential component, writing $C = (L/|n|)U \cdot e_T$ for the rest of the nonzero-frequency calculation. We keep the first two source entries unchanged, divide the third by $|n|$, and divide the fourth by $-L \det M_e$. We continue to write these normalized sources as $f_0, \dots, f_3$. With these conventions, the coefficients in (7.3) depend on n only through σ . Thus, $t = 1$ corresponds to $r = L/|n|$, which is the radius of the cap at cell frequency n , and the outer boundary $r = 1$ corresponds to $t = |n|/L$. Below, we take Cartesian derivatives in the rescaled coordinates. If

$$\begin{pmatrix} \mathbf{a} \\ \mathbf{b} \end{pmatrix} = H_\rho(\theta) \begin{pmatrix} A \\ B \end{pmatrix}, \quad H_\rho(\theta) = \frac{1}{1-\rho^2} \begin{pmatrix} 1-\rho \cos 2\theta & \rho \sin 2\theta \\ \rho \sin 2\theta & 1+\rho \cos 2\theta \end{pmatrix},$$

then the four equations for (A, B, C, S) are

$$(7.3) \quad \begin{aligned} f_0 &= \mathcal{R}S - 2A - 2\mathcal{R}B, \\ f_1 &= (I - \Pi)(DS - DB - \mathcal{R}A + 2B), \\ f_2 &= (I - \Pi)(\mathcal{R}C + i\sigma B - i\sigma S), \\ f_3 &= (I - \Pi)(D\mathbf{a} + \mathcal{R}\mathbf{b} + i\sigma t^2 C). \end{aligned}$$

Here, Π takes the angular average, so $I - \Pi$ removes the mode $m = 0$. The ellipse enters only through H_ρ , which is the identity at $\rho = 0$. Thus, with

$$(7.4) \quad X = S - B, \quad g_\pm = f_1 \pm if_0,$$

the first three equations, in which the ellipse does not occur, become on the nonzero angular modes

$$(7.5) \quad \begin{aligned} X &= i\sigma(f_2 - \mathcal{R}C), \\ (\mathcal{R} + 2i)Z &= (D + i\mathcal{R})X - g_+, \\ (\mathcal{R} - 2i)W &= (D - i\mathcal{R})X - g_-. \end{aligned}$$

The nonconstant entries of H_ρ are combinations of $e^{\pm 2i\theta}$, so they shift the angular mode by ± 2 and preserve its parity. The divisibility, axis, and boundary conditions also preserve parity. Hence, the closed scalar operator $\mathcal{L}_e$, including the mean and scale equations, splits as

$$(7.6) \quad \mathcal{L}_e = \mathcal{L}_e^{\text{od}} \oplus \mathcal{L}_e^{\text{ev, aug}}.$$

We include the angular means and the scale equation in the even summand. They couple to the other even components only through the modes $m = \pm 2$, and the resulting system is triangular.

7.3. The odd block on the cap. On odd angular modes, $\mathcal{R} \pm 2i$ are invertible because their multipliers are $i(m \pm 2)$. We solve (7.5) for Z and W and substitute them into the last equation of (7.3). Then, $u = |\mathcal{R}|C$, where $|\mathcal{R}|$ has multiplier $|m|$, satisfies

$$(7.7) \quad \mathcal{A}_\rho u = \phi(f), \quad \mathcal{A}_\rho = P\Delta_y - 2\rho\mathbf{H}(\partial_z^2 + \partial_{\bar{z}}^2)\mathbf{H} - (1 - \rho^2)I.$$

Here, $\phi(f)$ collects the terms depending on $f = (f_0, f_1, f_2, f_3)$ after substituting

$$Z = (\mathcal{R} + 2i)^{-1}((D + i\mathcal{R})X - g_+), \quad W = (\mathcal{R} - 2i)^{-1}((D - i\mathcal{R})X - g_-),$$

with $X = i\sigma(f_2 - \mathcal{R}C)$ from (7.5). The angular operators in (7.7) are

$$P = p(-i\partial_\theta), \quad p_m = \frac{m^2}{m^2 - 4}, \quad \mathbf{H} = \text{sgn}(-i\partial_\theta).$$

The nonconstant coefficients of H_ρ couple the equation for mode m to modes $m \pm 2$. Here and below, u_m denotes the coefficient of $e^{im\theta}$ in the angular Fourier series of u , and likewise $f_{i,m}$ denotes that coefficient of the source entry f_i . We obtain (7.7) from the identities

$$\begin{aligned} t^{-2}(D^2 + (2 - 2m)D + m(m - 2))u_{m-2}e^{im\theta} &= 4\partial_{\bar{z}}^2(u_{m-2}e^{i(m-2)\theta}), \\ t^{-2}(D^2 + (2m + 2)D + m(m + 2))u_{m+2}e^{im\theta} &= 4\partial_z^2(u_{m+2}e^{i(m+2)\theta}). \end{aligned}$$

The perturbation from the circle therefore involves the Cartesian operators ∂_z^2 and $\partial_{\bar{z}}^2$. The operators at the circle and ellipse act on the same smooth Cartesian functions with the same axis and boundary constraints. The divisibility of the compatible sources controls the quotients by z , $\bar{z}$, and $z\bar{z}$ in this substitution. Since at most three derivatives fall on f , we obtain, independently of all cutoffs,

$$(7.8) \quad \|\phi(f)\|_{H_y^s} \leq C_s \|f\|_{\mathbf{y}^{s+3}}.$$

Here, H_y^s denotes the Sobolev space of index s in the rescaled disk variable, and the norm on the right is that of the analytic scale of Section 3.3, restricted to the cell frequency n as in (7.45) below.

On the cap, which is the region $0 < t < 1$ in the rescaled variable, we let

$$\mathcal{L}_{\rho, \text{ax}} u = (\mathcal{A}_\rho u, \mathbf{j}_c u, \gamma_h u), \quad \gamma_h u = \Pi_{|m| \geq 3} u|_{t=1}.$$

Here, $j_c u$ records the coefficients $\alpha_{\pm 1}$ in $u_{\pm 1}(t) = \alpha_{\pm 1}t + O(t^3)$, while $\gamma_h u$ records the values of the higher odd modes on the cap boundary. The modes $m = \pm 1$ are the *center modes*, for which $p_{\pm 1} = -1/3$, while $|m| \geq 3$ are the *positive modes*, for which $p_m \geq 1$. At the circle, the center equation is $(\Delta_y + 3)c = -3\phi_c$, with ϕ_c the center component of $\phi(f)$. On each mode, this is the Bessel equation of order one and frequency $\sqrt{3}$. We let

$$u_0(t) = \frac{2}{\sqrt{3}}J_1(\sqrt{3}t), \quad v_0(t) = Y_1(\sqrt{3}t), \quad \mathcal{W} = u_0 v'_0 - u'_0 v_0$$

be its regular solution, its singular solution, and their Wronskian, so that $u_0(t) = t + O(t^3)$. The solution of the center equation with prescribed leading coefficient α is then given by the variation of constants formula

$$(7.9) \quad c(t) = \alpha u_0(t) - u_0(t) \int_0^t \frac{v_0(s)F(s)}{\mathcal{W}(s)} ds + v_0(t) \int_0^t \frac{u_0(s)F(s)}{\mathcal{W}(s)} ds.$$

Here, $F = -3\phi_c$. On $|m| \geq 3$, we solve the Dirichlet problem for $p_m \Delta_y - I$. The ellipticity is uniform in m because $1 \leq p_m \leq 9/5$. We call its solution operator the *Dirichlet Green map*. We denote their direct sum by $G_{0,\text{ax}}$. Thus, $G_{0,\text{ax}}(\phi, \alpha, a)$ solves the cap equation with interior source ϕ , center coefficients $\alpha = (\alpha_1, \alpha_{-1})$, and higher-mode boundary value a . For $\rho \neq 0$, the cap inverse is

$$(7.10) \quad G_{\rho,\text{ax}} = [I + G_{0,\text{ax}}(\mathcal{L}_{\rho,\text{ax}} - \mathcal{L}_{0,\text{ax}})]^{-1} G_{0,\text{ax}}.$$

Indeed, we have the estimate

$$\|(\mathcal{L}_{\rho,\text{ax}} - \mathcal{L}_{0,\text{ax}})u\|_{H^s} \leq C_{\text{dif}}(|\rho| + \rho^2) \|u\|_{H^{s+2}},$$

uniformly under restriction to $|m| \leq m_0$, which we call an *angular cutoff*. Decreasing ρ_0 makes the bracket in (7.10) invertible by a Neumann series at the base index. Cartesian difference quotients give the higher regularity estimates with the same ρ_0 . The prescribed axis conditions remain part of the domain, and the Cartesian compatibility and divisibility conditions remain part of the source space.

7.4. The odd block on the annulus. On the annulus $t \geq 1$, we write (7.7) as an evolution in the radial variable, using $D = t\partial_t$ as the derivative,

$$(7.11) \quad M_\rho D^2 u + K_\rho D u + N_\rho u - (1 - \rho^2)t^2 u = F,$$

on the scale of sequences

$$\mathfrak{h}^s = \left\{ (u_m)_{m \in 2\mathbb{Z}+1} : \sum_m \langle m \rangle^{2s} |u_m|^2 < \infty \right\}.$$

Here, u is the sequence of odd angular modes and $F = t^2 \phi(f)$. The matrices M_ρ and N_ρ are self-adjoint, and K_ρ is skew-adjoint. Each row couples only the adjacent odd modes m and $m \pm 2$. At the circle,

$$(M_0)_{mm} = p_m, \quad K_0 = 0, \quad (N_0)_{mm} = -p_m m^2.$$

We call $N_\rho u$ the *angular term*, since at the circle it is the multiplier $-p_m m^2$, and we call $(1 - \rho^2)t^2 u$ the *mass term*. Expanding the two off-diagonal entries of H_ρ , we obtain, for each fixed s ,

$$\|M_\rho - M_0\|_{\mathfrak{h}^s \rightarrow \mathfrak{h}^s} \leq C_s |\rho|,$$

$$\|K_\rho\|_{\mathfrak{h}^{s+1} \rightarrow \mathfrak{h}^s} \leq C_s |\rho|,$$

$$\|N_\rho - N_0\|_{\mathfrak{h}^{s+2} \rightarrow \mathfrak{h}^s} \leq C_s |\rho|.$$

Since $p_{\pm 1} = -1/3$ and $p_m \geq 1$ for the other odd m , the first estimate and the gap in the spectrum of M_0 show that M_ρ has exactly two negative directions for this choice of ρ_0 . Commuting the contour integrals that define the spectral projections with the weights $\langle m \rangle^s$, we obtain bounded analytic projections $P_\pm(\rho)$ onto the

positive and the negative spectral subspaces of every $\mathfrak{h}^s$. We choose an analytic family of orthogonal maps Q_ρ which transports the projections at ρ to those at the circle, so that

$$Q_\rho P_\pm(\rho) Q_\rho^* = P_\pm(0).$$

In the *fixed coordinates* given by Q_ρ , the spectral subspaces and their interface trace spaces are independent of ρ . We set

$$T_\rho = |M_\rho|^{1/2}, \quad J_\rho = \operatorname{sgn} M_\rho = P_+ - P_-, \quad \widehat{u} = T_\rho u.$$

Multiplying (7.11) by $|M_\rho|^{-1/2}$, we obtain the normalized equation

$$(7.12) \quad J_\rho D^2 \widehat{u} + B_\rho D \widehat{u} + C_\rho^\sharp \widehat{u} - (1 - \rho^2) t^2 A_\rho^\sharp \widehat{u} = \widetilde{f}.$$

Here,

$$\begin{aligned} B_\rho &= |M_\rho|^{-1/2} K_\rho |M_\rho|^{-1/2}, \quad C_\rho^\sharp = |M_\rho|^{-1/2} N_\rho |M_\rho|^{-1/2}, \\ A_\rho^\sharp &= |M_\rho|^{-1}, \quad \widetilde{f} = |M_\rho|^{-1/2} F. \end{aligned}$$

The two-dimensional negative spectral subspace gives the center component. In (7.12), the coupling between the positive and the negative subspaces contains no second-order term. The symmetry under the reflection $\mathcal{K}$, which sends the mode m to $-m$ and was introduced before (4.20), and the skew-adjointness of B_ρ also give

$$(7.13) \quad P_- J_\rho P_+ = 0, \quad P_- B_\rho P_- = 0.$$

The positive component. On the positive subspace, coercivity gives a boundary value problem on $1 \leq t \leq T$. We prescribe the value at $t = 1$ and the outer boundary datum at $t = T$, where $T = |n|/L$ at cell frequency n . We let $G_{+,T}$ be the solution operator for this problem, including its angular and mass terms. We denote its *positive quadratic form* by $\mathfrak{a}_{++,\rho,T}$. With the same normalization as the interior equations, the outer boundary condition at the circle is

$$(7.14) \quad (\mathcal{B}_{0,T} u_+)_m = 2\sqrt{p_m}(D+2)u_{+,m}(T), \quad |m| \geq 3.$$

For small ρ , the boundary operator at the ellipse is an analytic perturbation of (7.14) which couples each mode only to the adjacent ones. In particular, its conormal coefficient on the positive modes remains invertible, while its compression to the center modes is $O(\rho)$. We therefore separate the positive part from the center contribution and write, in the fixed coordinates,

$$(7.15) \quad \mathcal{B}_{\rho,T}(u_c + u_+) = \mathcal{B}_{++,\rho,T} u_+ + \mathcal{B}_{+-,\rho,T} u_c.$$

Here, $u_c = P_-(0) Q_\rho \widehat{u}$ and $u_+ = P_+(0) Q_\rho \widehat{u}$ are the center and the positive components of the unknown in the fixed coordinates. The balanced space for the boundary datum is

$$(7.16) \quad \mathfrak{T}_T^s = \mathfrak{h}_+^{s+1/2}, \quad \|b\|_{\mathfrak{T}_T^s} := T^{-1/2} \|b\|_{\mathfrak{h}^{s+1/2}}.$$

Here, $\mathfrak{h}_\pm^s = P_\pm(0) \mathfrak{h}^s$ are the positive and the negative subspaces of $\mathfrak{h}^s$ in the fixed coordinates, and the center contribution satisfies

$$(7.17) \quad \|\mathcal{B}_{+-,\rho,T} u_c\|_{\mathfrak{T}_T^s} \leq C_s |\rho| \|u_c\|_{\mathcal{C}_T^s}.$$

Here, writing $\xi = t^{1/2} u_c$, we measure the center solution by

$$\|u_c\|_{\mathcal{C}_T^s} = \sup_{1 \leq t \leq T} (\|\xi(t)\|_{\mathfrak{h}^s} + \|\xi'(t)\|_{\mathfrak{h}^s}).$$

Indeed, the negative spectral vectors are bounded in every $\mathfrak{h}^s$, the row couples only adjacent modes, and we have the bounds

$$|u_c(T)| \leq CT^{-1/2} \|u_c\|_{\mathcal{C}_T^s}, \quad |Du_c(T)| \leq CT^{1/2} \|u_c\|_{\mathcal{C}_T^s}.$$

We write $\Lambda_\theta = \langle -i\partial_\theta \rangle$ for the angular multiplier, which multiplies mode m by $\langle m \rangle$. We define the energy norm on the positive subspace by

$$(7.18) \quad \|u_+\|_{\mathcal{E}_T^s}^2 = \int_1^T \left(\|Du_+\|_{\mathfrak{h}^s}^2 + \|\Lambda_\theta u_+\|_{\mathfrak{h}^s}^2 + t^2 \|u_+\|_{\mathfrak{h}^s}^2 \right) \frac{dt}{t} + \|\mathrm{tr}_T u_+\|_{s+1/2}^2.$$

Here, $\mathrm{tr}_T u_+ = u_+(T)$ is the trace on the outer boundary. We let $G_{+,T}^0 q$ solve the positive equation with source q , zero inner value, and homogeneous outer boundary condition. We let $P_{+,T} a$ solve the homogeneous equation with inner value a and homogeneous outer boundary condition, and choose a bounded extension $E_T b$ with

$$(E_T b)(1) = 0, \quad \mathcal{B}_{++,\rho,T} E_T b = b.$$

Thus, the two extensions supply the inner and outer data separately, and we have

$$(7.19) \quad G_{+,T}(q, a, b) = P_{+,T} a + E_T b + G_{+,T}^0(q - \mathcal{N}_{++}(P_{+,T} a + E_T b)).$$

Here, $\mathcal{N}_{++}$ is the compression of (7.12) to the positive subspace in the fixed coordinates. In particular, $u_+ = G_{+,T}(q, a, b)$ solves

$$\mathcal{N}_{++} u_+ = q, \quad u_+(1) = a, \quad \mathcal{B}_{++,\rho,T} u_+ = b.$$

The form $\mathbf{a}_{++,\rho,T}$ satisfies, for some $c_+ > 0$ and uniformly for $T \geq 1$,

$$(7.20) \quad \mathbf{a}_{++,\rho,T}(u_+, u_+) \geq c_+ \int_1^T (\|Du_+\|^2 + \|\langle \mathcal{R} \rangle u_+\|^2 + t^2 \|u_+\|^2) \frac{dt}{t}.$$

Here, $\|\cdot\|$ is the norm of $\mathfrak{h}^0$, and $\langle \mathcal{R} \rangle$ is the multiplier Λ_θ . To verify this, we write

$$T_\rho^{-1} Q_\rho^* u_+ = h + \Gamma_\rho h, \quad \Pi_{|m| \geq 3} h = h, \quad \|\Gamma_\rho\|_{\mathfrak{h}^s \rightarrow \mathfrak{h}^s} \leq C_s |\rho|.$$

Here, $u = T_\rho^{-1} Q_\rho^* u_+$ is the positive component in the original coordinates. The spectral gap gives a positive lower bound for $\langle M_\rho Du, Du \rangle$, while we have the estimates

$$-\langle N_0 u, u \rangle \geq c_0 \|\Lambda_\theta h\|^2 - C\rho^2 \|h\|^2, \\ |\langle (N_\rho - N_0)u, u \rangle| + |\langle K_\rho Du, u \rangle| \leq C|\rho|(\|\Lambda_\theta h\|^2 + \|Du\|^2).$$

The positive t^2 -term absorbs the remaining L^2 term, and by density we obtain (7.20) on the complete sequence space, with no angular cutoff.

The center component. On the negative subspace, the substitution $u_c = t^{-1/2} \xi$ reduces (7.12) to two oscillators, which are separated by the reflection $\mathcal{K}$ and take the form

$$(7.21) \quad \xi'' + \Omega_\rho^2 \xi + t^{-2} V_{--, \rho} \xi = g.$$

Here, Ω_ρ^2 and $V_{--, \rho}$ are analytic 2×2 matrices, with Ω_ρ^2 positive. Its eigenvalues are the squares of the oscillator frequencies from Section 1.4, which coincide at the circle. We solve outward from $t = 1$, prescribing Cauchy data $a = (\xi(1), \xi'(1))$ and imposing no outer boundary condition. Writing $z = (\xi, \xi')$, the first-order system associated with (7.21) is

$$z' = \begin{pmatrix} 0 & I \\ -\Omega_\rho^2 - t^{-2} V_{--, \rho} & 0 \end{pmatrix} z + \begin{pmatrix} 0 \\ g \end{pmatrix}.$$

We let $\Phi_\rho(t, s)$ propagate its homogeneous solutions from s to t , with $\Phi_\rho(s, s) = I$, and set $\pi_1(\xi, \xi') = \xi$. The center solution u_c with source g is

$$(7.22) \quad V_{c,T}(a, g)(t) = t^{-1/2} \pi_1 \left[\Phi_\rho(t, 1)a + \int_1^t \Phi_\rho(t, s) \begin{pmatrix} 0 \\ g(s) \end{pmatrix} ds \right].$$

For the source of the center equation, we use the norm

$$\|g\|_{\mathcal{F}_{c,T}^s} = \int_1^T \|g(t)\|_{\mathfrak{h}^s} dt,$$

and we abbreviate $\mathcal{C}_T = \mathcal{C}_T^0$, $\mathcal{F}_{c,T} = \mathcal{F}_{c,T}^0$, and $\mathcal{E}_T = \mathcal{E}_T^0$. The energy of the oscillator and $\int_1^\infty t^{-2} dt < \infty$ imply

$$\|u_c\|_{\mathcal{C}_T} \leq C(|a| + \|g\|_{\mathcal{F}_{c,T}}), \quad T \geq 1,$$

because the eigenvalues of Ω_ρ^2 remain in one compact subinterval of $(0, \infty)$.

Coupling the two components. The remaining operators $\mathcal{R}_{+-}$ and $\mathcal{R}_{-+}$ in (7.12) couple the center and positive components. They have the form $t^{-1}B_\rho\partial_t + t^{-2}C_\rho^\#$ and satisfy

$$(7.23) \quad \|\mathcal{R}_{+-}u_c\|_{\mathcal{E}_T^*} \leq C|\rho| \|u_c\|_{\mathcal{C}_T}, \quad \|\mathcal{R}_{-+}u_+\|_{\mathcal{F}_{c,T}} \leq C|\rho| \|u_+\|_{\mathcal{E}_T}.$$

Indeed, the spectral projections are analytic in ρ and the coefficients couple only adjacent modes, so both terms carry a factor $O(\rho)$, while the Cauchy-Schwarz inequality applied to (7.18) leaves the integrable weights $\int_1^\infty t^{-2} dt$ and $\int_1^\infty t^{-3} dt$. To include the boundary term, we set

$$(7.24) \quad \tilde{\mathcal{R}}_{+-}u_c := (\mathcal{R}_{+-}u_c, -\mathcal{B}_{+-,\rho,T}u_c).$$

Then, (7.23) and (7.17) say, at every regularity level, that

$$\left\| \tilde{\mathcal{R}}_{+-}u_c \right\|_{(\mathcal{E}_T^s)^* \oplus \mathfrak{T}_T^s} \leq C_s |\rho| \|u_c\|_{\mathcal{C}_T^s}.$$

We now solve the coupled equations on the annulus. Substituting the positive solution into the center equation and then reconstructing the positive component, we obtain

$$(7.25) \quad \begin{aligned} u_c &= [I - V_{c,T}\mathcal{R}_{-+}G_{+,T}\tilde{\mathcal{R}}_{+-}]^{-1} V_{c,T}(\tilde{f}_- + \mathcal{R}_{-+}G_{+,T}\tilde{f}_+), \\ u_+ &= G_{+,T}(\tilde{f}_+ + \tilde{\mathcal{R}}_{+-}u_c). \end{aligned}$$

Here, $\tilde{f}_- = (a_c, g_c)$ consists of the center Cauchy data and interior source, and $\tilde{f}_+ = (q_+, a_+, b_+)$ consists of the positive interior source, inner value, and outer boundary datum. In these compositions, $V_{c,T}$ and $G_{+,T}$ use zero trace data when only an interior source is supplied; the coupling $\tilde{\mathcal{R}}_{+-}u_c$ supplies zero inner value. We impose the boundary condition (7.15) through the second component of (7.24). The correction inside the inverse contains one coupling from the center to the positive subspace and one in the reverse direction. Its norm is therefore at most $C\rho^2$, by (7.20)–(7.23) and $\int_1^\infty t^{-2} dt < \infty$. This argument also bounds the center contribution to the boundary operator.

The spaces on which the inverse (7.25) acts, and the estimates that we will use, are recorded in the following lemma.

Lemma 7.2. *With $\mathfrak{h}_\pm^s = P_\pm(0)\mathfrak{h}^s$ as in (7.16), we put*

$$\begin{aligned} \mathfrak{D}_{c,T}^s &= (\mathfrak{h}_-^s \times \mathfrak{h}_-^s) \oplus \mathcal{F}_{c,T}^s, \\ \mathfrak{D}_{+,T}^s &= (\mathcal{E}_T^s)^* \oplus \mathfrak{h}_+^{s+1/2} \oplus \mathfrak{T}_T^s. \end{aligned}$$

The three positive entries are, in order, the interior source in the dual of the energy space, the inner Dirichlet value, and the balanced datum on the outer boundary. For one $\rho_0 > 0$, every $0 \leq |\rho| \leq \rho_0$, and every $T \geq 2$, the operator

$$\mathcal{N}_{\rho,T}^{\text{ext}} : \mathcal{C}_T^s \oplus \mathcal{E}_T^s \longrightarrow \mathfrak{D}_{c,T}^s \oplus \mathfrak{D}_{+,T}^s$$

which records the equations, the center Cauchy data, the positive value at $t = 1$, and the boundary data $\mathcal{B}_{\rho,T}(u_c + u_+)$, is an invertible linear map. Its inverse is (7.25), with $\tilde{\mathcal{R}}_{+-}$ as in (7.24), and it satisfies

$$(7.26) \quad \|u_c\|_{\mathcal{C}_T^s} + \|u_+\|_{\mathcal{E}_T^s} \leq C_s \left(\|\tilde{f}_-\|_{\mathfrak{D}_{c,T}^s} + \|\tilde{f}_+\|_{\mathfrak{D}_{+,T}^s} \right).$$

The constants are independent of T and of every angular cutoff.

The lemma estimates the inverse before differentiation in ρ . For the derivatives ∂_ρ^q with $q \geq 1$, differentiating the phase of the center oscillator produces powers of $t - 1$, so the same bound need not be uniform in T without a loss of derivatives. The bounds in the analytic norms, which absorb these powers, are proved below, once the weight has been introduced.

Proof. At the base index, the Lax-Milgram lemma and (7.20) give the positive Green map with zero inner value and homogeneous outer boundary condition. The extensions $P_{+,T}a$ and $E_T b$ in (7.19) supply the prescribed inner value and outer boundary datum. For the center map, we use (7.22) and the energy estimate for the oscillator. For the coupling terms, including the boundary term, we have

$$(7.27) \quad \begin{aligned} \|\tilde{\mathcal{R}}_{+-}\|_{\mathcal{C}_T^s \rightarrow (\mathcal{E}_T^s)^* \oplus \mathfrak{T}_T^s} &\leq C_s |\rho|, \\ \|\mathcal{R}_{-+}\|_{\mathcal{E}_T^s \rightarrow \mathcal{F}_{c,T}^s} &\leq C_s |\rho|. \end{aligned}$$

Thus, the correction in (7.25) has norm at most $C_s \rho^2$. We decrease ρ_0 so that this norm is less than $1/2$ at the base index. The Neumann series then gives a right inverse with the asserted estimate, while uniqueness for the oscillator, coercivity of the positive form, and the boundary condition (7.15) give the left inverse.

For higher regularity, the contour formulas for $P_\pm$ and $T_\rho^{\pm 1}$, together with the differential equation satisfied by Q_ρ , give for each fixed s and q

$$(7.28) \quad \|\Lambda_\theta^s \partial_\rho^q Q_\rho \Lambda_\theta^{-s}\| + \|\Lambda_\theta^s \partial_\rho^q T_\rho^{\pm 1} \Lambda_\theta^{-s}\| \leq C_{s,q}.$$

We commute Λ_θ^s through the coefficients of (7.12), which couple adjacent modes. The diagonal angular term commutes with Λ_θ^s and remains in the coercive form. This gives (7.27) with C_s in place of C_0 . Finally, (7.17) and

$$\sqrt{T} \|u_+(T)\|_{\mathfrak{h}^s} \leq C_s \|u_+\|_{\mathcal{E}_T^s}$$

give the balanced value and conormal traces. $\square$

Radial differentiation requires additional care. Even for a homogeneous center solution $u_c = t^{-1/2}\xi$, we have

$$Du_c = -\frac{1}{2}t^{-1/2}\xi + t^{1/2}\xi',$$

so the bound for u_c does not give a bound for Du_c in the same center norm uniformly in T . The radial estimates used below must account for these powers of t after the analytic weight is introduced.

The restriction $T \geq 2$ in Lemma 7.2 keeps the two boundaries apart. At $T = 1$, they coincide, so the two trace data cannot be prescribed independently. In the application, T is the value $T_n = |n|/L$ at cell frequency n , and the cell frequencies with $T_n < 2$ form a finite set, which we treat separately below.

7.5. Matching at the interface. To match the cap and annulus solutions, we use the positive interface value a at $t = 1$ as an unknown in the fixed coordinates. Solving the cap problem (7.10) with this value determines the center Cauchy data. Together with a , these data determine the annular solution through (7.25). With the interior sources, prescribed axis coefficients, and outer boundary datum set to zero, we let $\mathcal{M}_{\rho,T}a$ be the cap conormal derivative minus the annular one. We let $d_{\rho,T}(f, b)$ be the mismatch for the prescribed data with $a = 0$. The source f and the outer boundary datum b contribute $d_{\rho,T}(f, b)$, while the unknown value a contributes $\mathcal{M}_{\rho,T}a$. Thus, matching means $\mathcal{M}_{\rho,T}a + d_{\rho,T}(f, b) = 0$, or

$$(7.29) \quad a = -\mathcal{M}_{\rho,T}^{-1}d_{\rho,T}(f, b).$$

To invert $\mathcal{M}_{0,T}$ uniformly in T , we let $u_{\text{in}}(a)$ and $u_{\text{out}}(a)$ be the solutions of the homogeneous positive equations at the circle on the cap and on the annulus, with the common value a at the interface and the homogeneous condition on the outer boundary. The outward normals have opposite signs at the interface, so Green's identity gives

$$(7.30) \quad \langle \mathcal{M}_{0,T}a, a \rangle = \mathbf{a}_{\text{in},0}(u_{\text{in}}, u_{\text{in}}) + \mathbf{a}_{\text{out},0,T}(u_{\text{out}}, u_{\text{out}}) \geq c \|a\|_{H^{1/2}}^2.$$

Here, $\mathbf{a}_{\text{in},0}$ and $\mathbf{a}_{\text{out},0,T}$ are the positive forms on the cap and annulus, the latter given by (7.20), and $H^{\pm 1/2} = \mathfrak{h}_+^{\pm 1/2}$. Both forms control the angular and mass terms, so the trace theorem at $t = 1$ bounds $\|a\|_{H^{1/2}}$ by either energy, uniformly in T . A collar extension of a and the Green estimates also bound $\mathcal{M}_{0,T}$ from $H^{1/2}$ to $H^{-1/2}$. The Lax-Milgram lemma on the trace space therefore gives

$$\left\| \mathcal{M}_{0,T}^{-1} \right\|_{H^{-1/2} \rightarrow H^{1/2}} \leq C, \quad T \geq 2.$$

At the circle, the center Cauchy data propagate without a condition on the outer boundary, and they do not affect $\mathcal{M}_{0,T}$. For $\rho \neq 0$, we let $\Lambda_{\text{in},\rho}$ and $\Lambda_{\text{out},\rho,T}^+$ denote the maps from the value a to the conormal derivative of the solution on the cap and on the positive part of the annulus, in the fixed coordinates. The couplings with the center are omitted from the latter, and we put all terms involving a coupling between the center and the positive subspace into $\mathcal{E}_{\text{ret},\rho,T}$. With the normal convention of (7.30), we then have

$$(7.31) \quad \mathcal{M}_{\rho,T} = \Lambda_{\text{in},\rho} - \Lambda_{\text{out},\rho,T}^+ + \mathcal{E}_{\text{ret},\rho,T}, \quad \mathcal{E}_{\text{ret},0,T} = 0.$$

To describe the coupling term, we let $\mathcal{C}_{\text{in},\rho}a$ be the center Cauchy trace of the homogeneous cap solution with value a . We write $G_{+,\rho,T}$ for (7.19) at parameter ρ , put $P_{+,\rho,T}a = G_{+,\rho,T}(0, a, 0)$, and define

$$\Xi_{\rho,T} = I - V_{c,T}\mathcal{R}_{-+}G_{+,\rho,T}\tilde{\mathcal{R}}_{+-},$$

which is invertible since its correction has norm at most $C\rho^2 < 1/2$. If we write $\Gamma_{N,+}$ for the conormal trace of the positive component at $t = 1$, then the last term in (7.31) is

$$(7.32) \quad \mathcal{E}_{\text{ret},\rho,T}a = -\Gamma_{N,+}G_{+,\rho,T}\tilde{\mathcal{R}}_{+-}\Xi_{\rho,T}^{-1}V_{c,T}(\mathcal{C}_{\text{in},\rho}a + \mathcal{R}_{-+}P_{+,\rho,T}a).$$

Both terms in the last parenthesis vanish at the circle. Away from it, the perturbation formula (7.10) gives $\|\mathcal{C}_{\text{in},\rho}\| \leq C_s|\rho|$, and (7.27) gives the same bound for $\mathcal{R}_{-+}P_{+,\rho,T}$, uniformly in T .

From the perturbation formula (7.10) and the trace theorem on the interface $t = 1$, we obtain

$$(7.33) \quad \|\Lambda_{\text{in},\rho} - \Lambda_{\text{in},0}\|_{\mathfrak{h}_+^{s+1/2} \rightarrow \mathfrak{h}_+^{s-1/2}} \leq C_s|\rho|.$$

The map Q_ρ and the trace operator in the fixed coordinates satisfy

$$(7.34) \quad \|Q_\rho - I\|_{\mathfrak{h}^s \rightarrow \mathfrak{h}^s} + \|\mathcal{T}_\rho - \mathcal{T}_0\|_{\mathfrak{h}_+^{s+1/2} \rightarrow \mathfrak{h}_+^{s-1/2}} \leq C_s|\rho|.$$

Here, $\mathcal{T}_\rho$ denotes the cap or annular Dirichlet-to-conormal map, $\Lambda_{\text{in},\rho}$ or $\Lambda_{\text{out},\rho,T}^+$, expressed in the fixed coordinates. On the positive subspace, the Green maps with homogeneous boundary data satisfy

$$G_{+,\rho,T}^0 - G_{+,0,T}^0 = G_{+,\rho,T}^0(\mathcal{N}_{++,0} - \mathcal{N}_{++,\rho})G_{+,0,T}^0,$$

which, combined with (7.20), the Poisson extensions, and the $O(\rho)$ change in the boundary operator, gives

$$(7.35) \quad \left\| \Lambda_{\text{out},\rho,T}^+ - \Lambda_{\text{out},0,T}^+ \right\|_{\mathfrak{h}_+^{s+1/2} \rightarrow \mathfrak{h}_+^{s-1/2}} \leq C_s |\rho|, \quad T \geq 2.$$

Explicitly, we have the bounds

$$\begin{aligned} \sqrt{T} \|u_+(T)\|_{\mathfrak{h}^s} &\leq C_s \|u_+\|_{\mathcal{E}_T^s}, \\ |u_c(T)| &\leq C_s T^{-1/2} \|u_c\|_{\mathcal{C}_T^s}, \quad |Du_c(T)| \leq C_s T^{1/2} \|u_c\|_{\mathcal{C}_T^s}. \end{aligned}$$

Each term in $\mathcal{E}_{\text{ret},\rho,T}$ contains two $O(\rho)$ factors. The first maps the positive component to the center through the cap trace or $\mathcal{R}_{-+}$. The second, $\tilde{\mathcal{R}}_{+-}$, maps back to the positive component and includes the boundary term (7.17). The intervening Green maps and $\Xi_{\rho,T}^{-1}$ are uniformly bounded. Hence,

$$(7.36) \quad \|\mathcal{E}_{\text{ret},\rho,T}\|_{\mathfrak{h}_+^{s+1/2} \rightarrow \mathfrak{h}_+^{s-1/2}} \leq C_s \rho^2, \quad T \geq 2.$$

Adding (7.33), (7.35), and (7.36), we obtain at every fixed index

$$(7.37) \quad \|\mathcal{M}_{\rho,T} - \mathcal{M}_{0,T}\|_{\mathfrak{h}_+^{s+1/2} \rightarrow \mathfrak{h}_+^{s-1/2}} \leq C_s |\rho|, \quad T \geq 2.$$

Here, the constants at higher regularity come from (7.28) and from the induction in the radial derivatives in the proof of Lemma 7.2. We can therefore invert $\mathcal{M}_{\rho,T}$ by a Neumann series at the base index. This gives (7.29) and completes the construction of the odd inverse for $T \geq 2$, with arbitrary compatible interior and boundary data. Substitution verifies that it is a right inverse. Uniqueness for the equations on the cap, for the oscillator, and for the coercive equation, together with the matching at the interface, shows that it is also a left inverse.

7.6. The analytic weight and the estimates for the odd block. We now estimate the odd inverse in the analytic scale of Section 3.3. After rescaling, its radial weight decreases outward and absorbs the growth of the center solution and its ρ -derivatives. We set $\lambda_n = \langle n \rangle$ and $T_n = |n|/L$. For $n \neq 0$, we factor out the axis value of the weight by writing

$$(7.38) \quad \omega_n(t) := \exp\{\Phi_\gamma(Lt/|n|, \lambda_n) - \Phi_\gamma(0, \lambda_n)\}.$$

Its logarithmic derivative is

$$(7.39) \quad d_n(t) := -\partial_t \log \omega_n(t) = \gamma \frac{L^2 t \lambda_n^2 / n^2}{\sqrt{1 + L^2 t^2 \lambda_n^2 / n^2}}, \quad 0 \leq d_n(t) \leq \gamma L \frac{\lambda_n}{|n|}.$$

For every fixed order q of differentiation in $D = t\partial_t$, the derivatives of d_n are bounded uniformly on the cell frequencies $|n| \geq 2L$, and those of td_n are bounded by $C_{q,L}\gamma(1+t)$. On the cap $0 \leq t \leq 1$, both ω_n and its inverse have uniformly bounded derivatives of each fixed order. On the positive part of the annulus, conjugation by ω_n makes the substitutions

$$\partial_t \mapsto \partial_t + d_n, \quad D \mapsto D + td_n,$$

and changes the form by at most $C_L\gamma$ times the radial and mass terms of (7.18). We choose γ so that this is less than a quarter of the coercivity margin. Thus, the positive form and its trace estimates remain uniform after conjugation. For the center component, we use the oscillator formula.

Since ω_n is decreasing and the fundamental matrix $\Phi_\rho(t, s)$ is uniformly bounded, we obtain from (7.22)

$$(7.40) \quad \omega_n(t) |V_{c,T_n}(a, g)(t)| \leq C \left(\omega_n(1) |a| + \int_1^t \omega_n(s) |g(s)| \, ds \right).$$

The same weight controls the derivatives in ρ of the center propagator Φ_ρ . We write

$$a_n = \frac{L\lambda_n}{|n|} \in [L, \sqrt{2}L].$$

For $t \geq t_0 \geq 1$, the rate in (7.39) is bounded below by γc_L , so that

$$(7.41) \quad \frac{\omega_n(t)}{\omega_n(t_0)} = \exp \left\{ - \int_{t_0}^t d_n(u) \, du \right\} \leq e^{-c_L \gamma (t-t_0)},$$

$$c_L := \min_{a \in [L, \sqrt{2}L]} \frac{a^2}{\sqrt{1+a^2}} > 0.$$

Differentiating the first-order system of the oscillator repeatedly by Duhamel's formula, and using the uniform bound on its undifferentiated propagator and the integrable coefficient t^{-2} , we obtain

$$(7.42) \quad \|\partial_\rho^q \Phi_\rho(t, t_0)\|_{\mathfrak{h}^s \rightarrow \mathfrak{h}^s} \leq C_{s,q} (1+t-t_0)^q, \quad t \geq t_0 \geq 1.$$

Here, t_0 denotes the initial radius and s the Sobolev index. Even a constant oscillator produces a factor $t-t_0$ when its frequency is differentiated. The weight absorbs this polynomial growth. Indeed, (7.41) and $(1+\tau)^q e^{-c\tau} \leq C_q (1+c^{-1})^q$ give

$$(7.43) \quad \sup_{t \geq t_0 \geq 1} (1+t-t_0)^q \frac{\omega_n(t)}{\omega_n(t_0)} \leq C_{q,L} \gamma^{-q}.$$

Setting

$$\|u_c\|_{\mathcal{C}_{T,\omega_n}^s} := \sup_{1 \leq t \leq T} \omega_n(t) (\|\xi(t)\|_{\mathfrak{h}^s} + \|\xi'(t)\|_{\mathfrak{h}^s}),$$

$$\|g\|_{\mathcal{F}_{c,T,\omega_n}^s} := \int_1^T \omega_n(t) \|g(t)\|_{\mathfrak{h}^s} \, dt, \quad u_c = t^{-1/2} \xi,$$

we obtain from (7.42), (7.43), and the formula (7.22)

$$(7.44) \quad \|\partial_\rho^q V_{c,T}(a, g)\|_{\mathcal{C}_{T,\omega_n}^s} \leq C_{s,q,L,\gamma} (\omega_n(1) |a| + \|g\|_{\mathcal{F}_{c,T,\omega_n}^s}).$$

The positive form, Q_ρ , $T_\rho^{\pm 1}$, and the coupling maps have weighted bounds for every fixed number of ρ -derivatives. Since the weight is independent of ρ , we can differentiate (7.25), including its Neumann series, to obtain the weighted version of (7.26) at each fixed order. The balanced value and conormal traces are included, and the bounds are uniform for $T \geq 2$.

At the interface, an $O(\rho)$ coupling occurs on either side of the center propagator. Its uniform bound therefore suffices without estimating $V_{c,T}(\rho) - V_{c,T}(0)$. At the outer boundary, the factor $\omega_n(T_n)$ agrees with the prescribed trace weight. Thus, all factors in (7.25) have weighted bounds independent of the cutoffs.

We write $\mathcal{X}_{y,n}^j$, $\mathcal{Y}_{y,n}^j$, and $\mathcal{T}_{y,n}^j$ for the norms at index j of the Cartesian domain, of the source, and of the balanced traces of (6.42), after the rescaling $t = |n|r/L$, defined with $\partial_r = (|n|/L)\partial_t$. Removing the common factor $e^{\Phi_\gamma(0,\lambda_n)}$, we obtain the equivalent norms at cell frequency n

$$(7.45) \quad e^{-2\Phi_\gamma(0,\lambda_n)} \|u_n\|_{\mathcal{X}_{\gamma,n}^s}^2 \simeq \sum_{j+\ell \leq s} \lambda_n^{2\ell} \|\omega_n u_n\|_{\mathcal{X}_{y,n}^j}^2,$$

$$e^{-2\Phi_\gamma(0,\lambda_n)} \|f_n\|_{\mathcal{Y}_{\gamma,n}^s}^2 \simeq \sum_{a+k \leq s} \lambda_n^{2k} \|\omega_n f_n\|_{\mathcal{Y}_{y,n}^a}^2.$$

Here, the boundary components are measured in $\mathcal{T}_{y,n}^a$. We write $K_{e,n}^{\text{od}}$ for the odd inverse at cell frequency n constructed above, and Tr_{phys} for the trace on the outer boundary. From the energy estimate, Cartesian difference quotients, and the trace theorem in the balanced norms, we obtain

$$(7.46) \quad \begin{aligned} & \|\omega_n K_{e,n}^{\text{od}} f_n\|_{\mathcal{X}_{y,n}^j} + \|\omega_n (T_n) \text{Tr}_{\text{phys}} K_{e,n}^{\text{od}} f_n\|_{\mathcal{T}_{y,n}^j} \\ & \leq C_j \sum_{a=0}^{j+4} \lambda_n^{j+4-a} \|\omega_n f_n\|_{\mathcal{Y}_{y,n}^a}. \end{aligned}$$

We denote the left side by $E_{j,n}^{(0)}$ and set $F_{a,n} = \|\omega_n f_n\|_{\mathcal{Y}_{y,n}^a}$, including all source trace entries. The base Green estimate and (7.8) give $E_{0,n}^{(0)} \leq C F_{4,n}$. Applying one more Cartesian difference quotient leaves the principal term in the coercive part. Each commutator contains either a disk derivative, carrying the rescaling factor $|n|/L \leq C_L \lambda_n$, or a term involving d_n , Dd_n , or its balanced trace. By (7.39) and (7.26), this gives

$$(7.47) \quad E_{j,n}^{(0)} \leq C_j (F_{j+4,n} + \lambda_n E_{j-1,n}^{(0)}), \quad j \geq 1,$$

whose iteration yields

$$E_{j,n}^{(0)} \leq C_j \sum_{a=4}^{j+4} \lambda_n^{j+4-a} F_{a,n},$$

which proves (7.46). The loss consists of three derivatives in (7.8) and one in the weighted annular and trace estimates, independently of the upper frequency cutoff.

We now turn to the derivatives in ρ . We define $E_{j,n}^{(q)}$ as $E_{j,n}^{(0)}$ with the solution and its outer trace replaced by those of $\partial_\rho^q K_{e,n}^{\text{od}} f_n$. Differentiating the equations, and using (7.44) for the center and (7.28) for the other factors, we obtain

$$(7.48) \quad \begin{aligned} E_{0,n}^{(q)} & \leq C_{q,\gamma} \left(F_{4+q,n} + \lambda_n \sum_{p < q} E_{0,n}^{(p)} \right), \\ E_{j,n}^{(q)} & \leq C_{j,q,\gamma} \left(F_{j+4+q,n} + \lambda_n E_{j-1,n}^{(q)} + \lambda_n \sum_{p < q} E_{j,n}^{(p)} \right), \quad j \geq 1, \quad q \geq 1. \end{aligned}$$

Here, the last term accounts for the factor λ_n , that is, one cell derivative, produced by each differentiated coefficient or trace. A double induction in (q, j) then gives

$$(7.49) \quad E_{j,n}^{(q)} \leq C_{j,q,\gamma} \sum_{a=0}^{j+4+q} \lambda_n^{j+4+q-a} F_{a,n}.$$

Note that all norms in this induction are weighted.

Restoring the common factor $e^{\Phi_\gamma(0,\lambda_n)}$ and multiplying (7.46) by λ_n^ℓ , with $j + \ell \leq s$, we obtain terms with at most $s + 4$ derivatives of the source, since

$$a + (\ell + j + 4 - a) = \ell + j + 4 \leq s + 4.$$

By a finite Cauchy-Schwarz inequality in (j, ℓ, a) and Parseval's identity, we therefore obtain

$$(7.50) \quad \sum_{|n| \geq 2L} \|K_{e,n}^{\text{od}} f_n\|_{\mathcal{X}_{\gamma,n}^s}^2 \leq C_s \sum_{|n| \geq 2L} \|f_n\|_{\mathcal{Y}_{\gamma,n}^{s+4}}^2.$$

The same multiplication, applied to (7.49), gives for every fixed q

$$(7.51) \quad \sum_{|n| \geq 2L} \|\partial_\rho^q K_{e,n}^{\text{od}} f_n\|_{\mathcal{X}_{\gamma,n}^s}^2 \leq C_{s,q,\gamma} \sum_{|n| \geq 2L} \|f_n\|_{\mathcal{Y}_{\gamma,n}^{s+4+q}}^2.$$

The remaining cell frequencies satisfy $T_n < 2$, or $|n| < 2L$, and form a finite set. At each such frequency, let $\mathcal{L}_{0,n}^{\text{od}} : \mathcal{D}_n^{s*} \rightarrow \mathcal{R}_n^{s*}$ be the closed Cartesian operator at the circle, including its trace operators, and let $\mathcal{L}_{\rho,n}^{\text{od}}$ be the corresponding operator at the ellipse on the same domain.

For $n \neq 0$, the inverse $K_{0,n}^{\text{od}}$ is obtained on the whole disk by solving the center Bessel equations with the prescribed axis coefficients and the positive equations with their Robin data. No interface is needed. We describe $n = 0$ below. Since the domain is unchanged, we have

$$(7.52) \quad \|\mathcal{L}_{\rho,n}^{\text{od}} - \mathcal{L}_{0,n}^{\text{od}}\|_{\mathcal{D}_n^{s*} \rightarrow \mathcal{R}_n^{s*}} \leq C_{n,s*} |\rho|,$$

and the inverse at cell frequency n is

$$(7.53) \quad K_{\rho,n}^{\text{od}} = [I + K_{0,n}^{\text{od}}(\mathcal{L}_{\rho,n}^{\text{od}} - \mathcal{L}_{0,n}^{\text{od}})]^{-1} K_{0,n}^{\text{od}}.$$

Taking the maximum over this finite set and decreasing ρ_0 , we make the norm of $K_{0,n}^{\text{od}}(\mathcal{L}_{\rho,n}^{\text{od}} - \mathcal{L}_{0,n}^{\text{od}})$ less than $1/2$ for every n in this set, so that the Neumann series in (7.53) converge.

For $n = 0$, we use the original variables. The second force equation (5.12) is $f_2 = LRC$, so on every nonzero odd angular mode

$$(7.54) \quad C_m = \frac{f_{2,m}}{iLm}.$$

The first two equations still give the last two equations in (7.5). We set

$$V = (-i\mathcal{R})^{-1}X.$$

The terms depending on V are

$$A = -iP(D+2)V, \quad B = -i(\mathcal{P}\mathcal{R} - 2\mathcal{P}\mathcal{R}^{-1}D)V.$$

All other terms depend only on $g_{\pm}$. The last equation gives (7.7) for $u = |\mathcal{R}|V$, with the mass term omitted. At the circle, the center satisfies the Laplace equation with prescribed coefficients of $re^{i\theta}$ and $re^{-i\theta}$ in its Taylor expansion on the axis. On $|m| \geq 3$, the condition on the outer boundary is of *Robin type*, that is, it prescribes a combination of the value and of the radial derivative, here with the boundary operator $D+2$ at $r=1$. Since the operators at the circle and at the ellipse have a common domain, the same perturbation argument gives the inverse for a small ellipse. We recover $X = (-i\mathcal{R})V$, then Z , W , A , B , and S from (7.5), keeping C as prescribed in (7.54).

Combining the construction above with (7.50) and (7.53), we obtain

$$(7.55) \quad K_e^{\text{od}} \mathcal{L}_e^{\text{od}} = I, \quad \mathcal{L}_e^{\text{od}} K_e^{\text{od}} = I, \quad \|K_e^{\text{od}} f\|_{\mathcal{X}_{\gamma}^s} \leq C_s \|f\|_{\mathcal{Y}_{\gamma}^{s+\mu_{\text{od}}}},$$

with C_s independent of the cell frequency. Note that the estimates above allow the conservative choice $\mu_{\text{od}} = 4$. Differentiating the finitely many Neumann formulas (7.53) and combining them with (7.51), we obtain in addition, for every fixed $q \geq 0$,

$$(7.56) \quad \|\partial_{\rho}^q K_e^{\text{od}} f\|_{\mathcal{X}_{\gamma}^s} \leq C_{s,q,\gamma} \|f\|_{\mathcal{Y}_{\gamma}^{s+4+q}}.$$

This completes the construction and the estimates for the odd block.

7.7. The even block. The even summand contains the angular means, the scale equation, and the affine modes $m = \pm 2$. Each multiplier in (7.5) vanishes on one exceptional mode, so these modes must be treated before $|m| \geq 4$. After solving for the means and scale, we determine $C_{\pm 2}$, then solve a coercive equation for $C_h = \sum_{m \in 2\mathbb{Z}, |m| \geq 4} C_m e^{im\theta}$. The nonsingular equations recover Z and W , leaving W_2 and Z_{-2} for the final radial integrations.

Means, scale, and radial integration. We begin with the algebraic equations. We call a vector *transverse* when it lies in the normal plane $e_T^\perp$, and *tangential* when it is parallel to the axis e_T . The datum h_{af} , which is the scale entry of the auxiliary data r_{aug} of (6.28), prescribes the scale $\dot{a}$ of the transverse component of the linear Taylor term of the variation through the equation

$$(7.57) \quad DH_{\text{af}}(1, 0)[\dot{a}, \dot{\tau}] = 4\dot{a}, \quad \dot{a} = \frac{1}{4}h_{\text{af}}.$$

Here, we temporarily allow the transverse scale a to vary independently of τ . The scalar constraint $H_{\text{af}}(a, \tau) = 2a^2 + |\tau|^2 - 2$ measures the error in the trace normalization (3.2). Its derivative is evaluated at $(a, \tau) = (1, 0)$. We then subtract the prescribed transverse linear term from the variation. The two gauge conditions prescribe the means of B and C , the mean condition on the scalar prescribes that of S , and averaging the first equation of (7.3) gives $A_0 = -f_{0,0}/2$, since the means of $\mathcal{R}S$ and $\mathcal{R}B$ vanish. We keep these known means in the right sides of the equations for $m = \pm 2$ below. From now on, the source denotes the remainder after the algebraic contributions have been subtracted.

For $n \neq 0$, we introduce the two integral operators

$$(7.58) \quad J_+q = t^{-2} \int_0^t sq(s) ds, \quad J_-q = t^2 \int_0^t q(s)s^{-3} ds,$$

which we call the *Hardy maps*. They solve the radial equations through $(D+2)J_+ = I$ and $(D-2)J_- = I$, the latter on functions with zero t^2 -coefficient. Integration from the axis selects zero coefficients for the homogeneous solutions t^{-2} and t^2 . The restriction on J_- also makes its integral converge. We verify below that the sources to which it is applied satisfy this restriction.

We use these maps in conjugated form. For the weight ω_n of (7.38), we put

$$(7.59) \quad J_{\pm, n}^\gamma := \omega_n J_\pm \omega_n^{-1}.$$

Conjugation replaces D by $D + td_n$, where d_n is the logarithmic derivative in (7.39), so that

$$(7.60) \quad \begin{aligned} (D + td_n + 2)J_{+, n}^\gamma &= I, \\ (D + td_n - 2)J_{-, n}^\gamma &= I \quad \text{on } \mathcal{Z}_{2, n}^\gamma. \end{aligned}$$

Here, $\mathcal{Z}_{2, n}^\gamma$ is the subspace of the weighted modes $m = \pm 2$ on which $[t^2](\omega_n^{-1}q) = 0$, where $[t^2]q$ denotes the coefficient of t^2 in the expansion of q at the axis, and the kernels are

$$\begin{aligned} J_{+, n}^\gamma q(t) &= t^{-2} \int_0^t s \frac{\omega_n(t)}{\omega_n(s)} q(s) ds, \\ J_{-, n}^\gamma q(t) &= t^2 \int_0^t s^{-3} \frac{\omega_n(t)}{\omega_n(s)} q(s) ds. \end{aligned}$$

Both weighted Hardy maps are bounded at every sufficiently high fixed Sobolev regularity in Cartesian coordinates, uniformly in n and T_n . Indeed, on $t \leq 1$, the weight and its inverse have uniformly bounded

derivatives, and we may write

$$J_- q(y) = \int_0^1 \tau^{-3} q(\tau y) \, d\tau.$$

Taking k Cartesian derivatives brings the factor τ^k under the integral, and the change of variables $y \mapsto \tau y$ in $L^2(\mathbb{D}^2)$ costs the factor τ^{-1} , so the k -th derivatives of $J_- q$ are bounded by $\int_0^1 \tau^{k-4} \, d\tau$ times those of q , and this integral is finite for $k > 3$. For $k = 4$, the fourth derivatives control the lower ones, since the only polynomials of degree at most three on the modes $m = \pm 2$ are quadratic, and their coefficients vanish on the subspace in question. For $k > 4$, we combine the estimate for the top derivatives with this H^4 bound. The same scaling calculation bounds J_+ at every order of differentiation, with τ in place of τ^{-3} . For $t \geq 1$, we split the integrals at $s = 1$. The contribution from the cap grows at most quadratically and is controlled by $\omega_n(t)$. For $1 \leq s \leq t$, (7.41) gives

$$t^2 s^{-3} \frac{\omega_n(t)}{\omega_n(s)} \leq (1 + t - s)^2 e^{-c_L \gamma(t-s)}.$$

Young's inequality then gives the bounds on the annulus. The exponential absorbs the polynomial factors from the radial measure and from any fixed number of derivatives, and local trace estimates give the balanced bounds at the endpoints. Since the weight is smooth in t^2 at the axis, $\omega_n^{\pm 1}$ preserves the jet and divisibility conditions. The extensions of the finitely many mean, scale, and affine polynomial data satisfy the same weighted estimates.

The exceptional tangential components. The multiplier $\mathcal{R} - 2i$ in (7.5) vanishes at $m = 2$, and $\mathcal{R} + 2i$ vanishes at $m = -2$, which is why these two modes were called exceptional in Section 6. At these two modes, the corresponding equation does not determine W_2 or Z_{-2} , but instead imposes an equation on X ,

$$(D + 2)X_2 = g_{-,2}, \quad (D + 2)X_{-2} = g_{+,-2}.$$

Applying J_+ and substituting $X = i\sigma(f_2 - \mathcal{R}C)$, we determine the two exceptional tangential components. The multiplier that does not vanish at each mode gives the other two components, and we obtain

$$(7.61) \quad \begin{aligned} C_2 &= -\frac{i}{2} f_{2,2} + \frac{\sigma}{2} J_+ g_{-,2}, \\ C_{-2} &= \frac{i}{2} f_{2,-2} - \frac{\sigma}{2} J_+ g_{+,-2}, \\ Z_2 &= \frac{(D-2)X_2 - g_{+,2}}{4i}, \quad W_{-2} = \frac{(D-2)X_{-2} - g_{-,-2}}{-4i}. \end{aligned}$$

These four components are now known. The divisibility of the source gives $g_{+,2} = O(t^4)$ and $g_{-,2} = O(t^2)$. Thus, $X_2 = J_+ g_{-,2}$ has a quadratic leading term, $(D-2)X_2 = O(t^4)$, and (7.61) gives $Z_2 = O(t^4)$. The same calculation at $m = -2$ gives $W_{-2} = O(t^4)$. Hence,

$$(7.62) \quad [t^2]Z_2 = [t^2]W_{-2} = 0,$$

so the right-hand sides to which J_- will be applied at the final step satisfy the required condition.

The high even modes. We now turn to the modes $|m| \geq 4$. We let Π_h be the projection onto these modes and set $C_h = \Pi_h C$. In the terms of (7.5) that depend on C_h , we substitute

$$(7.63) \quad \begin{aligned} (A, B) &= -i\sigma(\hat{A}, \hat{B}), \\ \hat{A} &= P(D+2)C_h, \\ \hat{B} &= PRC_h - 2PR^{-1}DC_h, \\ P &= \frac{\mathcal{R}^2}{\mathcal{R}^2 + 4}. \end{aligned}$$

Although we do not yet know W_2 and Z_{-2} , they do not enter this projection, since H_ρ sends W_2 only to the modes 0 and 2, and Z_{-2} only to the modes 0 and -2 . The projected last equation is therefore a closed equation for C_h . On these modes, we have $1 \leq p_m = m^2/(m^2 - 4) \leq 4/3$, and the resulting form $\mathfrak{b}_\rho$ satisfies

$$(7.64) \quad \begin{aligned} \operatorname{Re} \mathfrak{b}_\rho(C_h, C_h) &\geq \left(1 - 3\frac{\rho}{1-\rho}\right) \|(DC_h, \mathcal{R}C_h)\|_2^2 \\ &\quad + \text{nonnegative mass and Robin terms.} \end{aligned}$$

To prove (7.64), we integrate by parts, after the constant normalization of the equations already made in (7.3), and we obtain

$$(7.65) \quad \operatorname{Re} \mathfrak{b}_\rho(C_h, C_h) = \|(DC_h, \mathcal{R}C_h)\|_2^2 + \mathfrak{m}_{\rho,n}(C_h) + \mathfrak{r}_{\rho,T}(C_h) + \operatorname{Re} \mathfrak{e}_\rho(C_h).$$

Here, $\mathfrak{m}_{\rho,n} \geq 0$ is the mass contribution from $(1 - \rho^2)n^2/L^2$, and $\mathfrak{r}_{\rho,T} \geq 0$ is the squared boundary trace supplied by the Robin condition. Only $\mathfrak{e}_\rho$, the contribution of $H_\rho - I$, may have either sign. By (7.63), it satisfies

$$(7.66) \quad |\operatorname{Re} \mathfrak{e}_\rho(C_h)| \leq \|H_\rho - I\| \left\| (\hat{A}, \hat{B}) \right\|_2 \|(DC_h, \mathcal{R}C_h)\|_2.$$

Indeed,

$$\|H_\rho - I\| = \frac{\rho}{1-\rho}, \quad \left\| (\hat{A}, \hat{B}) \right\|_2 \leq 3 \|(DC_h, \mathcal{R}C_h)\|_2.$$

Here, the first identity follows from the eigenvalues $\rho(\rho \pm 1)/(1 - \rho^2)$ of $H_\rho - I$, and the second follows from mode $p_m \leq 4/3$ and $|m| \geq 4$. Together, (7.65)–(7.66) give (7.64). Its coefficient $1 - 3\rho/(1 - \rho)$ is positive for $0 < \rho < 1/4$, which is the range allowed by (7.1).

The prescribed trace on the outer boundary is $b = \Pi_h \mathfrak{a}(T)$. We let $E_T b$ be a Cartesian extension of it, bounded without loss of derivatives, and we define the Green map $G_{\rho,\sigma,T}^h$ for zero trace by

$$\mathfrak{b}_\rho(G_{\rho,\sigma,T}^h F, \psi) = \langle F, \psi \rangle \quad \text{for every zero-trace } \psi.$$

We write $\mathcal{A}_{\rho,n}^h$ for the operator in the projected last equation associated with $\mathfrak{b}_\rho$, and F_h for the source that remains after inserting (7.61) and projecting. The solution with prescribed outer trace is then

$$(7.67) \quad C_h = E_T b + G_{\rho,\sigma,T}^h (F_h - \mathcal{A}_{\rho,n}^h E_T b),$$

for arbitrary trace data. In the original disk coordinates, before rescaling $t = |n|r/L$, the operator has the following form after an invertible constant scaling of the equations,

$$(7.68) \quad \mathcal{A}_{\rho,n}^h u = \Pi_h [P\Delta_y u - 2\rho H(\partial_z^2 + \partial_{\bar{z}}^2)Hu - (1 - \rho^2)\frac{n^2}{L^2}u], \quad u = |\mathcal{R}|C_h.$$

To obtain the analytic estimates, we return to t and set $v_h = \omega_n C_h$. The form $\mathfrak{b}_{\rho,\gamma,n}$ of $\omega_n \mathcal{A}_{\rho,n}^h \omega_n^{-1}$, with its conjugated boundary operator, replaces every D by $D + td_n$. At the base index, we use the energy

$$(7.69) \quad \|v_h\|_{\mathcal{E}_{\text{ev},n}}^2 := \|(Dv_h, \mathcal{R}v_h)\|_2^2 + \|tv_h\|_2^2 + \|\text{tr}_T v_h\|_{\mathcal{T}_{\text{ev},\text{val}}}^2,$$

and at index j , we write $\mathcal{E}_{\text{ev},n}^j$ for the version obtained by commuting Cartesian derivatives through the form. Here, $\mathcal{T}_{\text{ev},\text{val}}$ is the balanced norm of the value on the outer boundary, which is the contribution of the value to the Robin form, and we write $\mathcal{T}_{\text{ev},n}^j$ for the balanced norm of the value and conormal traces at index j . We recover the conormal trace afterwards from the boundary equation. The coercivity margin that (7.64) leaves uniformly on $0 \leq \rho \leq \rho_0$ is

$$c_{\text{ev}} := 1 - 3 \frac{\rho_0}{1 - \rho_0} > 0.$$

Expanding $D + td_n$, integrating the cross term by parts, and using (7.39), we obtain, after retaining the nonnegative mass and outer boundary contributions,

$$(7.70) \quad \text{Re } \mathfrak{b}_{\rho,\gamma,n}(v_h, v_h) - \text{Re } \mathfrak{b}_{\rho}(v_h, v_h) \geq -C_L(\gamma + \gamma^2) \|v_h\|_{\mathcal{E}_{\text{ev},n}}^2.$$

Indeed, $td_n = O_L(\gamma t)$, with the same bound for every fixed number of D derivatives. On $t \leq 1$, the angular Poincaré inequality on the modes $|m| \geq 4$ controls the zeroth-order terms produced by this expansion, while on $t \geq 1$, the t^2 -mass term controls them. The boundary operator gains the term $T_n d_n(T_n) v_h(T_n)$, whose sign is favorable because the weight decreases outward, and its remaining change is $O_L(\gamma)$ from the balanced value space to the conormal space. We include the condition $C_L(\gamma + \gamma^2) < c_{\text{ev}}/4$ in the initial choice of γ , uniformly in n . Then,

$$(7.71) \quad \text{Re } \mathfrak{b}_{\rho,\gamma,n}(v_h, v_h) \geq \frac{c_{\text{ev}}}{2} \|v_h\|_{\mathcal{E}_{\text{ev},n}}^2, \quad |n| \geq 2L,$$

uniformly in the angular cutoff and in the cell frequency. The Lax-Milgram lemma, a weighted extension of the trace, and the boundary condition then give

$$(7.72) \quad \|v_h\|_{\mathcal{E}_{\text{ev},n}^j} + \|\text{Tr}_{\text{phys}} v_h\|_{\mathcal{T}_{\text{ev},n}^j} \leq C_j \left(\|\omega_n F_h\|_{(\mathcal{E}_{\text{ev},n}^j)^*} + \|\omega_n(T_n) b\|_{\mathcal{T}_{\text{ev},n}^j} \right).$$

Cartesian difference quotients applied to (7.71) give this bound at every regularity level j , uniformly in the angular cutoff and in the cell frequency.

Once C is known on every mode, we use (7.5) to recover Z and W at every mode on which the multipliers do not vanish,

$$(7.73) \quad \begin{aligned} Z_m &= \frac{(D-m)X_m - g_{+,m}}{i(m+2)} \quad (m \neq -2), \\ W_m &= \frac{(D+m)X_m - g_{-,m}}{i(m-2)} \quad (m \neq 2). \end{aligned}$$

The remaining affine components. It remains to determine W_2 and Z_{-2} . For this, we write $Z^G = \mathfrak{a} + i\mathfrak{b}$ and $W^G = \mathfrak{a} - i\mathfrak{b}$ for the complex combinations of $(\mathfrak{a}, \mathfrak{b})$. Multiplication by H_ρ gives

$$(7.74) \quad Z_k^G = \frac{Z_k - \rho W_{k+2}}{1 - \rho^2}, \quad W_k^G = \frac{W_k - \rho Z_{k-2}}{1 - \rho^2},$$

and hence

$$(7.75) \quad \begin{aligned} (D\mathfrak{a} + \mathcal{R}\mathfrak{b})_k &= \frac{1}{2(1-\rho^2)} [(D+k)Z_k + (D-k)W_k \\ &\quad - \rho(D-k)Z_{k-2} - \rho(D+k)W_{k+2}]. \end{aligned}$$

Substituting (7.75) into the last equation of (7.3) at $k = 2$ and $k = -2$, we are left with one unknown in each equation,

$$\begin{aligned}
 (7.76) \quad (D-2)W_2 &= 2(1-\rho^2)(f_{3,2} - i\sigma t^2 C_2) - (D+2)Z_2 \\
 &\quad + \rho(D-2)Z_0 + \rho(D+2)W_4, \\
 (D-2)Z_{-2} &= 2(1-\rho^2)(f_{3,-2} - i\sigma t^2 C_{-2}) - (D+2)W_{-2} \\
 &\quad + \rho(D+2)Z_{-4} + \rho(D-2)W_0.
 \end{aligned}$$

We check that both right-hand sides have zero t^2 -coefficient. Indeed, (7.62) removes that coefficient from $(D+2)Z_2$ and $(D+2)W_{-2}$, and the factor $D-2$ removes it from the terms containing $k=0$. The remaining terms vanish to fourth order by the divisibility $f_3 \in z\bar{z}C^\infty$ of the source, together with the bounds $C_{\pm 2} = O(t^2)$ and $W_4, Z_{-4} = O(t^4)$. Thus, J_- inverts $D-2$ on both right-hand sides, and

$$(7.77) \quad W_2 = J_- \text{ RHS}_2, \quad Z_{-2} = J_- \text{ RHS}_{-2}.$$

Here, $\text{RHS}_{\pm 2}$ denote the right-hand sides of (7.76). Finally, we recover the original coordinates from

$$(7.78) \quad A = \frac{Z+W}{2}, \quad B = \frac{Z-W}{2i}, \quad S = X+B.$$

Note that the reconstruction uses only nonvanishing angular multipliers and the two Hardy maps on the subspaces on which they are defined. It therefore preserves the divisibility of Z and W .

For $n=0$, the second force equation in the original variables determines C , and we obtain

$$C_m = \frac{f_{2,m}}{iLm} \quad (m \neq 0), \quad (D+2)X_2 = g_{-,2}, \quad (D+2)X_{-2} = g_{+,-2}.$$

We solve the high modes for $V_h = (-i\mathcal{R})^{-1}X_h$. Their equations are (7.63) with $\sigma=1$ and V_h in place of C_h . The last equation has no mass term, and the angular Poincaré inequality on $|m| \geq 4$ gives the coercivity in (7.64). We recover $X_h = (-i\mathcal{R})V_h$ and finish with (7.76)–(7.78), omitting the terms $t^2 C$. The mean of A remains $-f_{0,0}/2$, the means of B , C , and S keep their prescribed values, and the last equation imposes no condition on the means because it carries the factor $I - \Pi$.

In summary, the construction of the even inverse $K_e^{\text{ev, aug}}$ proceeds in the order

$$(7.79) \quad \text{mean/scale} \longrightarrow C_{\pm 2} \longrightarrow C_h \longrightarrow (Z, W)_{\text{nonsingular}} \longrightarrow (W_2, Z_{-2}).$$

Substitution gives $\mathcal{L}_e^{\text{ev, aug}} K_e^{\text{ev, aug}} = I$. For the other identity, we apply the same steps to an arbitrary element of the domain. Uniqueness for J_+ , the coercivity (7.64), the equations (7.73), and uniqueness for J_- with zero quadratic coefficient recover each component in turn, so that $K_e^{\text{ev, aug}} \mathcal{L}_e^{\text{ev, aug}} = I$ as well.

Estimates for the even inverse. We let $\mathcal{X}_{\text{ev}, y, n}^j$, $\mathcal{Y}_{\text{ev}, y, n}^j$, and $\mathcal{T}_{\text{ev}, y, n}^j$ be the rescaled Cartesian domain, source, and balanced trace norms, as in (7.45). We write $K_{e, n}^{\text{ev, aug}}$ for the even inverse at cell frequency n , and set

$$\begin{aligned}
 (7.80) \quad E_{j, n}^{\text{ev}, (q)} &:= \left\| \omega_n \partial_\rho^q K_{e, n}^{\text{ev, aug}} f_n \right\|_{\mathcal{X}_{\text{ev}, y, n}^j} \\
 &\quad + \left\| \omega_n (T_n) \text{Tr}_{\text{phys}} \partial_\rho^q K_{e, n}^{\text{ev, aug}} f_n \right\|_{\mathcal{T}_{\text{ev}, y, n}^j}, \\
 F_{a, n}^{\text{ev}} &:= \left\| \omega_n f_n \right\|_{\mathcal{Y}_{\text{ev}, y, n}^a}.
 \end{aligned}$$

The weighted Green estimate (7.72), the conjugated Hardy identities (7.60) and their order-zero bounds, and the reconstruction formulas (7.73), (7.77), and (7.78) give

$$(7.81) \quad E_{j, n}^{\text{ev}, (0)} \leq C_j (F_{j+3, n}^{\text{ev}} + \lambda_n E_{j-1, n}^{\text{ev}, (0)}), \quad j \geq 1, \quad E_{0, n}^{\text{ev}, (0)} \leq C F_{3, n}^{\text{ev}}.$$

Here, the passage from the source f to the scalar right-hand sides uses two derivatives, and the mass or balanced trace estimate uses one more, while every other map in (7.79) loses no derivatives. Iterating the recurrence, we obtain

$$(7.82) \quad E_{j,n}^{\text{ev},(0)} \leq C_j \sum_{a=3}^{j+3} \lambda_n^{j+3-a} F_{a,n}^{\text{ev}}.$$

Each derivative in ρ requires at most one additional derivative of the source. Indeed, $J_{\pm,n}^\gamma$ is independent of ρ , while the coefficients in (7.61), (7.63), (7.74), and (7.76) are analytic for $0 \leq \rho \leq \rho_0$. Differentiating the variational identity and then the triangular reconstruction formulas, we obtain, for $q \geq 1$,

$$(7.83) \quad \begin{aligned} E_{0,n}^{\text{ev},(q)} &\leq C_{q,\gamma} \left(F_{3+q,n}^{\text{ev}} + \lambda_n \sum_{p < q} E_{0,n}^{\text{ev},(p)} \right), \\ E_{j,n}^{\text{ev},(q)} &\leq C_{j,q,\gamma} \left(F_{j+3+q,n}^{\text{ev}} + \lambda_n E_{j-1,n}^{\text{ev},(q)} + \lambda_n \sum_{p < q} E_{j,n}^{\text{ev},(p)} \right), \quad j \geq 1, \quad q \geq 1. \end{aligned}$$

The last term accounts for the factor λ_n produced by each differentiated coefficient or trace, and a double induction in (q, j) then gives

$$(7.84) \quad E_{j,n}^{\text{ev},(q)} \leq C_{j,q,\gamma} \sum_{a=0}^{j+3+q} \lambda_n^{j+3+q-a} F_{a,n}^{\text{ev}}.$$

We sum these bounds over the cell frequencies as in the odd case. The even norms have the same form as (7.45). Restoring $e^{\Phi_\gamma(0, \lambda_n)}$ and multiplying (7.82) by λ_n^ℓ , with $j + \ell \leq s$, gives terms with at most $s + 3$ derivatives of the source, so the Cauchy-Schwarz inequality in the finite sums and Parseval's identity give

$$(7.85) \quad \sum_{|n| \geq 2L} \|K_{e,n}^{\text{ev},\text{aug}} f_n\|_{\mathcal{X}_{\gamma,n}^s}^2 \leq C_s \sum_{|n| \geq 2L} \|f_n\|_{\mathcal{Y}_{\gamma,n}^{s+3}}^2.$$

Likewise, (7.84) gives

$$(7.86) \quad \sum_{|n| \geq 2L} \|\partial_\rho^q K_{e,n}^{\text{ev},\text{aug}} f_n\|_{\mathcal{X}_{\gamma,n}^s}^2 \leq C_{s,q,\gamma} \sum_{|n| \geq 2L} \|f_n\|_{\mathcal{Y}_{\gamma,n}^{s+3+q}}^2.$$

On the finite set $|n| < 2L$, which includes $n = 0$, the weights $\omega_n^{\pm 1}$ are uniformly bounded on the cell, and at $n = 0$ we use the weight in the original variables instead. Applying (7.53) to each component, taking the maximum over this finite set, and decreasing the same ρ_0 once more, we obtain both estimates on these cell frequencies as well. Thus, we have the bounds

$$(7.87) \quad \begin{aligned} \|K_e^{\text{ev},\text{aug}} f\|_{\mathcal{X}_\gamma^s} &\leq C_{s,\gamma} \|f\|_{\mathcal{Y}_\gamma^{s+3}}, \\ \|\partial_\rho^q K_e^{\text{ev},\text{aug}} f\|_{\mathcal{X}_\gamma^s} &\leq C_{s,q,\gamma} \|f\|_{\mathcal{Y}_\gamma^{s+3+q}}. \end{aligned}$$

Hence, the even block loses at most $\mu_{\text{ev}} = 3$ derivatives of the source at $q = 0$, and at most $3 + q$ for its q -th derivative in ρ , independently of the cutoffs.

7.8. Returning to the Cartesian operator. We can now complete the proof of Proposition 7.1 by assembling the odd and even inverses and returning to the Cartesian variables through the transfer maps of Section 6.

Proof of Proposition 7.1. We let $\mathbf{b}_e$ be the constant reference associated with M_e , and let $\mathcal{S}_X$ and $\mathcal{S}_Y$ be the restrictions of $J_{X,\mathbf{b}_e}$ and $J_{Y,\mathbf{b}_e}^{\text{raw}}$ from (6.69) and (6.74) to the scalar variables and to the even and odd angular modes used here. We let ι_Y adjoin zero affine scale and auxiliary outer trace data, and we let p_Y remove those entries. Writing these auxiliary data as z , we have $\iota_Y F = (F, 0)$ and $p_Y(F, z) = F$, so that

$$(7.88) \quad p_Y \iota_Y = I, \quad \iota_Y p_Y = I \quad \text{on the zero-auxiliary subspace.}$$

We put

$$\widetilde{\mathcal{L}}_e := \mathcal{L}_e^{\text{od}} \oplus \mathcal{L}_e^{\text{ev, aug}}, \quad K_e^{\text{aug}} := K_e^{\text{od}} \oplus K_e^{\text{ev, aug}},$$

and on the domain $\mathcal{X}_{M_e}^\infty$, the identity (6.88) takes the form

$$(7.89) \quad \widetilde{\mathcal{L}}_e \mathcal{S}_X u = \iota_Y \mathcal{S}_Y L_e u.$$

Conversely, (7.57) and the condition on the outer boundary show that $K_e^{\text{aug}} \iota_Y \mathcal{S}_Y f$ has zero affine scale coordinate and zero auxiliary outer trace, so it belongs to $\mathcal{S}_X(\mathcal{X}_{M_e})$, and we define the Cartesian inverse by

$$(7.90) \quad V_e = \mathcal{S}_X^{-1} K_e^{\text{aug}} \iota_Y \mathcal{S}_Y.$$

Indeed, using (7.89) and the two inverse identities for K_e^{aug} , we obtain

$$\begin{aligned} \iota_Y \mathcal{S}_Y L_e V_e &= \widetilde{\mathcal{L}}_e K_e^{\text{aug}} \iota_Y \mathcal{S}_Y = \iota_Y \mathcal{S}_Y, \\ V_e L_e u &= \mathcal{S}_X^{-1} K_e^{\text{aug}} \widetilde{\mathcal{L}}_e \mathcal{S}_X u = u. \end{aligned}$$

Injectivity of $\iota_Y \mathcal{S}_Y$ gives $L_e V_e = I$ from the first identity, while the second gives $V_e L_e = I$.

We now count the derivatives lost in composing these maps. By (6.91)–(6.92), the maps on the domain lose no derivatives, while the map on the source loses ℓ_Y . The odd and even inverses act on complementary summands and lose at most four and three derivatives, respectively, so their direct sum loses at most four. Writing ℓ_{alg} for the finite shift in the Sobolev index of the maps to and from the scalar coordinates, noted before (7.3), and of the algebraic equations, we may take

$$\ell_e = \ell_Y + 4 + \ell_{\text{alg}} < \infty.$$

For q derivatives in ρ , (7.56) and (7.87) require at most $4 + q$ derivatives of the source in the odd block and $3 + q$ in the even block, while the derivatives of the coefficients in the finite transfer maps lose none. Leibniz' rule therefore gives

$$(7.91) \quad \|\partial_\rho^q V_e f\|_{\mathcal{X}_\gamma^s} \leq C_{s,q,\gamma} \|f\|_{\mathcal{Y}_\gamma^{s+\ell_Y+4+\ell_{\text{alg}}+q}}.$$

The analytic weight is the same on both sides. For q derivatives in ρ , the constant includes the factor γ^{-q} from (7.43).

All the remaining factors in (6.69)–(6.75) lose no derivatives, so at $q = 0$ we recover (7.2). Restoring the orientation amounts to conjugating by R_{α_0} , which is unitary on every Cartesian scale.

□

7.9. The case of circular cross-sections. The even and affine inverse extends to the circle with uniform bounds on $0 \leq \rho \leq \rho_0$. This requires coercivity on the high even modes and reconstruction of W_2 and Z_{-2} without division by ρ . The resulting uniformity lets us choose the constants before fixing ρ_* .

The matrix through which the ellipse enters the even block is the matrix H_ρ introduced before (7.3),

$$(7.92) \quad H_\rho(\theta) = \frac{1}{1-\rho^2} \begin{pmatrix} 1-\rho \cos 2\theta & \rho \sin 2\theta \\ \rho \sin 2\theta & 1+\rho \cos 2\theta \end{pmatrix}.$$

We write K_ρ^{ev} for the even inverse constructed above, now at ellipse parameter ρ instead of ρ_* , and $\mathfrak{X}^s$, $\mathfrak{Y}^s$, and $\mathfrak{T}^s$ for its Cartesian domain, its source space, and its space of balanced traces on the outer boundary. We also write $\mathbb{X}_{\text{ev},\mathfrak{h}}^s$ and $\mathbb{Y}_{\text{ev},\mathfrak{h}}^s$ for the corresponding block spaces of Section 6. Note that the norms of these block spaces include the prescribed orders of the cap value, of the conormal trace, and of the mismatch at the interface. The uniformity of the even and affine inverse up to the circle is the content of the following lemma.

Lemma 7.3. *Fix $0 < \rho_0 < 1/4$ and $L > 0$, together with the mean and scale equations and the chart with prescribed transverse first derivatives on the axis. For every $0 \leq \rho \leq \rho_0$, the circle included, the even inverse K_ρ^{ev} exists, it is a two-sided inverse of the even operator, and on the spaces $\mathfrak{X}^s$, $\mathfrak{Y}^{s+3}$, and $\mathfrak{T}^s$ above it satisfies*

$$(7.93) \quad \|K_\rho^{\text{ev}} f\|_{\mathfrak{X}^s} + \|\text{Tr}_{\text{phys}} K_\rho^{\text{ev}} f\|_{\mathfrak{T}^s} \leq C_{s,\rho_0,L} \|f\|_{\mathfrak{Y}^{s+3}}.$$

Here, Tr_{phys} denotes the balanced value and conormal traces on the outer boundary, as in (7.72). In the block norms, the same inverse loses no derivatives,

$$(7.94) \quad K_{\rho,\mathfrak{h}}^{\text{ev}} : \mathbb{Y}_{\text{ev},\mathfrak{h}}^s \longrightarrow \mathbb{X}_{\text{ev},\mathfrak{h}}^s, \quad \sup_{0 \leq \rho \leq \rho_0} \|K_{\rho,\mathfrak{h}}^{\text{ev}}\|_s < \infty.$$

All ρ -derivatives of fixed order obey the analogous uniform tame bounds.

Proof. The numerator matrix in (7.92) has trace 2 and determinant $1 - \rho^2$, so its eigenvalues are $1 \pm \rho$, and those of H_ρ are $(1 \mp \rho)^{-1}$. We therefore have

$$(7.95) \quad \frac{1}{1+\rho_0} I \leq H_\rho \leq \frac{1}{1-\rho_0} I,$$

and the form on the high even modes then satisfies

$$\begin{aligned} \text{Re } \mathfrak{b}_\rho(C_h, C_h) &\geq \frac{1-4\rho_0}{1-\rho_0} \|(DC_h, \mathcal{R}C_h)\|_2^2 \\ &\quad + \text{nonnegative mass and Robin terms.} \end{aligned}$$

Indeed, the coefficient $1 - 3\rho/(1 - \rho) = (1 - 4\rho)/(1 - \rho)$ decreases from 1 at the circle to a positive value at $\rho_0 < 1/4$. Thus, the Lax-Milgram bound, the outer trace extension, and the value and conormal estimates are uniform. For $n = 0$, the angular Poincaré inequality on $|m| \geq 4$ supplies coercivity without the mass term.

On the affine modes, the formulas (7.61) and (7.76) contain only the factors

$$1 - \rho^2, \quad \rho, \quad (1 - \rho^2)^{-1},$$

so there is no division by ρ . The Hardy maps (7.58) are independent of ρ . The prescribed first derivatives of the original variation on the axis exclude the homogeneous quadratic solutions for W_2 and Z_{-2} , while Cartesian smoothness excludes the singular solutions of $D + 2$. The scale equation remains (7.57). Therefore, no kernel appears on the affine modes at $\rho = 0$.

We obtain the estimates at higher spatial regularity from uniform elliptic difference quotients. Of the three derivatives of the source in (7.93), two are used to obtain the scalar right-hand sides, and one more in the mass or balanced trace estimate, as counted in (7.81). These orders are already part of the block norms. Each Green, Hardy, cap, and trace map is therefore bounded without loss, and we obtain (7.94).

Finally, we write $\mathcal{L}_\rho^{\text{ev}}$ for the even operator $\mathcal{L}_e^{\text{ev}, \text{aug}}$ at ellipse parameter ρ . This family is analytic in ρ on the scale, and differentiating the identity $K_\rho^{\text{ev}} \mathcal{L}_\rho^{\text{ev}} = I$ gives

$$\partial_\rho K_\rho^{\text{ev}} = -K_\rho^{\text{ev}} (\partial_\rho \mathcal{L}_\rho^{\text{ev}}) K_\rho^{\text{ev}},$$

and iterating this formula gives the bounds at every fixed order. A constant change of orientation acts by orthogonal changes of the coordinates in the disk and in the normal plane. These changes preserve the radial weight and the angular and cell cutoffs. At $\rho = 0$, the operator is independent of the orientation. $\square$

8. THE CENTER AND AFFINE EQUATIONS

We now construct an inverse for the center and affine equations at high cell frequencies. We first solve on the cap with the prescribed axis data, then use the resulting value and conormal derivative as initial data for the outward evolution on the annulus. No condition is imposed on these components at the outer boundary. The relation between this inverse and the minus diagonal block of (6.94) is stated in Proposition 8.1.

Throughout, $\Lambda = \langle D_\zeta \rangle$ multiplies the cell mode $e^{in\zeta}$ by $\lambda_n = \langle n \rangle$, and we fix the analytic widths $\sigma_0 > \sigma_- > 0$, the outer radius R of the annulus, the radius $2T$ of the cap in the rescaled variable, and the base index s_* . Of these, σ_0 is the width of the strip of analyticity of Section 3.3 on the axis, and σ_- is the width that remains at the outer boundary. All constants are uniform on the compact parameter set of Section 3. We choose first $0 < \gamma < \min\{\sigma_0, 1\}$, then $\delta_A > 0$, so that

$$(8.1) \quad \gamma R < \sigma_0 - \sigma_-, \quad C_{\text{wt}}(\gamma + \gamma^2) < \frac{c_{\text{wt}}}{4}, \quad C_{\text{cap}}\gamma < \frac{1}{8}, \quad C_*\sqrt{\delta_A} < \gamma.$$

We then choose an upper bound for the ellipse parameter

$$(8.2) \quad 0 < \rho_0 < \min \left\{ \frac{1}{4}, \rho_{\text{ax}}, \rho_{\text{ctr}}, \rho_{\text{chart}} \right\}.$$

The bounds ρ_{ax} , ρ_{ctr} , and ρ_{chart} ensure, respectively, cap invertibility, separation of the center roots from the other roots, and validity of the fixed-space identifications in Section 6. We then bound the distance from each state in $\mathfrak{U}_-$ to its constant ellipse by $\eta_* > 0$, and choose the dyadic cutoff J of (4.2), so that

$$(8.3) \quad \begin{aligned} C_A \eta_* &\leq M_B \delta_A, & C_c \eta_* &< \gamma, \\ C_{\text{cap}}(2^{-J} + \gamma + \eta_*) &< \frac{1}{2}, & C_{\text{sy1}}(\Lambda_J^{-1/2} + \chi_-(\eta_*)) &< \frac{1}{2}, \\ C_{\mathbb{N}} \eta_* &< \frac{1}{2}, \end{aligned}$$

where $\Lambda_J = \inf\{\langle n \rangle : |n| \geq 2^J\} \simeq 2^J$ and $\chi_-(t) \rightarrow 0$ as $t \rightarrow 0$. The first two conditions control the perturbations of the affine and center energies. The next two make the cap error and the coupling correction small, and the last permits the final Neumann inversion. These choices are independent of an upper Fourier cutoff and of the number N of periods of the field.

For a state $\mathbf{b} = (\rho, a, \varepsilon, p)$ and the constant ellipse $\mathbf{b}_\rho = (\rho, a_\rho, \varepsilon_\rho, p_\rho)$ of Section 6, where $a_\rho = a_{\rho, p_e}^0$, $\varepsilon_\rho = 0$, and $p_\rho = p_e$, we put

$$(8.4) \quad \|\mathbf{b} - \mathbf{b}_\rho\|_{\mathcal{B}^s} := \|a - a_\rho\|_{\mathcal{A}_\gamma^s} + |\varepsilon - \varepsilon_\rho| + |p - p_\rho|,$$

and we let $\mathfrak{U}_-$ consist of the states with $0 \leq \rho \leq \rho_0$ for which (8.4), taken at the index $s_* + k_c$, is less than η_* . Here, k_c is a fixed number of additional coefficient derivatives, chosen large enough for the cap estimates and independent of the solution index s . At $\rho = 0$, derivatives in ρ are defined by extending the coefficients analytically to a neighborhood of the endpoint. The orientation α_0 is a separate parameter, and the estimates are used only for $\rho \geq 0$.

8.1. Function spaces and the inverse for the minus block. We use the fixed block spaces and identifications of Section 6. We now specify the spaces for the cap solution, its Cauchy data, and the evolution on the annulus. For the cap, we define

$$(8.5) \quad \begin{aligned} X_{c,-}^s &= H_J^X \mathcal{X}_{-, \text{cap}}^s, \\ Y_{c,-}^s &= H_J^Y (\mathcal{Y}_{-, \text{bulk}}^s \oplus \mathfrak{J}_{-, \text{ax}}^s). \end{aligned}$$

Here, $\mathcal{X}_{-, \text{cap}}^s$ is the Cartesian cap domain, $\mathcal{Y}_{-, \text{bulk}}^s$ is its interior source space, and $\mathfrak{J}_{-, \text{ax}}^s$ contains the prescribed axis data. The interior components have H^{s+2} regularity in $X_{c,-}^s$ and H^s regularity in the source, as in Table 1. The auxiliary data $\mathfrak{J}_{-, \text{ax}}^s$ prescribe the axis coefficients of the regular center solutions and the admissible linear Taylor coefficients. In the normalized chart, the transverse first derivatives are fixed, while the tangential coefficient τ remains an unknown. The range $Y_{c,-}^s$ contains no interface datum.

We measure the outgoing Cauchy data of the cap solution in

$$(8.6) \quad Z_{c,-}^s = H_J^{Zc} \mathfrak{T}_{c,-,\gamma}^s, \quad \|(z_D, z_N)\|_{Z_{c,-}^s} \simeq \left\| \Lambda^{1/2} z_D \right\|_{H^s} + \left\| \Lambda^{-1/2} z_N \right\|_{H^s},$$

where $\mathfrak{T}_{c,-,\gamma}^s$ is the space of Cauchy data on the interface of the cap, projected by the spectral projections of the principal symbol and weighted by the analytic weight. The half-order factors are the balanced trace norms of Section 6. We write $Z_{c,-}^s = Z_{c,D}^s \oplus Z_{c,N}^s$ for the splitting into the value and the conormal component.

For the inverse, we add to the physical conormal derivative a boundary contribution determined by the interior source. The value is unchanged. We call this pair the *split Cauchy data*. To define the contribution, let

$$E_{c,\Sigma} : (Z_{c,N}^s)' \longrightarrow (\mathcal{Y}_{-, \text{bulk}}^s)'$$

be the extension of the trace on the cap built from (4.11). For $f_c = (f_{c, \text{bulk}}, j_{\text{ax}}) \in Y_{c,-}^s$, we define its boundary contribution by transposing this extension,

$$(8.7) \quad \langle \mathbf{M}_{c,\Sigma}^s f_c, \varphi \rangle := \langle f_{c, \text{bulk}}, E_{c,\Sigma} \varphi \rangle, \quad \mathbf{M}_{c,\Sigma}^s : Y_{c,-}^s \longrightarrow Z_{c,N}^s,$$

where the transpose uses the weighted source-dual pairing. It is not the adjoint for the unweighted Hilbert inner product. We let $j_N z = (0, z)$ include the conormal component in $Z_{c,-}^s$. The *physical Cauchy data*, consisting of the value and classical conormal derivative at the state, are related to the split data by

$$(8.8) \quad \begin{aligned} \Gamma_{c,-,\mathbf{b}}^{\text{sp}}(u_c, f_c) &:= \Gamma_{c,-,\mathbf{b}}^{\text{ph}} u_c + j_N \mathbf{M}_{c,\Sigma}^s f_c, \\ \Gamma_{c,-,\mathbf{b}}^{\text{ph}} u_c &= \Gamma_{c,-,\mathbf{b}}^{\text{sp}}(u_c, f_c) - j_N \mathbf{M}_{c,\Sigma}^s f_c. \end{aligned}$$

When $f_c = \mathcal{A}_{c,-,\mathbf{b}} u_c$, where $\mathcal{A}_{c,-,\mathbf{b}}$ is the operator on the cap at the state, we abbreviate the first expression to $\Gamma_{c,-,\mathbf{b}}^{\text{sp}} u_c$. Note that $\mathbf{M}_{c,\Sigma}^s$ is bounded at each regularity level s , and its contribution affects only the

conormal component, which has lower order in the balanced trace space. It therefore changes neither the principal symbol nor the estimates for the homogeneous problem on the cap. When no superscript is given, we mean the physical Cauchy data $\Gamma_{c,-,\mathbf{b}}^{\text{ph}}$.

On the annulus, we set

$$(8.9) \quad \begin{aligned} X_{a,-}^s &= H_J^X (X_{C,\gamma}^s \oplus X_{A,\Phi}^s \oplus \mathfrak{F}_{-,X}^s), \\ Y_{a,-}^s &= H_J^Y (Y_{C,\gamma}^s \oplus Y_{A,\Phi}^s \oplus \mathfrak{F}_{-,Y}^s). \end{aligned}$$

Here, the subscripts C and A denote the center and affine components. The remaining mean, scale, gauge, and quotient equations of Section 6 form a triangular system. Their unknowns belong to $\mathfrak{F}_{-,X}^s$ and their prescribed data to $\mathfrak{F}_{-,Y}^s$. The center unknown consists of the two oscillator variables and their radial derivatives, as in (7.21). Its norm is defined by an energy on these four coordinates that remains uniform when the two speeds coincide. Its source norm is of type L^1 in r with values in H^s . For the affine component, we collect the upper entries of the two chains as $x = (x_1, x_2)$ and the lower entries as $y = (y_1, y_2)$, as described after Table 1. We write $f = (f_1, f_2)$ and $g = (g_1, g_2)$ for the sources in their respective equations and use

$$(8.10) \quad \begin{aligned} \|(x, y)\|_{X_{A,\Phi}^s} &= \|(\Lambda^s \mu x, \Lambda^{s+1/2} y)\|_{\Phi}, \\ \|(f, g)\|_{Y_{A,\Phi}^s} &= \|(\Lambda^s \mu f, \Lambda^{s+1/2} g)\|_{\Phi}, \quad \mu^2 = \delta_A + \Lambda^{-1}. \end{aligned}$$

Here, $\|\cdot\|_{\Phi}$ denotes the norm on the annulus after multiplication by the weight $e^{\Phi_{\gamma}(r,\Lambda)}$, and the factors μ and $\Lambda^{1/2}$ are the weights in Table 1 of the upper and lower entries of the two affine chains. The weight $e^{\Phi_{\gamma}(r,\Lambda)}$ is the analytic weight W_{γ} of (3.15), with

$$(8.11) \quad \Phi_{\gamma}(r, \lambda) = \sigma_0 \lambda - \gamma(\sqrt{1 + r^2 \lambda^2} - 1).$$

We write $Z_{a,-}^s$ and $Z_{R,-}^s$ for the spaces of Cauchy data on the inner and on the outer boundary of the annulus, balanced as in (8.6), and

$$(8.12) \quad T_{-,\mathbf{b}} : Z_{c,-}^s \longrightarrow Z_{a,-}^s$$

for the order-zero map from cap to annular trace coordinates. For a field u smooth across the interface, its defining relation is

$$T_{-,\mathbf{b}} \Gamma_{c,-,\mathbf{b}}^{\text{ph}} u = \gamma_{-,\mathbf{b}} u,$$

where $\gamma_{-,\mathbf{b}}$ records the same value and conormal data in the annular coordinates. To obtain the physical mismatch, we subtract $\mathbf{M}_{c,\Sigma}^s f_c$ from the split conormal datum before applying $T_{-,\mathbf{b}}$. Keeping this contribution instead gives the split mismatch. The full domain and range spaces are

$$(8.13) \quad \mathbb{X}_{-,\text{ret}}^s = X_{c,-}^s \oplus X_{a,-}^s, \quad \mathbb{Y}_{-,\text{ret}}^s = Y_{c,-}^s \oplus Y_{a,-}^s \oplus Z_{a,-}^s.$$

The physical and the split mismatches are related on this range by

$$(8.14) \quad \begin{aligned} \mathfrak{P}_{-,\mathbf{b}}^s(f_c, f_a, g^{\text{ph}}) &= (f_c, f_a, g^{\text{ph}} - T_{-,\mathbf{b}} j_N \mathbf{M}_{c,\Sigma}^s f_c), \\ (\mathfrak{P}_{-,\mathbf{b}}^s)^{-1}(f_c, f_a, g^{\text{sp}}) &= (f_c, f_a, g^{\text{sp}} + T_{-,\mathbf{b}} j_N \mathbf{M}_{c,\Sigma}^s f_c). \end{aligned}$$

The map subtracts the cap source contribution from the physical mismatch, and its inverse adds it back. Both are bounded at index s and leave the highest-order terms unchanged because this contribution has lower order. In particular,

$$(8.15) \quad g^{\text{ph}} = 0 \implies g^{\text{sp}} = -T_{-,\mathbf{b}} j_N \mathbf{M}_{c,\Sigma}^s f_c,$$

which is in general nonzero when the interior source on the cap is nonzero.

Construction of the inverse. We construct the inverse by solving first on the cap and then on the annulus. The component estimates are proved in the next three subsections and collected in [Proposition 8.1](#). At the reference state, the cap operator and its outgoing physical Cauchy data are

$$(8.16) \quad \mathcal{A}_{c,-,\rho}^{\text{diag}} : X_{c,-}^s \longrightarrow Y_{c,-}^s, \quad \Gamma_{c,-,\rho}^{\text{ph}} G_{c,-,\rho}^{\text{diag}} : Y_{c,-}^s \longrightarrow Z_{c,-}^s,$$

with inverse $G_{c,-,\rho}^{\text{diag}}$ constructed below. Adding the cap source contribution gives the split trace

$$(8.17) \quad \Gamma_{c,-,\rho}^{\text{sp}} G_{c,-,\rho}^{\text{diag}} f_c := \Gamma_{c,-,\rho}^{\text{ph}} G_{c,-,\rho}^{\text{diag}} f_c + j_N M_{c,\Sigma}^s f_c.$$

We let $\mathcal{L}_{a,-,\mathbf{b}_\rho}$ be the reference center and affine operator on the annulus, and let $\gamma_{-,\rho}$ be its inner Cauchy trace. On each shell,

$$(8.18) \quad \mathcal{B}_{a,-,\rho} := (\mathcal{L}_{a,-,\mathbf{b}_\rho}, \gamma_{-,\rho}) : X_{a,-}^s \longrightarrow Y_{a,-}^s \oplus Z_{a,-}^s$$

records the interior equation and the incoming Cauchy data. Its inverse $G_{a,-,\rho}$ solves for arbitrary prescribed values of these data, with no condition at the outer boundary. The outgoing Cauchy data $\Gamma_{R,-,\rho} G_{a,-,\rho}$ on the outer boundary are bounded into $Z_{R,-}^s$.

We define the *coupled reference operator* by

$$(8.19) \quad N_{-,\rho}^{\text{diag}}(u_c, u_a) = (\mathcal{A}_{c,-,\rho}^{\text{diag}} u_c, \mathcal{L}_{a,-,\mathbf{b}_\rho} u_a, \gamma_{-,\rho} u_a - T_{-,\mathbf{b}_\rho} \Gamma_{c,-,\rho}^{\text{sp}}(u_c, \mathcal{A}_{c,-,\rho}^{\text{diag}} u_c)).$$

We write m_-^{sp} for its last component and $m_-^{\text{ph}} = \gamma_{-,\rho} u_a - T_{-,\mathbf{b}_\rho} \Gamma_{c,-,\rho}^{\text{ph}} u_c$. Then, [\(8.8\)](#) gives the identity

$$(8.20) \quad m_-^{\text{sp}} = m_-^{\text{ph}} - T_{-,\mathbf{b}_\rho} j_N M_{c,\Sigma}^s \mathcal{A}_{c,-,\rho}^{\text{diag}} u_c,$$

so that the physical Cauchy data of the two pieces agree if and only if

$$(8.21) \quad m_-^{\text{ph}} = 0 \iff m_-^{\text{sp}} = -T_{-,\mathbf{b}_\rho} j_N M_{c,\Sigma}^s f_c.$$

We allow the split mismatch to be prescribed independently. Solving on the cap and using its trace as part of the annular initial data gives the inverse

$$(8.22) \quad G_{-,\rho}^{\mathbb{N},\text{diag}}(f_c, f_a, g^{\text{sp}}) = \left(G_{c,-,\rho}^{\text{diag}} f_c, G_{a,-,\rho}(f_a, g^{\text{sp}} + T_{-,\mathbf{b}_\rho} [\Gamma_{c,-,\rho}^{\text{ph}} G_{c,-,\rho}^{\text{diag}} f_c + j_N M_{c,\Sigma}^s f_c]) \right).$$

Perturbing the coupled reference operator. We let ι_-^X and π_-^Y be the domain inclusion and range projection of the minus block. We take this diagonal block in physical coordinates and then convert its mismatch to split coordinates,

$$(8.23) \quad N_{-,\mathbf{b}}^{\text{ph}} := \pi_-^Y \mathbb{N}_{\mathbf{b}} \iota_-^X, \quad N_{-,\mathbf{b}} := \mathfrak{P}_{-,\mathbf{b}}^s N_{-,\mathbf{b}}^{\text{ph}}.$$

Here, $N_{-,\mathbf{b}}^{\text{ph}}$ is the minus diagonal block of [\(6.83\)](#) on all shells. At $\mathbf{b} = \mathbf{b}_\rho$, its split form is $N_{-,\rho}^{\text{diag}}$. We denote the perturbation by

$$(8.24) \quad \mathcal{R}_{-,\mathbf{b}}^{\mathbb{N}} := N_{-,\mathbf{b}} - N_{-,\rho}^{\text{diag}}.$$

Besides the interior perturbation computed below, this difference contains the change in conormal coordinates and in the mismatch map [\(8.14\)](#). For the mismatch map, we have

$$\|(\mathfrak{P}_{-,\mathbf{b}}^s - \mathfrak{P}_{-,\mathbf{b}_\rho}^s)F\|_{\mathbb{Y}_{-,\text{ret}}^s} \leq C_s \left[\delta_0(\mathbf{b}) \|F\|_{\mathbb{Y}_{-,\text{ret}}^s} + \delta_s(\mathbf{b}) \|F\|_{\mathbb{Y}_{-,\text{ret}}^{s*}} \right],$$

by boundedness of $M_{c,\Sigma}^s$ and of the change of trace coordinates, and the inverse maps satisfy the same estimate. Here, using (8.4), we set

$$\delta_0(\mathbf{b}) = \|\mathbf{b} - \mathbf{b}_\rho\|_{\mathcal{B}^{s_*+k_c}}, \quad \delta_s(\mathbf{b}) = \|\mathbf{b} - \mathbf{b}_\rho\|_{\mathcal{B}^{s+k_c}}.$$

These are the coefficient distances at the base and higher regularity indices. We exclude the terms from low to high and from high to low frequencies from $\mathcal{R}_{-, \mathbf{b}}^{\mathbb{N}}$, since they enter the global inverse separately, and we obtain

$$(8.25) \quad \|\mathcal{R}_{-, \mathbf{b}}^{\mathbb{N}} U\|_{\mathbb{Y}_{-, \text{ret}}^s} \leq C_s \left[\delta_0(\mathbf{b}) \|U\|_{\mathbb{X}_{-, \text{ret}}^s} + \delta_s(\mathbf{b}) \|U\|_{\mathbb{X}_{-, \text{ret}}^{s_*}} \right].$$

Indeed, the reconstruction, the extraction of the coordinates, the change of mismatch coordinates, and the identifications of the data spaces are bounded at every fixed index, and the map $\mathcal{T}_{\mathbf{b}, \rho}^{CN}$ of (6.46), which converts the reference Cauchy data to those at the state, satisfies

$$\|((\mathcal{T}_{\mathbf{b}, \rho}^{CN})^{-1} - I)z\|_{Z^s} \leq C_s (\delta_0(\mathbf{b}) \|z\|_{Z^s} + \delta_s(\mathbf{b}) \|z\|_{Z^{s_*}}).$$

Here, Z^s is the product of the balanced trace spaces on the interfaces. Combining this bound with the perturbation bound proved below, we obtain (8.25) on the product spaces. At the base index, the uniform bound for the reference inverse, the base case of (8.25), and the last condition in (8.3) imply

$$\|G_{-, \rho}^{\mathbb{N}, \text{diag}} \mathcal{R}_{-, \mathbf{b}}^{\mathbb{N}}\| < \frac{1}{2},$$

and a Neumann series gives the inverse

$$(8.26) \quad \begin{aligned} G_{-, \mathbf{b}}^{\mathbb{N}} &:= (I + G_{-, \rho}^{\mathbb{N}, \text{diag}} \mathcal{R}_{-, \mathbf{b}}^{\mathbb{N}})^{-1} G_{-, \rho}^{\mathbb{N}, \text{diag}} \\ &= G_{-, \rho}^{\mathbb{N}, \text{diag}} (I + \mathcal{R}_{-, \mathbf{b}}^{\mathbb{N}} G_{-, \rho}^{\mathbb{N}, \text{diag}})^{-1}, \quad N_{-, \mathbf{b}} G_{-, \mathbf{b}}^{\mathbb{N}} = G_{-, \mathbf{b}}^{\mathbb{N}} N_{-, \mathbf{b}} = I. \end{aligned}$$

The constraint correction. We let $\iota_-^Y F_- = (F_-, 0)$ include the minus source in the high-frequency range, with zero low-frequency entries when it is viewed in the full range. We let π_-^Y select the minus component. Taking the minus diagonal block of the map $\mathcal{S}_{Y, \mathbf{b}}$ of Section 6 gives

$$(8.27) \quad \mathcal{S}_{Y, -, \mathbf{b}} := \pi_-^Y \mathcal{S}_{Y, \mathbf{b}} \iota_-^Y,$$

and, for a range vector $F_- + j_{Z, -, \mathbf{b}} z_-$ with F_- having zero auxiliary entries, we have

$$\mathcal{S}_{Y, -, \mathbf{b}}(F_- + j_{Z, -, \mathbf{b}} z_-) = F_- - \pi_-^Y \mathbb{B}_{\mathbf{b}} z_- + j_{Z, -, \mathbf{b}} z_-,$$

the inverse being the same triangular formula with the sign reversed. Here, $j_{Z, -, \mathbf{b}}$ is the inclusion $j_{Z, \mathbf{b}}$ of (6.81), acting on the auxiliary coordinates of the minus block and compressed to its range, and z_- also denotes the inclusion of such a coordinate in the full auxiliary space $\mathcal{Z}_{\text{aux}, \mathfrak{h}}^s$. We put

$$(8.28) \quad \widehat{D}_{-, \mathbf{b}} := \mathcal{S}_{Y, -, \mathbf{b}} (\mathfrak{P}_{-, \mathbf{b}}^s)^{-1} N_{-, \mathbf{b}}, \quad \widehat{G}_{-, \mathbf{b}} := G_{-, \mathbf{b}}^{\mathbb{N}} \mathfrak{P}_{-, \mathbf{b}}^s \mathcal{S}_{Y, -, \mathbf{b}}^{-1}.$$

In the formula for $\widehat{D}_{-, \mathbf{b}}$, the range maps first recover the physical mismatch and then subtract the interior source $\pi_-^Y \mathbb{B}_{\mathbf{b}} z_-$ produced by extending the auxiliary data z_- . This is the map (6.82). At the reference, we set $\widehat{D}_{-, \rho} = \mathcal{S}_{Y, -, \mathbf{b}_\rho} (\mathfrak{P}_{-, \mathbf{b}_\rho}^s)^{-1} N_{-, \rho}^{\text{diag}}$ and $\widehat{G}_{-, \rho} = G_{-, \rho}^{\mathbb{N}, \text{diag}} \mathfrak{P}_{-, \mathbf{b}_\rho}^s \mathcal{S}_{Y, -, \mathbf{b}_\rho}^{-1}$. Multiplying out, we obtain

$$(8.29) \quad \widehat{D}_{-, \mathbf{b}} \widehat{G}_{-, \mathbf{b}} = I, \quad \widehat{G}_{-, \mathbf{b}} \widehat{D}_{-, \mathbf{b}} = I.$$

The interior response $\mathbb{B}_{\mathbf{b}}\mathbf{C}_{X,\mathbf{b}}$ of the constraint data enters (6.83) with a plus sign and the map (6.84) with a minus sign, so the two cancel in $\pi_+^Y \mathcal{S}_{Y,\mathbf{b}} \mathbb{N}_{\mathbf{b}} \iota_-^X$. Indeed, we have the identities

$$(8.30) \quad \begin{aligned} \pi_+^Y \mathcal{S}_{Y,\mathbf{b}} \mathbb{N}_{\mathbf{b}} \iota_-^X &= \pi_+^Y \iota_{Y,\mathbf{b}} \mathbf{J}_{Y,\mathbf{b}}^{\text{raw}} L_{\mathbf{b}} \mathbf{R}_{X,\mathbf{b}} \iota_-^X + j_{Z,+,\mathbf{b}} q_+^Z \mathbf{C}_{X,\mathbf{b}} \iota_-^X, \\ \pi_+^Y \mathbb{B}_{\mathbf{b}} \mathbf{C}_{X,\mathbf{b}} \iota_-^X - \pi_+^Y \mathbb{B}_{\mathbf{b}} \mathbf{C}_{X,\mathbf{b}} \iota_-^X &= 0. \end{aligned}$$

Here, q_+^Z is the projection onto the auxiliary data of the positive block and $j_{Z,+,\mathbf{b}}$ includes them, so the term $j_{Z,+,\mathbf{b}} q_+^Z \mathbf{C}_{X,\mathbf{b}} \iota_-^X$ remains in this off-diagonal block. The block is not part of $\widehat{D}_{-,\mathbf{b}}$ and is estimated separately.

Proposition 8.1. *For every $\mathbf{b} \in \mathfrak{U}_-$, the reference cap and annulus operators constructed above have two-sided inverses on their stated spaces. Their coupled inverse $G_{-,\mathbf{b}}^{\mathbb{N}}$ satisfies (8.26), and $\widehat{G}_{-,\mathbf{b}}$ is a two-sided inverse of $\widehat{D}_{-,\mathbf{b}}$ as in (8.29). For every $s \geq s_*$, the solution and its outgoing Cauchy data satisfy*

$$(8.31) \quad \begin{aligned} &\left\| \widehat{G}_{-,\mathbf{b}} F \right\|_{\mathbb{X}_{-,\text{ret}}^s} + \left\| \Gamma_{R,-,\mathbf{b}} \pi_a \widehat{G}_{-,\mathbf{b}} F \right\|_{Z_{R,-}^s} \\ &\leq C_s \left[\|F\|_{\mathbb{Y}_{-,\text{ret}}^s} + (1 + \|\mathbf{b} - \mathbf{b}_\rho\|_{\mathcal{B}^{s+k_0}}) \|F\|_{\mathbb{Y}_{-,\text{ret}}^{s_*}} \right], \end{aligned}$$

where π_a is the projection onto the annular component, $\Gamma_{R,-,\mathbf{b}}$ is the outgoing Cauchy trace on the outer boundary at the state, and k_0 is the fixed number of additional derivatives of the coefficients required in the estimate, independent of s . The inverses on the cap and on the annulus, together with their trace maps, satisfy the same estimate. On the cap, this holds for the physical Cauchy data, and then for the split ones by adding the bounded contribution $\mathbf{M}_{c,\Sigma}^s f_c$ in (8.8).

For each fixed $m \geq 1$, the state derivatives of every solution or trace map $Q_{-,\mathbf{b}} : E^s \rightarrow F^s$ satisfy the estimate below, for finite integers k_m and $0 \leq \mu_m \leq m$. We denote the m -th derivative in directions $h_1, \dots, h_m$ by $D_{\mathbf{b}}^m Q_{-,\mathbf{b}}[h_1, \dots, h_m]$. For $h = (\dot{\rho}, \dot{a}, \dot{\varepsilon}, \dot{p})$, we set

$$(8.32) \quad \|h\|_{\text{tan},s} = \|\dot{a}\|_{\mathcal{A}_\gamma^s} + |\dot{\rho}| + |\dot{\varepsilon}| + |\dot{p}|, \quad \|h\|_{\text{lo}} = \|h\|_{\text{tan},s_*+k_m}.$$

Then, we have

$$(8.33) \quad \begin{aligned} &\|D_{\mathbf{b}}^m Q_{-,\mathbf{b}}[h_1, \dots, h_m] q\|_{F^s} \\ &\leq C_{s,m} \left[\prod_i \|h_i\|_{\text{lo}} \|q\|_{E^{s+\mu_m}} \right. \\ &\quad + \sum_i \|h_i\|_{\text{tan},s+k_m} \prod_{j \neq i} \|h_j\|_{\text{lo}} \|q\|_{E^{s_*+\mu_m}} \\ &\quad \left. + (1 + \|\mathbf{b} - \mathbf{b}_\rho\|_{\mathcal{B}^{s+k_m}}) \prod_i \|h_i\|_{\text{lo}} \|q\|_{E^{s_*+\mu_m}} \right]. \end{aligned}$$

We may take $\mu_m = 0$ for the maps on the cap. All constants are uniform for $0 \leq \rho \leq \rho_0$, for both signs of the cell frequency, with or without a finite upper cutoff, and they contain no negative power of ρ . At real states, that is, at states with real coefficients, the operator, its inverse, and all trace maps carry real fields to real fields under the identifications with the fixed spaces.

The next three subsections prove the component estimates and complete the proof of Proposition 8.1. The correction from $\widehat{D}_{-,\mathbf{b}}$ to $\pi_-^Y \mathbb{D}_{\mathbf{b}} \iota_-^X$ will be made in Section 10, after estimating the couplings between the blocks. The low and high frequencies are then joined in the global inverse.

8.2. The cap. *The inverse at the circle.* We begin at the circle, where the inverse on the cap can be constructed explicitly on the spaces (8.5), with the prescribed axis conditions. In the center modes $m = \pm 1$, the scalar equation is the Bessel problem

$$(8.34) \quad (\partial_t^2 + t^{-1}\partial_t - t^{-2} + 3)c = F,$$

whose regular homogeneous solution

$$(8.35) \quad u_0(t) = \frac{2}{\sqrt{3}}J_1(\sqrt{3}t) = t + O(t^3)$$

has leading Cartesian coefficient one. With $v_0(t) = Y_1(\sqrt{3}t)$ and the Wronskian $W = u_0v'_0 - u'_0v_0 = C/t$, where C is a nonzero constant, the variation of constants formula

$$(8.36) \quad c(t) = \alpha u_0(t) - u_0(t) \int_0^t \frac{v_0(s)F(s)}{W(s)} ds + v_0(t) \int_0^t \frac{u_0(s)F(s)}{W(s)} ds$$

solves (8.34) with $c(t) = \alpha t + O(t^3)$ on the axis. We apply this formula separately to the coefficients of $e^{i\theta}$ and $e^{-i\theta}$. Since a Cartesian source on the modes $m = \pm 1$ vanishes to first order on the axis, the singular parts of the kernels in (8.36) act as Hardy operators of the type (7.58), and the remaining parts are smooth on the cap. The bounds for the Hardy operators and for the smooth parts therefore give the two Cartesian derivatives that the norm of $X_{c,-}^s$ in (8.5) requires. Regular homogeneous solutions are multiples of u_0 , which proves uniqueness for the prescribed leading coefficient α .

We reduce the affine chains on the exceptional modes $m = \pm 2$ to the two Hardy maps

$$(8.37) \quad J_+q = t^{-2} \int_0^t s q(s) ds, \quad J_-q = t^2 \int_0^t q(s) s^{-3} ds.$$

With $D = t\partial_t$, direct differentiation gives $(D+2)J_+q = q$ and $(D-2)J_-q = q$, the second on the subspace with vanishing t^2 -coefficient. Fixing the linear Taylor coefficients on the axis excludes the homogeneous solution t^2 . For arbitrary admissible coefficients, we first subtract a Cartesian polynomial with those coefficients and move its image under the operator to the source. We then apply J_- to the remainder. This polynomial extension and its image use only finitely many transverse coefficients and are bounded at every block index s . The formulas give both inverse identities.

Together with the mean and scale equations and the affine reconstruction in (7.79), (8.36) and (8.37) give the inverse at the circle. Pulling the family of small ellipses back to the fixed spaces, we obtain a family of operators analytic in ρ whose coefficients, in the coordinates given by the Hardy maps (8.37), contain only the factors

$$(8.38) \quad \rho, \quad 1 - \rho^2, \quad (1 - \rho^2)^{-1}.$$

A Neumann series about the inverse at the circle gives invertibility for sufficiently small ρ_0 , and the inverse and all its ρ -derivatives of fixed order are then uniformly bounded on $[0, \rho_0]$. The same holds uniformly in the *normalized cell frequency* $\kappa = 2^{-j}n$ of the j -th dyadic shell, which we let range over $1/2 \leq |\kappa| \leq 2$, when we replace 3 by $3\kappa^2$ and u_0 by $2J_1(\sqrt{3}|\kappa|t)/(\sqrt{3}|\kappa|)$. Note that the limit at the axis again has leading coefficient one. Complex conjugation exchanges the two signs of κ , and the Hardy maps and the remaining triangular factors depend analytically on κ , so compactness gives uniform bounds for the inverse of the operator with frozen coefficients and for the derivatives of its symbol used below.

The reference problem on the shells. Recall the dyadic shells Q_j of (6.2). We restrict a field on the j -th shell to

$$0 \leq r \leq 2TL2^{-j}, \quad t_j = 2^j r/L \in [0, 2T],$$

and pull it back to the fixed domain or range space, writing $\mathcal{R}_j^X$ and $\mathcal{R}_j^Y$ for these maps. We thus obtain the sequence spaces

$$(8.39) \quad \begin{aligned} \mathfrak{X}_{c,H}^s &= \{(U_j)_{j \geq J} : Q_j U_j = U_j\} \subset \ell_s^2(X^{\text{cap}}), \\ \mathfrak{Y}_{c,H}^s &= \{(F_j)_{j \geq J} : Q_j F_j = F_j\} \subset \ell_s^2(Y^{\text{cap}}). \end{aligned}$$

Here, X^{cap} and Y^{cap} are the fixed Cartesian domain and range spaces on the cap $[0, 2T]$, and ℓ_s^2 is the weighted direct sum of (6.54).

We let $\mathcal{P}_{a,\rho,j}^C z = G_{a,-,\rho}(0, z)$ on shell j . By (8.18), this solves the homogeneous reference annular equations with inner Cauchy data z . It extends a cap field to its annular complement by

$$(8.40) \quad \iota_{c,j}^{X,\text{gl}} U := \left(U, \mathcal{P}_{a,\rho,j}^C T_{-, \mathbf{b}_\rho} \Gamma_{c,-,\rho}^{\text{ph}} U, 0_{Z_j} \right).$$

The cap and annular fields then have matching physical Cauchy data at the reference, and if the source on the cap is nonzero, their split mismatch is instead given by (8.21). The zero auxiliary coordinate in (8.40) records the physical mismatch, and on the range it is converted by (8.14). We set $\iota_c^{X,\text{gl}} = \bigoplus_{j \geq J} \iota_{c,j}^{X,\text{gl}}$ and $\iota_c^Y F = (F, 0, 0)$ on the Cartesian range, and we write π_c^X and π_c^Y for the projections onto the cap coordinates. We extract the coordinates locally and reconstruct them through the full cap and annular field by setting

$$(8.41) \quad \begin{aligned} \mathcal{A}_X u &= (\mathcal{R}_j^X Q_j u)_{j \geq J}, \quad \mathcal{S}_X = \mathfrak{R}_{\text{sh},J,\rho}^X \iota_c^{X,\text{gl}}, \\ \mathcal{A}_Y F &= (\mathcal{R}_j^Y Q_j F)_{j \geq J}, \quad \mathcal{S}_Y = \mathfrak{R}_{\text{sh},J,\rho}^{Y,\text{raw}} \iota_c^Y, \end{aligned}$$

where (6.56) first glues each cap to its own annular complement and then sums the fields on the shells. We use the reference $\mathbf{b}_\rho$ because its value and normal extensions do not depend on ζ and preserve every Q_j . Since $\mathcal{A}_X = \pi_c^X \mathfrak{B}_{\text{sh},J,\rho}^X$ and $\mathcal{A}_Y = \pi_c^Y \mathfrak{B}_{\text{sh},J,\rho}^{Y,\text{raw}}$, the inverse identities in (6.56) give

$$(8.42) \quad \begin{aligned} \Pi_X &:= \mathcal{A}_X \mathcal{S}_X = I_{\mathfrak{X}_{c,H}}, & \Pi_Y &:= \mathcal{A}_Y \mathcal{S}_Y = I_{\mathfrak{Y}_{c,H}}, \\ \Pi_X^2 &= \Pi_X, & \Pi_Y^2 &= \Pi_Y, \\ \mathcal{S}_X \mathcal{A}_X &= I \text{ on } \text{Ran } \mathcal{S}_X, & \mathcal{S}_Y \mathcal{A}_Y &= I \text{ on } \text{Ran } \mathcal{S}_Y. \end{aligned}$$

As in Section 6, variable coefficients act after $\mathfrak{R}_{\text{sh},J,\rho}^X$ has joined the cap and annulus fields and reconstructed a global field.

The explicit inverse has coefficients frozen on the axis. It gives an approximate inverse, or *parametrix*, on the shells, with small errors on both sides at high cell frequency and for a sufficiently weak analytic weight. The following lemma applies to both cap blocks.

Lemma 8.2. *Let $\mathcal{C}(q, \kappa)$ be the operator with frozen coefficients on the cap $[0, 2T]$ of the minus block or of the positive block, after separation of the modes and identification with fixed Cartesian domain and range spaces. The parameter q lies in a compact set of finite jets, and $1/2 \leq |\kappa| \leq 2$ is the normalized cell frequency variable. Suppose that*

$$K(q, \kappa) \mathcal{C}(q, \kappa) = I_X, \quad \mathcal{C}(q, \kappa) K(q, \kappa) = I_Y,$$

that $(q, \kappa) \mapsto (\mathcal{C}, K)$ is analytic between the corresponding spaces at the base index s_* , and that for each finite order N used below we have the uniform bounds

$$(8.43) \quad \sup_{(q, \kappa)} \max_{|\alpha|+r \leq N} (\|\partial_q^\alpha \partial_\kappa^r \mathcal{C}(q, \kappa)\|_{X \rightarrow Y} + \|\partial_q^\alpha \partial_\kappa^r K(q, \kappa)\|_{Y \rightarrow X}) \leq M_N.$$

We require the same bounds for all added axis, Dirichlet, and balanced trace components. Let $q_{\mathbf{b}_\rho}$ be the jet of the reference ellipse with parameter ρ , whose coefficients do not depend on the cell variable, let

$$(8.44) \quad \mathbf{L}_{\rho, H}^{\text{sh}} := \mathfrak{B}_{\text{sh}, J, \rho}^{Y, \text{raw}} L_{\mathbf{b}_\rho} \mathfrak{R}_{\text{sh}, J, \rho}^X$$

be the reconstructed reference operator in the coordinates of the shells, and let

$$(8.45) \quad \mathbf{C}_\rho^{\text{cap}} := \text{Diag}_{j \geq J} \left(\pi_{c, j}^Y \mathbf{L}_{\rho, H}^{\text{sh}} \iota_{c, j}^{X, \text{gl}} \right) : \mathfrak{X}_{c, H}^{s_*} \longrightarrow \mathfrak{Y}_{c, H}^{s_*}$$

be its diagonal on the cap components. Here, $\pi_{c, j}^Y$ is the projection onto the cap coordinate of the j -th shell, and the inclusion $\iota_{c, j}^{X, \text{gl}}$ adds the annular field and the zero mismatch needed for the reconstruction. Thus, (8.45) evaluates each cap field only on its own collar.

For $2^j \leq |n| < 2^{j+1}$, we set $\kappa_{j, n} = 2^{-j}n$. Freezing the coefficients on the axis, we obtain the following Fourier multipliers on the sequence spaces,

$$(8.46) \quad \begin{aligned} (\mathbf{C}_\rho^{\text{fr}} U)_{j, n} &= \mathcal{C}(q_{\mathbf{b}_\rho}, \kappa_{j, n}) U_{j, n}, \\ (\mathbf{K}_\rho^{\text{fr}} F)_{j, n} &= K(q_{\mathbf{b}_\rho}, \kappa_{j, n}) F_{j, n}, \\ \widehat{\mathbf{C}}_\rho^{\text{fr}} &:= \Pi_Y \mathbf{C}_\rho^{\text{fr}} \Pi_X = \mathbf{C}_\rho^{\text{fr}}, \quad \widehat{\mathbf{K}}_\rho^{\text{fr}} := \Pi_X \mathbf{K}_\rho^{\text{fr}} \Pi_Y = \mathbf{K}_\rho^{\text{fr}}. \end{aligned}$$

Here, $U_{j, n}$ and $F_{j, n}$ denote the n -th cell Fourier coefficients of U_j and F_j . Then, the reference parametrix

$$(8.47) \quad \mathbf{K}_{\rho, H}^{(0)} := \widehat{\mathbf{K}}_\rho^{\text{fr}}$$

satisfies, on the sequence spaces at the base index,

$$(8.48) \quad \left\| \mathbf{K}_{\rho, H}^{(0)} \mathbf{C}_\rho^{\text{cap}} - I \right\|_{\mathfrak{X}_{c, H}^{s_*} \rightarrow \mathfrak{X}_{c, H}^{s_*}} + \left\| \mathbf{C}_\rho^{\text{cap}} \mathbf{K}_{\rho, H}^{(0)} - I \right\|_{\mathfrak{Y}_{c, H}^{s_*} \rightarrow \mathfrak{Y}_{c, H}^{s_*}} \leq C (2^{-J} + \gamma),$$

where C depends only on T , on the bounds (8.43), and on the compact set of coefficient jets. For the positive cap problem, the assumed operators include the Dirichlet datum on the interface. The outgoing balanced conormal derivative of $\mathbf{K}_{\rho, H}^{(0)}$ satisfies the corresponding estimate of order zero.

Proof. On Q_j , we evaluate the multiplier in (8.46) at each $\kappa_{j, n} = 2^{-j}n$, and the assumed bounds cover both signs of the cell frequency and the whole normalized frequency range $1/2 \leq |\kappa| \leq 2$.

After rescaling the equations and the unknowns, we find that each differential expression is a finite sum of terms of the form

$$(8.49) \quad a_\nu(2^{-j}Lt, t, \zeta)(t\partial_t)^{\alpha_\nu}(2^{-j}\partial_\zeta)^{\beta_\nu}, \quad \alpha_\nu + \beta_\nu \leq 2,$$

acting between the fixed Cartesian spaces. Note that the axis and boundary operators keep the orders of Table 1. Since the reference coefficients are smooth in the radial variable, Taylor's formula on $0 \leq t \leq 2T$ gives

$$(8.50) \quad a_\nu(2^{-j}Lt, t, \zeta) = a_\nu(0, t, \zeta) + 2^{-j}Lt \int_0^1 (\partial_r a_\nu)(s2^{-j}Lt, t, \zeta) \, ds.$$

The remainder is bounded by $C_T 2^{-j}$, and the same calculation applies to the finite axis equations. The spectral projections depend analytically on the coefficients through their contour integrals, so their remainders obey the same bound.

The reference coefficients do not depend on ζ , hence preserve each Fourier mode and each dyadic shell, giving

$$(8.51) \quad \widehat{\mathbf{K}}_\rho^{\text{fr}} \widehat{\mathbf{C}}_\rho^{\text{fr}} = \Pi_X, \quad \widehat{\mathbf{C}}_\rho^{\text{fr}} \widehat{\mathbf{K}}_\rho^{\text{fr}} = \Pi_Y,$$

and we have the corresponding identities for the projections

$$(8.52) \quad \Pi_Y \mathbf{C}_\rho^{\text{fr}} - \mathbf{C}_\rho^{\text{fr}} \Pi_X = 0, \quad \Pi_X \mathbf{K}_\rho^{\text{fr}} - \mathbf{K}_\rho^{\text{fr}} \Pi_Y = 0.$$

The bounds on the symbols and Parseval's identity give

$$(8.53) \quad \|\widehat{\mathbf{C}}_\rho^{\text{fr}}\|_{\mathfrak{X}_{c,H}^{s*} \rightarrow \mathfrak{Y}_{c,H}^{s*}} + \|\widehat{\mathbf{K}}_\rho^{\text{fr}}\|_{\mathfrak{Y}_{c,H}^{s*} \rightarrow \mathfrak{X}_{c,H}^{s*}} \leq C,$$

so the inverse of the operator with frozen coefficients satisfies both inverse identities on these spaces. In particular, we have

$$(8.54) \quad \|\widehat{\mathbf{K}}_\rho^{\text{fr}} \widehat{\mathbf{C}}_\rho^{\text{fr}} - \Pi_X\| + \|\widehat{\mathbf{C}}_\rho^{\text{fr}} \widehat{\mathbf{K}}_\rho^{\text{fr}} - \Pi_Y\| = 0.$$

We now compare the reference operator on the cap with the multiplier frozen on the axis. On the j -th shell, the difference is

$$(8.55) \quad \mathcal{E}_{\rho,j} := \pi_{c,j}^Y \mathbf{L}_{\rho,H}^{\text{sh}} \iota_{c,j}^{X,\text{gl}} - \mathcal{C}(q_{\mathbf{b}_\rho}, 2^{-j} D_\zeta) Q_j : Q_j X^{\text{cap}} \longrightarrow Q_j Y^{\text{cap}}.$$

Since $L_{\mathbf{b}_\rho}$ is independent of the cell variable, there are no terms between shells. Taylor's estimate (8.50) bounds the radial error by $C_T 2^{-j}$. The reconstruction joins the matching annular field to the cap before the operator acts. Its commutators belong to $\mathcal{E}_{\rho,j}$ and satisfy the same bound at the reference. Conjugation by the analytic weight contributes $C_T \gamma$, since $r\Lambda = O_T(1)$ on the scaled cap. The trace theorem in the Cartesian variables on the cap gives the same bounds for the added value and balanced conormal components. Parseval's identity then gives

$$(8.56) \quad \|\mathbf{C}_\rho^{\text{cap}} - \widehat{\mathbf{C}}_\rho^{\text{fr}}\|_{\mathfrak{X}_{c,H}^{s*} \rightarrow \mathfrak{Y}_{c,H}^{s*}} \leq C_T(2^{-j} + \gamma).$$

Using (8.51), we factor the two errors as

$$(8.57) \quad \begin{aligned} \mathbf{K}_{\rho,H}^{(0)} \mathbf{C}_\rho^{\text{cap}} - I &= \widehat{\mathbf{K}}_\rho^{\text{fr}} (\mathbf{C}_\rho^{\text{cap}} - \widehat{\mathbf{C}}_\rho^{\text{fr}}), \\ \mathbf{C}_\rho^{\text{cap}} \mathbf{K}_{\rho,H}^{(0)} - I &= (\mathbf{C}_\rho^{\text{cap}} - \widehat{\mathbf{C}}_\rho^{\text{fr}}) \widehat{\mathbf{K}}_\rho^{\text{fr}}. \end{aligned}$$

The bounds (8.53) and (8.56) now give (8.48), and applying the trace map to these identities, we obtain the estimate for the outgoing balanced conormal derivative. $\square$

Perturbation at a general state. For a general state, the coefficients may mix cell frequencies. We therefore retain both the cap and annulus fields when comparing with the reference, and set

$$(8.58) \quad \mathbf{L}_{\mathbf{b},H}^{\text{sh}} := \mathfrak{B}_{\text{sh},J,\rho}^{Y,\text{raw}} L_{\mathbf{b}} \mathfrak{R}_{\text{sh},J,\rho}^X, \quad \mathbf{R}_{\mathbf{b},H}^{\text{sh}} := \mathbf{L}_{\mathbf{b},H}^{\text{sh}} - \mathbf{L}_{\rho,H}^{\text{sh}}.$$

For $\alpha, \beta \in \{c, a\}$, its blocks are

$$(8.59) \quad \mathcal{R}_{\mathbf{b},jk}^{\alpha\beta} := \pi_{\alpha,j}^Y \mathbf{R}_{\mathbf{b},H}^{\text{sh}} \iota_{\beta,k}^X : Q_k X_{\beta,k}^s \longrightarrow Q_j Y_{\alpha,j}^s.$$

Here, $\iota_{\beta,k}^X$ is the inclusion of the component β of the k -th shell in the space of separate fields (6.54), $\pi_{\alpha,j}^Y$ is the projection onto the component α of the j -th shell of the range, and $X_{\beta,k}^s$ and $Y_{\alpha,j}^s$ are the corresponding

summands. The first superscript is the output region and the second is the input region; for example, $\mathcal{R}_{\mathbf{b},jk}^{ca}$ sends an annular field on shell k to a cap source on shell j . We write $\text{Ins}_{k,\rho}^{\text{br}}(U_{c,k}, U_{a,k})$ for the reference insertion of a cap field and its annular complement. Note that this insertion is defined at all radii and that a mismatch is allowed. The interior operator is then

$$(8.60) \quad \mathcal{R}_{\mathbf{b},jk}^{\text{bulk}} U_k = \mathcal{R}_j^Y Q_j (L_{\mathbf{b}} - L_{\mathbf{b}_\rho}) \text{Ins}_{k,\rho}^{\text{br}}(U_{c,k}, U_{a,k}).$$

The coefficient acts on the full cap and annulus pair. A cap input contributes only within its own collar. Outside that collar, the input belongs to the annular block $\beta = a$ of (8.59). We write $\mathcal{Y}_{\text{br},H}^{s,\text{raw}}$ for the weighted direct sum of the Cartesian source spaces on the caps and on the annuli, which is the counterpart on the range of $\mathcal{X}_{\text{br},H}^s$ in (6.54).

We abbreviate

$$(8.61) \quad \delta_0(\mathbf{b}) := \|\mathbf{b} - \mathbf{b}_\rho\|_{\mathcal{B}^{s_*+k_c}}, \quad \delta_s(\mathbf{b}) := \|\mathbf{b} - \mathbf{b}_\rho\|_{\mathcal{B}^{s+k_c}},$$

which measure the distance from the state to the constant ellipse at the base index and at the index s . Here, k_c is the fixed number of additional derivatives of the coefficients required in the estimate, independent of s . The numbers k_A , k_* , and k below have the same meaning in their respective estimates. Expanding the two reconstruction maps, the coefficients of the state, and the reduction of the source, we separate the difference into

$$(8.62) \quad \mathbf{R}_{\mathbf{b},H}^{\text{sh}} = R_{\mathbf{b}}^{\text{hi}} + R_{\mathbf{b}}^{\text{nb}} + R_{\mathbf{b}}^{\text{bd}} + R_{\mathbf{b}}^{\text{res}} + R_{\mathbf{b}}^{\text{triv}} + R_{\mathbf{b}}^{\text{syn}} + R_{\mathbf{b}}^{\text{ph}}.$$

The following table lists the contributions. Each is estimated on the reconstructed field by coefficient differences, bounded coordinate maps, and the tame product estimate. The cutoff term is separated because it becomes small with the distance to the reference, rather than with increasing J .

Term	Contribution
R^{hi}	High-frequency coefficient terms on the diagonal of the shells.
R^{nb}	Terms between distinct shells $j, k > J$, including cap-annulus terms.
R^{bd}	Terms meeting the cutoff, where $j = J$ or $k = J$.
R^{res}	Change in the reduction of the source.
R^{triv}	Change in the identifications with fixed spaces.
R^{syn}	Commutators of the reference reconstruction with $L_{\mathbf{b}} - L_{\mathbf{b}_\rho}$.
R^{ph}	Changes in the weight, the value and conormal coordinates after reconstruction, and the conormal extensions.

On each shell, we write (d, h_ρ) for the reference Cauchy data, $(d, n_{\mathbf{b}})$ for the physical data at the state, and $f_{c,j}$ for the cap interior source. The split data are

$$(8.63) \quad \begin{pmatrix} d \\ r_{\mathbf{b}}^{\text{sp}} \end{pmatrix} := \begin{pmatrix} d \\ n_{\mathbf{b}} \end{pmatrix} + j_N \mathbf{M}_{c,\Sigma}^s f_{c,j},$$

and the physical and the reference Cauchy data are related by

$$(8.64) \quad \begin{pmatrix} d \\ n_{\mathbf{b}} \end{pmatrix} = T_{\mathbf{b},\rho,j} \begin{pmatrix} d \\ h_\rho \end{pmatrix}, \quad T_{\mathbf{b},\rho,j} = \begin{pmatrix} I & 0 \\ K_{\mathbf{b},\rho,j} & E_{\mathbf{b},\rho,j} \end{pmatrix}, \quad E_{\mathbf{b},\rho,j} = e_{\mathbf{b}} e_\rho^{-1}.$$

Here, $T_{\mathbf{b},\rho,j}$ is $\mathcal{T}_{\Sigma,\mathbf{b},\rho}^{CN}$ from (6.46) on interface j . The normal coefficients $e_{\mathbf{b}}$ and e_{ρ} are uniformly invertible, while the tangential term $K_{\mathbf{b},\rho,j} = K_{\Sigma,\mathbf{b},\rho}$ has order zero between the balanced spaces. Thus, the inverse is

$$T_{\mathbf{b},\rho,j}^{-1} = \begin{pmatrix} I & 0 \\ -E_{\mathbf{b},\rho,j}^{-1} K_{\mathbf{b},\rho,j} & E_{\mathbf{b},\rho,j}^{-1} \end{pmatrix}.$$

To recover the reference conormal derivative, we therefore subtract the boundary contribution $\mathbf{M}_{c,\Sigma}^s f_{c,j}$ before applying the inverse, which gives

$$(8.65) \quad \begin{pmatrix} d \\ h_{\rho} \end{pmatrix} = T_{\mathbf{b},\rho,j}^{-1} \left(\begin{pmatrix} d \\ r_{\mathbf{b}}^{\text{sp}} \end{pmatrix} - j_N \mathbf{M}_{c,\Sigma}^s f_{c,j} \right).$$

If we applied $T_{\mathbf{b},\rho,j}^{-1}$ directly to the split Cauchy data, we would miss this source term. Since $T_{\mathbf{b},\rho,j}$ need not commute with Q_j , we include its terms between different shells in $R^{\text{bd}} + R^{\text{nb}} + R^{\text{ph}}$.

At $\mathbf{b} = \mathbf{b}_{\rho}$, the independence of the cell variable makes every term between different shells and every term at the cutoff in (8.62) vanish. On the reconstructed field, Parseval's identity and the tame product estimate (3.20) give

$$(8.66) \quad \|\mathbf{R}_{\mathbf{b},H}^{\text{sh}} U\|_{\mathcal{Y}_{\text{br},H}^{s,\text{raw}}} \leq C_s \left[\delta_0(\mathbf{b}) \|U\|_{\mathcal{X}_{\text{br},H}^s} + \delta_s(\mathbf{b}) \|U\|_{\mathcal{X}_{\text{br},H}^{s*}} \right].$$

This includes the operator $R_{\mathbf{b}}^{\text{bd}}$ at the cutoff, extended by zero away from the J -th shell, for which we have the estimate

$$(8.67) \quad \|R_{\mathbf{b}}^{\text{bd}} U\|_{\mathcal{Y}_{\text{br},H}^{s,\text{raw}}} \leq C_s \left[\delta_0(\mathbf{b}) \|U\|_{\mathcal{X}_{\text{br},H}^s} + \delta_s(\mathbf{b}) \|U\|_{\mathcal{X}_{\text{br},H}^{s*}} \right].$$

The factor 2^{-J} in (8.56) comes from freezing the radial coefficients. It does not apply to the cutoff estimate (8.67).

The terms coupling low and high frequencies also satisfy the perturbative bound

$$(8.68) \quad \begin{aligned} & \|P_J^Y L_{\mathbf{b}} H_J^X u_H\|_{Y^s} + \|H_J^Y L_{\mathbf{b}} P_J^X u_L\|_{Y^s} \\ & \leq C_{J,s} \left[\delta_0(\mathbf{b}) (\|u_H\|_{X^s} + \|u_L\|_{X^s}) + \delta_s(\mathbf{b}) (\|u_H\|_{X^{s*}} + \|u_L\|_{X^{s*}}) \right]. \end{aligned}$$

Here, u_H and u_L are arbitrary fields, of which the display keeps the high and the low cell frequencies. The two terms also vanish at the reference, whose coefficients do not depend on the cell variable.

For the cap of the minus block, we take the parameter

$$(8.69) \quad q_{-, \rho} = (\rho, j_y^1 a_{\rho}(0), \varepsilon_{\rho}, p_{\rho}, P_{-, \rho}^D(0), P_{-, \rho}^R(0)),$$

where $j_y^1 a_{\rho}(0)$ records the value and first derivatives in y of a_{ρ} at the axis, and $P_{-, \rho}^D(0)$ and $P_{-, \rho}^R(0)$ are the spectral projections (6.21) of the reference at the axis. We take the inverse $K_{-, \rho}^{\text{fr}}(\kappa)$ built from the Bessel and Hardy formulas above. Applying Lemma 8.2 with $q_{\mathbf{b}, \rho} = q_{-, \rho}$, we obtain the reference operator on the cap

$$(8.70) \quad \mathcal{A}_{c,-,\rho}^{\text{diag}} := \mathbf{C}_{\rho}^{\text{cap}} : \mathfrak{X}_{c,H}^{s*} \longrightarrow \mathfrak{Y}_{c,H}^{s*}.$$

Here, as in Proposition 8.1, we identify the spaces $X_{c,-}^s$ and $Y_{c,-}^s$ with the sequence spaces by $\mathcal{A}_X$ and $\mathcal{A}_Y$ in (8.41). The bounds on the symbols and Parseval's identity define the operator at every $s \geq s_*$, and we prove the Neumann estimate first at the base index. Note that the operator is the diagonal on the caps of the reconstructed reference problem, and not the separate compression $H_J^Y \mathcal{C}_{-, \mathbf{b}} H_J^X$. With $K_{c,-,\rho}^{(0)} := \mathbf{K}_{\rho,H}^{(0)}$, (8.48) gives

$$(8.71) \quad \|K_{c,-,\rho}^{(0)} \mathcal{A}_{c,-,\rho}^{\text{diag}} - I\| + \|\mathcal{A}_{c,-,\rho}^{\text{diag}} K_{c,-,\rho}^{(0)} - I\| \leq C_{\text{cap}} (2^{-J} + \gamma) < \frac{1}{2}.$$

We write the two errors as

$$R_X^\rho = K_{c,-,\rho}^{(0)} \mathcal{A}_{c,-,\rho}^{\text{diag}} - I, \quad R_Y^\rho = \mathcal{A}_{c,-,\rho}^{\text{diag}} K_{c,-,\rho}^{(0)} - I.$$

The operator

$$(8.72) \quad G_{c,-,\rho}^{\text{diag}} := (I + R_X^\rho)^{-1} K_{c,-,\rho}^{(0)} = K_{c,-,\rho}^{(0)} (I + R_Y^\rho)^{-1}$$

is a two-sided inverse of (8.70). The trace theorem in the Cartesian variables, with the half orders in (8.6), bounds the outgoing physical Cauchy data, and adding the boundary contribution $M_{c,\Sigma}^s f_c$ in (8.17) bounds the split Cauchy data at the same index. Note that the value on the interface is determined by this solution and is not prescribed.

The terms at the state on the diagonal, between different shells, at the cutoff, and between the cap and the annulus remain in the perturbation (8.60) and (8.62). Once the reference solutions on the cap and annulus have been coupled, we include these perturbations through the Neumann formula (8.26) to obtain the inverse at the state.

8.3. The annulus. We now derive the annular equations from the linearization (5.7) at $\mathbf{b}$ to obtain energy estimates for the center and affine components. For the center equation, we construct a *symmetrizer*, a positive matrix defining its energy. For the affine chains, we estimate the coefficient that takes the upper entries into the equations for the lower entries. For a variation (U, S) , we recall the frame (6.12),

$$p = \mathcal{D}v, \quad q_1 = \mathcal{R}v, \quad T = L^{-1} D_\varepsilon v, \quad \mathbb{B} = (p, q_1, T),$$

and, on the annulus where $\mathbb{B}$ is a frame, we write $U = \mathbb{B}c_{\text{fr}}$, where c_{fr} is the vector of its three frame coefficients. With the homogeneous cell derivative $\dot{D}_\varepsilon U = U_\zeta + \varepsilon e_z \times U$ of (6.14), we form the matrices

$$\Gamma_{\mathcal{D}} = \mathbb{B}^{-1} \mathcal{D} \mathbb{B}, \quad \Gamma_{\mathcal{R}} = \mathbb{B}^{-1} \mathcal{R} \mathbb{B}, \quad \Gamma_\varepsilon = \mathbb{B}^{-1} \dot{D}_\varepsilon \mathbb{B},$$

which record differentiation of the frame. For example, $\mathcal{D}U = \mathbb{B}(\mathcal{D}c_{\text{fr}} + \Gamma_{\mathcal{D}} c_{\text{fr}})$, and similarly for $\mathcal{R}$ and $\dot{D}_\varepsilon$. The frame derivatives multiply the unknown at order zero. The terms that differentiate it give the principal symbol with frozen coefficients.

We pass from symbols to operators in the cell variable in two ways. If an operator-valued symbol has the Fourier expansion $a(\zeta, \xi) = \sum_k a_k(\xi) e^{ik\zeta}$ in the cell variable, we set

$$(8.73) \quad \begin{aligned} (\text{Op}(a)u)_n &= \sum_m a_{n-m}(m) u_m, \\ (\text{Op}^w(a)u)_n &= \sum_m a_{n-m} \left(\frac{n+m}{2} \right) u_m, \end{aligned}$$

with the radial coordinate as a parameter. The first is the *standard quantization*, and the second is the *Weyl quantization*, which has the advantage that a Hermitian symbol gives a symmetric operator. Note that all symbols act on the finite-dimensional space of the center and affine components. Then, $\text{Op}^w(a) - \text{Op}(a)$ has the kernel

$$(8.74) \quad a_{n-m} \left(\frac{n+m}{2} \right) - a_{n-m}(m)$$

in the cell modes, and the mean value theorem introduces a factor $|n-m|$ and one derivative in the frequency variable, which lowers the order of the symbol by one.

The center and affine parts of the principal symbol with frozen coefficients are described in the following lemma.

Lemma 8.3. *The domain and range spectral projections of the principal symbol at $\mathbf{b}$, expressed on the fixed spaces of the annulus, have the following properties.*

- (i) *The four nonzero center roots correspond to an invariant spectral subspace. In the fixed coordinates on this subspace, we write $c = (c_1, c_2, c_3, c_4)$ for the center unknown. The principal part of the center evolution is*

$$(8.75) \quad \partial_r c = i \operatorname{Op}^w(\mathbf{G}_{\mathbf{b}}|\xi|)c, \quad \|\mathbf{G}_{\mathbf{b}} - \mathbf{G}_{\mathbf{b}_\rho}\|_{\text{low}} \leq C \|\mathbf{b} - \mathbf{b}_\rho\|_{\mathcal{B}^{s_*+k_c}}.$$

Here, $\|\cdot\|_{\text{low}}$ is a coefficient norm at a fixed lower regularity index. At the constant reference state, this system is equivalent to two oscillator equations

$$(8.76) \quad \xi'' + \Omega_\rho^2 \xi + r^{-2} V_\rho \xi = g, \quad 0 < c_\Omega I \leq \Omega_\rho^2 \leq C_\Omega I,$$

uniformly for $0 \leq \rho \leq \rho_0$. In the first-order variables $z = (\xi, \xi')$, we take

$$(8.77) \quad \mathbf{J}_\rho = \begin{pmatrix} 0 & I \\ -\Omega_\rho^2 & 0 \end{pmatrix}, \quad S_e = \begin{pmatrix} \Omega_\rho^2 & 0 \\ 0 & I \end{pmatrix}.$$

Then, $S_e(-i\mathbf{J}_\rho)$ is Hermitian and $cI \leq S_e \leq CI$ holds uniformly, including at the circle, where the speeds coincide.

- (ii) *The zero root gives the two affine chains, whose jets on the axis are prescribed. In the coordinates of the fixed spaces, its matrix of highest order is a direct sum of copies of*

$$(8.78) \quad J_A = \begin{pmatrix} 0 & I \\ 0 & 0 \end{pmatrix}, \quad J_A^2 = 0,$$

after possibly exchanging the two coordinates of a chain. Thus, $J_A(x, y) = (y, 0)$ for the upper and lower pairs (x, y) . If $B_{21, \mathbf{b}} x$ denotes the contribution of the upper pair to the equation for the lower pair, then

$$(8.79) \quad \|B_{21, \mathbf{b}}\|_{\mathcal{A}_\gamma^{s_*+k_A}} \leq C_A \|\mathbf{b} - \mathbf{b}_\rho\|_{\mathcal{B}^{s_*+k_A}} \leq M_B \delta_A,$$

where the last inequality is the first condition imposed in (8.3).

- (iii) *The remaining terms in the center and affine equations, including the couplings between them, have order zero, and their norm at the base index is at most $\chi_-(\mathbf{b})$. With $\mathbf{G}_e := \mathbf{G}_{\mathbf{b}_\rho}$ and with $\chi_c(\mathbf{b})$ bounding $S_e(\mathbf{G}_{\mathbf{b}} - \mathbf{G}_e) - (\mathbf{G}_{\mathbf{b}} - \mathbf{G}_e)^* S_e$, which measures the failure of self-adjointness in the energy inner product, we have*

$$(8.80) \quad \chi_c(\mathbf{b}) + \chi_-(\mathbf{b}) \leq C \|\mathbf{b} - \mathbf{b}_\rho\|_{\mathcal{B}^{s_*+k_*}},$$

together with the tame estimates in which at most one coefficient is measured in a higher regularity norm.

Proof. The frame identities reduce the part of highest order to the principal symbol determined by the frozen Gram matrix $\mathbb{B}^* \mathbb{B}$, whose normalized matrices are (6.19), and we take the domain and range spectral projections on the same contours around the center roots and around zero. By the resolvent identity, these projections, the identifications with the fixed spaces, and the reduced matrices depend analytically on the coefficients of the Gram matrix, so the estimates (8.75) and (8.80) follow, including their tame versions.

For the constant ellipse, (7.21) and the sign convention for the radial mass give (8.76), and in the first-order variables $z = (\xi, \xi')$ its matrix of highest order is $\mathbf{J}_\rho$. The eigenvalues of Ω_ρ^2 lie in a fixed compact subset of $(0, \infty)$, and we verify the identity for the symmetrizer by multiplying out (8.77). This verification uses the matrix of the oscillator directly and requires no choice of eigenvectors. An affine solution whose cross-section rotates along the axis adds no cell derivative of highest order. Indeed, simultaneous rotation of the disk and of the normal plane gives

$$(8.81) \quad \mathcal{U}_\alpha^{-1} \partial_\zeta \mathcal{U}_\alpha = \partial_\zeta + \alpha'(\zeta)(\hat{J} - \mathcal{R}_y),$$

where $\mathcal{U}_\alpha$ is the rotation by the angle $\alpha(\zeta)$ of the disk and of the normal plane together, $\hat{J}$ is its generator on the components, and $\mathcal{R}_y$ is the angular derivative in the disk. The second term has order zero on the finite-dimensional center space, so it contributes only to the lower-order part of the radial evolution.

For the zero root, the affine solutions at zero curvature of Section 1.4 give two chains of length two, and substituting into (6.19), in the coordinates given by the Hardy maps (8.37), we obtain (8.78). At the constant reference state, whose matrix M_{ρ, p_e} is constant in ζ , the lower left coefficient vanishes by the parity in the angle, and the maps that extract and reconstruct the chains are analytic, so the mean value theorem in Banach spaces gives the first bound in (8.79). Note that the matrix $B_{21, \mathbf{b}}$ need not be nilpotent, but its smallness suffices for the estimate with the modified energy (8.89) below. $\square$

We next estimate the initial value problem on the annulus uniformly at high cell frequency. We begin with the center equation, keeping all four modes together, and write it as

$$(8.82) \quad \partial_r c = i \operatorname{Op}^w(\mathbf{G}_\mathbf{b}(r, \zeta, \omega) |\xi|) c + B_\mathbf{b} c + f_c, \quad \omega = \xi / |\xi|,$$

where $B_\mathbf{b}$ collects the terms of order zero. The four center modes have a uniformly positive symmetrizer, and we have the estimates

$$(8.83) \quad cI \leq S_e \leq CI, \quad S_e \mathbf{G}_e = (S_e \mathbf{G}_e)^*, \quad \|\mathbf{G}_e^{-1}\| \leq C.$$

The energy defined by this symmetrizer is that of the first-order oscillator system. It remains uniformly positive when the speeds coincide because the eigenvalues of Ω_ρ^2 stay in a compact subset of $(0, \infty)$.

For a smooth cutoff Δ_j to a dyadic band of cell frequencies, we let $\lambda_j = \min\{\langle n \rangle : n \in \operatorname{supp} \Delta_j\} \simeq 2^j$ and put

$$(8.84) \quad c_j = e^{\Phi_\gamma(r, D_\zeta)} \Delta_j c, \quad E_j = \langle S_e c_j, c_j \rangle, \quad \mathcal{F}_j = \|e^{\Phi_\gamma(r, D_\zeta)} \Delta_j f_c\|_{S_e},$$

where $\|g\|_{S_e} = \langle S_e g, g \rangle^{1/2}$ is the norm defined by S_e , and we differentiate E_j in r . The reference term of highest order has zero real part, since $S_e \mathbf{G}_e$ is Hermitian. Since the damping $d_\gamma(r, \lambda) = -\partial_r \Phi_\gamma(r, \lambda)$ of (3.21) increases with λ , the weight contributes at most $-2d_\gamma(r, \lambda_j) E_j$. The part of $\mathbf{G}_\mathbf{b} - \mathbf{G}_e$ that fails to be self-adjoint in the S_e inner product contributes at most $C_1 \chi_c \lambda_j E_j$. The commutators with the dyadic projections and with the coefficients, as well as the difference between the Weyl and the standard quantization, all carry the factor in (8.74), so Lemma 6.1 bounds them at order zero. The terms coming from the frame, from the rotation, and from the principal symbol also have order zero, by Lemma 8.3. Collecting in $\mathcal{H}_j$ the terms with a higher regularity coefficient norm and the solution at the base index, we obtain

$$(8.85) \quad \frac{d}{dr} E_j \leq (C_0 + C_1 \chi_c \lambda_j - 2d_\gamma(r, \lambda_j)) E_j + 2E_j^{1/2} \mathcal{F}_j + \mathcal{H}_j,$$

where the bounds for $\mathcal{H}_j$ are summable with the Sobolev weights, and

$$(8.86) \quad d_\gamma(r, \lambda) = \frac{\gamma r \lambda^2}{\sqrt{1 + r^2 \lambda^2}}$$

is the damping of the weight, as in (3.21). If $C_1\chi_c < 2\gamma$, the change of variable $t = r\lambda$ gives

$$(8.87) \quad \int_0^R [C_1\chi_c\lambda - 2d_\gamma(r, \lambda)]_+ dr = \int_0^{R\lambda} \left[C_1\chi_c - \frac{2\gamma t}{\sqrt{1+t^2}} \right]_+ dt \leq C \frac{\chi_c^2}{\gamma},$$

so the positive part of the growth rate is integrable uniformly in λ . Indeed, for $0 \leq a < 2\gamma$, integration over $0 \leq t < \infty$ gives

$$\int_0^\infty \left[a - \frac{2\gamma t}{\sqrt{1+t^2}} \right]_+ dt = 2\gamma - \sqrt{4\gamma^2 - a^2} = \frac{a^2}{2\gamma + \sqrt{4\gamma^2 - a^2}} \leq \frac{a^2}{2\gamma}.$$

We apply this with $a = C_1\chi_c$. The same bound holds when the integration starts at a point inside the cap instead of at $r = 0$. Grönwall's inequality and summation over the dyadic blocks then give the estimates for the initial value problem and for the outgoing Cauchy data with no loss of derivatives.

For the upper and lower pairs $x = (x_1, x_2)$ and $y = (y_1, y_2)$ of (8.10), the full affine equations are

$$(8.88) \quad \begin{aligned} \partial_r x &= \Lambda y + a_{11}x + a_{12}y + f, \\ \partial_r y &= B_{21}x + \Lambda^{-1}a_{21}x + a_{22}y + g, \end{aligned}$$

where the coefficients a_{ij} have order zero and $\|B_{21}\|_{\text{low}} \leq M_B \delta_A$. On one dyadic shell, we use the *modified energy*

$$(8.89) \quad E_{A,\lambda} = \mu_\lambda^2 |x|^2 + \lambda |y|^2, \quad \mu_\lambda^2 = \delta_A + \lambda^{-1},$$

whose weights are those of the affine rows of Table 1. In the variables $\tilde{x} = \mu_\lambda x$ and $\tilde{y} = \sqrt{\lambda}y$, we bound the two off-diagonal coefficients by

$$(8.90) \quad \mu_\lambda \sqrt{\lambda} = \sqrt{1 + \delta_A \lambda}, \quad \frac{\sqrt{\lambda} \|B_{21}\|}{\mu_\lambda} \leq C \sqrt{\delta_A \lambda},$$

and all other terms have order zero, so the unweighted energy grows at rate at most $C(1 + \sqrt{\delta_A \lambda})$. Including the damping (8.86) and repeating the preceding integration, we obtain

$$(8.91) \quad \int_0^R \left[C \sqrt{\delta_A \lambda} - 2d_\gamma(r, \lambda) \right]_+ dr \leq C \frac{\delta_A}{\gamma},$$

provided that $C_* \sqrt{\delta_A} < \gamma$. We have therefore obtained the estimate for the affine component in the norms (8.10). The required inequality explains why δ_A must be fixed before η_* .

The order-zero coupling between the center and affine components is not directly controlled by the modified energy. We remove its leading part by a change of variables. For either sign of the cell frequency, the principal matrix is

$$(8.92) \quad A_0 = \text{diag}(A_C, J_A), \quad J_A^2 = 0, \quad \|A_C^{-1}\| + \|A_C^{-2}\| \leq C,$$

where A_C is the center matrix. We write C_{CA} for the coefficient taking affine variables into the center equation and C_{AC} for the coefficient taking center variables into the affine equation. We set

$$(8.93) \quad \begin{aligned} X &= -A_C^{-1}C_{CA} - A_C^{-2}C_{CA}J_A, \\ Y &= C_{AC}A_C^{-1} + J_A C_{AC}A_C^{-2}. \end{aligned}$$

Multiplying out and using $J_A^2 = 0$, we obtain

$$(8.94) \quad A_C X - X J_A = -C_{CA}, \quad J_A Y - Y A_C = -C_{AC}.$$

We write the original center and affine unknown as $(c, x, y) = \mathcal{T}\tilde{U}$, where $\tilde{U}$ is the new unknown and

$$(8.95) \quad \mathcal{T} = I + \text{Op} \left(\Lambda^{-1} \begin{pmatrix} 0 & X \\ Y & 0 \end{pmatrix} \right).$$

This change of variables cancels the two couplings to leading order. Indeed, with the convention (8.73), the conjugated off-diagonal terms of order zero are $A_C X - X J_A + C_{CA}$ and $J_A Y - Y A_C + C_{AC}$, both of which are zero by (8.94). Every remaining term of the composition contains either the factor Λ^{-1} of (8.95) or a frequency difference from (8.74), so the moment estimate of Lemma 6.1 leaves a remainder $R_{\mathbf{b},H}^{(-1)}$ of order minus one. In the norms (8.10), the four ratios of weights that occur for a map of order minus one are

$$(8.96) \quad \frac{\lambda^{-1}}{\mu_\lambda} \leq \lambda^{-1/2}, \quad \frac{\lambda^{-1}}{\lambda^{1/2}} \leq \lambda^{-3/2}, \quad \mu_\lambda \lambda^{-1} \leq C \lambda^{-1}, \quad \lambda^{1/2} \lambda^{-1} = \lambda^{-1/2},$$

and we conclude that

$$(8.97) \quad \left\| R_{\mathbf{b},H}^{(-1)} U \right\|_{Y_{a,-}^s} \leq C_s \left[(\Lambda_J^{-1/2} + \chi_-) \|U\|_{X_{a,-}^s} + \|\mathbf{b} - \mathbf{b}_\rho\|_{\mathcal{B}^{s+k}} \|U\|_{X_{a,-}^{s*}} \right].$$

The change of variables and its inverse satisfy the same estimate on the Cauchy data.

We let $K_{0,\mathbf{b}}$ be the direct sum of the inverses of the initial value problems for the center, affine, and remaining auxiliary components in these variables. The bounds (8.87) and (8.91) give a solution for every source and every incoming Cauchy datum. The remaining error is small by (8.3) and (8.97), so the inverse is

$$(8.98) \quad G_{a,-,\mathbf{b}} = (I + K_{0,\mathbf{b}} R_{\mathbf{b},H}^{(-1)})^{-1} K_{0,\mathbf{b}} = K_{0,\mathbf{b}} (I + R_{\mathbf{b},H}^{(-1)} K_{0,\mathbf{b}})^{-1},$$

with the changes of variables undone and the triangular equations for the remaining components included.

We first impose a finite upper cutoff in (8.98), and the uniform energy estimates then give a subsequence converging weak-* in the weighted energy space of type L^∞ in r with values in H^{s*} . The equation bounds the radial derivative one index lower, so the unknown of the first-order system converges in $C_w([0, R]; H^{s*-1})$, the space of weakly continuous functions of r , and its value and balanced conormal coordinates therefore converge at both endpoints. We pass the equation to the limit by testing against smooth functions with finitely many cell modes. For zero data, the center and affine energies vanish, and then so do the finite coordinates, so the solution is unique and the limit does not depend on the subsequence. Uniqueness also identifies u_a with $G_{a,-,\mathbf{b}}(\mathcal{L}_{a,-,\mathbf{b}} u_a, \gamma_{-,\mathbf{b}} u_a)$ for every $u_a \in X_{a,-}^s$, where $\mathcal{L}_{a,-,\mathbf{b}}$ and $\gamma_{-,\mathbf{b}}$ are the center and affine equations on the annulus and the Cauchy trace on its inner boundary at the state $\mathbf{b}$, as in (8.18) at the reference. This proves the left inverse identity, and the right identity follows from (8.98).

8.4. The inverse identities and the tame estimates. We now verify that the constructions on the cap and on the annulus give the asserted inverse of the coupled problem. For $F = (f_c, f_a, g^{\text{sp}})$, their inverse identities give

$$(8.99) \quad N_{-,\rho}^{\text{diag}} G_{-,\rho}^{\mathbb{N},\text{diag}} F = (f_c, f_a, g^{\text{sp}} + T_{-,\mathbf{b}_\rho} \Gamma_{c,-,\rho}^{\text{sp}} G_{c,-,\rho}^{\text{diag}} f_c - T_{-,\mathbf{b}_\rho} \Gamma_{c,-,\rho}^{\text{sp}} G_{c,-,\rho}^{\text{diag}} f_c) = F.$$

Conversely, if we apply (8.22) to $N_{-,\rho}^{\text{diag}}(u_c, u_a)$, the Cauchy datum for the problem on the annulus reduces to

$$(8.100) \quad \gamma_{-,\rho} u_a - T_{-,\mathbf{b}_\rho} \Gamma_{c,-,\rho}^{\text{sp}} u_c + T_{-,\mathbf{b}_\rho} \Gamma_{c,-,\rho}^{\text{sp}} G_{c,-,\rho}^{\text{diag}} \mathcal{A}_{c,-,\rho}^{\text{diag}} u_c = \gamma_{-,\rho} u_a,$$

and the left inverse identities on the cap and on the annulus then recover u_c and u_a . This proves both inverse identities at the reference. The Neumann formulas (8.26), with the range factors in (8.28), then give (8.29).

The estimate on the cap, the center and affine energies, and (8.97) bound the reference inverse $G_{-, \rho}^{\mathbb{N}, \text{diag}}$, and we combine that bound with (8.66) in the Neumann formula (8.26). Applying the bounded range factors $\mathfrak{P}_{-, \mathbf{b}}^s$ and $\mathcal{S}_{Y, -, \mathbf{b}}^{-1}$, we then obtain (8.31). The estimate (8.68) will be used when the low and high frequencies are coupled.

For the state derivatives, we work on the reconstructed space. Each derivative of $\mathcal{R}_{-, \mathbf{b}}^{\mathbb{N}}$ places at most one cell derivative on the solution. To see this for the center equation, let $L_{\mathbf{b}}^c$ be the operator obtained by moving the homogeneous terms of (8.82) to the left, so that

$$L_{\mathbf{b}}^c c = \partial_r c - i \text{Op}^w(\mathbf{G}_{\mathbf{b}}|\xi|)c - B_{\mathbf{b}}c = f_c.$$

Holding f_c fixed and differentiating gives

$$(8.101) \quad L_{\mathbf{b}}^c c^{(m)} = - \sum_{\emptyset \neq I \subset \{1, \dots, m\}} (D_{\mathbf{b}}^{|I|} L_{\mathbf{b}}^c [h_I]) c^{(m-|I|)} [h_{I^c}],$$

where $c^{(m)}$ denotes the m -th derivative of the solution c with respect to $\mathbf{b}$, h_I the tangent directions with indices in I , and I^c the complement of I . Every factor $D_{\mathbf{b}}^{|I|} L_{\mathbf{b}}^c$ has at most one cell derivative, so induction from the energy estimate loses at most one Sobolev index per differentiation in $\mathbf{b}$, which gives $\mu_m \leq m$. The reference inverse on the cap, the traces, the finite equations, and the changes of variables (8.95) have tame derivatives with respect to $\mathbf{b}$ of fixed order, so Leibniz' rule applied to (8.22) and to the resolvent factors in (8.26), together with the bounded range factors in (8.28), proves (8.33). The weight is fixed, so these derivatives use the same strip of analyticity.

Complex conjugation in Cartesian variables pairs (m, n) with $(-m, -n)$, preserves the contour around the four center roots, and exchanges the paired affine chains and cell frequencies. The changes of variables, the Neumann series, and the triangular equations commute with this conjugation, and, by uniqueness, so do the inverses and the traces.

The estimates depend on the separation of the four center roots from zero, and they do not require these roots to be separated from one another. The parameter δ_A is fixed, the factors in (8.38) remain bounded for $0 \leq \rho \leq \rho_0 < 1/4$, and the symmetrizer at the circle remains positive when the center speeds coincide. Thus, the bounds are uniform through $\rho = 0$. For compatible Cartesian sources, Proposition 6.2 supplies the fixed loss of the source before this inverse is applied. This completes the proof of Proposition 8.1.

9. THE ELLIPTIC BOUNDARY PROBLEM

We solve the elliptic part of the high-frequency problem by coupling second-order equations on the cap and annulus through their values and conormal derivatives on the common interface. As in Section 8, we first invert the operator $\widehat{D}_{+, \mathbf{b}}$ defined below. We relate it to the positive diagonal block of the full operator in Section 10.

For this *transmission problem*, we use a coercive quadratic form on the pair of fields, including both parities of the positive cap modes. At energy regularity, corresponding to $s = 0$ below, an interior source acts on test functions with zero boundary values and need not determine a physical conormal derivative. We therefore prescribe the boundary functionals supplied by the form. At higher regularity, we recover the physical conormal and Robin data by subtracting the boundary contributions of the sources.

Throughout, we fix $0 < \rho_0 < 1/4$, the phase of (3.15),

$$(9.1) \quad \Phi_\gamma(r, \lambda) = \sigma_0 \lambda - \gamma(\sqrt{1 + r^2 \lambda^2} - 1), \quad W_\gamma = e^{\Phi_\gamma(r, \Lambda)},$$

and the projection H_J of (4.2) onto the high cell frequencies. For $\mathbf{b} = (\rho, a, \varepsilon, p)$, we let $\mathbf{b}_\rho = (\rho, a_\rho, \varepsilon_\rho, p_\rho)$ be the constant ellipse with the same ellipse parameter, as in Section 6, and we put

$$(9.2) \quad \eta_+(\mathbf{b}) = \|a - a_\rho\|_{\mathcal{A}_\gamma^{s_* + k_+}} + |\varepsilon - \varepsilon_\rho| + |p - p_\rho|.$$

Here, k_+ is the fixed number of additional derivatives of the coefficients required in the estimates, independent of s , as k_c in Section 8. At index s , we use the norm

$$(9.3) \quad \|\mathbf{b} - \mathbf{b}_\rho\|_{\mathcal{B}^s} := \|a - a_\rho\|_{\mathcal{A}_\gamma^s} + |\varepsilon - \varepsilon_\rho| + |p - p_\rho|.$$

These norms measure the distance from the constant ellipse in the components a , ε , and p . We choose a closed set of coefficients and J such that

$$(9.4) \quad C(\gamma + \gamma^2) + C\eta_* < \frac{c_+}{2}, \quad 0 < \rho_0 < \frac{1}{4},$$

where η_* bounds $\eta_+(\mathbf{b})$ on this set and c_+ is the coercivity constant of the positive form (7.20) at the constant ellipse. These choices are independent of an upper Fourier cutoff and of the number N of periods of the field.

9.1. Function spaces and the inverse for the positive block. On the charts of the cap and of the annulus, we set

$$(9.5) \quad \begin{aligned} V_{c,+}^s &= H_J^X \mathcal{V}_{+, \text{cap}}^s, & Y_{c,+}^s &= H_J^Y \mathcal{Y}_{+, \text{cap}}^s, \\ V_{a,+}^s &= H_J^X \mathcal{E}_{+, a, \gamma}^s, & Y_{a,+}^s &= H_J^Y (\mathcal{E}_{+, a, \gamma}^s)^*_{\text{bulk}}. \end{aligned}$$

Here, $\mathcal{V}_{+, \text{cap}}^s$ and $\mathcal{Y}_{+, \text{cap}}^s$ are the weighted spaces of the positive modes on the cap, whose orders are given below, and $\mathcal{E}_{+, a, \gamma}^s$ is the weighted energy space of the positive block on the annulus. The space $(\mathcal{E}_{+, a, \gamma}^0)^*_{\text{bulk}}$ is the dual of the subspace of $\mathcal{E}_{+, a, \gamma}^0$ whose fields have zero value on both boundaries. At index s , we increase the Sobolev order of the source norm by s , from H^{-1} to H^{s-1} , keeping the analytic weight fixed. This defines $(\mathcal{E}_{+, a, \gamma}^s)^*_{\text{bulk}}$. We call the spaces V on which the quadratic form is defined the *form spaces*, and write Y for the corresponding source spaces. The cap and annular form spaces have the same regularity. On the cap, the local orders before applying the weight are

$$(9.6) \quad V_{c,+}^s \simeq H_+^{s+1}(\Omega_c), \quad Y_{c,+}^s \simeq H_+^{s-1}(\Omega_c).$$

Here, Ω_c is the scaled cap and $+$ restricts to the positive angular modes. At $s = 0$, the cap form space is Cartesian H^1 , its source space H^{-1} is the dual of the zero-Dirichlet subspace, and the annular form space is weighted H^1 , as in Table 1. For the operator domains, $\gamma_D u$ denotes the value on the interface. We use Z_D^s for this datum, of order $H^{s+1/2}$, and Z_B^s for the outer Robin datum, of order $H^{s-1/2}$, with the analytic weight included. As domains of the operator, we take

$$(9.7) \quad \begin{aligned} X_{c,+}^s &= \{u \in V_{c,+}^s : P_{c,+} \mathbf{b} u \in Y_{c,+}^s, \gamma_D u \in Z_D^s\}, \\ X_{a,+}^s &= \{u \in V_{a,+}^s : P_{a,+} \mathbf{b} u \in Y_{a,+}^s, \gamma_D u \in Z_D^s, \mathcal{R}_{+, \mathbf{b}}^{\text{sp}}(u, P_{a,+} \mathbf{b} u) \in Z_B^s\}. \end{aligned}$$

Here, $P_{c,+} \mathbf{b} u_c = f_c$ and $P_{a,+} \mathbf{b} u_a = f_a$ are the interior equations of the positive block on the cap and annulus. The operator $\mathcal{R}_{+, \mathbf{b}}^{\text{sp}}$ records the Robin datum using the weak conormal residual supplied by the quadratic form. For smooth fields, this is the physical Robin datum plus the boundary contribution of the interior

source, as in the split Cauchy data (8.8). We give its weak definition below. Thus, the operator domains impose regularity on the equation and traces in addition to the form regularity H^{s+1} . In particular,

$$(9.8) \quad X_{c,+}^0 \oplus X_{a,+}^0 \hookrightarrow V_{c,+}^0 \oplus V_{a,+}^0 \hookrightarrow L_{\text{bulk}}^2.$$

We call a source a *full source* if it acts canonically on Poisson extensions of boundary values. An index is *regular* if every source at that index has this property. For example, $s \geq 1$ is regular because the interior sources belong to $H^{s-1} \subset L^2$. At $s = 0$, we instead use the extension of functionals in (9.10) below.

The spaces Z_D^s , Z_N^s , and Z_B^s contain the value on the interface, the residual on the interface, and the residual on the outer boundary, respectively. At regular indices, the latter two also contain the physical conormal and Robin data, which are related to the residuals by the change of data below. A *Robin datum* prescribes a linear combination of the conormal derivative and the value on the outer boundary, as required by the boundary condition of the positive block. The norms of these spaces are

$$(9.9) \quad \|z\|_{Z_D^s} \simeq \left\| \Lambda^{1/2} z \right\|_{H^s}, \quad \|z\|_{Z_N^s} \simeq \left\| \Lambda^{-1/2} z \right\|_{H^s}, \quad \|z\|_{Z_B^s} \simeq \left\| \Lambda^{-1/2} z \right\|_{H^s},$$

with the analytic weight on the traces and the spectral projections of the principal symbol understood, as in (8.6). These are the trace orders associated with a solution in H^{s+1} , namely $H^{s+1/2}$ for the value and $H^{s-1/2}$ for the conormal or Robin data.

Data for the weak problem. To formulate the weak problem at $s = 0$, we extend the interior sources to test fields with nonzero boundary values. We write γ_Σ^c and γ_Σ^a for the traces on the interface and γ_∂^a for the trace on the outer boundary. We let $\mathcal{P}_\Sigma^c$, $\mathcal{P}_\Sigma^a$, and $\mathcal{P}_\partial^a$ be their localized Poisson extensions from Section 4, with disjoint supports at the interface and at the outer boundary. We set

$$(9.10) \quad \begin{aligned} R_c^0 &:= I - \mathcal{P}_\Sigma^c \gamma_\Sigma^c, \\ R_a^0 &:= I - \mathcal{P}_\Sigma^a \gamma_\Sigma^a - \mathcal{P}_\partial^a \gamma_\partial^a, \\ \text{Ext}_c f_c(v_c) &:= f_c(R_c^0 v_c), & \text{Ext}_a f_a(v_a) &:= f_a(R_a^0 v_a). \end{aligned}$$

The source f_c acts on test fields on the cap with zero value on the interface, and f_a acts on test fields on the annulus with zero value at both boundaries. Formula (9.10) extends these functionals to the full form spaces. For a test pair $V = (v_c, v_a)$ with $\gamma_\Sigma^c v_c = \gamma_\Sigma^a v_a$, the resulting functional is uniquely given by

$$(9.11) \quad \text{Ext}_c f_c(v_c) + \text{Ext}_a f_a(v_a) + \langle g_N^{\text{sp}}, \gamma_\Sigma V \rangle + \langle V_{B,b}^{-1} g_B^{\text{sp}}, \gamma_\partial^a v_a \rangle.$$

Here, $\gamma_\Sigma V$ denotes the common value of v_c and v_a on the interface. The coefficient $V_{B,b}$ multiplies the outward conormal derivative in the physical Robin boundary operator, and is invertible. Its inverse converts the prescribed boundary datum to the normalization in Green's identity. The last two terms prescribe the boundary data for the weak problem. We call these independent functionals the *split residuals* at the interface and outer boundary. We relate them to the physical conormal derivatives below. Conjugating the splitting (9.11) by the isomorphisms that shift the index of the scale, we obtain the same convention at every s .

If u_c satisfies its interior equation, its weak split residual is

$$(9.12) \quad \langle \mathcal{N}_{c,\Sigma}^{\text{sp}}(u_c, f_c), \varphi \rangle := \mathfrak{c}_{+,b}(u_c, \mathcal{P}_\Sigma^c \varphi) - \text{Ext}_c f_c(\mathcal{P}_\Sigma^c \varphi),$$

where $\mathbf{c}_{+, \mathbf{b}}$ is the form on the cap at the state. Writing $\mathbf{b}_{+, \mathbf{b}}^{\text{int}}$ for the interior part of the annular form, with the outer Robin boundary term omitted, we set

$$\begin{aligned}\langle \mathcal{N}_{a, \Sigma}^{\text{sp}}(u_a, f_a), \varphi \rangle &= \mathbf{b}_{+, \mathbf{b}}^{\text{int}}(u_a, \mathcal{P}_{\Sigma}^a \varphi) - \text{Ext}_a f_a(\mathcal{P}_{\Sigma}^a \varphi), \\ \langle \mathcal{N}_{a, \partial}^{\text{sp}}(u_a, f_a), \psi \rangle &= \mathbf{b}_{+, \mathbf{b}}^{\text{int}}(u_a, \mathcal{P}_{\partial}^a \psi) - \text{Ext}_a f_a(\mathcal{P}_{\partial}^a \psi).\end{aligned}$$

For the chosen extensions, the subtracted terms vanish by (9.10). Since the form and the extensions are bounded, these residuals are functionals of order $H^{-1/2}$. The interior equation also shows that the residual is unchanged if we add a test function with zero trace to the extension. This defines the split residual even when the source does not determine a physical conormal derivative.

Recovery of physical boundary data. For full sources ℓ_c and ℓ_a at a regular index, we define their boundary contributions, as in Section 8, by the transposes

$$\begin{aligned}(9.13) \quad \langle \mathbf{M}_{c, \Sigma}^s \ell_c, \varphi \rangle &:= \langle \ell_c, \mathcal{P}_{\Sigma}^c \varphi \rangle, \\ \langle \mathbf{M}_{a, \Sigma}^s \ell_a, \varphi \rangle &:= \langle \ell_a, \mathcal{P}_{\Sigma}^a \varphi \rangle, \\ \langle \mathbf{M}_{a, \partial}^s \ell_a, \psi \rangle &:= \langle \ell_a, \mathcal{P}_{\partial}^a \psi \rangle.\end{aligned}$$

Here, $\mathbf{M}^s = (\mathcal{P})'$ is the transpose with respect to the weighted dualities. It differs from the unweighted Hilbert adjoint. By Green's identity, the split residuals are related to the outward physical conormal derivatives by

$$\begin{aligned}(9.14) \quad \mathcal{N}_{c, \Sigma}^{\text{sp}} &= \gamma_{N, \mathbf{b}}^{c, \text{out}} u_c + \mathbf{M}_{c, \Sigma}^s \ell_c, \\ \mathcal{N}_{a, \Sigma}^{\text{sp}} &= \gamma_{N, \mathbf{b}}^{a, \text{out}} u_a + \mathbf{M}_{a, \Sigma}^s \ell_a, \\ \mathcal{N}_{a, \partial}^{\text{sp}} &= \gamma_{N, \mathbf{b}}^{a, \partial, \text{out}} u_a + \mathbf{M}_{a, \partial}^s \ell_a.\end{aligned}$$

Here, $\gamma_{N, \mathbf{b}}^{a, \partial, \text{out}}$ is the conormal derivative on the outer boundary of the annulus. We write $\mathbf{V}_{B, \mathbf{b}}$ and $\mathbf{F}_{B, \mathbf{b}}$ for the coefficients of the conormal derivative and of the value in the operator on the outer boundary. For the unnormalized boundary operator $\mathcal{B}_{++, \mathbf{b}}$ of the positive block on the outer boundary at the state, whose coefficient of the conormal derivative we denote by $\mathbf{E}_{\mathbf{b}}$, this gives $\mathbf{V}_{B, \mathbf{b}} = \mathbf{E}_{\mathbf{b}}$, while $\mathbf{V}_{B, \mathbf{b}} = I$ after normalization by that coefficient. Wherever the physical conormal derivative is defined, we write

$$\mathcal{R}_{++, \mathbf{b}}^{\text{ph}} u_a := \mathbf{V}_{B, \mathbf{b}} \gamma_{N, \mathbf{b}}^{a, \partial, \text{out}} u_a + \mathbf{F}_{B, \mathbf{b}} \gamma_{\partial}^a u_a,$$

and, at every regularity level, we define the *boundary operator in split form* by

$$(9.15) \quad \mathcal{R}_{++, \mathbf{b}}^{\text{sp}}(u_a, f_a) := \mathbf{V}_{B, \mathbf{b}} \mathcal{N}_{a, \partial}^{\text{sp}}(u_a, f_a) + \mathbf{F}_{B, \mathbf{b}} \gamma_{\partial}^a u_a.$$

When s is a regular index and the source is full, we have

$$(9.16) \quad \mathcal{R}_{++, \mathbf{b}}^{\text{sp}}(u_a, \ell_a) = \mathcal{R}_{++, \mathbf{b}}^{\text{ph}} u_a + \mathbf{V}_{B, \mathbf{b}} \mathbf{M}_{a, \partial}^s \ell_a,$$

with the factors in the displayed order.

We order the data as $F = (\ell_c, \ell_a, g_D, g_N, g_B)$, where g_D is the jump of the values on the interface, g_N is the datum of the conormal condition on the interface, and g_B is the datum on the outer boundary. In this order, we define the map from physical to split residuals at a regular index by

$$(9.17) \quad \mathfrak{S}_{+, \mathbf{b}}^{\text{res}, s}(\ell_c, \ell_a, g_D, g_N^{\text{ph}}, g_B^{\text{ph}}) := (\ell_c, \ell_a, g_D, g_N^{\text{ph}} + \mathbf{M}_{c, \Sigma}^s \ell_c + \mathbf{M}_{a, \Sigma}^s \ell_a, g_B^{\text{ph}} + \mathbf{V}_{B, \mathbf{b}} \mathbf{M}_{a, \partial}^s \ell_a),$$

which adds the source contributions to the physical boundary data to obtain the split residuals. It leaves the sources and the jump of the values unchanged. Subtracting the same contributions gives the inverse. Both interface contributions have a plus sign because the conormal derivatives point outward on both sides. At

the index $s = 0$, there is no such map $\mathfrak{S}_{+, \mathbf{b}}^{\text{res}, 0}$ on the data alone, and (9.11) fixes the split coordinates there directly.

The bounds for the Poisson extensions and for the coefficient of the conormal derivative in the outer boundary operator give tame estimates for $(\mathfrak{S}_{+, \mathbf{b}}^{\text{res}, s})^{\pm 1}$ and their derivatives with respect to $\mathbf{b}$, from index s to index s . The range bounds in Proposition 9.1 include the norms of $\mathfrak{S}_{+, \mathbf{b}}^{\text{res}, s}$ and its inverse. The transpose of this map is triangular as well. Indeed, with $(v_c, v_a, \lambda_\Sigma, \lambda_B)$ denoting the elements of the dual of the range that are paired with $(\ell_c, \ell_a, g_N, g_B)$, its nontrivial components on the sources are

$$(9.18) \quad \begin{aligned} v_c &\longmapsto v_c + \mathcal{P}_\Sigma^c \lambda_\Sigma, \\ v_a &\longmapsto v_a + \mathcal{P}_\Sigma^a \lambda_\Sigma + \mathcal{P}_\partial^a V'_{B, \mathbf{b}} \lambda_B. \end{aligned}$$

Note that the Hilbert adjoint with respect to the weighted inner products differs from this transpose by the Riesz maps of the weighted spaces. Given a projection $P_{Y, +}^{\text{ph}}$ on the range in physical coordinates, we transport it to the split coordinates by

$$(9.19) \quad P_{Y, +}^{\text{sp}} = \mathfrak{S}_{+, \mathbf{b}}^{\text{res}, s} P_{Y, +}^{\text{ph}} (\mathfrak{S}_{+, \mathbf{b}}^{\text{res}, s})^{-1}.$$

This projection includes a component from the source to the trace, which cannot be removed without an additional identity for these boundary contributions.

We now collect the solution and data spaces as

$$(9.20) \quad \begin{aligned} \mathbb{V}_{+, \text{ret}}^s &= V_{c, +}^s \oplus V_{a, +}^s, \\ \mathbb{X}_{+, \text{ret}}^s &= X_{c, +}^s \oplus X_{a, +}^s, \\ \mathbb{Y}_{+, \text{ret}}^s &= Y_{c, +}^s \oplus Y_{a, +}^s \oplus Z_D^s \oplus Z_N^s \oplus Z_B^s. \end{aligned}$$

If $\gamma_\Sigma^a u_a - \gamma_\Sigma^c u_c = g_D$, then the pair

$$(u_c, u_a) - (0, \mathcal{P}_\Sigma^a g_D)$$

has matching values on the interface, because $\gamma_\Sigma^a \mathcal{P}_\Sigma^a g_D = g_D$. Such pairs form the *form domain*

$$(9.21) \quad \mathbb{V}_{+, 0}^0 := \{(v_c, v_a) \in V_{c, +}^0 \oplus V_{a, +}^0 : \gamma_j^D v_a = \gamma_j^D v_c\},$$

where γ_j^D is the trace of the value on the interface, at the radius r_j . The Robin condition is imposed through the form and is therefore not an additional constraint on the form domain. We also write

$$(9.22) \quad Z_+^s = H_J^{Z+} \mathfrak{T}_{+, \gamma}^s$$

for the product of the balanced value and conormal trace spaces on the interface and outer boundary. Here, we retain the traces of each field separately, rather than only their jump or sum in the prescribed data. These separate traces are needed when coupling the positive and minus blocks. The space $\mathfrak{T}_{+, \gamma}^s$ includes the analytic weight, as in (8.6), and H_J^{Z+} is its projection onto the high cell frequencies.

For fields u_c and u_a on the cap and on the annulus at a regular index, we first define the operator using the physical values and conormal derivatives at the state,

$$(9.23) \quad N_{+, \mathbf{b}}^{\text{cur}, \text{ph}}(u_c, u_a) = (P_{c, +, \mathbf{b}} u_c, P_{a, +, \mathbf{b}} u_a, \gamma_j^D u_a - \gamma_j^D u_c, \nabla_{a, \mathbf{b}}^{\text{out}} u_a + \nabla_{c, \mathbf{b}}^{\text{out}} u_c, \mathcal{B}_{+, \mathbf{b}} u_a),$$

where $\nabla_{c,\mathbf{b}}^{\text{out}} = \gamma_{N,\mathbf{b}}^{c,\text{out}}$ and $\nabla_{a,\mathbf{b}}^{\text{out}} = \gamma_{N,\mathbf{b}}^{a,\text{out}}$ are the outward conormal derivatives from the cap and annulus. Replacing the physical boundary data by the weak residuals gives an operator defined at every index,

$$(9.24) \quad \begin{aligned} N_{+,\mathbf{b}}^{\text{cur,sp}}(u_c, u_a) = & (P_{c,+,\mathbf{b}} u_c, P_{a,+,\mathbf{b}} u_a, \gamma_j^D u_a - \gamma_j^D u_c, \\ & \mathcal{N}_{a,\Sigma}^{\text{sp}}(u_a, P_{a,+,\mathbf{b}} u_a) + \mathcal{N}_{c,\Sigma}^{\text{sp}}(u_c, P_{c,+,\mathbf{b}} u_c), \\ & \mathcal{R}_{++,\mathbf{b}}^{\text{sp}}(u_a, P_{a,+,\mathbf{b}} u_a)). \end{aligned}$$

At the index $s = 0$, we regard the two interior sources as functionals on test fields with zero boundary trace before using them in $\mathcal{N}^{\text{sp}}$. At regular indices with full sources, we relate the two operators by

$$(9.25) \quad N_{+,\mathbf{b}}^{\text{cur,sp}} = \mathfrak{S}_{+,\mathbf{b}}^{\text{res},s} N_{+,\mathbf{b}}^{\text{cur,ph}}.$$

Return to the reference coordinates. We let ι_+^X and π_+^Y denote the inclusion of the positive block and the projection onto it, as in Section 6, and we let $\mathcal{R}_{CN,\mathbf{b},\rho}$ be the isomorphism of the range from (6.80). It acts by $(\mathcal{T}_{\mathbf{b},\rho}^{CN})^{-1}$ on the pair of the value and the conormal derivative and by the identity on the other components. To recover the physical conormal data, we first subtract the source contributions using (9.17). We then express these data in the reference coordinates using $\mathcal{R}_{CN,\mathbf{b},\rho}$. At regular indices, this gives

$$(9.26) \quad N_{+,\mathbf{b}} := \pi_+^Y \mathbb{N}_{\mathbf{b}} \iota_+^X = \mathcal{R}_{CN,\mathbf{b},\rho} N_{+,\mathbf{b}}^{\text{cur,ph}} = \mathcal{R}_{CN,\mathbf{b},\rho} (\mathfrak{S}_{+,\mathbf{b}}^{\text{res},s})^{-1} N_{+,\mathbf{b}}^{\text{cur,sp}}.$$

At the index $s = 0$, we work instead with the operator $N_{+,\mathbf{b}}^{\text{cur,sp}}$ defined by the form. We apply the conversion in (9.26) to the reconstructed field on the cap and on the annulus, and we do not commute it with the shell projections Q_j .

We write ι_+^Y for the inclusion of the positive range. In addition to the conversion $\mathfrak{S}_{+,\mathbf{b}}^{\text{res},s}$ between physical and split data, we use the positive diagonal block of the map $\mathcal{S}_{Y,\mathbf{b}}$ from Section 6, defined by

$$(9.27) \quad \mathcal{S}_{Y,+,\mathbf{b}} := \pi_+^Y \mathcal{S}_{Y,\mathbf{b}} \iota_+^Y.$$

This diagonal block is defined even if the full map does not preserve the positive summand. For a positive range vector $F_+ + j_{Z,+,\mathbf{b}} z_+$, where F_+ has zero auxiliary entries and z_+ also denotes its inclusion in the full auxiliary space, we have

$$\mathcal{S}_{Y,+,\mathbf{b}}(F_+ + j_{Z,+,\mathbf{b}} z_+) = F_+ - \pi_+^Y \mathbb{B}_{\mathbf{b}} z_+ + j_{Z,+,\mathbf{b}} z_+,$$

and its inverse has the same formula with a plus sign before $\pi_+^Y \mathbb{B}_{\mathbf{b}} z_+$. We put

$$(9.28) \quad \widehat{D}_{+,\mathbf{b}} := \mathcal{S}_{Y,+,\mathbf{b}} N_{+,\mathbf{b}}, \quad \widehat{G}_{+,\mathbf{b}} := G_{+,\mathbf{b}}^{\mathbb{N}} \mathcal{S}_{Y,+,\mathbf{b}}^{-1},$$

where $G_{+,\mathbf{b}}^{\mathbb{N}} = N_{+,\mathbf{b}}^{-1}$ is constructed below.

The component taking positive unknowns to minus equations remains in the augmented operator. Formula (6.83) gives

$$(9.29) \quad \begin{aligned} \pi_-^Y \mathcal{S}_{Y,\mathbf{b}} \mathbb{N}_{\mathbf{b}} \iota_+^X &= \pi_-^Y \iota_{Y,\mathbf{b}} \mathbf{J}_{Y,\mathbf{b}}^{\text{raw}} L_{\mathbf{b}} \mathbf{R}_{X,\mathbf{b}} \iota_+^X + j_{Z,-,\mathbf{b}} q_-^Z C_{X,\mathbf{b}} \iota_+^X, \\ \pi_-^Y \mathbb{B}_{\mathbf{b}} C_{X,\mathbf{b}} \iota_+^X - \pi_-^Y \mathbb{B}_{\mathbf{b}} C_{X,\mathbf{b}} \iota_+^X &= 0. \end{aligned}$$

The off-diagonal block of $\mathbb{D}_{\mathbf{b}} = \mathcal{S}_{Y,\mathbf{b}} \mathbb{N}_{\mathbf{b}}$ is estimated once both diagonal blocks are inverted. The projection q_-^Z selects the auxiliary data of the minus block, and this contribution remains after the two interior source corrections cancel.

Proposition 9.1 (The inverse of $\widehat{D}_{+,b}$). *For the coefficients fixed in (9.4) and every regular index s at which the boundary contributions (9.13) are defined,*

$$(9.30) \quad \widehat{D}_{+,b} : \mathbb{X}_{+,ret}^s \longrightarrow \mathbb{Y}_{+,ret}^s$$

is an isomorphism. We write $\widehat{G}_{+,b} = \widehat{D}_{+,b}^{-1}$, and we let $\Gamma_{+,b} : \mathbb{X}_{+,ret}^s \rightarrow Z_+^s$ be the map that records the remaining balanced traces in the space (9.22). For every $s \geq s_$,*

$$(9.31) \quad \left\| \widehat{G}_{+,b} F \right\|_{\mathbb{X}_{+,ret}^s} + \left\| \Gamma_{+,b} \widehat{G}_{+,b} F \right\|_{Z_+^s} \leq C_s \left[\|F\|_{\mathbb{Y}_{+,ret}^s} + (1 + \|b - b_\rho\|_{\mathcal{B}^{s+k_+}}) \|F\|_{\mathbb{Y}_{+,ret}^{s_*}} \right],$$

for arbitrary $F = (f_c, f_a, g_D, g_N, g_B)$ in $\mathbb{Y}_{+,ret}^s$, written in the normalized physical coordinates, that is, in the coordinates of the range of $\widehat{D}_{+,b}$. The data supplied to the inverse of the form at $s = 0$ are

$$F^{sp} = \mathfrak{S}_{+,b}^{res,s} \mathcal{R}_{CN,b,\rho}^{-1} \mathcal{S}_{Y,+,b}^{-1} F,$$

and (9.31) includes the bounds for all three factors on the range, with g_D , g_N , and g_B independent data at a regular index.

For every $m \geq 1$, there are finite integers $k_{+,m}$ and $\mu_{+,m}$, independent of s , for which either $Q_{+,b} = \widehat{G}_{+,b}$ or $Q_{+,b} = \Gamma_{+,b} \widehat{G}_{+,b}$ satisfies

$$(9.32) \quad \begin{aligned} & \|D_b^m Q_{+,b} [h_1, \dots, h_m] F\|_{Z^s} \\ & \leq C_{s,m} \left[\prod_i \|h_i\|_{lo} \|F\|_{\mathbb{Y}_{+,ret}^{s+\mu_{+,m}}} \right. \\ & \quad + \sum_i \|h_i\|_{tan,s+k_{+,m}} \prod_{j \neq i} \|h_j\|_{lo} \|F\|_{\mathbb{Y}_{+,ret}^{s_*+\mu_{+,m}}} \\ & \quad \left. + (1 + \|b - b_\rho\|_{\mathcal{B}^{s+k_{+,m}}}) \prod_i \|h_i\|_{lo} \|F\|_{\mathbb{Y}_{+,ret}^{s_*+\mu_{+,m}}} \right], \end{aligned}$$

where $Z^s = \mathbb{X}_{+,ret}^s$ for the solution map and $Z^s = Z_+^s$ for the trace map, and

$$(9.33) \quad \|h\|_{tan,s} = \|\dot{a}\|_{\mathcal{A}_\gamma} + |\dot{\rho}| + |\dot{\epsilon}| + |\dot{p}|, \quad \|h\|_{lo} = \|h\|_{tan,s_*+k_{+,m}}.$$

The constants in (9.31)–(9.32) are uniform for $0 \leq \rho \leq \rho_0$ and both signs of the frequency. They apply to the full high-frequency space and to every finite upper cutoff, with no factor $1/\rho$. At real states, the operator, its inverse, and the trace map preserve the real subspaces under the fixed identifications.

To prove Proposition 9.1, we first use coercivity to invert $N_{+,b}^{cur,sp}$ for split data at $s = 0$. Higher regularity then allows us to recover the inverse in physical coordinates.

9.2. Coercivity at the circle and the boundary condition. We first prove coercivity with the physical boundary conditions. In the homogeneous equations of highest order, the interior sources and their boundary contributions vanish. The conversion (9.17) therefore leaves the boundary data unchanged, so the same computation applies to the split residuals.

The computation includes the endpoint $\rho = 0$. In the scalar coordinates of (7.3), the matrix through which the ellipse enters is

$$(9.34) \quad H_\rho(\theta) = \frac{1}{1 - \rho^2} \begin{pmatrix} 1 - \rho \cos 2\theta & \rho \sin 2\theta \\ \rho \sin 2\theta & 1 + \rho \cos 2\theta \end{pmatrix},$$

and its eigenvalues are $(1 \mp \rho)^{-1}$, so

$$(9.35) \quad \frac{1}{1 + \rho_0} I \leq H_\rho \leq \frac{1}{1 - \rho_0} I.$$

We bound the terms coming from $H_\rho - I$, whose nonconstant coefficients shift the angular mode by two, by $3\rho_0/(1 - \rho_0)$ times the form of highest order, so that

$$(9.36) \quad \Re \mathbf{b}_\rho(v, v) \geq c_{\rho_0} \|v\|_{\mathcal{E}_+^0}^2, \quad c_{\rho_0} = \frac{1 - 4\rho_0}{1 - \rho_0} > 0.$$

Here, $\mathbf{b}_\rho$ is the quadratic form of the positive block on the annulus at the constant ellipse $\mathbf{b}_\rho$, and $\mathcal{E}_+^s$ is the energy scale of the positive modes on the annulus without the weight, so that $\mathcal{E}_{+,a,\gamma}^s$ in (9.5) is its weighted version. The restriction $\rho_0 < 1/4$ gives $c_{\rho_0} > 0$, and all fixed-order ρ -derivatives of H_ρ are uniformly bounded. On the cap, the odd positive modes are $|m| \geq 3$ and the even positive modes are $|m| \geq 4$, and their forms with zero Dirichlet data are strictly positive. The Lax-Milgram lemma therefore gives a solution for an arbitrary source and an arbitrary value on the interface, and tangential difference quotients together with the equation give its regularity. The modes $m = \pm 1$ and the affine chains fixed on the axis do not enter this problem.

Dividing the boundary operator by its invertible coefficient of the conormal derivative, we obtain

$$(9.37) \quad \mathbf{E}_\mathbf{b}^{-1} \mathcal{B}_{++,\mathbf{b}} u = (\nabla_\mathbf{b} + \mathbf{S}_\mathbf{b}) u = h,$$

where $\nabla_\mathbf{b} u$ is the outward conormal derivative, $\mathbf{S}_\mathbf{b}$ acts on the boundary value of u , and $h = \mathbf{E}_\mathbf{b}^{-1} g_B^{\text{ph}}$ is the normalized prescribed datum. We write $\mathbf{S}_e$ for $\mathbf{S}_\mathbf{b}$ at the constant ellipse $\mathbf{b}_\rho$. For a constant $s_B > 0$, we have

$$(9.38) \quad \mathbf{S}_e = \mathbf{S}_e^*, \quad \mathbf{S}_e \geq s_B I.$$

To verify the normalization of the conormal derivative and of the Robin condition, we set $D = r\partial_r$, so that the positive reference operator on the annulus takes the form

$$(9.39) \quad \mathcal{L}_{++,\rho} = D^2 + B_{++,\rho} D + C_{++,\rho} - V_{++,\rho}(r), \quad B_{++,\rho}^* = -B_{++,\rho}.$$

Here, $B_{++,\rho}$, $C_{++,\rho}$, and $V_{++,\rho}(r)$ act on the sequence of positive angular modes. The coefficients $B_{++,\rho}$ and $C_{++,\rho}$ are independent of r . We then set

$$(9.40) \quad \begin{aligned} \nabla_\rho &= D + \frac{1}{2} B_{++,\rho}, & \mathcal{P}_{++,\rho} &= -\mathcal{L}_{++,\rho} = -\nabla_\rho^2 + W_\rho, \\ W_\rho &= V_{++,\rho} - C_{++,\rho} + \frac{1}{4} B_{++,\rho}^2. \end{aligned}$$

Writing the boundary operator in the same variables, in terms of the radial derivative and the value, we have

$$(9.41) \quad \mathcal{B}_{++,\rho} u = \mathbf{E}_\rho D u + \mathbf{F}_\rho u, \quad \mathbf{S}_\rho = \mathbf{E}_\rho^{-1} \mathbf{F}_\rho - \frac{1}{2} B_{++,\rho}.$$

Substituting the Fourier modes at the circle, we obtain

$$(9.42) \quad \mathbf{E}_0 = 2 \operatorname{diag}_{|m| \geq 3} \sqrt{p_m}, \quad \mathbf{S}_0 = 2I.$$

Here, $p_m > 0$ is the diagonal mass of the angular mode m at the circle, as in (7.7). Note that its lower bound for $|m| \geq 3$ is the gap used in (9.36).

Making the same substitution before the conjugations by the mass and by the maps (6.67), we obtain $\mathbf{S}_\rho = \mathbf{S}_\rho^*$. Since these conjugations depend analytically on ρ , the invertibility of $\mathbf{E}_0$ and the lower bound $\mathbf{S}_0 = 2I$ in (9.42) persist for small ρ . After decreasing ρ_0 if necessary, we obtain

$$(9.43) \quad \|\mathbf{E}_\rho^{\pm 1}\| \leq C, \quad \mathbf{S}_\rho \geq s_B I, \quad 0 \leq \rho \leq \rho_0.$$

This verifies (9.38) for the normalized boundary operator, and integrating in the measure dr/r , we obtain

$$(9.44) \quad \Re \int \langle \mathcal{P}_{++,\rho} u, u \rangle \frac{dr}{r} = \int \left(\|\nabla_\rho u\|^2 + \langle W_\rho u, u \rangle \right) \frac{dr}{r} - \Re [\langle \nabla_\rho u, u \rangle]_{r_j}^{r_b}.$$

At the outer radius r_b , we substitute $(\nabla_\rho + \mathbf{S}_\rho)u = \mathbf{E}_\rho^{-1}b$ into the boundary term, which turns it into the positive quadratic form in (9.38), minus the functional associated with the prescribed data. The estimates for the coefficients in Section 4, with the full distance (9.2), also give

$$(9.45) \quad \|\mathbf{E}_b^{\pm 1}\| \leq C, \quad \|\mathbf{S}_b - \mathbf{S}_e\|_{Z_D^{s*} \rightarrow Z_N^{s*}} \leq C\eta_+(\mathbf{b}).$$

We now conjugate by the analytic weight. We set $v = W_\gamma u$ and let ν_b be the coefficient of the outward radial derivative ∂_r in the conormal derivative ∇_b on the outer boundary. Writing $\nu_{b,\gamma} = W_\gamma \nu_b W_\gamma^{-1}$, we set

$$(9.46) \quad d_{\gamma,b} = -\nu_{b,\gamma} \partial_r \Phi_\gamma(r_b, \Lambda).$$

At the reference ellipse, the coefficient is independent of ζ , and this is a nonnegative Fourier multiplier. At a general state, its change is included in the coefficient perturbation. The weighted boundary condition is

$$(9.47) \quad (\nabla_{b,\gamma} + d_{\gamma,b} + \mathbf{S}_{b,\gamma})v = h_\gamma, \quad \|d_{\gamma,b}q\|_{Z_N^s} \leq C\gamma \|q\|_{Z_D^s}.$$

Here, $\nabla_{b,\gamma}$ and $\mathbf{S}_{b,\gamma}$ denote the conjugated conormal operator with the displayed radial weight term removed and the conjugated Robin coefficient, respectively. Thus, $W_\gamma \nabla_b W_\gamma^{-1} = \nabla_{b,\gamma} + d_{\gamma,b}$, and $h_\gamma = W_\gamma h$. The sign of the added boundary term and the cancellation of derivatives of d_γ can be seen first in the scalar model. If $(\partial_r + d + S)v(r_b) = h$ and the inner value is zero, integration by parts gives

$$(9.48) \quad \Re \int [-(\partial_r + d)^2 + \lambda^2] v \bar{v} dr = \int (|\partial_r v|^2 + (\lambda^2 - d^2) |v|^2) dr + \langle Sv(r_b), v(r_b) \rangle - \Re \langle h, v(r_b) \rangle.$$

The two terms involving d' cancel. The same cancellation holds for the conormal operator, with $D = r\partial_r$ and the measure dr/r . Relative to the energy form, the remaining terms coming from the weight are bounded by $C(\gamma + \gamma^2)$ and the perturbations of the coefficients by $C\eta_+$. Writing $\mathbf{b}_{+,b,\gamma}$ for the quadratic form of the positive block on the annulus at the state, conjugated by the weight, we thus obtain

$$(9.49) \quad \Re \mathbf{b}_{+,b,\gamma}(v, v) \geq [c_+ - C(\gamma + \gamma^2) - C\eta_+(\mathbf{b})] \|v\|_{\mathcal{E}_+^0}^2 \geq \frac{c_+}{2} \|v\|_{\mathcal{E}_+^0}^2.$$

9.3. Solving the boundary problems on the annulus and cap. On the annulus, we consider first the boundary value problem

$$(9.50) \quad u \mapsto (P_{a,+,\mathbf{b}} u, \gamma_j^D u, \mathcal{R}_{++,\mathbf{b}}^{\text{sp}}(u, P_{a,+,\mathbf{b}} u)) = (f, g, g_B^{\text{sp}}),$$

with independent source f , inner value g , and outer split residual g_B^{sp} . We write $u = \mathcal{P}_\Sigma^a g + v$, using the localized Poisson extension of Section 4. Then v has zero inner value and interior source $f - P_{a,+,\mathbf{b}} \mathcal{P}_\Sigma^a g$. The extension is supported away from the outer boundary and is bounded on the weighted scale by (4.10). The outer datum is unchanged, with $\mathbf{V}_{B,\mathbf{b}}^{-1} g_B^{\text{sp}}$ paired with the outer value in the boundary functional (9.11). At $s = 0$, f acts on test fields on the annulus with zero value on both boundaries.

From the coercivity (9.49) and the Lax-Milgram lemma, we obtain a unique solution for every (f, g, g_B^{sp}) , and at $s = 0$, Green's identity defines its split residual on the inner boundary. At a regular index with full sources, we recover the physical conormal derivative from that residual by subtracting $\mathbf{M}_{a,\Sigma}^s \ell_a$ as in (9.14). In weighted coordinates, the exact identity is

$$(9.51) \quad W_\gamma \nabla_b u|_{r_j} = (W_\gamma \nabla_b W_\gamma^{-1})v|_{r_j}.$$

For each side of the interface, the added radial term is $-\nu_{\mathbf{b},\gamma}\partial_r\Phi_\gamma(r_j,\Lambda)v$ with that side's outward conormal coefficient. This sign differs between the inner and outer boundaries of the annulus. The conormal derivative has the balanced order in (9.9).

On the scaled cap, V_+^{cap} is the Cartesian form space H^1 of the positive modes, $V_{+,0}^{\text{cap}} = \ker \gamma_D$, and $X_+^{\text{cap},s}$ and $Y_+^{\text{cap},s}$ are the corresponding scales of solutions of order H^{s+1} and of sources of order H^{s-1} . We specify the frozen coefficients by the jet

$$(9.52) \quad q_{+,\mathbf{b}}(\zeta) = (\rho, j_y^1 a(0, \zeta), \varepsilon, p, P_{+,\mathbf{b}}^D(0, \zeta), P_{+,\mathbf{b}}^R(0, \zeta)).$$

Here, the last two entries are the spectral projections of Section 6 on the domain and on the range, evaluated on the axis. The entry $j_y^1 a(0, \zeta)$ records the value and first derivatives of a in the disk variables on the axis, as in (8.69). The coefficients of the cap operator depend analytically on these entries through the constructions of Section 4. The positive block contains only the odd modes with $|m| \geq 3$ and the even modes with $|m| \geq 4$, and the term in the cell frequency is nonnegative and enters only as a parameter. Together, these give the coercivity of the form at $s = 0$, uniformly in the normalized cell frequency $\kappa = 2^{-j}n$ with $1/2 \leq |\kappa| \leq 2$. Writing $\mathfrak{c}_+(q, \kappa; \cdot, \cdot)$ for the form on the cap with the coefficients frozen at the jet q and at the cell frequency κ , and $P_+(q, \kappa)$ for the corresponding operator, we have

$$(9.53) \quad \Re \mathfrak{c}_+(q, \kappa; v, v) \geq \frac{c_{\rho_0}}{2} \|v\|_{V_+^{\text{cap}}}^2, \quad v \in V_{+,0}^{\text{cap}}.$$

For an interior source f and a prescribed value g on the interface, we write $u = \mathcal{P}_D g + v$, where $\mathcal{P}_D$ is the Poisson extension from the interface into the cap. The remaining problem is

$$P_+(q, \kappa)v = f - P_+(q, \kappa)\mathcal{P}_D g, \quad \gamma_D v = 0.$$

The coercivity (9.53) and the Lax-Milgram lemma give a unique $v \in V_{+,0}^{\text{cap}}$. Adding back $\mathcal{P}_D g$ gives the solution $u \in V_+^{\text{cap}}$. Elliptic regularity, uniform in the parameter, then yields

$$(9.54) \quad \begin{aligned} \|u\|_{X_+^{\text{cap},s}} + \|\mathcal{N}_{c,\Sigma}^{\text{sp}}(u, P_+(q, \kappa)u)\|_{Z_N^s} &\simeq \|u\|_{H^{s+1}} + \|\mathcal{N}_{c,\Sigma}^{\text{sp}}(u, P_+(q, \kappa)u)\|_{H^{s-1/2}} \\ &\leq C_s (\|P_+(q, \kappa)u\|_{H^{s-1}} + \|\gamma_D u\|_{H^{s+1/2}}). \end{aligned}$$

At $s = 0$, the boundary term in (9.54) is the split residual in $H^{-1/2}$. For $s \geq 1$, we apply tangential difference quotients to the remainder with zero Dirichlet data, and the estimate for the Dirichlet problem and the equation then give regularity H^{s+1} . Green's identity and (9.14) give both the split residual and the physical conormal derivative in $H^{s-1/2}$. The finitely many additional equations, namely the finite triangular system of Section 6 and the conditions on the axis, enter through bounded triangular factors and do not change these orders. All constants are uniform on the compact set of jets and on the normalized frequency range $1/2 \leq |\kappa| \leq 2$.

We conclude that the operator with frozen coefficients

$$(9.55) \quad \mathcal{C}_+^{\text{op}}(q, \kappa)u := (P_+(q, \kappa)u, \gamma_D u) : X_+^{\text{cap},s} \longrightarrow Y_+^{\text{cap},s} \oplus Z_D^s$$

has a two-sided inverse

$$(9.56) \quad K_+^{\text{cap}}(q, \kappa) := (\mathcal{C}_+^{\text{op}}(q, \kappa))^{-1}.$$

Differentiating the weak equation and applying (9.54), we obtain analyticity and a bound for every seminorm (8.43) of the operator-valued symbols $(q, \kappa) \mapsto \mathcal{C}_+^{\text{op}}(q, \kappa)$ and $(q, \kappa) \mapsto K_+^{\text{cap}}(q, \kappa)$, including the component

that gives the balanced split residual on the interface. At regular indices, the estimate for $M_{c,\Sigma}^s$ also bounds the seminorms of the physical conormal derivative.

We measure the solution and the source on the cap of the positive block in the shell spaces

$$(9.57) \quad \mathfrak{X}_{c,+,H}^s := \bigoplus_{j \geq J}^{\ell_s^2} Q_j X_+^{\text{cap},s}, \quad \mathfrak{Y}_{c,+,H}^s := \bigoplus_{j \geq J}^{\ell_s^2} Q_j Y_+^{\text{cap},s},$$

with local orders H^{s+1} and H^{s-1} . Before reconstructing the field, we use the spectral projections and the maps (6.67) to send each summand $Q_j X_+^{\text{cap},s}$, which is defined at the constant ellipse $\mathbf{b}_\rho$, to the corresponding summand at the state $\mathbf{b}$. The positive cap field retains its form regularity H^{s+1} , whereas the minus block uses H^{s+2} . For the off-diagonal terms, we apply the projection on the range after the differential operator, where the identity (6.24) leaves at most one derivative on the cap.

For the prescribed jump of the values on the interface, we set

$$(9.58) \quad \mathfrak{Z}_{D,H}^s := \{(d_j)_{j \geq J} : Q_j d_j = d_j\} \subset \ell_s^2(Z_D^{\text{cap}}).$$

Here, Z_D^{cap} is the space of values on the interface of the scaled cap. With $\mathcal{A}_D$ denoting the map that extracts the value on the interface on each shell and $\mathcal{S}_D$ denoting its inclusion in the auxiliary trace space $\mathcal{Z}_{\text{aux},H}^s$ of (6.54) at the reference, we have

$$(9.59) \quad \mathcal{A}_D \mathcal{S}_D = I_{\mathfrak{Z}_{D,H}^s}, \quad \Pi_D := \mathcal{S}_D \mathcal{A}_D, \quad \Pi_D^2 = \Pi_D.$$

Both maps retain the value at the interface radius of its own shell.

The reference coefficients are independent of the cell variable, so the positive cap operator acts separately on each cell mode. We define

$$(9.60) \quad \mathbf{C}_{+, \rho}^{\text{cap}}(U_j)_{j,n} := \mathcal{C}_+^{\text{op}}(q_{+, \mathbf{b}_\rho}, 2^{-j}n) U_{j,n} : \mathfrak{X}_{c,+,H}^s \longrightarrow \mathfrak{Y}_{c,+,H}^s \oplus \mathfrak{Z}_{D,H}^s.$$

Its inverse is the direct sum of $K_+^{\text{cap}}(q_{+, \mathbf{b}_\rho}, 2^{-j}n)$. We use this operator as a local model for regularity. The terms at the state that couple different shells, or the cap and the annulus, act only after the full reference reconstruction.

9.4. Matching across the interface and higher regularity. In the coordinates of the shells, $N_{+, \mathbf{b}}$ is the restriction of $\mathbb{N}_{\mathbf{b}}$ from (6.83) to the positive block. We write $\text{Ins}_{k,\rho}^{\text{br}}(U_{c,k}, U_{a,k})$ for the reference insertion of a cap field and its annular complement, with a mismatch allowed, so that the interior operator is

$$(9.61) \quad \mathcal{N}_{+, \mathbf{b}, jk}^{\text{bulk}}(U_{c,k}, U_{a,k}) = \mathcal{R}_j^Y Q_j L_{\mathbf{b}} \text{Ins}_{k,\rho}^{\text{br}}(U_{c,k}, U_{a,k}).$$

The cap and annular fields together cover all radii, so the coefficients of the state act before Q_j . The inverse map $(\mathfrak{S}_{+, \mathbf{b}}^{\text{res},s})^{-1}$ then subtracts the boundary contributions of the sources, and (9.26) returns the physical value and conormal derivative to the reference coordinates.

Existence at energy regularity. We first solve (9.24) with split data. We let

$$F^{\text{sp}} = (f_c, f_a, g_D, g_N^{\text{sp}}, g_B^{\text{sp}}) \in \mathbb{Y}_{+, \text{ret}}^0$$

have arbitrary, independent entries in the splitting (9.11). We subtract $(0, \mathcal{P}_{\Sigma}^a g_D)$ from the unknown pair and its image under $N_{+, \mathbf{b}}^{\text{cur}, \text{sp}}$ from the prescribed data. The new pair then belongs to (9.21). The remaining conormal and Robin data act continuously on this form domain. Adding the weak identities on the cap and on the annulus, we obtain (9.11), and for regular fields, (9.14) expresses this identity in physical coordinates. The

two outward conormal terms at the interface cancel. With the corresponding opposite outward orientations, the radial weight terms in (9.51) cancel as well, while at the outer boundary, (9.48) gives a positive Robin term. Thus, the form on the pair with matching values on the interface satisfies

$$(9.62) \quad \Re \mathfrak{b}_{+, \mathbf{b}, \gamma}^{\text{glued}}((v_c, v_a), (v_c, v_a)) \geq \frac{c_+}{2} \|(v_c, v_a)\|_{\mathbb{V}_{+, \text{ret}}^0}^2.$$

The extended sources and prescribed residuals define a continuous functional on the form domain. The Lax-Milgram lemma therefore gives an inverse of $N_{+, \mathbf{b}}^{\text{cur}, \text{sp}}$ for all five independent data at $s = 0$. If these data vanish, the sources and their boundary contributions are zero, so the homogeneous split and physical conditions agree, and the coercivity (9.62) then forces both fields to vanish. This proves injectivity at the weak level.

Regularity across the interface. To prove regularity, we need an estimate for the coupled problem that also controls the trace on the interface. The homogeneous equation of highest order has $\ell_c = \ell_a = 0$, so the boundary contributions vanish and the conversion between physical and split data is the identity there. Up to the invertible reductions of the principal symbol, the scalar principal symbol of the positive block with frozen coefficients is, by (6.35),

$$(9.63) \quad q_\sigma(d, \eta) = a_\sigma d^2 + 2b_\sigma(\eta)d + c_\sigma(\eta) = \xi^* G_{\sigma, \mathbf{b}}^{-1} \xi, \quad \sigma \in \{c, a\}, \quad \xi = (d, \eta),$$

where d is the normal frequency variable after flattening the interface, η collects the tangential and cell parameters, and $G_{\sigma, \mathbf{b}}$ is the Gram matrix (6.33) of the frame on the cap for $\sigma = c$ and on the annulus for $\sigma = a$. Positivity of the inverse Gram matrix gives

$$(9.64) \quad a_\sigma > 0, \quad \Delta_\sigma(\eta) := a_\sigma c_\sigma(\eta) - b_\sigma(\eta)^2 \geq c|\eta|^2$$

on the normalized sphere of parameters. The quadratic formula is first understood for real frequencies and then extended polynomially in d . On each side, we use a normal coordinate increasing outward. Of the two roots

$$d_\sigma^\pm = \frac{-b_\sigma(\eta) \pm i\sqrt{\Delta_\sigma(\eta)}}{a_\sigma},$$

the root d_σ^- has negative imaginary part and gives the solution that decays into that side of the domain. After the same invertible normalization, the symbols of the maps from the value to the outward conormal derivative are $\Lambda_\sigma(\eta) = \sqrt{\Delta_\sigma(\eta)}$. On the decaying amplitudes, the jump of the values and the sum of the conormal derivatives therefore have the symbol

$$(9.65) \quad \mathcal{B}_{\text{tr}}(\eta) = \begin{pmatrix} -I & I \\ \Lambda_c(\eta) & \Lambda_a(\eta) \end{pmatrix}, \quad \text{Schur } \mathcal{B}_{\text{tr}} = \Lambda_c + \Lambda_a \geq c|\eta|I.$$

Here, Schur $\mathcal{B}_{\text{tr}}$ is the *Schur complement* of the upper left entry, that is, the operator $\Lambda_a - \Lambda_c(-I)^{-1}I$ that remains on the second row after the first row is used to eliminate the first unknown. The bound on the Schur complement says that this transmission problem satisfies the *complementing condition*, that is, the algebraic condition on the boundary symbol under which the interface conditions determine the decaying solutions uniformly. For the finite-dimensional system, the same argument applies to the positive Hermitian matrices which map the value to the conormal derivative. At the outer boundary, the Robin symbol of highest order $\mathbf{E}_\mathbf{b}\Lambda_a$ is invertible by (9.45). At the reference, the nonnegative damping in (9.47) strengthens the bound. Its change at the state is included in the coefficient perturbation. The term $\mathbf{S}_\mathbf{b}$ supplies coercivity at low frequencies, and it does not enter the principal symbol.

Explicitly, for a prescribed value jump g_D and sum of conormal derivatives g_N , the two decaying amplitudes w_c, w_a satisfy

$$-w_c + w_a = g_D, \quad \Lambda_c w_c + \Lambda_a w_a = g_N.$$

Thus, $w_c = (\Lambda_c + \Lambda_a)^{-1}(g_N - \Lambda_a g_D)$ and $w_a = w_c + g_D$. This is the inverse controlled by (9.65).

For higher regularity, we use this bound for the frozen coefficients. We choose cap and annular patches with common tangential coordinates, flatten the interface, and take cutoffs $\chi \prec \tilde{\chi}$ in such a patch, that is, cutoffs with $\tilde{\chi} = 1$ on the support of χ . We then apply the same tangential difference quotient Δ_h^τ to both fields and test with $-\Delta_h^\tau(\chi^2 \Delta_h^\tau U)$, which preserves the equality of the interface values. We control the interior terms of highest order by the estimate (9.54) on the cap and by its counterpart on the annulus. The terms on the interface combine into $\mathcal{B}_{\text{tr}}$ from (9.65), whose Schur bound controls the common value and the sum of the conormal derivatives.

We then freeze the coefficients at the center of each patch. Once the patch is small enough, the oscillation of the coefficients over the patch is less than a quarter of the coercivity constant of the interior form and of the lower bound on the Schur complement in (9.65). We absorb the resulting errors into the left-hand side. The commutators with χ , with the flattening map, and with the lower-order coefficients contain at most one derivative of U , and we bound them by the energy norm on the enlarged patch. Since $a_\sigma \geq c > 0$ in (9.64), the equations recover the remaining normal derivatives. Summing over a finite partition of unity, we obtain the estimates for higher regularity. At $s = 0$, the forms and the source and Poisson extensions are bounded. They control each split residual by the energy norm of the field and the norm of its source in the dual of the interior. The trace theorem for the value and the bounded Robin coefficients control the other terms in the graph norm. Thus, on smooth elements,

$$(9.66) \quad \|U\|_{\mathbb{X}_{+, \text{ret}}^0} + \|\Gamma_{+, \mathbf{b}} U\|_{Z_+^0} \leq C(\|N_{+, \mathbf{b}}^{\text{cur}, \text{sp}} U\|_{\mathbb{Y}_{+, \text{ret}}^0} + \|U\|_{\mathbb{V}_{+, \text{ret}}^0}).$$

The boundary terms at $s = 0$ are split residuals in $H^{-1/2}$. At higher regularity, tangential differentiation and the equations give the remaining derivatives. Formula (9.14) then recovers the physical conormal derivatives. No derivative of a trace occurs on the right-hand side. The reference reconstruction of the shells precedes this estimate, so the coefficients of the state which couple different shells act on a full field and are controlled by the commutator bound of (6.10) together with the product estimate (3.20).

For smooth data, we take nested finite-dimensional subspaces of the smooth form domain with matching values on the interface. The *Galerkin approximations*, that is, the solutions of the variational problem on these subspaces, obey the coercive bound (9.62) and converge weakly in $\mathbb{V}_{+, 0}^0$ to the Lax-Milgram solution, and the bound for the weak residual gives (9.66) for this limit. For higher regularity, we apply the localized argument with difference quotients to the limiting variational equation. After flattening and localization, the quotients are admissible test functions.

For arbitrary split data, we take smooth approximations $F_k^{\text{sp}} \rightarrow F^{\text{sp}}$ in the scale of the range, and the energy and graph estimates bound the corresponding solutions U_k uniformly in $\mathbb{X}_{+, \text{ret}}^0$, split residuals included. Weak compactness, the trace theorem for the value, and the reconstruction maps (6.56) pass the constraint on the values, both interior equations, and the split boundary conditions to the limit. Uniqueness in the energy space identifies that limit with the weak solution, and we obtain

$$(9.67) \quad \|(v_c, v_a)\|_{\mathbb{X}_{+, \text{ret}}^0} + \|\Gamma_{+, \mathbf{b}}(v_c, v_a)\|_{Z_+^0} \leq C\|F^{\text{sp}}\|_{\mathbb{Y}_{+, \text{ret}}^0}.$$

Thus, the weak solution belongs to the component graph spaces and has the prescribed residuals. We write $G_{+,b}^{\text{cur,sp}} = (N_{+,b}^{\text{cur,sp}})^{-1}$ for the solution map at $s = 0$.

At regular indices with full sources, the map $\mathfrak{S}_{+,b}^{\text{res},s}$ converts the physical data to split data, and we set

$$G_{+,b}^{\text{cur,ph}} := G_{+,b}^{\text{cur,sp}} \mathfrak{S}_{+,b}^{\text{res},s},$$

so (9.26) gives

$$(9.68) \quad \begin{aligned} G_{+,b}^{\text{N}} &:= G_{+,b}^{\text{cur,sp}} \mathfrak{S}_{+,b}^{\text{res},s} \mathcal{R}_{CN,b,\rho}^{-1} = N_{+,b}^{-1}, \\ N_{+,b} G_{+,b}^{\text{N}} &= I, & G_{+,b}^{\text{N}} N_{+,b} &= I. \end{aligned}$$

Construction gives the right inverse identity, and uniqueness on the operator domain gives the left inverse identity. In (9.68), $\mathcal{R}_{CN,b,\rho}^{-1}$ first converts reference trace coordinates to physical data at the state, $\mathfrak{S}_{+,b}^{\text{res},s}$ adds the source contributions, and $G_{+,b}^{\text{cur,sp}}$ solves for the cap and annular fields. The bound on the range for $G_{+,b}^{\text{N}}$ includes the norm of $\mathfrak{S}_{+,b}^{\text{res},s}$, and the bound for the adjoint includes its transpose (9.18). Multiplying (9.28) and (9.68) out, we obtain

$$(9.69) \quad \widehat{D}_{+,b} \widehat{G}_{+,b} = I, \quad \widehat{G}_{+,b} \widehat{D}_{+,b} = I.$$

This proves (9.30) on the fixed domain and range of the operator.

Tame estimates. It remains to prove the tame estimates at higher regularity. We commute Cartesian and tangential derivatives through the conjugated reference operator, retaining every term containing d_γ . We also keep on the left-hand side the product of the derivatives of highest order with the low-frequency part of the coefficients of the state. For every $0 < \delta < c_+/4$, the remainders satisfy

$$(9.70) \quad \|R_s(v)\|_{(\mathcal{E}_+^s)^*} + \|R_s^\partial(v)\|_{Z_N^s} \leq \delta \|v\|_{\mathcal{E}_+^s} + C_{s,\delta} \|v\|_{\mathcal{E}_+^{s-1}} + C_s (1 + \|b - b_\rho\|_{B^{s+k_+}}) \|v\|_{\mathcal{E}_+^{s*}}.$$

Here, $R_s(v)$ collects the interior commutators produced by differentiating $P_{a,+} v$, and $R_s^\partial(v)$ collects the corresponding commutators with the boundary operator (9.15), in both cases after conjugation by W_γ . In each term, at least one derivative has fallen on a coefficient of the state. At most one coefficient is measured in a higher regularity norm, and it multiplies the solution at the base index. If a derivative falls on a low-frequency coefficient, one fewer derivative remains on the solution, and interpolation gives the middle term. Once the tangential derivatives and the split residuals are estimated, we use the boundary contributions of the sources to recover the physical conormal derivatives at regular indices. The equations then give the remaining normal regularity, namely Cartesian H^{s+1} on the cap, and the graph regularity on the annulus defined by the weighted energy and by the dual of the interior. All terms coming from the weight remain in the operator of highest order, so the derivatives of d_γ require no condition $C_s \gamma < c_+$ depending on the index. We absorb δ and induct on s , starting with the estimate at the base index and commuting the two transmission conditions at their balanced orders, which proves (9.31).

The domain and the range are fixed when we differentiate in b , and the inverse identity gives

$$(9.71) \quad D_b \widehat{G}_{+,b}[h] = -\widehat{G}_{+,b} (D_b \widehat{D}_{+,b}[h]) \widehat{G}_{+,b}.$$

Iterating this formula, we express $D_b^m \widehat{G}_{+,b}$ as a finite sum of alternating inverse factors and derivatives of $\widehat{D}_{+,b}$. The product estimate (3.20) measures at most one factor at the higher regularity index and the others at the base index. Together with (9.70), this gives (9.32). Derivatives of the trace map add only factors of fixed order. In (9.71), $D_b \widehat{D}_{+,b}$ includes the derivatives of $(\mathfrak{S}_{+,b}^{\text{res},s})^{-1}$, and transposition reverses their order as in (9.18). The width of the strip of analyticity is unchanged.

Finally, we obtain uniformity at the circle from (9.35) and (9.36). The reconstruction on the cap for the exceptional modes depends on ρ only through

$$(9.72) \quad \rho, \quad 1 - \rho^2, \quad (1 - \rho^2)^{-1}.$$

Since the Hardy maps are independent of ρ , all fixed-order parameter derivatives are bounded on $[0, \rho_0]$. A unitary rotation restores a constant transverse orientation without division by ρ . The Cartesian coefficients, the quadratic form of the Robin condition on the outer boundary, the weight, the contours, and the Poisson extensions preserve real data under the fixed identifications. By uniqueness, so do $\widehat{G}_{+,b}$ and its outgoing trace. Thus, the inverse loses no derivatives in the block norms. As in Section 8, the loss in converting a Cartesian source to these norms is accounted for separately by Proposition 6.2.

We have proved (9.67) at $s = 0$ for split data and the isomorphism at regular indices for physical data, together with the tame bounds. This completes the proof of Proposition 9.1.

10. COUPLING ESTIMATES AND DIAGONAL INVERSES

We estimate the off-diagonal blocks of the high-frequency operator $D_{H,b}$ in (6.94) and use these estimates to correct the diagonal inverses. The operators $\widehat{D}_{\pm,b}$ inverted in Propositions 8.1 and 9.1 can differ from the diagonal blocks $\pi_\alpha^Y \mathbb{D}_b \iota_\alpha^X$ of the augmented operator because the change of source variables couples the blocks. We account for this difference before combining the inverses. The off-diagonal terms of highest order cancel by (6.26), leaving the lower-order terms estimated below.

10.1. Interior terms and terms with one angular derivative. For each remaining interior term, we compare the cell-frequency weights of its input and output, including the powers introduced by cell derivatives. At a fixed cell frequency n , with $\lambda = \lambda_n$ and $\mu = \mu_n$ as in (6.3), Table 4 lists the leading terms and their weight ratios. The order in the table counts cell derivatives on the input. For example, a first-order term taking a center variable to the positive source costs one factor λ_n ; the source weight λ_n^{-1} cancels this factor. We estimate angular and radial derivatives separately below.

Term	Order	Ratio of the weights
from C to Y_+	1	$\lambda^{-1}(\lambda C)/ C = 1$
from X_+ to Y_C	1	$(\lambda u_+)/(\lambda u_+) = 1$
from X_+ to $Y_{A,y}$	0	$\lambda^{1/2} u_+ /(\lambda u_+) = \lambda^{-1/2}$
from X_+ to $Y_{A,x}$	0	$\mu u_+ /(\lambda u_+) \leq C_{\delta_A} \lambda^{-1}$
from $X_{A,x}$ to Y_+	0	$\lambda^{-1} x /(\mu x) \leq C_{\delta_A} \lambda^{-1/2}$
from $X_{A,x}$ to Y_+	1	$ x /(\mu x) \leq \delta_A^{-1/2}$
from $X_{A,y}$ to Y_+	1	$ y /(\lambda^{1/2} y) = \lambda^{-1/2}$

TABLE 4. The leading interior terms remaining between different block components after the principal cancellation, their order, and the ratio of the weights of the two components at cell frequency n , with $\lambda = \lambda_n$ and $\mu = \mu_n$.

In this table, C denotes the four center coordinates of (8.82), and Y_C is their source space. The symbols X_+ and Y_+ denote the positive component on the domain and range sides, and $X_{A,x}$, $X_{A,y}$, $Y_{A,x}$, and $Y_{A,y}$ are

the upper and lower entries (x, y) of the affine chains on the two sides. The quantities $|C|$, $|u_+|$, $|x|$, and $|y|$ denote the sizes of the corresponding Fourier coefficients, the weights are those of Table 1, and C_{δ_A} is a constant depending only on δ_A . Every ratio is bounded, so these terms lose no powers of the cell frequency. Some terms have order zero in the cell variable but contain one angular derivative. We estimate them next. We set $\Lambda_\theta = \langle \mathcal{R} \rangle$, the operator which multiplies the angular mode m by $\langle m \rangle$, and we write $\mathcal{M}_{\mathbf{b}}$ for an order-zero factor, or a product of such factors, used to normalize the scalar equation. These include the inverse square root of the radial mass, as in (7.12), and the fixed-space maps (6.67). We let $U_{R,\mathbf{b}}$ be the map (6.67) from the reference range to the range at the state, so that

$$(10.1) \quad P_{-,\mathbf{b}}^R = U_{R,\mathbf{b}} P_{-,*}^R U_{R,\mathbf{b}}^{-1}.$$

Here, $P_{-,*}^R$ is the reference projection P_* of (6.67) for the range projection of the minus block. We use the angular smoothing of this reference projection, together with the tame estimates for the maps (6.67) in Proposition 6.2, to obtain, for each $k \geq 0$,

$$(10.2) \quad \|P_{-,\mathbf{b}}^R f\|_{H_\theta^{s+k}} \leq C_{s,k} \left[\|f\|_{H_\theta^s} + \|\mathbf{b} - \mathbf{b}_\rho\|_{\mathcal{B}^{s+k+k_{\text{sm}}(k)}} \|f\|_{H_\theta^{s*}} \right],$$

where H_θ^s denotes the Sobolev space of order s in the angular variable, with the other variables held fixed, and where $k_{\text{sm}}(k)$ is a finite shift in the regularity of the coefficients, independent of s . The state $\mathbf{b}_\rho$ is the constant ellipse introduced in Section 6, written in the fixed coordinates, and $\|\mathbf{b} - \mathbf{b}_\rho\|_{\mathcal{B}^s}$ is the norm (8.4). The constant is uniform on the ball of states in the norm at the base index s_* , and the higher regularity norm of the state appears explicitly in the estimate. Differentiating (10.1), we obtain

$$(10.3) \quad \begin{aligned} & \|D_{\mathbf{b}}^m P_{-,\mathbf{b}}^R [\mathbf{h}] f\|_{H_\theta^{s+k}} \\ & \leq C_{s,k,m} \left\{ \mathcal{H}_{0,m}(\mathbf{h}) \|f\|_{H_\theta^s} + \mathcal{H}_{s+k,m}(\mathbf{h}) \|f\|_{H_\theta^{s*}} \right. \\ & \quad \left. + \|\mathbf{b} - \mathbf{b}_\rho\|_{\mathcal{B}^{s+k+k_m}} \mathcal{H}_{0,m}(\mathbf{h}) \|f\|_{H_\theta^{s*}} \right\}, \end{aligned}$$

with the products of tangent norms of (6.1). Here and below, we write $\mathbb{X}_{\pm,\mathfrak{q}}^s := \mathbb{X}_{\pm,\text{ret}}^s$ and $\mathbb{Y}_{\pm,\mathfrak{q}}^s := \mathbb{Y}_{\pm,\text{ret}}^s$ for the block spaces of the two high-frequency blocks, defined in (8.13) and (9.20). We use this notation throughout the remaining linear analysis. To absorb an angular derivative on the datum, we also use smoothing on the input side of the projection. More precisely, $P_{-,\mathbf{b}}^R \mathcal{M}_{\mathbf{b}}$ must extend from H_θ^{s-1} to the minus source space at index s . Since $\Lambda_\theta : H_\theta^s \rightarrow H_\theta^{s-1}$, this gives the bounded map

$$(10.4) \quad P_{-,\mathbf{b}}^R \mathcal{M}_{\mathbf{b}} \Lambda_\theta : H_\theta^s \longrightarrow \mathbb{Y}_{-,\mathfrak{q}}^s$$

into the combined center and affine source space, with tame estimates for its derivatives with respect to $\mathbf{b}$. For the positive component, the dual norm includes an inverse angular derivative, and we write

$$(10.5) \quad \Lambda_\theta^{-1} P_{+,\mathbf{b}}^R \mathcal{M}_{\mathbf{b}} \Lambda_\theta = \mathcal{M}_{\mathbf{b}} + \Lambda_\theta^{-1} [\mathcal{M}_{\mathbf{b}}, \Lambda_\theta] - \Lambda_\theta^{-1} P_{-,\mathbf{b}}^R \mathcal{M}_{\mathbf{b}} \Lambda_\theta.$$

Since $\mathcal{M}_{\mathbf{b}}$ has order zero with smooth coefficients, the commutator $[\mathcal{M}_{\mathbf{b}}, \Lambda_\theta]$ also has angular order zero. Estimate (10.4) bounds the last term. Thus, the source norms absorb the angular derivative without loss of a power of the cell frequency. In particular, Table 4 still applies, and the norm of the source of the center component requires no additional power of λ_n .

Table 5 summarizes the interior estimates, all of which use the weights of Table 1.

Table 5: The interior maps between block components and the reason why each of them is bounded in the block norms.

Map	Why it is bounded
compatible quotient	The reconstruction (6.27) of a compatible source from its quotients is a multiplication operator in the disk variables, so it is diagonal in the cell frequency, and it involves no Hardy map.
frame, maps (6.67), and projections	On the annulus, the lower bound on the determinant shows that the inverse $\mathbb{B}^{-1}$ of the frame (6.12) depends analytically on the state. Derivatives of the frame enter through (6.15). The spectral projections (6.21), defined by fixed contours, and the maps (6.67) have finite moments in the sense of Lemma 6.1.
scalar differential source	After projection onto the Cartesian range, (6.28) has no off-diagonal part of highest order, by (6.26). Derivatives of coefficients and projections lower the order by one.
inversion of the factors $\mathcal{R} \pm 2i$ on the odd and on the high even modes	We invert the factors $\mathcal{R} \pm 2i$ only on the modes on which they do not vanish, listed after Table 2, and we measure the positive output in the dual of the form space.
affine equations at the exceptional modes	We solve for the independent forcings at $m = \pm 2$ using the Hardy maps (8.37), which act in the radial variable alone and are invariant under dilations. The weights μ and $\lambda^{1/2}$ occur on both sides in Table 4.
terms with one angular derivative	The projection of the minus block is smoothing, and the dual of the positive form includes the inverse derivative, by (10.4)–(10.5).
extraction of the center	The angular smoothing used in (10.4) controls this projection. It has order zero in the cell variable. Its radial output is measured in the L^1 source norm of the center equation in Section 8.
interior equations of the positive block	The positive form maps the form space of the pairs of restrictions to the cap and to the annulus to its dual, with one derivative on each argument, and the weights λ and λ^{-1} of Table 1 then bound the map in the cell frequency.
value and first derivatives on the axis, and affine mean	We use the entries λ^{-1} and λ^{-2} of Table 1, together with the localized extension operators of Proposition 4.1.
mean, scale, gauge, and chart conditions	The system is triangular, with finitely many angular modes and invertible diagonal entries. The coefficients may depend on ζ , and a derivative on a coefficient leaves one fewer on the unknown.
analytic weight	Conjugation by the analytic weight adds the multiplier d_γ of (3.21) to the radial derivative. Its diagonal part remains in the corresponding diagonal block. Its commutator with a spectral projection has a kernel which contains the factor $d_\gamma(r, \lambda_n) - d_\gamma(r, \lambda_m)$, as in (6.10), so it is controlled by the moment of order one, and it vanishes at the circle, where the projections do not depend on ζ .

For the entries on the axis, E_0 extends a prescribed value and E_1 extends prescribed first derivatives, using the localized extensions of Proposition 4.1. They satisfy

$$(10.6) \quad \|E_0 b\|_{\mathcal{A}_\gamma^s} \leq C_s \|e^{\sigma_0 \Lambda} b\|_{H^{s-1}}, \quad \|E_1 b\|_{\mathcal{A}_\gamma^s} \leq C_s \|e^{\sigma_0 \Lambda} b\|_{H^{s-2}}.$$

At cell frequency n , the extension is supported in a region of radius comparable to λ_n^{-1} . A derivative of the cutoff $\phi(\lambda_n y)$ therefore gives a factor λ_n , already accounted for by the index shifts in (10.6).

10.2. Coupling between the two blocks on the cap. To estimate the coupling on the cap, we use the isomorphisms (8.16) and (9.60) for the reference cap operators. At the state $\mathbf{b}$, we instead invert the operators $N_{-, \mathbf{b}}$ and $N_{+, \mathbf{b}}$ of Sections 8 and 9, which couple the cap and the annulus. We denote the outward conormal derivative on the interface by $\Gamma_{c,+,N,\mathbf{b}} u := \nabla_{c,\mathbf{b}}^{\text{out}} u|_{r=r_j}$. The reference operators and the traces at the state act as follows, where $\Gamma_{c,-,\mathbf{b}} = \Gamma_{c,-,\mathbf{b}}^{\text{ph}}$ is the physical Cauchy trace of (8.8)

$$(10.7) \quad \begin{aligned} \mathcal{A}_{c,-,\rho}^{\text{diag}} : X_{c,-}^s &\xrightarrow{\sim} Y_{c,-}^s, & \Gamma_{c,-,\mathbf{b}} : X_{c,-}^s &\longrightarrow Z_{c,-}^s, \\ \mathcal{C}_{+, \rho}^{\text{cap}} : \mathfrak{X}_{c,+,H}^s &\xrightarrow{\sim} \mathfrak{Y}_{c,+,H}^s \oplus \mathfrak{Z}_{D,H}^s, & \Gamma_{c,+,N,\mathbf{b}} : X_{c,+}^s &\longrightarrow Z_N^s. \end{aligned}$$

The two reference operators on the cap use different data. In the minus block, the interior source and the axis and affine conditions determine the outgoing Cauchy data, so the value on the interface cannot be prescribed independently. In the positive block, we prescribe the value on the interface and recover the outward conormal derivative. The source spaces reflect this distinction.

We let $\mathcal{C}_{\mathbf{b}}^c$ be the Cartesian operator on the cap at the state. The map $\iota_{c,\alpha}^X$ includes a cap field from block α , and $\pi_{c,\beta}^Y$ selects the cap equations of block β . Here, $\alpha, \beta \in \{-, +\}$; the first sign in $E_{\beta\alpha}^c$ denotes the output block and the second the input block. We define the *off-diagonal operators on the cap* by

$$(10.8) \quad \begin{aligned} E_{-, \mathbf{b}}^c &= \pi_{c,-}^Y P_{-, \mathbf{b}}^R \mathcal{C}_{\mathbf{b}}^c P_{+, \mathbf{b}}^D \iota_{c,+}^X : X_{c,+}^s \longrightarrow Y_{c,-}^s, \\ E_{+, \mathbf{b}}^c &= \pi_{c,+}^Y P_{+, \mathbf{b}}^R \mathcal{C}_{\mathbf{b}}^c P_{-, \mathbf{b}}^D \iota_{c,-}^X : X_{c,-}^s \longrightarrow Y_{c,+}^s \oplus Z_D^s. \end{aligned}$$

On each dyadic shell, with the cap rescaled to a fixed size as in Table 1, the spaces of the minus block are $X_{c,-}^s = H^{s+2}$ and $Y_{c,-}^s = H^s$, and the spaces of the positive block are $X_{c,+}^s = H^{s+1}$ and $Y_{c,+}^s = H^{s-1}$, with the Cartesian compatibility conditions imposed. By (6.24), the off-diagonal terms of second order cancel. Hence, $E_{-, \mathbf{b}}^c$ has order one and maps H^{s+1} to H^s , while $E_{+, \mathbf{b}}^c$ maps H^{s+2} to H^{s-1} . The estimates for the inverse frame, the contour integrals (6.21), and the coefficients show that both families in (10.8) depend analytically on the state in these operator norms. They vanish at the circle, by the argument given for (10.32) below.

The traces in (10.7) are bounded. In the minus block, we compose the trace with $T_{-, \mathbf{b}} : Z_{c,-}^s \rightarrow Z_{a,-}^s$ from (8.19), without using its inverse. In the positive block, the extension of the value and its outward conormal derivative are part of the operator (9.23). Each composition remains in its own diagonal block, and so do the identity terms in the transmission conditions for the value and for the conormal derivative.

10.3. Coupling at the interface and outer boundary. We now include the conditions on the interface and on the outer boundary. The corresponding off-diagonal maps act between the spaces

$$(10.9) \quad \begin{aligned} E_{-, \mathbf{b}} &: X_{c,+}^s \oplus X_{a,+}^s \longrightarrow Y_{c,-}^s \oplus Y_{a,-}^s \oplus Z_{a,-}^s, \\ E_{+, \mathbf{b}} &: X_{c,-}^s \oplus X_{a,-}^s \longrightarrow Y_{c,+}^s \oplus Y_{a,+}^s \oplus Z_D^s \oplus Z_N^s \oplus Z_B^s. \end{aligned}$$

We first collect the operators on the interface and on the outer boundary, before the spectral projections (6.21) are applied. For a pair (u_c, u_a) of a field on the cap and a field on the annulus, we put

$$(10.10) \quad \mathcal{R}_{\text{ret}, \mathbf{b}}(u_c, u_a) = \begin{pmatrix} \gamma_{-, \mathbf{b}} u_a - T_{-, \mathbf{b}} \Gamma_{c, -, \mathbf{b}} u_c \\ \gamma_j^D u_a - \gamma_j^D u_c \\ \nabla_{a, \mathbf{b}}^{\text{out}} u_a + \nabla_{c, \mathbf{b}}^{\text{out}} u_c \\ \mathcal{B}_{+, \mathbf{b}} u_a \end{pmatrix},$$

whose rows are the mismatch of the incoming Cauchy data of the minus block, the jump of the values, the sum of the outward conormal derivatives, and the condition on the outer boundary. Here, $\gamma_{-, \mathbf{b}}$ is the Cauchy trace of the minus block on the inner boundary of the annulus at the state, and $\gamma_{-, \rho}$ in (8.18) is the corresponding trace at the reference. If ι_α^X denotes the inclusion of the pair $u_\alpha = (u_{c, \alpha}, u_{a, \alpha})$ of a field on the cap and a field on the annulus into the space of separate fields, the four off-diagonal terms are

$$(10.11) \quad \begin{aligned} E_{-, Z, \mathbf{b}} u_+ &= \pi_{Z_{a, -}}^Y P_{-, \mathbf{b}}^R \mathcal{R}_{\text{ret}, \mathbf{b}} P_{+, \mathbf{b}}^D \iota_+^X u_+, \\ (E_{+, D, \mathbf{b}}, E_{+, N, \mathbf{b}}, E_{+, B, \mathbf{b}}) u_- &= (\pi_D^Y, \pi_N^Y, \pi_B^Y) P_{+, \mathbf{b}}^R \mathcal{R}_{\text{ret}, \mathbf{b}} P_{-, \mathbf{b}}^D \iota_-^X u_-. \end{aligned}$$

Here, $\pi_{Z_{a, -}}^Y$ selects the first row, the mismatch of the incoming Cauchy data in $Z_{a, -}^s$. The projections π_D^Y , π_N^Y , and π_B^Y select the remaining three rows, in Z_D^s , Z_N^s , and Z_B^s , respectively. These are all the noninterior terms in (10.9).

The value and conormal spaces satisfy

$$(10.12) \quad \|g_D\|_{Z_D^s} \simeq \|\lambda^{1/2} g_D\|_{H^s}, \quad \|g_N\|_{Z_N^s} + \|g_B\|_{Z_B^s} \simeq \|\lambda^{-1/2} (g_N, g_B)\|_{H^s},$$

where λ acts on the coefficient at cell frequency n as multiplication by λ_n , so an operator of order one from the value to the conormal derivative has the weighted ratio

$$(10.13) \quad \frac{\lambda^{-1/2} \lambda |g_D|}{\lambda^{1/2} |g_D|} = 1.$$

We write the conormal derivative at the state as

$$(10.14) \quad \nabla_{\mathbf{b}} u = E_{\mathbf{b}} D_r u + F_{\mathbf{b}} u.$$

Here, D_r is the radial derivative, $E_{\mathbf{b}}$ is its coefficient, and $F_{\mathbf{b}}$ contains the tangential and zeroth-order terms. Thus, $F_{\mathbf{b}}$ acts on the boundary value of u . After the spectral projections (6.21) are applied, the off-diagonal term of highest order cancels by (6.24), and (10.13) bounds the remaining terms of first order, while the derivatives of $E_{\mathbf{b}}$, of the frames on the traces, and of the projections are multipliers of order zero. The term from the center to the positive component depends only on the value and can be written as

$$(10.15) \quad (\nabla_{\mathbf{b}} z_+)_{C \rightarrow +} = \mathcal{V}_{+, C, \mathbf{b}} z_C,$$

with no normal derivative of z_C . Here, z_+ and z_C are the positive component and the center component of the unknown, the left side denotes the part of the conormal operator of the positive block which acts on the center component, and $\mathcal{V}_{+, C, \mathbf{b}}$ is a coefficient depending analytically on the state. The ratio of the weights of its components is at most λ^{-1} , and the analytic coefficient $\mathcal{V}_{+, C, \mathbf{b}}$ vanishes at the circle. We include this term in the component Z_N^s of $E_{+, \mathbf{b}}$.

Writing $E_{\mathbf{b}, \gamma} = W_\gamma E_{\mathbf{b}} W_\gamma^{-1}$, conjugation by the analytic weight adds the radial term $E_{\mathbf{b}, \gamma} d_\gamma(r, \Lambda) g_D$, where

$$(10.16) \quad d_\gamma(r, \lambda) = \frac{\gamma r \lambda^2}{\sqrt{1 + r^2 \lambda^2}}, \quad |d_\gamma| \leq \gamma \lambda,$$

as in (9.51). The sign is carried by $E_{\mathbf{b},\gamma}$, which uses the outward normal on the side under consideration. The multiplier d_γ is defined in (3.21). Its bound by $\gamma\lambda$ and the ratio (10.13) control the cell-frequency weights of the added term. The coefficient estimates control $E_{\mathbf{b},\gamma}$. At the interface, the two outward normals have opposite orientations. For matching values and coefficients, the added terms therefore cancel in Green's identity.

For the condition on the outer boundary, the off-diagonal term of (10.11) is

$$(10.17) \quad E_{+-,B,\mathbf{b}} := \pi_B^Y E_{+,-,\mathbf{b}} : \mathbb{X}_{-,\mathfrak{q}}^s \longrightarrow Z_B^s, \quad E_{-+,B,\mathbf{b}} = 0,$$

since the boundary term from the positive block to itself already belongs to the diagonal block $D_{+,\mathbf{b}}$ defined in (10.44) below, while the minus system has no independent condition on the outer boundary. The bounds in the weighted trace norms follow from (8.18)–(8.31) and (9.31). The localized Poisson extension (4.8), whose decay rate depends on the frequency, and the exponential decay (4.10), whose rate $(1-\gamma)\lambda_{mn}$ is positive for $\gamma < 1$, then give all moments for (10.17). We first estimate the trace and then apply the frequency-dependent extension, since an extension with fixed width would not give the required weighted bounds. These bounds use only $\widehat{G}_{-,\mathbf{b}}$ and $\widehat{G}_{+,\mathbf{b}}$, so they are available before we correct the diagonal inverses.

10.4. Fourier estimates across the cutoff. We use $P^X = P_J^X$ and $P^Y = P_J^Y$ for the projections onto $|n| < 2^J$, and $H^X = I - P^X$, $H^Y = I - P^Y$ for their complements. The three blocks of (6.94) estimated here are

$$A_{\mathbf{b}} = P^Y \mathbb{D}_{\mathbf{b}} P^X, \quad B_{\mathbf{b}} = P^Y \mathbb{D}_{\mathbf{b}} H^X, \quad C_{\mathbf{b}} = H^Y \mathbb{D}_{\mathbf{b}} P^X.$$

Thus, $A_{\mathbf{b}}$ acts on the low frequencies, $B_{\mathbf{b}}$ takes high-frequency unknowns to low-frequency equations, and $C_{\mathbf{b}}$ acts in the reverse direction. We write $A_e = A_{\mathbf{b}_e}$ for the low block at the constant reference state. For a large enough shift k_0 in the regularity of the coefficients, we put

$$(10.18) \quad d_0(\mathbf{b}) := \|\mathbf{b} - \mathbf{b}_e\|_{\mathcal{B}^{s_*+k_0}},$$

which measures the distance to the constant reference state at the base index. This distance controls both the change in the low block and its coupling to the high frequencies.

Lemma 10.1. *For every J , there are finite constants $C_{L,J}$ and $C_{X,J}$ and a finite shift $k \leq k_0$ such that*

$$(10.19) \quad \|A_{\mathbf{b}} - A_e\|_{P^X \mathbb{X}_{\mathfrak{q}}^{s_*} \rightarrow P^Y \mathbb{Y}_{\mathfrak{q}}^{s_*}} \leq C_{L,J} d_0(\mathbf{b}),$$

$$(10.20) \quad \|B_{\mathbf{b}}\|_{H^X \mathbb{X}_{\mathfrak{q}}^{s_*} \rightarrow P^Y \mathbb{Y}_{\mathfrak{q}}^{s_*}} + \|C_{\mathbf{b}}\|_{P^X \mathbb{X}_{\mathfrak{q}}^{s_*} \rightarrow H^Y \mathbb{Y}_{\mathfrak{q}}^{s_*}} \leq C_{X,J} d_0(\mathbf{b}).$$

For every $s \geq s_$, every u_L in the range of P^X , and every u_H in the range of H^X ,*

$$(10.21) \quad \|(A_{\mathbf{b}} - A_e)u_L\|_{\mathbb{Y}_{\mathfrak{q}}^s} \leq C_{J,s} [d_0(\mathbf{b}) \|u_L\|_{\mathbb{X}_{\mathfrak{q}}^s} + \|\mathbf{b} - \mathbf{b}_e\|_{\mathcal{B}^{s+k}} \|u_L\|_{\mathbb{X}_{\mathfrak{q}}^{s_*}}],$$

$$(10.22) \quad \|B_{\mathbf{b}}u_H\|_{\mathbb{Y}_{\mathfrak{q}}^s} \leq C_{J,s} [d_0(\mathbf{b}) \|u_H\|_{\mathbb{X}_{\mathfrak{q}}^s} + \|\mathbf{b} - \mathbf{b}_e\|_{\mathcal{B}^{s+k}} \|u_H\|_{\mathbb{X}_{\mathfrak{q}}^{s_*}}],$$

$$(10.23) \quad \|C_{\mathbf{b}}u_L\|_{\mathbb{Y}_{\mathfrak{q}}^s} \leq C_{J,s} [d_0(\mathbf{b}) \|u_L\|_{\mathbb{X}_{\mathfrak{q}}^s} + \|\mathbf{b} - \mathbf{b}_e\|_{\mathcal{B}^{s+k}} \|u_L\|_{\mathbb{X}_{\mathfrak{q}}^{s_*}}].$$

These estimates hold for all interior, cap, quotient, axis, interface, conormal, and outer boundary equations, and every derivative with respect to $\mathbf{b}$ of fixed order obeys the corresponding tame estimate.

The proof uses the bounds at low frequency for fixed J , the product estimate (3.20), and Fourier estimates across the cutoff. In particular, it does not depend on the high-frequency inverse or on the global inverse, so we may use (10.20) before those inverses are constructed.

Proof. Since P restricts the cell frequencies to a finite set, every cell derivative is bounded on its range, with a constant depending on J . We estimate the radial derivatives in the norms of the center and affine components of Section 8, and we measure the positive equation in the form space and its dual, with the value and conormal traces measured with the weights $\lambda^{1/2}$ and $\lambda^{-1/2}$ of Table 1. On the cap, the minus block acts from H^{s+2} to H^s and the positive block acts from H^{s+1} to H^{s-1} , as in Table 1. The quotient, mean, scale, gauge, chart, and trace operators all have fixed order. Since $\mathbb{D}_{\mathbf{b}}$ already includes the extraction of the scalar differential source (6.28), its cancellations of highest order are measured before P is applied, and the product estimate (3.20) on this finite list of operators proves (10.19) and (10.21).

At the constant reference state, the coefficients and reductions do not depend on the cell variable, so that

$$(10.24) \quad \mathbb{D}_{\mathbf{b}_e} P^X = P^Y \mathbb{D}_{\mathbf{b}_e}, \quad B_{\mathbf{b}_e} = 0, \quad C_{\mathbf{b}_e} = 0,$$

and we need only estimate the coupling between low and high frequencies in $\mathbb{D}_{\mathbf{b}} - \mathbb{D}_{\mathbf{b}_e}$. Consider an operator $c(\zeta)D_\zeta^q$, which sends the input at cell frequency m to the output at cell frequency n with the coefficient $c_{n-m}m^q$, where c_k are the Fourier coefficients of c . If $|n| < 2^J \leq |m|$, then either $|m| < 2^{J+1}$, or $|n - m| \geq |m| - |n| \geq |m|/2$, and in both cases

$$(10.25) \quad \langle m \rangle^q \leq C_{J,q} \langle n - m \rangle^q.$$

The factor $\langle n - m \rangle^q$ is absorbed by the Fourier coefficients of the derivatives of c , so the derivatives of the coefficient control the cell derivatives on the input. For the coupling in the opposite direction, $|m| < 2^J$, and the derivative on the input is bounded by $C_{J,q}$. Consider a map from a component α to a component β with the weights $w_\alpha = d_\alpha^X$ and $w_\beta = d_\beta^Y$ of Table 1, and let $q_{\beta\alpha}$ be the number of derivatives in ζ applied to the input. The inequality (6.4) then gives

$$(10.26) \quad \mathbf{1}_{\{|n| < 2^J\}} \mathbf{1}_{\{|m| \geq 2^J\}} \frac{w_\beta(n)}{w_\alpha(m)} \langle m \rangle^{q_{\beta\alpha}} \leq C_J \langle n - m \rangle^{k_{\beta\alpha}},$$

with the same estimate in the reverse direction. Here, $k_{\beta\alpha}$ is a finite exponent depending only on the two components, obtained from the exponents of (6.4) and from $q_{\beta\alpha}$. For example, the first row of Table 4 has $w_\alpha = 1$, $w_\beta(n) = \lambda_n^{-1}$, and $q_{\beta\alpha} = 1$, so its ratio across the cutoff is $\lambda_n^{-1}\lambda_m$, controlled by a power of $\langle n - m \rangle$. Radial derivatives are estimated separately in the norms of the center, of the affine component, of the positive form, and of the cap. Multiplication by a coefficient of the state is a convolution in the cell frequency, and the weight $e^{\Phi_r(n)}$ is submultiplicative by (3.18). Hence, the weighted kernel (6.6) of each coupling term is bounded by the weighted Fourier coefficients of the coefficient, times the polynomial factor of (10.26). Summing the rows and the columns as in Schur's test then proves (10.20), (10.22), and (10.23) for each component. Since P and H are fixed, the tame product estimate also applies after differentiation in the parameters. $\square$

Once A_e is inverted, (10.19) also gives the invertibility of the low block near the reference state. Indeed, if

$$(10.27) \quad \|A_e^{-1}(A_{\mathbf{b}} - A_e)\| < 1,$$

then $A_{\mathbf{b}} = A_e(I + A_e^{-1}(A_{\mathbf{b}} - A_e))$ is invertible by a Neumann series.

10.5. The off-diagonal estimate. We use the constant ellipse $\mathbf{b}_\rho$ as the reference for the coefficient differences. We let $a_o(\rho) = a_{\rho, p_e}^0$ and $M_o(\rho) = M_{\rho, p_e}$ denote its unperturbed solution and its matrix, whose axes do not rotate with ζ , and we let $M(p) = M_{\rho, p}$ denote the periodic matrix (2.33) with the parameters p of the state. For a finite shift $k_\times$ in the regularity of the coefficients, fixed with the estimate (10.30) below, we put

$$(10.28) \quad \eta_{\text{circ}}(\mathbf{b}) := \|a - a_o(\rho)\|_{\mathcal{A}_\gamma^{s_*+k_\times}} + |\varepsilon| + \|M(p) - M_o(\rho)\|_{\mathcal{A}_\gamma^{s_*+k_\times}}, \quad \chi(\mathbf{b}) = |\rho| + \eta_{\text{circ}}(\mathbf{b}).$$

Thus, $\chi(\mathbf{b})$ measures the distance from the state to the circle, and the off-diagonal blocks will be bounded by this distance.

In the fixed coordinates, we define the off-diagonal maps at high frequency by

$$E_{\beta\alpha, \mathbf{b}} = P_{\beta, \mathbf{b}}^R H^Y \mathbb{L}_{\mathbf{b}} H^X P_{\alpha, \mathbf{b}}^D, \quad \alpha \neq \beta,$$

where H^X and H^Y denote H_J on the domain and on the range, with the cap coordinates of the two high-frequency blocks and the finitely many coordinates of the triangular system included. We set $\mathbb{A}_{\mathbf{b}} := \iota_{Y, \mathbf{b}} \mathbf{J}_{Y, \mathbf{b}}^{\text{raw}} L_{\mathbf{b}} R_{X, \mathbf{b}}$, we let q_β^Z denote the projection onto the auxiliary data of the range of the component β , and we write $j_{Z, \beta, \mathbf{b}} := \iota_\beta^Y \pi_\beta^Y j_{Z, \mathbf{b}}$ on that component. With this notation, (6.86) gives

$$(10.29) \quad E_{\beta\alpha, \mathbf{b}} = \pi_\beta^Y \mathbb{A}_{\mathbf{b}} \iota_\alpha^X + j_{Z, \beta, \mathbf{b}} q_\beta^Z C_{X, \mathbf{b}} \iota_\alpha^X.$$

The subtraction in (6.84) removes $\mathbb{B}_{\mathbf{b}} C_{X, \mathbf{b}}$, the interior response to the constraint data in (6.83). The second term of (10.29) is the projection of $E_{\beta\alpha, \mathbf{b}}$ onto the auxiliary data of the component β , so the estimate (10.30) below bounds it without using a diagonal inverse.

Proposition 10.2 (The off-diagonal estimate). *Fix $0 \leq \rho \leq \rho_0 < 1/4$, the strength γ of the analytic weight, the parameter δ_A of (6.3), and the cutoff index J . On a sufficiently small ball of states, the off-diagonal maps are bounded from index s to index s , and for every $s \geq s_*$ they satisfy*

$$(10.30) \quad \begin{aligned} & \|E_{-+, \mathbf{b}} u_+ \|_{\mathbb{Y}_{-, \mathfrak{h}}^s} + \|E_{+-, \mathbf{b}} u_- \|_{\mathbb{Y}_{+, \mathfrak{h}}^s} \\ & \leq C_s \left[\chi(\mathbf{b}) (\|u_+ \|_{\mathbb{X}_{+, \mathfrak{h}}^s} + \|u_- \|_{\mathbb{X}_{-, \mathfrak{h}}^s}) \right. \\ & \quad \left. + \|\mathbf{b} - \mathbf{b}_\rho\|_{\mathcal{B}^{s+k_\times}} (\|u_+ \|_{\mathbb{X}_{+, \mathfrak{h}}^{s_*}} + \|u_- \|_{\mathbb{X}_{-, \mathfrak{h}}^{s_*}}) \right]. \end{aligned}$$

For each number m of derivatives with respect to $\mathbf{b}$, there is a finite shift k_m , independent of s , such that

$$(10.31) \quad \begin{aligned} & \|D_{\mathbf{b}}^m E_{-+, \mathbf{b}} [\mathbf{h}] u_+ \|_{\mathbb{Y}_{-, \mathfrak{h}}^s} + \|D_{\mathbf{b}}^m E_{+-, \mathbf{b}} [\mathbf{h}] u_- \|_{\mathbb{Y}_{+, \mathfrak{h}}^s} \\ & \leq C_{s, m} \left\{ \mathcal{H}_{0, m}(\mathbf{h}) (\|u_+ \|_{\mathbb{X}_{+, \mathfrak{h}}^s} + \|u_- \|_{\mathbb{X}_{-, \mathfrak{h}}^s}) \right. \\ & \quad \left. + \mathcal{H}_{s, m}(\mathbf{h}) (\|u_+ \|_{\mathbb{X}_{+, \mathfrak{h}}^{s_*}} + \|u_- \|_{\mathbb{X}_{-, \mathfrak{h}}^{s_*}}) \right. \\ & \quad \left. + \|\mathbf{b} - \mathbf{b}_\rho\|_{\mathcal{B}^{s+k_m}} \mathcal{H}_{0, m}(\mathbf{h}) (\|u_+ \|_{\mathbb{X}_{+, \mathfrak{h}}^{s_*}} + \|u_- \|_{\mathbb{X}_{-, \mathfrak{h}}^{s_*}}) \right\}. \end{aligned}$$

The constants are uniform for $0 \leq \rho \leq \rho_0$, for both signs of the frequency, and for every upper Fourier cutoff, including the case without a cutoff. The operators commute with complex conjugation, and

$$(10.32) \quad E_{-+, \mathbf{b}_0} = E_{+-, \mathbf{b}_0} = 0.$$

Proof. We first cancel the off-diagonal part of highest order by (6.25)–(6.26), and only then extract the differential source, so that the resulting scalar equation has order two by (6.32) and (6.35). We estimate the remaining interior terms using Table 4, (10.4)–(10.5), and Table 5. For the terms on the cap, on the interface,

and on the boundary, we use (10.8) and (10.9)–(10.17). Combining these bounds with Lemma 6.1, we obtain the tame estimate from index s to index s .

It remains to include the cutoffs and the derivatives with respect to $\mathbf{b}$. On the interior components, we write $\mathbb{L}_{\mathbf{b}} = Q_{\mathbf{b}} + \mathbb{R}_{\mathbf{b}}$, where $Q_{\mathbf{b}}$ now denotes the operator with the principal symbol (6.20), and where $\mathbb{R}_{\mathbf{b}}$ collects the terms of lower order, which come from the derivatives of the frame (6.15) and from the passage from symbols to operators. We set

$$(10.33) \quad C_{\beta, \mathbf{b}}^R = [P_{\beta, \mathbf{b}}^R, H^Y], \quad C_{\alpha, \mathbf{b}}^D = [H^X, P_{\alpha, \mathbf{b}}^D].$$

Since $P_{\beta}^R H^Y = H^Y P_{\beta}^R + C_{\beta}^R$ and $H^X P_{\alpha}^D = P_{\alpha}^D H^X + C_{\alpha}^D$, the off-diagonal term of highest order decomposes as

$$(10.34) \quad P_{\beta}^R H^Y Q_{\mathbf{b}} H^X P_{\alpha}^D = H^Y P_{\beta}^R Q_{\mathbf{b}} P_{\alpha}^D H^X + H^Y P_{\beta}^R Q_{\mathbf{b}} C_{\alpha}^D + C_{\beta}^R Q_{\mathbf{b}} P_{\alpha}^D H^X + C_{\beta}^R Q_{\mathbf{b}} C_{\alpha}^D.$$

The principal symbol of the first term vanishes by (6.26). The operator itself can still contain lower-order terms, produced when derivatives fall on the projections. Each of the other three terms contains a commutator between a cutoff and a projection. We include all these terms, together with $P_{\beta}^R H^Y \mathbb{R}_{\mathbf{b}} H^X P_{\alpha}^D$, in $\mathfrak{R}_{\beta\alpha, \mathbf{b}}$. Thus,

$$(10.35) \quad P_{\beta}^R H^Y \mathbb{L}_{\mathbf{b}} H^X P_{\alpha}^D = \mathfrak{R}_{\beta\alpha, \mathbf{b}},$$

and Lemma 6.1, applied to the finite moments of the cutoffs, of the projections, and of the coefficients, gives, for every m ,

$$(10.36) \quad \begin{aligned} & \|D_{\mathbf{b}}^m \mathfrak{R}_{\beta\alpha, \mathbf{b}}[\mathbf{h}] u_{\alpha}\|_{\mathbb{Y}_{\beta, \mathfrak{h}}^s} \\ & \leq C_{s, m} \left\{ \mathcal{H}_{0, m}(\mathbf{h}) \|u_{\alpha}\|_{\mathbb{X}_{\alpha, \mathfrak{h}}^s} + \mathcal{H}_{s, m}(\mathbf{h}) \|u_{\alpha}\|_{\mathbb{X}_{\alpha, \mathfrak{h}}^{s*}} + \|\mathbf{b} - \mathbf{b}_{\rho}\|_{\mathcal{B}^{s+k_m}} \mathcal{H}_{0, m}(\mathbf{h}) \|u_{\alpha}\|_{\mathbb{X}_{\alpha, \mathfrak{h}}^{s*}} \right\}. \end{aligned}$$

For the cutoffs H^X and H^Y in (10.33), which are indicator functions of sets of cell frequencies, the commutator is supported where the input and output cell frequencies lie on opposite sides of the cutoff. The estimates (10.25)–(10.26) then control derivatives in ζ on the input by derivatives of the coefficient, with constants depending on the cutoff J . This control differs from the gain of one order for the smooth dyadic cutoffs in (6.10). Since the cutoffs are fixed, derivatives with respect to $\mathbf{b}$ are again commutators of the same form, and (10.34)–(10.36) apply after differentiation.

The cancellation of highest order for $\alpha \neq \beta$ holds on the whole ball of states. Differentiating the symbol identity, we obtain

$$(10.37) \quad D_{\mathbf{b}}^m (P_{\beta, \mathbf{b}}^R Q_{\mathbf{b}} P_{\alpha, \mathbf{b}}^D)[\mathbf{h}] = 0 \quad (m \geq 0).$$

This is an identity for symbols. It applies to the sum of the terms in Leibniz' rule, so we keep them together and then estimate the lower-order terms arising in the corresponding operators. Likewise, the symbol $s_{G_{\mathbf{b}}}$ of the differential source component in (6.34) satisfies the quadratic identity (6.35), so

$$(10.38) \quad D_{\mathbf{b}}^m [s_{G_{\mathbf{b}}} \ell_{1, G_{\mathbf{b}}}] [\mathbf{h}] = D_{\mathbf{b}}^m (\xi^* G_{\mathbf{b}}^{-1} \xi) [\mathbf{h}] e_C^*.$$

The differentiated terms have the same orders as those in Table 4, including the terms on the cap, on the traces, and on the outer boundary.

It remains to estimate the compositions of these maps. For two composable families $S_{\mathbf{b}}$ and $T_{\mathbf{b}}$ satisfying (10.31), Leibniz' rule gives

$$(10.39) \quad D_{\mathbf{b}}^m(T_{\mathbf{b}}S_{\mathbf{b}})[\mathbf{h}] = \sum_{I \subset \{1, \dots, m\}} D_{\mathbf{b}}^{|I|} T_{\mathbf{b}}[\mathbf{h}_I] D_{\mathbf{b}}^{m-|I|} S_{\mathbf{b}}[\mathbf{h}_{I^c}].$$

We use the estimate at index s for one factor and the estimate at the base index s_* for the others. The products of lower regularity tangent norms give $\mathcal{H}_{0,m}$. Measuring one tangent direction at a higher regularity index and summing over that choice gives $\mathcal{H}_{s,m}$. Applying this to the finite list of factors in the interior, on the cap, and on the boundary, we obtain (10.31), with at most one unknown, coefficient, or tangent direction measured in a higher regularity norm.

At the constant circle $\mathbf{b}_0$, the coefficients are independent of ζ . The operator preserves each angular mode and parity, the four-dimensional center space, and the complementary positive subspace. Thus, the off-diagonal maps vanish there, giving (10.32). Since the off-diagonal maps depend analytically on the state and vanish at $\mathbf{b}_0$, the mean value formula bounds them by the distance $\chi(\mathbf{b})$ to the circle times the bounds for their derivatives, which proves (10.30). The constants remain uniform when the center speeds coincide. Indeed, we use one contour around the four center roots, which is separated from the affine and positive roots, and we do not select individual eigenvectors. The gap of the positive form of Section 9 also stays positive at $\rho = 0$. We treat the orientation α_0 of the ellipse as a parameter of the unperturbed solutions, so every constant is uniform through $\rho = 0$.

Finally, we let $\mathbf{c}_{X,\alpha}$ and $\mathbf{c}_{Y,\beta}$ denote complex conjugation on the complexified spaces. The Cartesian operators, weight (6.5), cap operators, Poisson extensions, and traces preserve real data. The cutoff H_J is invariant under $n \mapsto -n$, the contours in (6.21) are invariant under conjugation, and the exceptional equations at $m = \pm 2$ form a conjugate pair. The transfer maps of Proposition 6.2, which commute with complex conjugation, therefore give, for real $\mathbf{b}$,

$$(10.40) \quad E_{\beta\alpha, \mathbf{b}} \mathbf{c}_{X,\alpha} = \mathbf{c}_{Y,\beta} E_{\beta\alpha, \mathbf{b}}.$$

Finite sums, the products in (10.39), and differentiation in real tangent directions preserve (10.40), which completes the proof. $\square$

10.6. Correcting the diagonal inverses. We now compare the separately constructed operators $\hat{D}_{\alpha, \mathbf{b}}$ with the diagonal blocks $D_{\alpha, \mathbf{b}}$ of the full augmented operator. Their difference is the correction $K_{\alpha, \mathbf{b}}$ defined below. For $\alpha \in \{-, +\}$, we define the projections of the auxiliary variables onto the auxiliary data of the component α and onto the auxiliary data at low frequency by

$$(10.41) \quad Q_{\alpha}^Z := \text{aux}_{Y, \mathbf{b}} \iota_{\alpha}^Y \pi_{\alpha}^Y J_{Z, \mathbf{b}}, \quad Q_P^Z := \text{aux}_{Y, \mathbf{b}} P^Y J_{Z, \mathbf{b}}, \quad I_Z = Q_P^Z + Q_{-}^Z + Q_{+}^Z.$$

Here, I_Z is the identity on the auxiliary variables, and the last identity expresses that the low frequencies and the two high-frequency blocks exhaust the range. On $\text{Ran } Q_{\alpha}^Z$, the map $J_{Z, \alpha, \mathbf{b}} = \iota_{\alpha}^Y \pi_{\alpha}^Y J_{Z, \mathbf{b}}$ of (10.29) is the inclusion into the auxiliary variables of this component, so that on the full auxiliary space

$$(10.42) \quad \iota_{\alpha}^Y \pi_{\alpha}^Y J_{Z, \mathbf{b}} = J_{Z, \alpha, \mathbf{b}} Q_{\alpha}^Z.$$

We write α^c for the complement of α , which includes both the other high-frequency block and the auxiliary variables at low frequency. We set

$$(10.43) \quad \begin{aligned} C_{\alpha^c, \mathbf{b}} &:= (I_Z - Q_\alpha^Z) C_{X, \mathbf{b}} \iota_\alpha^X, \\ B_{\alpha^c, \mathbf{b}} &:= \pi_\alpha^Y \mathbb{B}_{\mathbf{b}} (I_Z - Q_\alpha^Z), \\ K_{\alpha, \mathbf{b}} &:= B_{\alpha^c, \mathbf{b}} C_{\alpha^c, \mathbf{b}} : \mathbb{X}_{\alpha, \mathbf{b}}^s \longrightarrow \mathbb{Y}_{\alpha, \mathbf{b}}^s. \end{aligned}$$

Here, $C_{\alpha^c, \mathbf{b}}$ extracts the constraint data outside block α , and $B_{\alpha^c, \mathbf{b}}$ returns their extension's interior source to block α . Their composition is the correction $K_{\alpha, \mathbf{b}}$. The map $\mathbb{B}_{\mathbf{b}}$ is defined by (6.82); the frame matrix (6.12) is denoted by $\mathbb{B}$ without a subscript.

The following lemma shows that $D_{\alpha, \mathbf{b}} = \widehat{D}_{\alpha, \mathbf{b}} - K_{\alpha, \mathbf{b}}$. We estimate $K_{\alpha, \mathbf{b}}$ in the base norm and use its smallness to correct the inverse of $\widehat{D}_{\alpha, \mathbf{b}}$.

Lemma 10.3. *For $\alpha \in \{-, +\}$, the diagonal block $D_{\alpha, \mathbf{b}}$ of the augmented operator and the separately constructed operator $\widehat{D}_{\alpha, \mathbf{b}}$ satisfy the identity*

$$(10.44) \quad D_{\alpha, \mathbf{b}} := \pi_\alpha^Y \mathbb{D}_{\mathbf{b}} \iota_\alpha^X = \widehat{D}_{\alpha, \mathbf{b}} - K_{\alpha, \mathbf{b}}.$$

At the base index,

$$(10.45) \quad \|K_{\alpha, \mathbf{b}}\| \leq M_B (K_Z \chi(\mathbf{b}) + K_{Z, J} d_0(\mathbf{b})), \quad d_0(\mathbf{b}) := \|\mathbf{b} - \mathbf{b}_e\|_{\mathcal{B}^{s_* + k_0}},$$

where M_B bounds $B_{\alpha^c, \mathbf{b}}$, K_Z bounds the part of $C_{\alpha^c, \mathbf{b}}$ in the other high-frequency block, by (10.30), and $K_{Z, J}$ bounds its part at low frequency, by the argument of (10.20). If

$$(10.46) \quad q_{\text{corr}} := \max_{\alpha \in \{-, +\}} \|\widehat{G}_{\alpha, \mathbf{b}} K_{\alpha, \mathbf{b}}\| < 1,$$

then the diagonal block has the two-sided inverse

$$(10.47) \quad G_{\alpha, \mathbf{b}} = (I - \widehat{G}_{\alpha, \mathbf{b}} K_{\alpha, \mathbf{b}})^{-1} \widehat{G}_{\alpha, \mathbf{b}} = \widehat{G}_{\alpha, \mathbf{b}} (I - K_{\alpha, \mathbf{b}} \widehat{G}_{\alpha, \mathbf{b}})^{-1}.$$

These inverses, together with the traces of the solutions which appear in (8.31) and (9.31), obey the same bounds as $\widehat{G}_{\alpha, \mathbf{b}}$, without loss of derivatives, and the corresponding tame bounds for derivatives with respect to $\mathbf{b}$ of every order.

Proof. We write $C_\alpha := C_{X, \mathbf{b}} \iota_\alpha^X$ and $\mathbb{A}_{\alpha\alpha} := \pi_\alpha^Y \mathbb{A}_{\mathbf{b}} \iota_\alpha^X$. We take the α diagonal block of (6.83) and apply the diagonal block $\mathcal{S}_{\alpha\alpha}$ of the map in (6.89), which subtracts $\pi_\alpha^Y \mathbb{B}_{\mathbf{b}} Q_\alpha^Z C_\alpha$, the response of the interior to the auxiliary data of the component α alone. Using (10.42), we obtain

$$\widehat{D}_{\alpha, \mathbf{b}} = \mathbb{A}_{\alpha\alpha} + \pi_\alpha^Y \mathbb{B}_{\mathbf{b}} (I_Z - Q_\alpha^Z) C_\alpha + J_{Z, \alpha, \mathbf{b}} Q_\alpha^Z C_\alpha,$$

while (6.86) gives

$$\pi_\alpha^Y \mathbb{D}_{\mathbf{b}} \iota_\alpha^X = \mathbb{A}_{\alpha\alpha} + J_{Z, \alpha, \mathbf{b}} Q_\alpha^Z C_\alpha.$$

Their difference is $K_{\alpha, \mathbf{b}}$, which proves (10.44).

We estimate the two parts of the correction separately. Projecting (10.29) onto the auxiliary variables, we obtain the part of $C_{\alpha^c, \mathbf{b}}$ in the other high-frequency component, whose norm at the base index is bounded by $K_Z \chi(\mathbf{b})$, by (10.30). For the part at low frequency, (6.87) identifies $Q_P^Z C_{X, \mathbf{b}} \iota_\alpha^X$ with the auxiliary projection of $P^Y \mathbb{D}_{\mathbf{b}} H^X$, which vanishes at $\mathbf{b}_e$ because the reference is diagonal in the cell frequency. For fixed J , the Fourier estimate (10.26) then bounds this projection by $K_{Z, J} d_0(\mathbf{b})$, as in (10.20). Finally, (6.82) and Proposition 6.2 bound $B_{\alpha^c, \mathbf{b}}$, which proves (10.45).

Factoring

$$D_{\alpha, \mathbf{b}} = \widehat{D}_{\alpha, \mathbf{b}}(I - \widehat{G}_{\alpha, \mathbf{b}}K_{\alpha, \mathbf{b}}) = (I - K_{\alpha, \mathbf{b}}\widehat{G}_{\alpha, \mathbf{b}})\widehat{D}_{\alpha, \mathbf{b}}$$

proves (10.47) and both inverse identities. The tame bounds follow from the resolvent identity for the Neumann series and from (10.39), using the direct estimates for $B_{\alpha\alpha^c, \mathbf{b}}$ and $C_{\alpha^c\alpha, \mathbf{b}}$ at every order of differentiation. $\square$

Thus, the construction proceeds in the order

$$(10.48) \quad \widehat{G}_{\pm} \longrightarrow \{E_{\mp\pm}, Q_P^Z C_X \iota_{\pm}^X\} \longrightarrow G_{\pm} \longrightarrow G_H.$$

The estimates for the off-diagonal maps and low-frequency auxiliary data therefore precede the correction of the diagonal inverses. Here, G_H denotes the inverse of the full high-frequency block $D_{H, \mathbf{b}}$ in (6.94); it must also account for the couplings E_{-+} and E_{+-} . We construct it after the low-frequency reference inverse. The low block $A_{\mathbf{b}}$ requires no such correction.

Finally, the composition used in the Neumann series at high frequency is

$$(10.49) \quad \mathbb{X}_{-, \mathfrak{q}}^{s_*} \xrightarrow{E_{+-, \mathbf{b}}} \mathbb{Y}_{+, \mathfrak{q}}^{s_*} \xrightarrow{D_{+, \mathbf{b}}^{-1}} \mathbb{X}_{+, \mathfrak{q}}^{s_*} \xrightarrow{E_{-+, \mathbf{b}}} \mathbb{Y}_{-, \mathfrak{q}}^{s_*} \xrightarrow{D_{-, \mathbf{b}}^{-1}} \mathbb{X}_{-, \mathfrak{q}}^{s_*},$$

with each factor acting between the indicated spaces without loss of derivatives. Combining (10.30) with the bounds for the diagonals at the same index, and using (6.11), we obtain

$$(10.50) \quad \|D_{-, \mathbf{b}}^{-1} E_{-+, \mathbf{b}} D_{+, \mathbf{b}}^{-1} E_{+-, \mathbf{b}}\| \leq K_- K_+ K_{\times}^2 \chi(\mathbf{b})^2.$$

Here, $K_{\times}$ is the constant of (10.30) at the base index s_* , and $K_{\pm}$ bound the diagonal block inverses (10.47) under (10.46). This estimate requires only the operator norms of the diagonal inverses. No bounds for their symbols or moments are needed. For their derivatives with respect to $\mathbf{b}$, we use

$$D(D_{\alpha, \mathbf{b}}^{-1})[h] = -D_{\alpha, \mathbf{b}}^{-1}(DD_{\alpha, \mathbf{b}}[h])D_{\alpha, \mathbf{b}}^{-1}$$

and estimate the resulting compositions on the same block spaces.

11. THE LOW-FREQUENCY INVERSE AT A FIXED ELLIPSE

For the low-frequency construction in Section 12, we need an inverse without loss of derivatives in the block norms. We obtain this bound directly in the block variables, first at the circle and then at nearby constant ellipses. The Cartesian estimate of Section 7 loses derivatives of the source, so it does not give the bound needed here. The resulting block estimate also governs the choice of the reference ellipse parameter ρ_* .

We fix the base index s_* , the transfer maps of Section 6, and the identifications (6.67) of the trace spaces with fixed products. We use the projections onto the cell frequencies (4.2), as in (6.94),

$$H_J = \mathbf{1}_{\{|n| \geq 2^J\}}, \quad P_J = I - H_J = \mathbf{1}_{\{|n| < 2^J\}},$$

with the versions P_J^X and P_J^Y on the domain and on the range. We apply these projections in the block variables, that is, after the transfer maps (6.69) and (6.74) have been applied to the domain and to the source. We let $\mathbb{D}_{\rho, e} := \mathbb{D}_{\mathbf{b}_{\rho}}$ be the augmented operator (6.85) at the constant ellipse $\mathbf{b}_{\rho}$ with ellipse parameter ρ from Section 6, acting between the fixed block spaces, and we set

$$(11.1) \quad A_{\rho, J} = P_J^Y \mathbb{D}_{\rho, e} P_J^X,$$

which is the low block of [Section 10](#) at the constant ellipse. Here, $A_{\rho,J} = A_{\mathbf{b}_\rho}$ in the notation of [\(6.94\)](#), with the cutoff J made explicit, and the low block A_e of [Section 10](#) is $A_{\rho_*,J}$. At the constant ellipse, the identities [\(6.87\)–\(6.88\)](#) for the operator $\mathbb{D}_{\rho,e}$ and the identities [\(6.72\)](#) for the transfer maps read

$$\begin{aligned}
 \text{aux}_{Y,\rho,e} \mathbb{D}_{\rho,e} U &= C_{X,\rho,e} U, \\
 \mathbb{D}_{\rho,e} J_{X,\rho,e} u &= \iota_{Y,\rho,e} J_{Y,\rho,e}^{\text{raw}} L_{\rho,e} u, \\
 R_{X,\rho,e} J_{X,\rho,e} &= I, \quad C_{X,\rho,e} J_{X,\rho,e} = 0, \\
 J_{X,\rho,e} R_{X,\rho,e} U &= U \quad \text{when } C_{X,\rho,e} U = 0.
 \end{aligned}
 \tag{11.2}$$

Here, the subscript ρ, e means that the transfer maps of [Section 6](#) and the linearization are evaluated at $\mathbf{b} = \mathbf{b}_\rho$. Thus, $L_{\rho,e} = L_{\mathbf{b}_\rho}$ and $J_{X,\rho,e} = J_{X,\mathbf{b}_\rho}$, and similarly for $R_{X,\rho,e}$, $C_{X,\rho,e}$, $\iota_{Y,\rho,e}$, $J_{Y,\rho,e}^{\text{raw}}$, and $\text{aux}_{Y,\rho,e}$.

Lemma 11.1. *For every $J < \infty$, there are finite constants $K_{0,J}$ and C_J , a number $\rho_{L,J} > 0$, and, for every $s \geq s_*$ and every integer $q \geq 0$, a finite constant $K_{J,s,q}$, such that the following hold.*

(i) *The low block $A_{0,J}$ at the circle has a two-sided inverse $G_{0,J}$, and*

$$\|G_{0,J} F\|_{\mathbb{X}_{\mathfrak{h}}^{s*}} \leq K_{0,J} \|F\|_{\mathbb{Y}_{\mathfrak{h}}^{s*}}.
 \tag{11.3}$$

(ii) *In the block norms at the base index,*

$$\|A_{\rho,J} - A_{0,J}\|_{P_J^X \mathbb{X}_{\mathfrak{h}}^{s*} \rightarrow P_J^Y \mathbb{Y}_{\mathfrak{h}}^{s*}} \leq C_J |\rho|, \quad 0 \leq \rho \leq \rho_{L,J}.
 \tag{11.4}$$

(iii) *After requiring $K_{0,J} C_J \rho_{L,J} \leq 1/2$, the formula*

$$\begin{aligned}
 G_{\rho,J} &= [I + G_{0,J}(A_{\rho,J} - A_{0,J})]^{-1} G_{0,J} \\
 &= G_{0,J} [I + (A_{\rho,J} - A_{0,J}) G_{0,J}]^{-1}
 \end{aligned}
 \tag{11.5}$$

defines, for every $0 \leq \rho \leq \rho_{L,J}$, a two-sided inverse of $A_{\rho,J}$, with

$$\|G_{\rho,J}\|_{s_*} \leq 2K_{0,J}, \quad A_{\rho,J} G_{\rho,J} = I, \quad G_{\rho,J} A_{\rho,J} = I.
 \tag{11.6}$$

Here, $\|\cdot\|_{s_}$ denotes the operator norm between the block spaces $P_J^X \mathbb{X}_{\mathfrak{h}}^{s*}$ and $P_J^Y \mathbb{Y}_{\mathfrak{h}}^{s*}$ at the base index, in the direction in which the operator acts. Moreover, for every $s \geq s_*$ and every integer $q \geq 0$, we have*

$$\sup_{0 \leq \rho \leq \rho_{L,J}} \|\partial_\rho^q G_{\rho,J} F\|_{\mathbb{X}_{\mathfrak{h}}^{s*}} \leq K_{J,s,q} \|F\|_{\mathbb{Y}_{\mathfrak{h}}^{s*}},
 \tag{11.7}$$

and the family also commutes with complex conjugation.

Recall the family [\(2.33\)](#). In [\(2.32\)](#), α_0 is the orientation and δ the amplitude of the tilt. We fix α_0 before constructing the reference inverse, and the constant ellipses of the lemma have $\delta = 0$. We choose the constants $K_{0,J}$ and C_J on this family of constant ellipses. The family [\(2.33\)](#) with $\delta \neq 0$ enters through the perturbation estimate of [Lemma 10.1](#), and δ is chosen in the resulting neighborhood.

Proof. At the circle, every component of $\mathbb{D}_{0,e}$ preserves the cell frequency and angular mode. We fix a cell frequency n and write a block source as

$$F_n^{\mathfrak{h}} = (F_{c,n}^{\mathfrak{h}}, F_{A,n}^{\mathfrak{h}}, F_{+,n}^{\mathfrak{h}}, F_{\text{cap},n}^{\mathfrak{h}}, F_{\text{fin},n}).$$

The entries are, in order, the center source, the two affine sources, the positive source, the cap data, and the remaining mean, scale, gauge, chart, and quotient data. Their weights are listed in [Table 1](#); each source entry includes its prescribed axis or boundary data. We solve the corresponding equations as follows.

Component	Data	Solution procedure
Center	Interior source and prescribed first Taylor coefficients.	Bessel formulas (8.34)–(8.36) for nonzero cell frequency.
Affine	The two Hardy forcings and the prescribed affine coefficient.	Hardy maps (8.37) on $ m = 2$.
Positive	Source in the dual of the form space, interface data, and Robin data.	Coercivity (9.36) and (9.49), with the even-mode bound (7.94).
Cap	Cartesian interior and trace data.	The cap constructions in Sections 8 and 9, with the prescribed axis conditions.
Remaining	Mean, scale, gauge, chart, and compatible quotient data.	A triangular system with nonzero diagonal entries.

The center is kept as one four-dimensional block because its two speeds coincide at the circle. At zero cell frequency, we use (7.54) for the tangential component and the radial equations following it, together with the even reconstruction (7.76)–(7.78) with the mass terms omitted. The reconstruction (3.10) recovers the Cartesian source entries from the quotient variables at the orders of the block norms.

The block source is the pair $(Z, \Psi_{\mathbf{b}}Z)$ in (6.28). Here, Z contains the compatible quotient data, and $\Psi_{\mathbf{b}}Z$ is the scalar differential source. The block norm controls both entries at the same index. We convert a Cartesian source to this pair once, by (6.92), before the inverse at the circle is applied.

Substituting the explicit formulas for the components into the equations, we obtain the two identities $A_{0,J}G_{0,J} = I$ and $G_{0,J}A_{0,J} = I$ at cell frequency n on smooth data. The prescribed Taylor coefficients on the axis, the coercive form, and the triangular finite system give uniqueness, so density extends the identities to the block spaces at cell frequency n . Thus, for each n and s ,

$$(11.8) \quad \|G_{0,n}F_n^{\mathbf{b}}\|_{\mathbb{X}_{\mathbf{b},n}^s} \leq C_{n,s} \|F_n^{\mathbf{b}}\|_{\mathbb{Y}_{\mathbf{b},n}^s}.$$

Here, $G_{0,n}$ is the inverse at cell frequency n constructed above, $\mathbb{X}_{\mathbf{b},n}^s$ and $\mathbb{Y}_{\mathbf{b},n}^s$ are the subspaces of $\mathbb{X}_{\mathbf{b}}^s$ and $\mathbb{Y}_{\mathbf{b}}^s$ at cell frequency n , and the constant $C_{n,s}$ does not depend on any cutoff in the angular modes. We assemble these inverses by

$$G_{0,J}F = \sum_{|n| < 2^J} (G_{0,n}F_n^{\mathbf{b}})e^{in\zeta}.$$

Parseval's identity and the maximum of $C_{n,s}$ over $|n| < 2^J$ give $K_{0,J} < \infty$ and (11.3). The cutoff J restricts only the cell frequency. The component estimates control all radial and angular variables without further truncation. Since the operator at the circle is diagonal in the cell frequency, this construction is independent of the high-frequency inverse. The off-diagonal blocks between the minus and positive components also vanish there, by (10.32).

Before fixing J , we choose ρ_0 so that the bounds for the determinant, the contours, the form, the cap, the quotients, and the traces hold for $0 \leq \rho \leq \rho_0$. The Cartesian inverse of Section 7 is used only for $0 < \rho < \rho_0$. Thus, the choice of ρ_0 does not depend on J . We claim that, for every J , s , and q ,

$$(11.9) \quad \sup_{0 \leq \rho \leq \rho_0} \|\partial_{\rho}^q A_{\rho,J}\|_{P_J^X \mathbb{X}_{\mathbf{b}}^s \rightarrow P_J^Y \mathbb{Y}_{\mathbf{b}}^s} < \infty,$$

and we verify this component by component. For the center block, the graph norm on the maximal domain controls the radial derivative, and all cell derivatives are bounded on $|n| < 2^J$. The spectral projection

onto the four center modes is given by a contour integral, and the resolvent identity bounds it and all its ρ -derivatives as operators of order zero. The coefficients of the affine Hardy maps are analytic and carry the weights of (7.94). The positive form and its derivatives map the form space to its dual, while the value, conormal, and Robin data use the balanced trace norms. On the cap, the minus operator maps H^{s+2} to its source space H^s , whereas the positive operator maps H^{s+1} to H^{s-1} , with the value and the split residual of Section 9 in $H^{s+1/2}$ and $H^{s-1/2}$. Both cap estimates include the Cartesian compatibility and axis conditions. The remaining quotient, source, mean, scale, chart, and gauge operators are analytic maps of fixed order. These bounds control every component of $A_{\rho,J}$, and the estimates for the angular multipliers make them uniform in the angular mode.

With $C_J = \sup_{0 \leq \rho \leq \rho_0} \|\partial_\rho A_{\rho,J}\|_{s_*}$, we obtain (11.4) from the fundamental theorem of calculus in the Banach space of bounded operators. We may then define

$$\rho_{L,J} := \min \left\{ \rho_0, \frac{1}{2 \max\{1, K_{0,J} C_J\}} \right\},$$

so that both Neumann series in (11.5) converge. Since $G_{0,J}[(A_{\rho,J} - A_{0,J})G_{0,J}]^k = [G_{0,J}(A_{\rho,J} - A_{0,J})]^k G_{0,J}$ for every $k \geq 0$, the two series define the same operator $G_{\rho,J}$, and composing with $A_{\rho,J} = A_{0,J} + (A_{\rho,J} - A_{0,J})$ on either side proves (11.6).

To control higher regularity, we use the estimate below, with $E_{J,s}$ independent of $0 \leq \rho \leq \rho_{L,J}$.

$$(11.10) \quad \|u\|_{\mathbb{X}_{\natural}^s} \leq E_{J,s} (\|A_{\rho,J} u\|_{\mathbb{Y}_{\natural}^s} + \|u\|_{\mathbb{X}_{\natural}^{s*}}).$$

The components are coupled in $A_{\rho,J}$, so we commute the equation $A_{\rho,J} u = F$ with the derivatives that enter the block norms and estimate the commutators. The identity (6.26) cancels the off-diagonal terms of highest order, and by Proposition 10.2 a derivative falling on a frame, on a projection, on a reduction of the source, or on a coefficient lowers the block index by one, so each commutator is bounded at the index below. Induction from s_* and interpolation give (11.10). Applying it to $u = G_{\rho,J} F$ and using (11.6) for the base-index term proves (11.7) for $q = 0$.

For $q \geq 1$, we fix the datum F , set $u = G_{\rho,J} F$, and differentiate $A_{\rho,J} u = F$ in ρ to obtain

$$A_{\rho,J} \partial_\rho^q u = - \sum_{k=1}^q \binom{q}{k} (\partial_\rho^k A_{\rho,J}) \partial_\rho^{q-k} u$$

whose right-hand side is bounded in $\mathbb{Y}_{\natural}^s$ by (11.9) and by the estimates for $\partial_\rho^{q-k} u$ with $k \geq 1$. Applying $G_{\rho,J}$ and the case $q = 0$ of (11.7) to this right-hand side, we obtain the estimate for each q by induction on q . Finally, complex conjugation preserves all the kernels, contours, spaces, and finite-dimensional operators. Thus, $\overline{G_{\rho,J} F}$ and $G_{\rho,J} \overline{F}$ solve the same equation and are equal by uniqueness. $\square$

The block inverse agrees with the Cartesian inverse of Section 7 where both are defined, as we now verify.

Remark 11.2. The two inverses agree on compatible Cartesian data, after the return $R_{X,\rho,e}$ to Cartesian variables. We fix $0 < \rho \leq \rho_{L,J}$ with $\rho < \rho_0$ and let $V_{\rho,e}$ be the inverse V_e of Proposition 7.1 with ρ_* replaced by ρ . These restrictions ensure that both inverses are defined. Since the coefficients of $\mathbb{D}_{\rho,e}$ are independent of ζ , the operator is diagonal in the cell frequency,

$$\mathbb{D}_{\rho,e} P_J^X = P_J^Y \mathbb{D}_{\rho,e}, \quad P_J^Y \mathbb{D}_{\rho,e} (I - P_J^X) = 0, \quad (I - P_J^Y) \mathbb{D}_{\rho,e} P_J^X = 0,$$

so (11.1) is the restriction of $\mathbb{D}_{\rho,e}$ to the low cell frequencies. We let f be a smooth Cartesian datum at low frequency and put

$$F = \iota_{Y,\rho,e} J_{Y,\rho,e}^{\text{raw}} f, \quad P_J^Y F = F.$$

If $U = G_{\rho,J} F$, then $U = P_J^X U$ and $A_{\rho,J} U = F$, so $\mathbb{D}_{\rho,e} U = P_J^Y \mathbb{D}_{\rho,e} U = F$ by the diagonality above. Since F has vanishing auxiliary data, the first and the last identities in (11.2) give

$$C_{X,\rho,e} U = \text{aux}_{Y,\rho,e} F = 0, \quad U = J_{X,\rho,e} R_{X,\rho,e} U.$$

Applying the second identity in (11.2) to $u = R_{X,\rho,e} U$, we obtain $\iota_{Y,\rho,e} J_{Y,\rho,e}^{\text{raw}} L_{\rho,e} R_{X,\rho,e} U = \mathbb{D}_{\rho,e} U = F = \iota_{Y,\rho,e} J_{Y,\rho,e}^{\text{raw}} f$. Since $J_{Y,\rho,e}^{\text{raw}}$ has the left inverse (6.75) on smooth elements and $p_{Y,\mathbf{b}} \iota_{Y,\mathbf{b}} = I$ by (6.68), the map $\iota_{Y,\rho,e} J_{Y,\rho,e}^{\text{raw}}$ is injective on smooth elements. Hence, $L_{\rho,e} R_{X,\rho,e} U = f$, and uniqueness in Proposition 7.1 gives $R_{X,\rho,e} U = V_{\rho,e} f$. We therefore obtain

$$(11.11) \quad G_{\rho,J} \iota_{Y,\rho,e} J_{Y,\rho,e}^{\text{raw}} f = J_{X,\rho,e} V_{\rho,e} f, \quad R_{X,\rho,e} G_{\rho,J} \iota_{Y,\rho,e} J_{Y,\rho,e}^{\text{raw}} f = V_{\rho,e} f.$$

The Cartesian estimate (7.2) loses ℓ_e derivatives of the source and requires $\rho > 0$. The block estimate in Lemma 11.1 was proved directly and has neither restriction. Both inverses here concern a constant ellipse. The family (2.33) agrees with this reference only at $\delta = 0$.

12. COMBINING THE INVERSES

We now combine the preceding inverses to invert the augmented operator $\mathbb{D}_{\mathbf{b}}$. We first couple the two high-frequency blocks, then include the low frequencies by a Schur complement. We estimate state and parameter derivatives in the block spaces before returning to Cartesian variables.

Throughout, we apply the projections P and H of (6.94) in the block variables, that is, only after the transfer maps of Section 6 have been applied to the domain and to the source. Thus, every block acts between fixed spaces. We recall from (10.28) the distance from the circle,

$$(12.1) \quad \begin{aligned} \eta_{\text{circ}}(\mathbf{b}) &:= \|a - a_o(\rho)\|_{\mathcal{A}_\gamma^{s_*+k_\times}} + |\varepsilon| + \|M(p) - M_o(\rho)\|_{\mathcal{A}_\gamma^{s_*+k_\times}}, \\ \chi(\mathbf{b}) &:= |\rho| + \eta_{\text{circ}}(\mathbf{b}). \end{aligned}$$

Here, $a_o(\rho)$ and $M_o(\rho)$ are the unperturbed solution and the matrix of the constant ellipse $\mathbf{b}_\rho$, whose axes do not rotate with ζ , and $k_\times$ is the shift in the regularity of the coefficients fixed with the estimate (10.30). We use χ in the estimates at high frequency. For the perturbation at low frequency below, we instead measure the distance from the constant reference state $\mathbf{b}_e$.

12.1. The high-frequency inverse. We first solve $D_{H,\mathbf{b}}(u_-, u_+) = (f_-, f_+)$. The unknown u_- contains the center and affine fields, and u_+ contains the positive fields; each includes its cap and annular components. The operator of (6.94) has the block form

$$(12.2) \quad D_{H,\mathbf{b}} = \begin{pmatrix} D_{-,\mathbf{b}} & E_{-+,\mathbf{b}} \\ E_{+-,\mathbf{b}} & D_{+,\mathbf{b}} \end{pmatrix}.$$

All four entries belong to the same augmented operator $\mathbb{D}_{\mathbf{b}}$. The off-diagonal blocks are bounded from index s to index s by Proposition 10.2. Under (10.46), the correction (10.44)–(10.47) turns the inverses of Propositions 8.1 and 9.1 into the diagonal inverses used below.

Lemma 12.1. *Suppose that*

$$\|D_{-,b}^{-1}\| \leq K_-, \quad \|D_{+,b}^{-1}\| \leq K_+, \quad \|E_{-,b}\| + \|E_{+,b}\| \leq K_\times \chi(b)$$

in the block norms at the base index, and let

$$(12.3) \quad \mathcal{R}_{-,b} := D_{-,b}^{-1} E_{-,b} D_{+,b}^{-1} E_{+,b}.$$

Solving the second row for u_+ and substituting it into the first gives

$$(I - \mathcal{R}_{-,b})u_- = D_{-,b}^{-1}(f_- - E_{-,b}D_{+,b}^{-1}f_+).$$

If $\|\mathcal{R}_{-,b}\| < 1$, the operator on the left is invertible by a Neumann series. Thus, $D_{H,b}$ has a two-sided inverse $G_{H,b}$, given for $f = (f_-, f_+)$ by

$$(12.4) \quad u_- = (I - \mathcal{R}_{-,b})^{-1} D_{-,b}^{-1}(f_- - E_{-,b}D_{+,b}^{-1}f_+),$$

$$(12.5) \quad u_+ = D_{+,b}^{-1}(f_+ - E_{+,b}u_-),$$

$$(12.6) \quad G_{H,b}f = (u_-, u_+).$$

On the block spaces, this inverse obeys tame estimates without loss of derivatives, and each order of differentiation with respect to b requires only a finite increase in the regularity of the source.

Proof. As in (10.49), the factors in (12.3) act between the spaces

$$\mathbb{X}_{-,b}^s \xrightarrow{E_{+,-}} \mathbb{Y}_{+,b}^s \xrightarrow{D_{+,b}^{-1}} \mathbb{X}_{+,b}^s \xrightarrow{E_{-+}} \mathbb{Y}_{-,b}^s \xrightarrow{D_{-,b}^{-1}} \mathbb{X}_{-,b}^s,$$

so each term of the Neumann series acts on the same block space and preserves the index. No Cartesian transfer map enters this composition. The smallness assumption gives convergence of the Neumann series at the base index. We establish its higher regularity bounds below. Substituting (12.4) and (12.5) into (12.2), we obtain the right inverse identity. For the left inverse identity, we fix $w = (w_-, w_+)$ and subtract $G_{H,b}D_{H,b}w$ from w . Since $D_{H,b}G_{H,b} = I$, the difference $v = (v_-, v_+)$ solves the homogeneous system $D_{H,b}v = 0$. Applying $D_{+,b}^{-1}$ to the second row gives $v_+ = -D_{+,b}^{-1}E_{+,b}v_-$, and inserting this into the first row and applying $D_{-,b}^{-1}$ gives $(I - \mathcal{R}_{-,b})v_- = 0$, so $v_- = 0$ and then $v_+ = 0$. For higher regularity, we commute derivatives through the equations and use the strict smallness condition at the base index to absorb the highest-order term. Only the source or one coefficient is measured in a higher regularity norm. The derivatives with respect to b are estimated below. $\square$

12.2. The low-frequency block. We write $A_e = A_{b_e}$ for the low block at the constant reference state, as in Section 10. By Lemma 11.1 at $\rho = \rho_*$, which applies since $\rho_* < \rho_{L,J}$ by (13.2) below,

$$(12.7) \quad A_e^{-1} = G_{\rho_*,J} : P^Y \mathbb{Y}_{\natural}^{s_*} \longrightarrow P^X \mathbb{X}_{\natural}^{s_*}$$

is a two-sided inverse and loses no derivatives. Under the condition (10.27), we therefore have the two-sided inverse of the low block

$$(12.8) \quad A_b^{-1} = [I + A_e^{-1}(A_b - A_e)]^{-1} A_e^{-1}.$$

For F_L in the range of P^Y , we write $u_L = A_b^{-1}F_L$. Commuting derivatives through $A_b u_L = F_L$ and applying (10.21) gives the tame bound at each Sobolev index without loss of derivatives.

12.3. The Schur complement at low frequencies. For $U = u_L + u_H$ and $F = F_L + F_H$, where the subscripts denote the low and high projections, the equation $\mathbb{D}_{\mathbf{b}}U = F$ is

$$A_{\mathbf{b}}u_L + B_{\mathbf{b}}u_H = F_L, \quad C_{\mathbf{b}}u_L + D_{H,\mathbf{b}}u_H = F_H.$$

The second equation gives $u_H = G_{H,\mathbf{b}}(F_H - C_{\mathbf{b}}u_L)$. Substituting this into the first eliminates the high-frequency unknown. The operator that remains on the low-frequency component is the *Schur complement*

$$(12.9) \quad S_{L,\mathbf{b}} := A_{\mathbf{b}} - B_{\mathbf{b}}G_{H,\mathbf{b}}C_{\mathbf{b}} = A_{\mathbf{b}}(I - \mathcal{T}_{L,\mathbf{b}}), \quad \mathcal{T}_{L,\mathbf{b}} := A_{\mathbf{b}}^{-1}B_{\mathbf{b}}G_{H,\mathbf{b}}C_{\mathbf{b}}.$$

Proposition 12.2 (The global inverse). *Let $G_{H,\mathbf{b}}$ be the inverse of Lemma 12.1. Assume (10.27) and $\|\mathcal{T}_{L,\mathbf{b}}\| \leq q_L < 1$ in the block norms at the base index. Then*

$$(12.10) \quad S_{L,\mathbf{b}}^{-1} = (I - \mathcal{T}_{L,\mathbf{b}})^{-1}A_{\mathbf{b}}^{-1},$$

and

$$(12.11) \quad G_{\mathbf{b}} = \begin{pmatrix} S_{L,\mathbf{b}}^{-1} & -S_{L,\mathbf{b}}^{-1}B_{\mathbf{b}}G_{H,\mathbf{b}} \\ -G_{H,\mathbf{b}}C_{\mathbf{b}}S_{L,\mathbf{b}}^{-1} & G_{H,\mathbf{b}} + G_{H,\mathbf{b}}C_{\mathbf{b}}S_{L,\mathbf{b}}^{-1}B_{\mathbf{b}}G_{H,\mathbf{b}} \end{pmatrix}$$

maps (F_L, F_H) to (u_L, u_H) . Equivalently, we solve in the order

$$u_L = S_{L,\mathbf{b}}^{-1}(F_L - B_{\mathbf{b}}G_{H,\mathbf{b}}F_H), \quad u_H = G_{H,\mathbf{b}}(F_H - C_{\mathbf{b}}u_L).$$

It is a two-sided inverse in the block variables, so that

$$\mathbb{D}_{\mathbf{b}}G_{\mathbf{b}} = I, \quad G_{\mathbf{b}}\mathbb{D}_{\mathbf{b}} = I.$$

Uniformly on every ball of states on which the norms in the smallness conditions at the base index are bounded away from one, we have

$$(12.12) \quad \|G_{\mathbf{b}}F\|_{\mathbb{X}_{\mathbf{b}}^s} \leq C_s \left[\|F\|_{\mathbb{Y}_{\mathbf{b}}^s} + (1 + \|\mathbf{b} - \mathbf{b}_e\|_{\mathcal{B}^{s+k}}) \|F\|_{\mathbb{Y}_{\mathbf{b}}^{s*}} \right].$$

Here, k is a fixed shift large enough for the diagonal inverse, off-diagonal, and low-frequency estimates used in the construction.

Proof. The strict smallness assumption makes the Neumann series in (12.10) converge on the low-frequency space at the base index. Multiplying (12.11) and (6.94) in both orders verifies the two inverse identities directly on the full block spaces.

For the regularity at index s , we write $w = (I - \mathcal{T}_{L,\mathbf{b}})^{-1}g$ for g in the low-frequency space and use the equation $w = g + \mathcal{T}_{L,\mathbf{b}}w$. The tame estimates for $A_{\mathbf{b}}^{-1}$, $B_{\mathbf{b}}$, $G_{H,\mathbf{b}}$, and $C_{\mathbf{b}}$ give

$$\|w\|_s \leq \|g\|_s + q_L \|w\|_s + C_{J,s}(1 + \|\mathbf{b} - \mathbf{b}_e\|_{\mathcal{B}^{s+k}}) \|w\|_{s_*}, \quad q_L < 1.$$

Here, $\|\cdot\|_s$ denotes the norm of $P^X \mathbb{X}_{\mathbf{b}}^s$. The inverse at the base index controls $\|w\|_{s_*}$, and the strict smallness condition absorbs $q_L \|w\|_s$. Inserting the result into the four blocks of (12.11), we obtain (12.12). $\square$

12.4. Derivatives of the inverse. Differentiating the center equation (8.82) with respect to the state and parameters produces the source in (8.101), with at most one cell derivative of the solution. Accordingly, we estimate derivatives of the inverse between fixed Sobolev spaces with a finite loss that depends on the number of differentiations.

Lemma 12.3. *For each $m \geq 1$, the families of blocks in (6.94) and (12.2), together with the inverses $D_{\pm, \mathbf{b}}^{-1}$ and $A_{\mathbf{b}}^{-1}$ and the Neumann series $(I - \mathcal{R}_{-, \mathbf{b}})^{-1}$ and $(I - \mathcal{T}_{L, \mathbf{b}})^{-1}$, are m times Fréchet differentiable between block spaces whose indices differ by a finite shift. There are finite integers k_m and μ_m , independent of the Sobolev index s , such that*

$$\begin{aligned}
 & \|D_{\mathbf{b}}^m G_{\mathbf{b}}[h_1, \dots, h_m] F\|_{\mathbb{X}_{\mathfrak{h}}^s} \\
 & \leq C_{s, m} \left[\left(\prod_{i=1}^m \|h_i\|_{\tan, s_* + k_m} \right) \|F\|_{\mathbb{Y}_{\mathfrak{h}}^{s + \mu_m}} \right. \\
 & \quad + \sum_{i=1}^m \|h_i\|_{\tan, s + k_m} \prod_{j \neq i} \|h_j\|_{\tan, s_* + k_m} \|F\|_{\mathbb{Y}_{\mathfrak{h}}^{s_* + \mu_m}} \\
 & \quad \left. + (1 + \|\mathbf{b} - \mathbf{b}_e\|_{\mathcal{B}^{s + k_m}}) \left(\prod_{i=1}^m \|h_i\|_{\tan, s_* + k_m} \right) \|F\|_{\mathbb{Y}_{\mathfrak{h}}^{s_* + \mu_m}} \right].
 \end{aligned}
 \tag{12.13}$$

At most m derivatives of the source are lost in the inverse of the center equation, and the remaining components and traces contribute only a fixed finite loss.

Proof. For any of the families of operators $T_{\mathbf{b}}$ inverted above, we have on smooth elements

$$D(T_{\mathbf{b}}^{-1})[h] = -T_{\mathbf{b}}^{-1}(DT_{\mathbf{b}}[h])T_{\mathbf{b}}^{-1}, \tag{12.14}$$

$$D(I - T_{\mathbf{b}})^{-1}[h] = (I - T_{\mathbf{b}})^{-1}(DT_{\mathbf{b}}[h])(I - T_{\mathbf{b}})^{-1}. \tag{12.15}$$

To justify these derivatives, we use the resolvent identity

$$T_{\mathbf{b}+h}^{-1} - T_{\mathbf{b}}^{-1} = -T_{\mathbf{b}+h}^{-1}(T_{\mathbf{b}+h} - T_{\mathbf{b}})T_{\mathbf{b}}^{-1},$$

and we subtract the linear terms in (12.14) and (12.15). The uniform higher-index bounds for the inverses, together with the remainder estimate for the Fréchet derivative of $T_{\mathbf{b}}$, which costs a fixed finite number of derivatives, control the error at an index lower by that fixed amount. By induction, we obtain differentiability at every finite order.

For the oscillatory center component, we differentiate the center equation $L_{\mathbf{b}}^c u = f$ in the directions $h_1, \dots, h_q$, as in (8.101), and, at order q , Leibniz' rule gives

$$L_{\mathbf{b}}^c D^q u = - \sum_{\emptyset \neq I \subset \{1, \dots, q\}} D^{|I|} L_{\mathbf{b}}^c[h_I] D^{q-|I|} u[h_{I^c}]. \tag{12.16}$$

Here, the source f is held fixed, h_I collects the directions h_i with $i \in I$, and I^c is the complement of I in $\{1, \dots, q\}$. Differentiation in the parameters changes the coefficients of $L_{\mathbf{b}}^c$ but does not add a second factor D_{ζ} . Every differentiated center operator therefore contains at most one cell derivative of the solution. Repeated use of the energy estimate for the undifferentiated center equation, as in the proof of Proposition 8.1, requires one additional index on the source at each differentiation, and hence at most q at order q . The change of variables (8.95) for the affine component obeys the same bound. For the positive form, one derivative falls on each argument. For the cap, we use the Sobolev scales of Table 1, from H^{s+2} to H^s for the minus block

and from H^{s+1} to H^{s-1} for the positive block, while the quotient and trace maps retain their orders. Finally, [Proposition 10.2](#) bounds the differentiated off-diagonal maps without loss of derivatives.

Differentiating (12.6), (12.8), and (12.10) repeatedly, we obtain finite sums of products in which inverse factors alternate with derivatives of operators. The strict smallness conditions bound the inverse factors. The tame product estimate (3.20) measures one of $F, h_1, \dots, h_m, \mathbf{b} - \mathbf{b}_e$ in a higher regularity norm and the others at the base index. This gives (12.13). $\square$

12.5. Return to Cartesian variables and real data. We recover the Cartesian inverse by converting the source to block variables, applying the block inverse, and returning the solution to Cartesian variables. The source conversion occurs only once, including in each differentiated term.

Lemma 12.4. *Let $\ell = \ell_Y$ be the number of derivatives lost when converting the Cartesian source to block variables, as defined in (6.66). Then*

$$(12.17) \quad V_{\mathbf{b}} = R_{X,\mathbf{b}} G_{\mathbf{b}} \iota_{Y,\mathbf{b}} J_{Y,\mathbf{b}}^{\text{raw}}$$

satisfies both inverse identities in the Cartesian variables,

$$L_{\mathbf{b}} V_{\mathbf{b}} = I, \quad V_{\mathbf{b}} L_{\mathbf{b}} = I,$$

and, in every m -th derivative of $V_{\mathbf{b}}$ with respect to $\mathbf{b}$, the total loss of derivatives on the Cartesian source is $\ell + \mu_m$. The transfer map $J_{Y,\mathbf{b}}^{\text{raw}}$ of the source occurs once in each differentiated term, so its loss does not accumulate to $(m+1)\ell + \mu_m$.

Proof. We fix a smooth compatible Cartesian datum f , and we set

$$(12.18) \quad \widehat{f} = \iota_{Y,\mathbf{b}} J_{Y,\mathbf{b}}^{\text{raw}} f, \quad W = G_{\mathbf{b}} \widehat{f}.$$

The inclusion $\iota_{Y,\mathbf{b}}$ assigns zero to the auxiliary entries, so $\text{aux}_{Y,\mathbf{b}} \widehat{f} = 0$. From the identity (6.87) for the auxiliary equations and $\mathbb{D}_{\mathbf{b}} G_{\mathbf{b}} = I$, we therefore obtain

$$(12.19) \quad C_{X,\mathbf{b}} W = \text{aux}_{Y,\mathbf{b}} \mathbb{D}_{\mathbf{b}} W = \text{aux}_{Y,\mathbf{b}} \widehat{f} = 0,$$

while (6.72) yields

$$(12.20) \quad J_{X,\mathbf{b}} R_{X,\mathbf{b}} W = W.$$

Using (6.90) and the identity $p_{Y,\mathbf{b}} \iota_{Y,\mathbf{b}} = I$ of (6.68), we obtain

$$(12.21) \quad \begin{aligned} J_{Y,\mathbf{b}}^{\text{raw}} L_{\mathbf{b}} V_{\mathbf{b}} f &= p_{Y,\mathbf{b}} \mathbb{D}_{\mathbf{b}} J_{X,\mathbf{b}} R_{X,\mathbf{b}} W \\ &= p_{Y,\mathbf{b}} \mathbb{D}_{\mathbf{b}} W = p_{Y,\mathbf{b}} \widehat{f} = J_{Y,\mathbf{b}}^{\text{raw}} f, \end{aligned}$$

which, by injectivity of $J_{Y,\mathbf{b}}^{\text{raw}}$ on the scale of compatible Cartesian sources, proves $L_{\mathbf{b}} V_{\mathbf{b}} = I$. For the left inverse identity, we take any smooth u satisfying the constraints, and then compute

$$(12.22) \quad \begin{aligned} V_{\mathbf{b}} L_{\mathbf{b}} u &= R_{X,\mathbf{b}} G_{\mathbf{b}} \iota_{Y,\mathbf{b}} J_{Y,\mathbf{b}}^{\text{raw}} L_{\mathbf{b}} u \\ &= R_{X,\mathbf{b}} G_{\mathbf{b}} \mathbb{D}_{\mathbf{b}} J_{X,\mathbf{b}} u = R_{X,\mathbf{b}} J_{X,\mathbf{b}} u = u. \end{aligned}$$

Boundedness and the shifted identities (6.76)–(6.77) extend both inverse identities to the closures of smooth elements in the indicated norms.

Differentiating (12.17), we find that every term of Leibniz' rule carries exactly one factor $D^q J_{Y,\mathbf{b}}^{\text{raw}}$ on the Cartesian source, and that the derivatives of $\iota_{Y,\mathbf{b}}$ and of $R_{X,\mathbf{b}}$ lose no derivatives. By Proposition 6.2, differentiation of the source map changes its analytic coefficients but leaves its differential order ℓ unchanged. Combining these bounds with (12.13), we obtain the stated loss on the source. $\square$

We next show that the inverses preserve real data.

Lemma 12.5. *For real $\mathbf{b}$, the block and the Cartesian inverses preserve the real subspaces, that is, the fixed subspaces of the complex conjugations on the block spaces and on the Cartesian spaces.*

Proof. We let $\mathbf{c}_X$ and $\mathbf{c}_Y$ denote the conjugations on $\mathbb{X}_{\mathfrak{d}}^s$ and $\mathbb{Y}_{\mathfrak{d}}^s$ induced by Cartesian complex conjugation. For a scalar Fourier series, this conjugation acts by $(cu)_{m,n} = \overline{u_{-m,-n}}$. In paired scalar or quotient coordinates, it also exchanges the entries paired by complex conjugation, such as $Z = A + iB$ and $W = A - iB$ in Section 7. The operators of the components, the spectral projections (6.21) on fixed contours, the quotient maps, the traces, and the transfer maps commute with $\mathbf{c}_X$ and $\mathbf{c}_Y$. So do the projections P and H , whose symbols are even in the cell frequency, and the blocks $A_{\mathbf{b}}$, $B_{\mathbf{b}}$, $C_{\mathbf{b}}$, and $D_{H,\mathbf{b}}$. If $T\mathbf{c}_X = \mathbf{c}_Y T$ and T has a two-sided inverse, then uniqueness implies $T^{-1}\mathbf{c}_Y = \mathbf{c}_X T^{-1}$, and this relation is preserved under finite products and norm-convergent Neumann series. Applying this successively to (12.6), (12.8), (12.10), and (12.17) proves the claim. $\square$

13. UNIFORM ESTIMATES FOR THE INVERSE

We now choose the constants so that the inverses constructed in Sections 11 and 12 apply on one neighborhood of the reference state. We first fix the frequency cutoff J , then the ellipse parameter ρ_* , and finally the neighborhood. The resulting Cartesian inverse and its tame estimates are stated in Theorem 13.2.

13.1. The choice of the constants. The low-frequency bounds may depend on J , so the order of these choices matters. We record it together with the four smallness conditions needed for the inverse.

Proposition 13.1 (Choice of the constants). *There are an integer $J < \infty$, an ellipse parameter $\rho_* > 0$, and an open neighborhood $\mathfrak{U}_{\text{inv}}$ of $\mathbf{b}_e$, chosen in the order below. We call the norm $\|\cdot\|_{\mathfrak{B}^{s_*+k_0}}$ of (10.18) the base norm. The closure of $\mathfrak{U}_{\text{inv}}$ in the base norm lies inside $\{0 < \rho < \rho_0\}$ and inside the image of the preliminary neighborhood $\mathcal{U}_0$ of Proposition 5.1 under the normalized chart of Section 3.1.*

- (i) *We fix the analytic widths $\sigma_0 > \sigma_- > 0$ of Section 8, the orientation α_0 and the parameter λ_* of the constant reference state of Section 6, and a compact neighborhood in the finite-dimensional parameters. All constants of the reference inverse of Section 7 may depend on the orientation α_0 .*
- (ii) *We choose the strength γ of the analytic weight and then the parameter δ_A of (6.3), so that the center and affine damping conditions (8.1) and the positive-form condition (9.4) remain strict. These conditions are uniform up to the circle.*
- (iii) *We choose $\rho_0 < 1/4$ so that the bounds for the determinant, the contours around the center roots, the positive form, the cap, the quotients, and the traces hold for $0 \leq \rho \leq \rho_0$. The Cartesian inverse of Proposition 7.1 is used only for $0 < \rho < \rho_0$, as explained in Section 11.*

- (iv) We choose a preliminary neighborhood in the state and curvature and take the cell cutoff J large enough for (8.3). All subsequent constants may depend on J . We then invert the low block at the circle by Lemma 11.1, obtain its constants $K_{0,J}$ and C_J , and put

$$(13.1) \quad \rho_{L,J} := \min \left\{ \rho_0, \frac{1}{2 \max\{1, K_{0,J} C_J\}} \right\} > 0.$$

- (v) The constants for $\widehat{G}_\pm$ and for the off-diagonal blocks are uniform for $0 \leq \rho \leq \rho_0$, and the part at low frequency of the correction $K_{\alpha,\mathbf{b}}$ of (10.43) vanishes at the constant reference state, because the reference is diagonal in the cell frequency. We may therefore choose

$$(13.2) \quad 0 < \rho_* < \rho_{L,J}$$

so that first the two norms $\|\widehat{G}_{\alpha,\mathbf{b}_e} K_{\alpha,\mathbf{b}_e}\|$, $\alpha \in \{-, +\}$, of (10.46), and then the norm of the operator $\mathcal{R}_{-, \mathbf{b}_e}$ of (12.3) at the reference state, are strictly below one.

- (vi) Only after J and ρ_* are fixed do we shrink the final neighborhood $\mathfrak{U}_{\text{inv}}$ in the state, in the unperturbed solution, and in the curvature.

The four conditions below control, respectively, the diagonal correction, the coupling of the two high-frequency blocks, the perturbation of the low block, and the coupling between low and high frequencies. On the closure of $\mathfrak{U}_{\text{inv}}$, there are constants $q_{\text{corr}}^*, q_H^*, q_A^*, q_L^* < 1$ such that

$$(13.3) \quad \max_{\alpha \in \{-, +\}} \|\widehat{G}_{\alpha,\mathbf{b}} K_{\alpha,\mathbf{b}}\| \leq q_{\text{corr}}^*,$$

$$(13.4) \quad \|D_{-, \mathbf{b}}^{-1} E_{-, \mathbf{b}} D_{+, \mathbf{b}}^{-1} E_{+, \mathbf{b}}\| \leq q_H^*,$$

$$(13.5) \quad \|A_e^{-1} (A_{\mathbf{b}} - A_e)\| \leq q_A^*,$$

$$(13.6) \quad \|A_{\mathbf{b}}^{-1} B_{\mathbf{b}} D_{H, \mathbf{b}}^{-1} C_{\mathbf{b}}\| \leq q_L^*.$$

All four norms are taken in the block variables at the base index s_* , and the constants may depend on the already fixed J , but not on the upper Fourier cutoff or on the number N of periods of the field.

Proof. The bounds for $\widehat{G}_\pm$ are uniform for $0 \leq \rho \leq \rho_0$ before ρ_* is chosen. They follow from Lemma 7.3 for the even and affine components, the contour around the four center roots, and the positive form estimate. The off-diagonal blocks $E_{-, \mathbf{b}}$ and $E_{+, \mathbf{b}}$ vanish at the circle by (10.32), and on the preliminary neighborhood they obey the bound $K_\chi \chi(\mathbf{b})$ of Proposition 10.2 in the distance $\chi(\mathbf{b}) = |\rho| + \eta_{\text{circ}}(\mathbf{b})$ from the circle of (10.28). Once J is fixed, Lemma 11.1 supplies $K_{0,J}$ and C_J , and with them $\rho_{L,J} > 0$. At the constant reference state, the part of $K_{\alpha,\mathbf{b}_e}$ at low frequency is zero and its part in the other high-frequency component is $O(\rho_*)$, by the two bounds in the proof of Lemma 10.3. We shrink ρ_* until (13.3) holds at $\mathbf{b}_e$ with $q_{\text{corr}}^* < 1$. This is the hypothesis (10.46) of Lemma 10.3, so the diagonal blocks of the full operator have the inverses (10.47). We shrink ρ_* again until (13.4) holds at $\mathbf{b}_e$ with $q_H^* < 1$. The perturbation $A_{\mathbf{b}} - A_e$ of the low block, the part at low frequency of the correction $K_{\alpha,\mathbf{b}}$, and the coupling maps $B_{\mathbf{b}}$ and $C_{\mathbf{b}}$ between low and high frequencies depend continuously on $\mathbf{b}$ and vanish at $\mathbf{b} = \mathbf{b}_e$, by (10.19), (10.20), and the proof of Lemma 10.3. Hence, shrinking the final neighborhood $\mathfrak{U}_{\text{inv}}$ preserves the first two conditions (13.3) and (13.4) and makes (13.5) and (13.6) strict.

For an explicit choice, we fix J and take $A \geq 2$ to dominate $\rho_{L,J}^{-1}$ and the finitely many base-index constants in the four bounds below. These are the constants for $\widehat{G}_\pm$, for the off-diagonal blocks, for the auxiliary

equations, for the reference at low frequency, for the perturbation, and for the coupling between low and high frequencies. We also take A large enough to bound every Lipschitz constant in $\mathbf{b}$ obtained from the mean value theorem on the preliminary neighborhood. We then set

$$t = (2^{12}A^{12})^{-1}.$$

We choose $0 < \rho_* \leq t$ and shrink the final neighborhood $\mathfrak{U}_{\text{inv}}$ in the base norm $\|\cdot\|_{\mathcal{B}^{s_*+k_0}}$ of (10.18), in which d_0 is measured, until on its closure $\chi(\mathbf{b}) \leq \rho_* + t$ and $d_0(\mathbf{b}) \leq \min\{t, \rho_*/2\}$. The bound on d_0 keeps $\rho \geq \rho_*/2 > 0$, while the bound on χ gives $\chi(\mathbf{b}) \leq 2t$. From (10.45), (10.50), and (10.19)–(10.20), we then obtain

$$q_{\text{corr}} \leq 3A^3t < \frac{1}{2}, \quad q_H \leq 16A^4t^2 < \frac{1}{2}, \quad q_A \leq A^2t < \frac{1}{2}.$$

Here and in the next display, q_{corr} , q_H , q_A , and q_L denote the four norms on the left of (13.3)–(13.6) at the state $\mathbf{b}$. Since $q_{\text{corr}} < 1/2$, the Neumann series in (10.47) bounds each inverse of a diagonal block by $2A$. Since $q_H < 1/2$, the formulas (12.4)–(12.5) then bound the inverse at high frequency by $16A$, and since $q_A < 1/2$, the Neumann series in (12.8) bounds $A_{\mathbf{b}}^{-1}$ by $2A$. Hence

$$q_L \leq (2A)(At)(16A)(At) = 32A^4t^2 < \frac{1}{2}.$$

This choice of $t > 0$ makes all four norms strictly less than one, with the constants fixed in the stated order. The argument gives $\rho_* > 0$, but no effective lower bound for its value. $\square$

We recall that N is the number of periods of the field and is unrelated to J . We prove the theorem below on $\mathfrak{U}_{\text{inv}}$, in which ε ranges over an interval about 0, and we solve the nonlinear equations there before setting $\varepsilon = 1/N$.

13.2. The inverse at a general state. On the neighborhood just chosen, the Cartesian linearization has a two-sided inverse with smooth tame dependence on the state. The next theorem gives the estimates needed for the nonlinear problem.

We use the norms $\|h\|_{\text{tan},s}$ of the tangent directions and the products $\mathcal{H}_{0,m}$ and $\mathcal{H}_{s,m}$ of (6.1), with the finite integers k_m supplied by the theorem below. Thus, $\mathcal{H}_{0,m}$ uses lower regularity norms of all tangent directions, while each term of $\mathcal{H}_{s,m}$ uses a higher regularity norm of one direction. For $m = 0$, the conventions are $\mathcal{H}_{0,0} = 1$ and $\mathcal{H}_{s,0} = 0$.

Theorem 13.2 (The uniform inverse). *After the choices of Proposition 13.1, the following hold on $\mathfrak{U}_{\text{inv}}$.*

(i) *For every $\mathbf{b} \in \mathfrak{U}_{\text{inv}}$, the augmented block operator $\mathbb{D}_{\mathbf{b}}$ has a two-sided inverse*

$$G_{\mathbf{b}} : \mathbb{Y}_{\mathfrak{h}}^{\infty} \longrightarrow \mathbb{X}_{\mathfrak{h}}^{\infty}, \quad \mathbb{D}_{\mathbf{b}}G_{\mathbf{b}} = I, \quad G_{\mathbf{b}}\mathbb{D}_{\mathbf{b}} = I.$$

Here, $\mathbb{X}_{\mathfrak{h}}^{\infty} = \bigcap_{s \geq s_} \mathbb{X}_{\mathfrak{h}}^s$ and $\mathbb{Y}_{\mathfrak{h}}^{\infty} = \bigcap_{s \geq s_*} \mathbb{Y}_{\mathfrak{h}}^s$.*

(ii) *For every $m \geq 0$, there are finite integers k_m and μ_m such that, for every $s \geq s_*$,*

$$\begin{aligned} & \|D_{\mathbf{b}}^m G_{\mathbf{b}}[h_1, \dots, h_m]F\|_{\mathbb{X}_{\mathfrak{h}}^s} \\ & \leq C_{s,m} \left[\mathcal{H}_{0,m} \|F\|_{\mathbb{Y}_{\mathfrak{h}}^{s+\mu_m}} + \mathcal{H}_{s,m} \|F\|_{\mathbb{Y}_{\mathfrak{h}}^{s_*+\mu_m}} \right. \\ & \quad \left. + (1 + \|\mathbf{b} - \mathbf{b}_e\|_{\mathcal{B}^{s+k_m}}) \mathcal{H}_{0,m} \|F\|_{\mathbb{Y}_{\mathfrak{h}}^{s_*+\mu_m}} \right]. \end{aligned} \tag{13.7}$$

Here, $D_{\mathbf{b}}^m G_{\mathbf{b}}[h_1, \dots, h_m]$ is the m -th derivative of $\mathbf{b} \mapsto G_{\mathbf{b}}$ in the tangent directions $h_1, \dots, h_m$, as in (8.33), and $C_{s,m}$ is a finite constant, which may depend on s , m , and J . One may take $\mu_0 = 0$. Each μ_m is finite, and no assertion uniform in m is needed.

(iii) On the compatible Cartesian scale, we define

$$(13.8) \quad V_{\mathbf{b}} := R_{X,\mathbf{b}} G_{\mathbf{b}} \iota_{Y,\mathbf{b}} J_{Y,\mathbf{b}}^{\text{raw}}.$$

Here, $J_{Y,\mathbf{b}}^{\text{raw}}$ converts a compatible Cartesian source to the block source of (6.74), $\iota_{Y,\mathbf{b}}$ adjoins zero auxiliary data, and $R_{X,\mathbf{b}}$ is the map (6.70) from the block variables back to the constrained Cartesian domain. For every compatible Cartesian source f ,

$$(13.9) \quad C_{X,\mathbf{b}} G_{\mathbf{b}} \iota_{Y,\mathbf{b}} J_{Y,\mathbf{b}}^{\text{raw}} f = \text{aux}_{Y,\mathbf{b}} \iota_{Y,\mathbf{b}} J_{Y,\mathbf{b}}^{\text{raw}} f = 0,$$

so the block solution $W = G_{\mathbf{b}} \iota_{Y,\mathbf{b}} J_{Y,\mathbf{b}}^{\text{raw}} f$ satisfies the constraint $C_{X,\mathbf{b}} W = 0$. By (6.72), $W = J_{X,\mathbf{b}} R_{X,\mathbf{b}} W$ lies in the image of $J_{X,\mathbf{b}}$, and $V_{\mathbf{b}} f = R_{X,\mathbf{b}} W$ lies in the constrained Cartesian domain. Moreover,

$$L_{\mathbf{b}} V_{\mathbf{b}} = I, \quad V_{\mathbf{b}} L_{\mathbf{b}} = I,$$

and the analogue of (13.7) is

$$(13.10) \quad \begin{aligned} & \|D_{\mathbf{b}}^m V_{\mathbf{b}}[h_1, \dots, h_m] f\|_{\mathcal{X}_{\gamma}^s} \\ & \leq C_{s,m} \left[\mathcal{H}_{0,m} \|f\|_{\mathcal{Y}_{\gamma}^{s+\ell+\mu_m}} + \mathcal{H}_{s,m} \|f\|_{\mathcal{Y}_{\gamma}^{s_*+\ell+\mu_m}} \right. \\ & \quad \left. + (1 + \|\mathbf{b} - \mathbf{b}_e\|_{\mathcal{B}^{s+k_m}}) \mathcal{H}_{0,m} \|f\|_{\mathcal{Y}_{\gamma}^{s_*+\ell+\mu_m}} \right]. \end{aligned}$$

Converting the Cartesian source to block variables loses ℓ derivatives, as defined in (6.66). This loss occurs only once, regardless of the number of terms of the Neumann series or of derivatives with respect to $\mathbf{b}$.

(iv) Let $\mathbf{c}_X$ and $\mathbf{c}_Y$ denote complex conjugation on the block spaces $\mathbb{X}_{\mathbf{q}}^s$ and $\mathbb{Y}_{\mathbf{q}}^s$, and let $\mathbf{c}_X^{\text{raw}}$ and $\mathbf{c}_Y^{\text{raw}}$ denote complex conjugation on the Cartesian spaces $\mathcal{X}_{\gamma}^s$ and $\mathcal{Y}_{\gamma}^s$. For real $\mathbf{b}$,

$$(13.11) \quad \mathbb{D}_{\mathbf{b}} \mathbf{c}_X = \mathbf{c}_Y \mathbb{D}_{\mathbf{b}}, \quad G_{\mathbf{b}} \mathbf{c}_Y = \mathbf{c}_X G_{\mathbf{b}}, \quad V_{\mathbf{b}} \mathbf{c}_Y^{\text{raw}} = \mathbf{c}_X^{\text{raw}} V_{\mathbf{b}}.$$

In particular, $(\mathbf{b}, f) \mapsto V_{\mathbf{b}} f$ is a smooth tame map between these scales, in the sense of Section 4, and it commutes with complex conjugation.

Proof. The condition (13.3) is the hypothesis (10.46) of Lemma 10.3, which gives the inverses of the diagonal blocks of the full operator. The remaining conditions (13.4), (13.5), and (13.6) are precisely the hypotheses of Lemma 12.1 and Proposition 12.2, which give the block inverse, both of its inverse identities, and (12.12), which is (13.7) for $m = 0$. We obtain the estimates for the derivatives from Lemma 12.3. The two inverse identities in Cartesian variables follow from Lemma 12.4, which also shows that converting the Cartesian source loses ℓ derivatives only once. The preservation of real data follows from Lemma 12.5. The construction is independent of an upper Fourier cutoff and of the number N of periods of the field. The identity (13.9) is (12.19), and the block inverse is (12.11). $\square$

14. SOLUTIONS FOR SMALL CURVATURE

We now apply Hamilton's Nash-Moser theorem to the map constructed in [Section 5](#). This gives a family of solutions for all small curvature parameters ε . We then set $\varepsilon = 1/N$ to obtain a solution for every sufficiently large N .

14.1. Verifying the hypotheses of the Nash-Moser theorem. The orientation $\alpha_0 \notin (\pi/2)\mathbb{Z}$ was fixed in [Section 6](#). We now fix $\rho = \rho_*$, the cutoff J , and the neighborhood $\mathfrak{U}_{\text{inv}}$ supplied by [Proposition 13.1](#). We construct the branch for λ near the value $\lambda_* \in (0, 1/2)$ fixed in [Section 6](#).

We choose δ so that the family (2.33) lies in $\mathfrak{U}_{\text{inv}}$, the neighborhood of the constant ellipse $\mathbf{b}_e = (\rho_*, a_e, 0, p_e)$ with $\delta = 0$. Along this family, the distance $\|\mathbf{b} - \mathbf{b}_e\|_{\mathcal{B}^s}$ of [Section 6](#) is the sum of the distance $\|a - a_e\|_{\mathcal{A}_\gamma^s}$ and the parameter distance $|p - p_e|$, and both are small once δ and $|\lambda - \lambda_*|$ are small. Indeed, by (2.32), we have $\alpha_\lambda - \alpha_0 = \delta(\cos \zeta + \lambda \sin 2\zeta)$, where $\cos \zeta$ and $\sin 2\zeta$ are entire, so M_λ and M_λ^{-1} converge to their constant values in every weighted analytic norm as $\delta \rightarrow 0$, uniformly for bounded λ . Since $\mathfrak{U}_{\text{inv}}$ is open, we may therefore choose $\delta_* \neq 0$ and $\eta_0 > 0$ so that every state $\mathbf{b} = (\rho_*, a_{\rho_*, p}^0, 0, p)$ with $\delta = \delta_*$ and $|\lambda - \lambda_*| < 2\eta_0$ belongs to it. We now fix this δ_* .

Using the identifications of [Section 4](#) with the function space at M_{λ_*} , we obtain fixed tame Fréchet spaces E, F and a map

$$(14.1) \quad \widehat{\mathcal{F}} : \mathcal{U} \subset \mathbb{R}_\varepsilon \times \mathbb{R}_\lambda \times E \longrightarrow F, \quad (\varepsilon, \lambda, x) \longmapsto \widehat{\mathcal{F}}(\varepsilon, \lambda, x),$$

which is (5.5) with $\rho = \rho_*$, $\delta = \delta_*$, and α_0 fixed. Here, E and F are the constrained domain and the compatible range of [Section 3](#), graded by $E^s = \mathcal{X}_\gamma^s$ and $F^s = \mathcal{Y}_\gamma^s$, and x is the correction coordinate in (5.1). Reconstructing the chart gives the tangential coefficient τ , the remainder $\tilde{w}$ whose value and first derivatives vanish on the axis, and the scalar correction $s = w - w_{M_\lambda}$. We take $\mathcal{U}$ to consist of the triples $(\varepsilon, \lambda, x)$ in the balls $\mathcal{U}_0$ of [Proposition 5.1](#) whose reconstructed states lie in $\mathfrak{U}_{\text{inv}}$. A zero of $\widehat{\mathcal{F}}$ is a solution of (2.20)–(2.23) with all gauge and boundary conditions imposed. By [Lemma 2.2](#) and (5.6), we have

$$(14.2) \quad \widehat{\mathcal{F}}(0, \lambda, 0) = 0$$

for λ near λ_* .

We check the three hypotheses of the Nash-Moser theorem on an open neighborhood of $(0, \lambda_*, 0)$.

(i) $\widehat{\mathcal{F}}$ is smooth with every derivative tame.

(ii) The derivative at each point,

$$(14.3) \quad \mathcal{L}_\beta := D_x \widehat{\mathcal{F}}(\beta) : E \longrightarrow F, \quad \beta = (\varepsilon, \lambda, x),$$

is an invertible linear map.

(iii) Its inverse

$$(14.4) \quad V_\beta = \mathcal{L}_\beta^{-1} : F \longrightarrow E$$

depends smoothly and tamely on β .

Property (i) follows from [Proposition 5.1\(iii\)](#). Restricting the state to $\rho = \rho_*$ and $p = p(\delta_*, \lambda)$ puts a neighborhood of $(0, \lambda_*, 0)$ in the preimage of $\mathfrak{U}_{\text{inv}}$ and identifies $\mathcal{L}_\beta$ with $L_{\mathbf{b}}$ of (5.7) and V_β with the Cartesian inverse (13.8). Here, $\beta = (\varepsilon, \lambda, x)$ is the chart triple, whereas $\mathbf{b} = (\rho_*, a, \varepsilon, p(\delta_*, \lambda))$ contains the physical

state a reconstructed from x . These operators are related through the fixed-space identifications of [Section 4](#). Thus, (ii) and (iii) are the inverse identities and the tame bound (13.10) of [Theorem 13.2](#).

In (iii), the loss of derivatives may depend on the number of differentiations in β . At each fixed order, the loss must be finite and independent of the Sobolev index. The differentiability argument of [Lemma 12.3](#) applies after restricting to this parameter family. In the present notation, its first identity is

$$(14.5) \quad D_\beta V_\beta[h] = -V_\beta(D_\beta \mathcal{L}_\beta[h])V_\beta$$

and its iterates express $D_\beta^m V_\beta$, up to signs, as a finite sum of products

$$(14.6) \quad V_\beta \mathcal{L}_\beta^{(|I_1|)}[h_{I_1}] V_\beta \cdots \mathcal{L}_\beta^{(|I_j|)}[h_{I_j}] V_\beta$$

over the ordered partitions $I_1, \dots, I_j$ of $\{1, \dots, m\}$. Here, $\mathcal{L}_\beta^{(k)} = D_\beta^k \mathcal{L}_\beta$, and h_{I_ℓ} is the list of directions whose indices belong to I_ℓ . Each set in the partition is nonempty. The difference identity used in [Lemma 12.3](#) justifies these as Fréchet derivatives. The estimate (13.10) gives the required finite loss at each fixed order.

14.2. Existence and smoothness of the solutions. In the normalized chart of [Section 3.1](#), we reconstruct the mapping $\overline{\mathbb{D}^2} \times \mathbb{T} \rightarrow \mathbb{R}^3$ as

$$(14.7) \quad v_{\varepsilon, \lambda}(y, \zeta) = P_{M_\lambda, \tau_{\varepsilon, \lambda}}(\zeta)y + \tilde{v}_{\varepsilon, \lambda}(y, \zeta),$$

$$(14.8) \quad P_{M, \tau} = a(\tau)\iota M + e_y \otimes \tau, \quad a(\tau) = \sqrt{1 - \frac{|\tau|^2}{2}},$$

$$(14.9) \quad j_y^1 \tilde{v}_{\varepsilon, \lambda}|_{y=0} = 0,$$

as in (3.1) and (3.3), with $e_y = e_T$ tangent to the limiting straight axis. The factor $a(\tau)$ ensures that adding the term $e_y \otimes \tau$ preserves the trace normalization (3.2). The condition (14.9) says that $\tilde{v}_{\varepsilon, \lambda}$ and its first derivatives in y vanish on the axis. Thus, the first term in (14.7) is the linear term in the Taylor expansion of $v_{\varepsilon, \lambda}$ at $y = 0$. This linear Taylor term will determine the pressure Hessian on the axis and, through it, the symmetry group.

Theorem 14.1. *There are $\varepsilon_H > 0$ and*

$$0 < \eta_\lambda < \min \left\{ \eta_0, \lambda_*, \frac{1}{2} - \lambda_* \right\}, \quad I = [\lambda_* - \eta_\lambda, \lambda_* + \eta_\lambda] \subset (0, 1/2),$$

such that for $|\varepsilon| < \varepsilon_H$ and $\lambda \in I$, there is a solution

$$(14.10) \quad \widehat{\mathcal{F}}(\varepsilon, \lambda, x_{\varepsilon, \lambda}) = 0, \quad x_{0, \lambda} = 0,$$

unique in the neighborhood of $(0, \lambda_, 0)$ given by the Nash-Moser theorem. The map $(\varepsilon, \lambda) \mapsto x_{\varepsilon, \lambda}$ is smooth with every derivative tame. At every fixed index s and for every fixed $j \geq 0$,*

$$(14.11) \quad \left\| \partial_\lambda^j x_{\varepsilon, \lambda} \right\|_{E^s} \leq C_{s, j} |\varepsilon|, \quad \lambda \in I.$$

After reconstruction in (14.7), we have

$$(14.12) \quad \|\tau_{\varepsilon, \lambda}\|_{C^2(\mathbb{T})} + \|\tilde{v}_{\varepsilon, \lambda}\|_{C^2(\overline{\mathbb{D}^2} \times \mathbb{T})} \leq C_{\text{sol}} |\varepsilon|,$$

uniformly for $\lambda \in I$.

Proof. We include the parameters among both the inputs and outputs, leaving them unchanged. With $q = (\varepsilon, \lambda)$, we set

$$(14.13) \quad \mathcal{G}(q, x) = (q, \widehat{\mathcal{F}}(q, x)).$$

The derivative acts by

$$D\mathcal{G}(q, x)(\dot{q}, \dot{x}) = (\dot{q}, D_q \widehat{\mathcal{F}}(q, x)\dot{q} + \mathcal{L}_{(q, x)}\dot{x}),$$

where $\mathcal{L}_{(q, x)} = D_x \widehat{\mathcal{F}}(q, x)$ is (14.3). Solving first for the parameter variation and then for $\dot{x}$ gives its two-sided inverse

$$(14.14) \quad (D\mathcal{G}(q, x))^{-1}(\dot{q}, \dot{f}) = \left(\dot{q}, V_{(q, x)}[\dot{f} - D_q \widehat{\mathcal{F}}(q, x)\dot{q}] \right).$$

The inverse retains the parameter variation $\dot{q}$ and subtracts its contribution from the residual before applying $V_{(q, x)}$. It is smooth with every derivative tame by (i) and (iii). The Nash-Moser theorem in Hamilton's form [23] therefore applies to $\mathcal{G}$ at $(0, \lambda_*, 0)$, where $\widehat{\mathcal{F}}$ vanishes by (14.2). We evaluate the local inverse at $(q, 0)$, and since $\mathcal{G}$ preserves q , the preimage is $(q, x_{\varepsilon, \lambda})$, which gives (14.10) with $x_{0, \lambda} = 0$ by uniqueness. We then shrink the parameter neighborhood to a rectangle $\{|\varepsilon| < \varepsilon'\} \times \{|\lambda - \lambda_*| < \eta'\}$.

Differentiating (14.10) in ε and using (ii), we obtain

$$(14.15) \quad \partial_\varepsilon x_{\varepsilon, \lambda} = -V_{(\varepsilon, \lambda, x_{\varepsilon, \lambda})} \partial_\varepsilon \widehat{\mathcal{F}}(\varepsilon, \lambda, x_{\varepsilon, \lambda}).$$

At each fixed index, we bound the right-hand side uniformly on a smaller closed rectangle, so the identity

$$x_{\varepsilon, \lambda} = \int_0^\varepsilon \partial_t x_{t, \lambda} \, dt,$$

which uses $x_{0, \lambda} = 0$, gives (14.11) for $j = 0$. For $j \geq 1$, we also differentiate in λ . Since $\partial_\lambda^j x_{0, \lambda} = 0$ and $\partial_\varepsilon \partial_\lambda^j x_{\varepsilon, \lambda}$ is bounded on the same rectangle, integration in ε gives the corresponding estimate. We take ε_H and η_λ to be the half-widths of that rectangle, decreased so that $\eta_\lambda < \min\{\eta_0, \lambda_*, \frac{1}{2} - \lambda_*\}$, which ensures that I is compactly contained in $(0, 1/2)$. Finally, we choose the Sobolev index sufficiently high that each component embeds into C^2 . We reconstruct $\tau_{\varepsilon, \lambda}$ and $\tilde{v}_{\varepsilon, \lambda}$ through (14.7). The chart sends $x = 0$ to $\tau = 0$ and $\tilde{v} = 0$. Its tame bounds, applied at a sufficiently high index, therefore give (14.12) from (14.11). $\square$

Setting $\varepsilon = N^{-1}$ gives a solution for every sufficiently large integer N .

Corollary 14.2. *Let*

$$(14.16) \quad N_{\text{an}} = 1 + \lceil \varepsilon_H^{-1} \rceil.$$

For every integer $N \geq N_{\text{an}}$ and every $\lambda \in I$, the choice $\varepsilon = N^{-1}$ gives an exact 2π -periodic solution of the system on one period cell which satisfies (14.12) with right-hand side C_{sol}/N .

Proof. Any such N exceeds ε_H^{-1} , so $0 < N^{-1} < \varepsilon_H$ and we may substitute $\varepsilon = N^{-1}$ in Theorem 14.1. The estimate holds throughout $|\varepsilon| < \varepsilon_H$, so the conclusion holds for every such N , without passing to a subsequence. $\square$

15. CLOSING THE TORUS AND COMPUTING ITS SYMMETRIES

We join the cells to construct an equilibrium in $\mathbb{R}^3$ and prove [Theorem 1.1](#). We first use tubular coordinates to show that the resulting map is globally injective. We then compute the Hessian of the pressure on the magnetic axis and use it to determine the symmetry group.

We fix the interval I , the branch $(\varepsilon, \lambda) \mapsto x_{\varepsilon, \lambda}$ of [Theorem 14.1](#) and its map $v_{\varepsilon, \lambda} : \overline{\mathbb{D}^2} \times \mathbb{T} \rightarrow \mathbb{R}^3$ of (14.7), and recall that

$$(15.1) \quad 0 < \rho = \rho_* < \frac{1}{4}, \quad \delta = \delta_* \neq 0, \quad \lambda > 0, \quad \alpha_0 \notin (\pi/2)\mathbb{Z},$$

for every $\lambda \in I \subset (0, 1/2)$. These parameters were fixed in [Sections 6, 13 and 14](#). All constants below are uniform on I . We use the C^2 bound (14.12), the smooth tame dependence on the parameters, and the vanishing of the remainder and its first derivatives on the axis in (14.9). Throughout, $|\cdot|$ denotes the Euclidean norm of a vector or the operator norm of a matrix, and for maps on $Q = \overline{\mathbb{D}^2} \times \mathbb{T}$ we take derivatives in (y, ζ) and use the norm $\|f\|_{C^m} = \sup \sum_{j \leq m} |D^j f|$, so that $|D_y f| \leq \|f\|_{C^1}$.

For an integer N with $N^{-1} < \varepsilon_H$, we set

$$(15.2) \quad \varepsilon = N^{-1}, \quad \mathcal{R} = NL, \quad \zeta = N\phi,$$

and define

$$(15.3) \quad X_{N, \lambda}(y, \phi) = R_\phi(\mathcal{R}e_x + v_{\varepsilon, \lambda}(y, N\phi)),$$

$$(15.4) \quad P_{N, \lambda} \circ X_{N, \lambda} = \psi_a - |y|^2,$$

$$(15.5) \quad B_{N, \lambda} \circ X_{N, \lambda} = \partial_\theta X_{N, \lambda} = D_y X_{N, \lambda}(-y_2, y_1),$$

for $y \in \overline{\mathbb{D}^2}$ and $\phi \in \mathbb{T}$. Here, R_ϕ is the rotation about the z -axis through angle ϕ , and ψ_a is the prescribed pressure on the magnetic axis. Since $v_{\varepsilon, \lambda}$ is 2π -periodic in ζ and N is an integer, (15.3) defines a map on Q . The remaining formulas define $P_{N, \lambda}$ and $B_{N, \lambda}$ on $\Omega_{N, \lambda} = X_{N, \lambda}(Q)$ once we have proved injectivity of $X_{N, \lambda}$. The function spaces of [Section 3](#) give the smoothness asserted in [Theorem 1.1](#). We write

$$(15.6) \quad v_0(y, \zeta) = \iota M_\lambda(\zeta)y, \quad q = v_{\varepsilon, \lambda} - v_0,$$

for the unperturbed solution at zero curvature from [Section 2](#) and its correction. By (14.7) and (14.8), we have $q = (a(\tau) - 1)\iota M_\lambda y + (e_y \otimes \tau)y + \tilde{v}$, and since $|a(\tau) - 1| \leq |\tau|^2/2$ for $|\tau| \leq 1$, (14.12) implies

$$(15.7) \quad \|q\|_{C^2(Q)} \leq C\varepsilon.$$

The proof of the following theorem occupies the remainder of this section.

Theorem 15.1. *There is $0 < \varepsilon_{\text{geom}} \leq \varepsilon_H$, uniform for $\lambda \in I$, such that whenever $0 < \varepsilon = N^{-1} < \varepsilon_{\text{geom}}$, the map $X_{N, \lambda}$ is a smooth embedding of a closed solid torus and $(B_{N, \lambda}, P_{N, \lambda})$ is a smooth solution of (1.1) on $\Omega_{N, \lambda}$, including the boundary condition. Its regular pressure levels are nested embedded tori, its field vanishes exactly on the round magnetic axis, and its extended Euclidean stabilizer is C_N . For fixed N , the family depends smoothly and nontrivially on λ , modulo reparametrizations and the action (1.3).*

We begin with the Jacobian and injectivity estimates needed to define the physical fields.

15.1. The Jacobian and the zero set of the magnetic field. We use Cartesian disk coordinates, which remain regular on the axis. We set

$$(15.8) \quad K_\varepsilon(v) = \det(v_{y_1}, v_{y_2}, D_\varepsilon v),$$

where, by (2.11), $D_\varepsilon v = v_\zeta + \varepsilon Av + Le_y$ and $Av = e_z \times v$. At zero curvature and the unperturbed solution, the third column contributes only through Le_y , since $(v_0)_\zeta = \iota M'_\lambda y$ lies in the plane of the first two. With the ordering of ι fixed as in Section 2, where $\det(e_x, e_z, e_y) = -1$, we obtain

$$(15.9) \quad K_0(v_0) = -L \det M_\lambda = -L\sqrt{1 - \rho^2}.$$

We set

$$\begin{aligned} m_0 &= \sup_\zeta |M_\lambda(\zeta)|, & m_1 &= \sup_\zeta |M'_\lambda(\zeta)|, \\ \eta &= \|q\|_{C^1}, & \eta_D &= \eta + \varepsilon(m_0 + \eta), \end{aligned}$$

so that η_D bounds $|D_\varepsilon v - D_0 v_0| = |q_\zeta + \varepsilon A(v_0 + q)|$ on Q . Expanding the determinant multilinearly in $v = v_0 + q$ and bounding each of the seven remaining terms by Hadamard's inequality, we obtain the estimate

$$(15.10) \quad |K_\varepsilon(v) - K_0(v_0)| \leq E(\eta, \varepsilon) := 2m_0(L + m_1)\eta + m_0^2\eta_D + (L + m_1)\eta^2 + 2m_0\eta\eta_D + \eta^2\eta_D.$$

On $r > 0$, we have $\partial_\theta = \mathcal{R}$ and $\partial_\psi = -(2r^2)^{-1}\mathcal{D}$, while $\det(\mathcal{D}v, \mathcal{R}v, W) = r^2 \det(v_{y_1}, v_{y_2}, W)$ for every W , so

$$(15.11) \quad J_\varepsilon = \det(v_\psi, v_\theta, D_\varepsilon v) = -\frac{1}{2}K_\varepsilon(v).$$

Differentiating (15.3) in ϕ and using that A commutes with R_ϕ together with $Ae_x = e_y$ and $\mathcal{R} = NL$, we obtain

$$(15.12) \quad (X_{N,\lambda})_\phi = NR_\phi D_\varepsilon v, \quad \det(X_{y_1}, X_{y_2}, X_\phi) = NK_\varepsilon(v).$$

Since $\eta \leq C\varepsilon$ by (15.7), small ε gives

$$(15.13) \quad E(\eta, \varepsilon) < \frac{L}{2}\sqrt{1 - \rho^2}.$$

Together with $K_0(v_0) = -L\sqrt{1 - \rho^2}$, the bound (15.10) keeps $K_\varepsilon(v)$ below $-\frac{L}{2}\sqrt{1 - \rho^2}$ on Q , so K_ε has a fixed sign. From (15.11), we then obtain

$$(15.14) \quad J_\varepsilon \geq \frac{L}{4}\sqrt{1 - \rho^2} > 0$$

on $r > 0$. By (15.12), the map $X_{N,\lambda}$ is a local diffeomorphism at every point of Q , the axis included.

Since $D_y X_{N,\lambda} = R_\phi D_y v$ with R_ϕ an isometry, $D_y v = \iota M_\lambda + D_y q$ with $|D_y q| \leq \eta$, and M_λ has smallest singular value $\sqrt{1 - \rho}$ by (2.33), we obtain

$$(15.15) \quad |B_{N,\lambda} \circ X_{N,\lambda}| = |D_y v(-y_2, y_1)| \geq (\sqrt{1 - \rho} - \eta)|y|.$$

For small ε , this bound is positive when $y \neq 0$. At $y = 0$, the vector $(-y_2, y_1)$ vanishes. Thus, the zero set of $B_{N,\lambda}$ is exactly the image of the disk axis.

15.2. Global injectivity. To pass from local invertibility to global injectivity, we use tubular coordinates to rule out intersections between different parts of the torus. A tangential displacement changes the angular coordinate by an amount of order its size divided by NL . Multiplying this correction by N therefore leaves a bound independent of N .

We decompose the cell map into its normal and tangential components,

$$v = \nu u + t e_y, \quad u = (v \cdot e_x, v \cdot e_z), \quad t = v \cdot e_y.$$

Here, t is the tangential displacement. The rescaled radial variable used in the linear analysis is no longer needed. We assume for now that $\mathcal{R} > 2 \|v\|_{C^0}$, which holds once ε is small, and set

$$(15.16) \quad h(y, \zeta) = \arctan \frac{t}{\mathcal{R} + u_1},$$

$$(15.17) \quad \nu_1(y, \zeta) = \sqrt{(\mathcal{R} + u_1)^2 + t^2} - \mathcal{R}, \quad \nu_2(y, \zeta) = u_2,$$

$$(15.18) \quad \gamma(y, \zeta) = \zeta + Nh(y, \zeta).$$

The assumption gives $\mathcal{R} + u_1 > \mathcal{R}/2 > 0$ and $|t| < \mathcal{R}/2$, so h is well defined, with values in $(-\pi/4, \pi/4)$. In the cylindrical frame $e_r(\beta) = R_\beta e_x$, $e_\phi(\beta) = R_\beta e_y$ of $\mathbb{R}^3$, we have $(\mathcal{R} + u_1)e_x + te_y = (\mathcal{R} + \nu_1)e_r(h)$, which rewrites (15.3) as

$$(15.19) \quad X_{N,\lambda} = (\mathcal{R} + \nu_1)e_r(\beta) + \nu_2 e_z, \quad \beta = \phi + h(y, N\phi).$$

On the universal cover $\overline{\mathbb{D}^2} \times \mathbb{R}$ in ζ , we introduce the map

$$(15.20) \quad \Psi(y, \zeta) = (M_\lambda(\gamma(y, \zeta))^{-1} \nu(y, \zeta), \gamma(y, \zeta)).$$

The map Ψ records the corrected cell angle $\gamma = N\beta$ and the normal position expressed in the ellipse coordinates at that angle. At $q = 0$, we have $t = h = 0$, $\nu = M_\lambda(\zeta)y$, and $\Psi(y, \zeta) = (y, \zeta)$, while in general $\nu_1 = u_1 + O(t^2/\mathcal{R})$, $\nu_2 = u_2$, and $|h| \leq 2|t|/\mathcal{R}$, where $t = q \cdot e_y$. Since $N/\mathcal{R} = L^{-1}$ by (15.2), the function Nh and its first derivatives are bounded by a constant times $\|q\|_{C^1}$, and

$$\|\nu - M_\lambda(\zeta)y\|_{C^1} + \|\gamma - \zeta\|_{C^1} \leq C \|q\|_{C^1}$$

holds with C independent of N . The formula (2.33) bounds M_λ in C^2 , uniformly for $\lambda \in I$, and $\det M_\lambda = \sqrt{1 - \rho^2} > 0$. Thus, M_λ^{-1} has the same uniform regularity. Composing with it, we obtain a constant C_{tub} , again independent of N , with

$$(15.21) \quad \|D\Psi - I\|_{C^0(\overline{\mathbb{D}^2} \times \mathbb{R})} \leq C_{\text{tub}} \|q\|_{C^1}.$$

Once the right-hand side is below one, we obtain injectivity of Ψ . Indeed, the domain is convex, so that $\Psi(p) - \Psi(p') = (p - p') + \int_0^1 (D\Psi - I)(p' + s(p - p'))(p - p') ds$, whence

$$(15.22) \quad |\Psi(p) - \Psi(p')| \geq (1 - C_{\text{tub}} \|q\|_{C^1}) |p - p'|.$$

Suppose $X_{N,\lambda}(y, \phi) = X_{N,\lambda}(y', \phi')$. Comparing cylindrical coordinates in (15.19) and using that $\mathcal{R} + \nu_1 > 0$, we find $\nu = \nu'$ and $\beta \equiv \beta'$ modulo 2π . Since h is 2π -periodic in ζ , shifting the lift of ϕ by 2π shifts β by 2π , so we may take $\beta = \beta'$, and then $\gamma = N\beta$ and $\gamma' = N\beta'$ agree as well. Now, (15.20)–(15.22) give $(y, \zeta) = (y', \zeta')$, so the two points of Q coincide. We therefore have an injective immersion of a compact manifold with boundary, hence a homeomorphism onto its image and a smooth embedding of Q .

15.3. Reconstruction of the MHS equations. On $r > 0$, we use the flux coordinates (ψ, θ, ϕ) with $\psi = \psi_a - r^2$, in which (15.12) gives

$$(15.23) \quad J_{N,\lambda} := \det(X_\psi, X_\theta, X_\phi) = NJ_\varepsilon,$$

so that (2.6) can be written as

$$(15.24) \quad (\operatorname{div} B_{N,\lambda}) \circ X_{N,\lambda} = \frac{1}{NJ_\varepsilon} \partial_\theta (NJ_\varepsilon).$$

Writing

$$Q_{N,\lambda} = B_{N,\lambda} \times \operatorname{curl} B_{N,\lambda} + \nabla P_{N,\lambda}$$

for the force residual and using $X_\phi = NR_\phi D_\varepsilon v$, we may write (2.1)–(2.3) as

$$(15.25) \quad (Q_{N,\lambda} \circ X_{N,\lambda}) \cdot X_\theta = 0,$$

$$(15.26) \quad (Q_{N,\lambda} \circ X_{N,\lambda}) \cdot X_\psi = -F_{1,\varepsilon},$$

$$(15.27) \quad (Q_{N,\lambda} \circ X_{N,\lambda}) \cdot X_\phi = NF_{2,\varepsilon}.$$

The solution of $\widehat{\mathcal{F}} = 0$ from Section 14 is a zero of the operator H_ε of (2.20)–(2.23), with all gauge and boundary conditions imposed. The pair $(v_{\varepsilon,\lambda}, w)$ reconstructed in (3.3), with scalar component w , is smooth in y , and (2.24) recovers the integrated system on $r > 0$. Thus, Lemma 2.1 gives

$$F_{1,\varepsilon} = F_{2,\varepsilon} = 0, \quad \partial_\theta J_\varepsilon = 0,$$

on $0 < r \leq 1$. There, X_ψ, X_θ, X_ϕ span $\mathbb{R}^3$ by (15.14) and (15.23), so (15.25)–(15.27) force $Q_{N,\lambda} = 0$ and (15.24) forces $\operatorname{div} B_{N,\lambda} = 0$. We have thus proved that

$$(15.28) \quad B_{N,\lambda} \times \operatorname{curl} B_{N,\lambda} + \nabla P_{N,\lambda} = 0, \quad \operatorname{div} B_{N,\lambda} = 0,$$

away from the axis. Both identities extend across it. Indeed, the right-hand sides of (15.4) and (15.5) are smooth on Q and $X_{N,\lambda}$ is a diffeomorphism onto $\Omega_{N,\lambda}$, so both expressions in (15.28) are continuous and vanish on the dense complement of the axis. Finally, $(-y_2, y_1)$ is tangent to $\{|y| = 1\}$, so (15.5) gives

$$(15.29) \quad B_{N,\lambda} \cdot n = 0 \quad \text{on } \partial\Omega_{N,\lambda}, \quad \Omega_{N,\lambda} = X_{N,\lambda}(Q).$$

15.4. Pressure surfaces and the Hessian on the axis. Since $X_{N,\lambda}$ is an embedding, each set

$$(15.30) \quad X_{N,\lambda}(\{|y| = r\} \times \mathbb{T}), \quad 0 < r \leq 1,$$

is an embedded torus, and by (15.4) it is the pressure level $\{P_{N,\lambda} = \psi_a - r^2\}$. These tori foliate $\Omega_{N,\lambda}$ away from the axis and collapse smoothly to

$$(15.31) \quad \Gamma_N(\phi) = X_{N,\lambda}(0, \phi) = \mathcal{R}e_r(\phi),$$

a round circle of radius $\mathcal{R} = NL$ in the plane $z = 0$, since (14.7) and (14.9) give $v_{\varepsilon,\lambda}(0, \zeta) = 0$. Since $d(\psi_a - |y|^2) = -2y \cdot dy$, the critical set of the pressure is exactly Γ_N , as is the zero set of the field by (15.15).

The pressure Hessian on the axis determines the orientation of the cross-sections. We compute it using the linear Taylor term $P_{M_\lambda, \tau} y$ in (14.7), then divide its inverse by its trace to remove the scalar factor involving τ . We fix $p = \Gamma_N(\phi)$ and let H_ϕ be the Euclidean Hessian of $P_{N,\lambda}$ restricted to the plane normal to Γ_N , written in the orthonormal basis $(e_r(\phi), e_z)$ of that plane.

By (14.7) and (14.9), the remainder $\tilde{v}$ contributes neither a value nor a first derivative at $y = 0$, so $D_y v(0, \zeta) = P_{M_\lambda, \tau}(\zeta)$, and the same normalization gives $D_\varepsilon v(0, \zeta) = L e_y$. Hence, on the axis, we have

$$(15.32) \quad D_y X_{N, \lambda} = R_\phi(a(\tau)\iota M_\lambda + e_y \otimes \tau), \quad X_\phi = \mathcal{R}e_\phi(\phi),$$

where each function of ζ is evaluated at $N\phi$.

Let $w = q_0^1 e_r(\phi) + q_0^2 e_z = R_\phi \iota q_0$ be a normal vector at p , with coordinates $q_0 \in \mathbb{R}^2$ in the basis above. We realize w as a derivative of $X_{N, \lambda}$ by combining a disk variation with an angle variation, since the first identity in (15.32) alone also produces a tangential component. For $\xi \in \mathbb{R}^2$ and $s \in \mathbb{R}$, we have

$$D_y X_{N, \lambda} \xi + s X_\phi = R_\phi(a(\tau)\iota M_\lambda \xi + (\tau \cdot \xi + s \mathcal{R})e_y),$$

so the choice

$$\xi = (a(\tau)M_\lambda)^{-1}q_0, \quad s = -\frac{\tau \cdot \xi}{\mathcal{R}}$$

gives w . For two normal vectors w, w' at p , we let $Z = (\xi, s)$ and $Z' = (\xi', s')$ be the corresponding directions in (y, ϕ) , and we set $F = P_{N, \lambda} \circ X_{N, \lambda} = \psi_a - |y|^2$. Differentiating twice, we obtain

$$\partial_Z \partial_{Z'} F = \text{Hess } P_{N, \lambda}(w, w') + dP_{N, \lambda}(\partial_Z \partial_{Z'} X_{N, \lambda}),$$

where the second term vanishes because $dP_{N, \lambda} = 0$ on the axis, while F is a quadratic function of y alone, so that $\partial_Z \partial_{Z'} F = -2\xi \cdot \xi'$. Therefore, we have $\text{Hess } P_{N, \lambda}(w, w') = -2a(\tau)^{-2}(M_\lambda^{-1}q_0) \cdot (M_\lambda^{-1}q'_0)$. In the basis $(e_r(\phi), e_z)$, this can be written as

$$(15.33) \quad H_\phi = -2a(\tau)^{-2}M_\lambda^{-T}M_\lambda^{-1}.$$

We normalize the inverse Hessian to remove $a(\tau)^{-2}$. Since $\text{tr}(M_\lambda^T M_\lambda) = 2$ by (2.34), we have $(-H_\phi)^{-1} = \frac{1}{2}a(\tau)^2 M_\lambda M_\lambda^T$ and $\text{tr}((-H_\phi)^{-1}) = a(\tau)^2$, so we obtain the *normalized inverse Hessian*

$$(15.34) \quad \mathcal{S}_\phi := \frac{(-H_\phi)^{-1}}{\text{tr}((-H_\phi)^{-1})} = \frac{1}{2}M_\lambda(N\phi)M_\lambda(N\phi)^T.$$

Thus, $\mathcal{S}_\phi$ is independent of τ and $\tilde{v}$. It is also unchanged when $P_{N, \lambda}$ is replaced by $\kappa^2 P_{N, \lambda} + C$, which multiplies H_ϕ by κ^2 . Since M_λ is symmetric, (2.33) gives

$$\mathcal{S}_\phi = \frac{1}{2}R_{\alpha_\lambda(N\phi)} \begin{pmatrix} 1 + \rho & 0 \\ 0 & 1 - \rho \end{pmatrix} R_{-\alpha_\lambda(N\phi)},$$

whose eigenvalues $(1 \pm \rho)/2$ are distinct because $\rho = \rho_* > 0$. The eigenline of the larger eigenvalue is therefore well defined, with angle $\alpha_\lambda(N\phi)$ modulo π in the moving frame $(e_r(\phi), e_z)$.

15.5. Computing the stabilizer. We work with the extended stabilizer $\text{Stab}_\pm$ of (1.4), whose elements preserve $\Omega_{N, \lambda}$ and $P_{N, \lambda}$ and send $B_{N, \lambda}$ to $\pm B_{N, \lambda}$. The argument uses the pressure alone and therefore treats both signs at once.

Let $g \in \text{Stab}_\pm(\Omega_{N, \lambda}, B_{N, \lambda}, P_{N, \lambda})$. Since $P_{N, \lambda} \circ g^{-1} = P_{N, \lambda}$, the map g preserves the critical set of the pressure, which by Section 15.4 is the round circle Γ_N . An isometry preserving a round circle fixes its center, which here is the origin, so g is linear. It preserves the affine plane of the circle, giving $g e_z = \chi e_z$ for $\chi \in \{\pm 1\}$, and on that plane it is orthogonal, so that in the angular variable g acts by

$$(15.35) \quad \phi \mapsto \sigma \phi + c, \quad e_z \mapsto \chi e_z, \quad \sigma, \chi \in \{\pm 1\}.$$

We now use (15.35) to obtain a condition on α_λ . Since $g\Gamma_N(\phi) = \Gamma_N(\sigma\phi + c)$, we have $ge_r(\phi) = e_r(\sigma\phi + c)$, so g carries the normal plane at $\Gamma_N(\phi)$ to the normal plane at $\Gamma_N(\sigma\phi + c)$. In the bases $(e_r(\phi), e_z)$ and $(e_r(\sigma\phi + c), e_z)$, it acts by $D = \text{diag}(1, \chi)$. We obtain $H_{\sigma\phi+c} = DH_\phi D$ from the invariance of the pressure. Since $\text{tr}(DSD) = \text{tr}(S)$, the normalization in (15.34) is preserved by conjugation with D , giving $\mathcal{S}_{\sigma\phi+c} = D\mathcal{S}_\phi D$. Finally, $DR_\alpha D = R_{\chi\alpha}$, so conjugation by D replaces α_λ by $\chi\alpha_\lambda$ in (15.34). Since the eigenvalues are distinct, equality of these tensors is equivalent to equality of their principal eigenlines. With $\zeta = N\phi$ and $c_0 = Nc$, this gives

$$(15.36) \quad \alpha_\lambda(\sigma\zeta + c_0) \equiv \chi\alpha_\lambda(\zeta) \pmod{\pi}.$$

The following lemma determines when this relation can hold. Recall from (2.32) that

$$\alpha_\lambda(\zeta) = \alpha_0 + \delta_*(\cos \zeta + \lambda \sin 2\zeta).$$

We allow two values of λ so that the lemma also applies when comparing distinct equilibria.

Lemma 15.2. *Let $\lambda, \lambda' \in I$, let $\sigma, \chi \in \{\pm 1\}$, and let $c_0 \in \mathbb{R}$. If*

$$\alpha_{\lambda'}(\sigma\zeta + c_0) \equiv \chi\alpha_\lambda(\zeta) \pmod{\pi} \quad \text{for every } \zeta \in \mathbb{R},$$

then $\chi = 1$, $c_0 \in 2\pi\mathbb{Z}$, $\sigma = 1$, and $\lambda' = \lambda$.

Proof. The difference of the two sides is continuous and takes values in $\pi\mathbb{Z}$, so it is a constant πk , $k \in \mathbb{Z}$. Averaging over a period, we are left with $(1 - \chi)\alpha_0 = \pi k$, since the two terms in the brackets in (2.32) have zero mean. If $\chi = -1$, this says that $2\alpha_0 \in \pi\mathbb{Z}$, which (15.1) excludes. Hence, $\chi = 1$, and then $k = 0$.

With $\chi = 1$ and $k = 0$, division by $\delta \neq 0$ turns the hypothesis into

$$[\cos(\sigma\zeta + c_0) - \cos \zeta] + [\lambda' \sin(2\sigma\zeta + 2c_0) - \lambda \sin 2\zeta] = 0.$$

The terms of frequency one and frequency two must vanish separately. The first gives $\cos(\sigma\zeta + c_0) = \cos \zeta$ for every ζ , which for either sign of σ forces $c_0 \in 2\pi\mathbb{Z}$. The second then reads as $\lambda'\sigma \sin 2\zeta = \lambda \sin 2\zeta$, that is, $\lambda' = \sigma\lambda$. Since both parameters are positive, $\sigma = 1$ and $\lambda' = \lambda$. $\square$

We apply Lemma 15.2 to (15.36), whose two parameters are equal, and obtain

$$(15.37) \quad \sigma = \chi = 1, \quad c = \frac{2\pi j}{N},$$

for some integer j , since $c_0 = Nc$. Thus, $g = R_{2\pi j/N}$. Conversely, these rotations preserve both the pressure and the field. Indeed, $v_{\varepsilon, \lambda}$ is 2π -periodic in ζ , so (15.3) gives $X_{N, \lambda}(y, \phi + 2\pi/N) = R_{2\pi/N} X_{N, \lambda}(y, \phi)$. Moreover, the right-hand side of (15.4) does not depend on ϕ , and ∂_θ commutes with $R_{2\pi/N}$, so that (15.5) gives $(R_{2\pi/N})_* B_{N, \lambda} = B_{N, \lambda}$. Hence, we obtain

$$(15.38) \quad \text{Stab}_\pm(\Omega_{N, \lambda}, B_{N, \lambda}, P_{N, \lambda}) = C_N.$$

15.6. Distinct equilibria and non-isolation. We fix N and suppose that the equilibria at $\lambda, \lambda' \in I$ are equivalent under (1.3). Reparametrization leaves the physical fields and $\mathcal{S}_\phi$ unchanged, so it suffices to consider the similarity, amplitude rescaling, and pressure shift.

Such an operation maps the critical set of one pressure to that of the other, and by (15.31) both critical circles have radius NL , so the scale factor is one and we are left with an isometry g . As in Section 15.5, g acts on the axis by (15.35). The normalized inverse Hessian $\mathcal{S}_\phi$ in (15.34) is unchanged by the amplitude rescaling and the pressure shift. Applying the transformation rule from Section 15.5 gives

$$(15.39) \quad \alpha_{\lambda'}(\sigma\zeta + c_0) \equiv \chi\alpha_\lambda(\zeta) \pmod{\pi},$$

with $c_0 = Nc$ as before. Hence, Lemma 15.2 gives $\lambda' = \lambda$, so that distinct parameters give inequivalent equilibria under (1.3).

We recall the topology fixed in Section 1.1. For an integer $k \geq 3$, a configuration is a triple

$$(15.40) \quad (X, b, p) \in \text{Emb}^k(Q, \mathbb{R}^3) \times C^{k-1}(Q, \mathbb{R}^3) \times C^k(Q), \quad b = B \circ X, \quad p = P \circ X,$$

and we identify two configurations differing by a reparametrization

$$(15.41) \quad (X, b, p) \mapsto (X \circ \Phi, b \circ \Phi, p \circ \Phi), \quad \Phi \in \text{Diff}^k(Q),$$

or by (1.3). We write $\mathcal{M}_{\text{emb}}^k(Q)$ for the quotient, in which the embedded domain is part of the data, so the domains may move with λ .

By Theorem 14.1, for each $\varepsilon = N^{-1}$ the solution depends smoothly on λ and so does the reconstruction (15.3)–(15.5). For each k , derivatives of order at most k of W_γ^{-1} are bounded by a polynomial in $\langle n \rangle$ times $e^{-(\sigma_0 - \gamma)\langle n \rangle}$. Thus, removal of the weight is continuous into every fixed Sobolev space. Sobolev embedding controls each C^k norm by a sufficiently high Sobolev index. Composition with $\zeta = N\phi$ is continuous for fixed N . Hence, the curve

$$\lambda \mapsto [X_{N,\lambda}, B_{N,\lambda} \circ X_{N,\lambda}, P_{N,\lambda} \circ X_{N,\lambda}] \in \mathcal{M}_{\text{emb}}^k(Q)$$

is continuous, also for the smooth quotient topology of Section 1.1. It is injective by (15.39). For any interior $\lambda \in I$, choose distinct $\lambda_j \in I$ with $\lambda_j \rightarrow \lambda$. The corresponding classes are distinct by injectivity and converge to the class at λ by continuity. Thus, this equilibrium is not isolated in $\mathcal{M}_{\text{emb}}^k(Q)$.

15.7. Proof of the main theorem. We complete the proof of Theorem 15.1 and then deduce Theorem 1.1.

Proof of Theorem 15.1. The preceding arguments required the four strict inequalities

$$E(\eta, \varepsilon) < \frac{L}{2} \sqrt{1 - \rho^2}, \quad \eta < \sqrt{1 - \rho}, \quad \mathcal{R} > 2 \|v\|_{C^0}, \quad C_{\text{tub}} \eta < 1,$$

respectively in (15.13), (15.15), the construction of (15.16)–(15.17), and (15.22). All four hold after one uniform decrease of ε , since $\eta \leq C\varepsilon$ by (15.7), the constants m_0 , m_1 , and C_{tub} are independent of N , and $\|v\|_{C^0} \leq m_0 + \eta$ while $\mathcal{R} = L/\varepsilon$. We let $\varepsilon_{\text{geom}} \leq \varepsilon_H$ be the resulting threshold, uniform on I because every constant entering it is. Then, Sections 15.1 and 15.2 give the embedding of the closed solid torus, and Section 15.3 gives (15.28) with (15.29). Moreover, Section 15.4 gives the nested tori (15.30) and the round axis (15.31), on which the field vanishes and nowhere else by (15.15). We computed the stabilizer in (15.38). Finally, the branch is smooth in λ by Theorem 14.1, while (15.39) and Lemma 15.2 make the λ -family nonconstant modulo reparametrizations, similarities, and the amplitude action. $\square$

We set

$$(15.42) \quad N_0 = 1 + \lceil \max\{2, \varepsilon_H^{-1}, \varepsilon_{\text{geom}}^{-1}\} \rceil.$$

Since $\varepsilon_{\text{geom}} \leq \varepsilon_H$, this is at least the N_{an} of (14.16), so Corollary 14.2 and Theorem 15.1 apply to every integer $N \geq N_0$.

Proof of Theorem 1.1. The orientation α_0 and the ellipse parameter ρ_* were fixed in Sections 6 and 13, the amplitude δ_* was fixed in Section 14, and the interval $I \subset (0, 1/2)$ was fixed in Theorem 14.1. The integer N_0 is defined by (15.42). For $N \geq N_0$ and $\lambda \in I$, Corollary 14.2 permits $\varepsilon = N^{-1}$, and Theorem 15.1 supplies the embedding $X_{N,\lambda}$ and the smooth fields on $\Omega_{N,\lambda}$, which solve (1.1) by (15.28) and (15.29). Equations (15.30) and (15.31) give the nested pressure tori and the round axis (1.5), proving (i). The field vanishes precisely on this axis by (15.15) and (15.5), which proves (ii). Its extended stabilizer is exactly the group of rotations $R_{2\pi j/N}$ by (15.37) and (15.38). This proves (iii), including the absence of continuous symmetries and orientation-reversing elements. Finally, Theorem 14.1 gives smooth dependence on λ , and Section 15.6 proves that the curve (1.8) is continuous and injective in the quotient. Thus, the equilibria form a nontrivial family and are not isolated at interior parameter values, proving (iv). $\square$

16. LEAN FORMALIZATION

The main result of this paper, Theorem 1.1, has been verified in Lean 4 using its mathematical library Mathlib. The objective of this section is to explain the correspondence between the mathematical statement and its Lean formulation without assuming broad familiarity with Lean. We repeat the relevant mathematical definitions alongside their formal counterparts. The Lean code is available at the accompanying repository:

<https://github.com/lukasliehr/Grad-Conjecture>.

In this repository, the file `Showcase.lean` contains the main theorem in Lean. It is self-contained, imports only Mathlib, and defines every additional notion needed to state the result. The companion file `Showcase_WithProofs.lean` proves the same statements using the verified library. The distinction between the presentation and the proofs is explained in Subsection 16.9.

16.1. Main theorem. In this section, we recall the main theorem of the present paper and state its corresponding formalization in Lean. To do so, we recall the reference domain and the representatives used to compare equilibria on different embedded domains:

$$\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z}), \quad Q = \overline{\mathbb{D}^2} \times \mathbb{T}, \quad (X, b, p) = (X, B \circ X, P \circ X).$$

Here, $X : Q \rightarrow \mathbb{R}^3$ is an embedding and $\Omega = X(Q)$ is the closed solid torus. The notation $\mathcal{M}_{\text{emb}}^k(Q)$ denotes the space of such configurations modulo reference reparametrizations, spatial similarities, magnetic rescaling, and pressure shifts, with the quotient topology. Their corresponding Lean definitions are spelled out and explained in Section 16.3. The theorem we verified in Lean is the following (see also Theorem 1.1).

Theorem (Equilibria with exact cyclic symmetry). Fix $L > 0$. There exist $0 < a < b < 1/2$, an integer $N_0 \geq 1$, and representatives $(X_{N,\lambda}, b_{N,\lambda}, p_{N,\lambda})$ for $N \geq N_0$ and $\lambda \in I = [a, b]$ with the following

properties. Each $X_{N,\lambda}$ is a smooth embedding and there are fields $B_{N,\lambda}, P_{N,\lambda}$, smooth on a neighborhood of $\Omega_{N,\lambda} = X_{N,\lambda}(Q)$, whose pullbacks are $b_{N,\lambda}, p_{N,\lambda}$, and which satisfy

$$B \times \text{curl } B + \nabla P = 0, \quad \text{div } B = 0 \quad \text{in } \text{int } \Omega, \quad B \cdot n = 0 \quad \text{on } \partial\Omega.$$

Suppressing N, λ on the physical fields and domain, one has:

- (i) The critical set of P in Ω is the circle $\Gamma_N = \{NL(\cos \phi, \sin \phi, 0) : \phi \in \mathbb{T}\} \subset \text{int } \Omega$. Every nonempty regular pressure level is an embedded torus, and these levels smoothly foliate $\Omega \setminus \Gamma_N$.
- (ii) For $x \in \Omega$, one has $B(x) = 0$ if and only if $x \in \Gamma_N$.
- (iii) The extended Euclidean stabilizer, allowing a global reversal of B , consists exactly of the rotations $R_{2\pi j/N}$, $j = 0, \dots, N-1$, about the third coordinate axis.
- (iv) For each fixed $N \geq N_0$, the representatives depend smoothly on λ . The curve of their classes in $\mathcal{M}_{\text{emb}}^k(Q)$ is continuous and injective for every finite $k \geq 3$ and for $k = \infty$. Every member with $\lambda \in (a, b)$ is non-isolated in that moduli space.

The interval I and threshold N_0 are common to all four conclusions. The Lean statement also makes explicit the boundary regularity and the boundary leaf of the pressure foliation. We use the closed body $X(Q)$ throughout; the differential equations are imposed on its interior.

The formalization in Lean of the latter theorem takes the following form:

```

theorem equilibria_with_exact_cyclic_symmetry (L : ℝ) (hL : 0 < L) :
  ∃ a b : ℝ, 0 < a ∧ a < b ∧ b < 1 / 2 ∧
    ∃ N₀ : ℕ, 1 ≤ N₀ ∧ ∃ family : ℕ → Set.Icc a b → Representative,
      ∀ N : ℕ, N₀ ≤ N →
        SmoothFamily (Set.Icc a b) (family N) ∧
        (∀ lambda : Set.Icc a b, Equilibrium L N (family N lambda)) ∧
        (∀ k : ℕ, 3 ≤ k → ModuliCurve (.finite k) (Set.Icc a b) (family N)) ∧
        ModuliCurve .smooth (Set.Icc a b) (family N)

```

In the following subsections we provide a precise description of all clauses in the previous Lean theorem.

16.2. The reference domain and regularity. We recall that Lean records the type of each object. For example, writing $L : \mathbb{R}$ means mathematically that $L \in \mathbb{R}$, and $x : A \rightarrow B$ denotes a map X from A to B . A definition is introduced by **def** or **abbrev**. For reading the following Lean statements, an abbreviation can be treated as a definition. We begin with the types corresponding to $\mathbb{R}^3$, $\mathbb{R}^2$, $\mathbb{T}$, $\overline{\mathbb{D}^2}$, Q , and $\mathbb{T}^2$. In Lean, we define them as follows:

```

abbrev Vec := EuclideanSpace ℝ (Fin 3)
abbrev Plane := EuclideanSpace ℝ (Fin 2)
abbrev PeriodicCircle := AddCircle (2 * Real.pi)
abbrev ClosedDisk := {x : Plane // ||x|| ≤ 1}
abbrev ReferenceDomain := ClosedDisk × PeriodicCircle
abbrev Torus := PeriodicCircle × PeriodicCircle

```

The type `EuclideanSpace ℝ (Fin 3)` represents $\mathbb{R}^3$ endowed with its Euclidean norm, where `Fin 3` is the index set $\{0, 1, 2\}$. The type `AddCircle (2 * Real.pi)` is the quotient $\mathbb{R}/(2\pi\mathbb{Z})$. The notation `x : Plane // ||x|| ≤ 1` restricts the points of the plane to those satisfying $\|x\| \leq 1$. Thus, `ClosedDisk` is precisely the closed unit disk.

For derivatives on Q , we use the covering cylinder $C = \overline{\mathbb{D}^2} \times \mathbb{R}$ and the projection $\pi_C : C \rightarrow Q$ defined by $\pi_C(y, z) = (y, [z])$, where $[z]$ denotes the class of z in $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$. The projection onto the first two coordinates and the cylinder C are defined by

```
def planarPart (x : Vec) : Plane := WithLp.toLp 2 ![x 0, x 1]
def cylinder : Set Vec := {x | ||planarPart x|| ≤ 1}
```

In the above code, `x 0`, `x 1`, and `x 2` are the three coordinates of $x \in \mathbb{R}^3$. The expression `![x 0, x 1]` collects its first two coordinates, and `WithLp.toLp 2` regards this pair as an element of the Euclidean plane. Thus, `planarPart` is the map $(y, z) \mapsto y$, and `cylinder` is precisely the set $\{(y, z) : \|y\| \leq 1\}$. The covering projection π_C is represented in Lean by

```
def quotientPoint (x : Vec) (hx : x ∈ cylinder) :
  ReferenceDomain := (⟨planarPart x, hx⟩, (x 2 : PeriodicCircle))
```

The argument `hx` is a proof that $x \in C$, or equivalently that $\|\text{planarPart}(x)\| \leq 1$. The expression `⟨planarPart x, hx⟩` therefore gives a point of `ClosedDisk`. The second component, `(x 2 : PeriodicCircle)`, takes the third coordinate modulo 2π . Together, these components give a point of $Q = \overline{\mathbb{D}^2} \times \mathbb{T}$.

For $f : Q \rightarrow E$, where E has a specified zero element, define

$$\tilde{f}(x) = \begin{cases} f(\pi_C(x)), & x \in C, \\ 0, & x \notin C. \end{cases}$$

Thus, $\tilde{f}$ is a function on $\mathbb{R}^3$ whose restriction to C is the lift of f . In Lean, we define the periodic lift as follows:

```
def periodicLift {E : Type*} [Zero E]
  (f : ReferenceDomain → E) (x : Vec) : E := by
  classical
  exact if hx : x ∈ cylinder then
    f (quotientPoint x hx)
  else
    0
```

The notation `E : Type*` allows the target E to be inferred from f , and `[Zero E]` specifies its zero element. The keywords `by` and `exact` introduce a construction and give its resulting value. The command `classical` permits the use of classical logic. Here it allows the case distinction $x \in C$ or $x \notin C$, without requiring a procedure/algorithm that decides membership in C . In `if hx : x ∈ cylinder then`, the name `hx` records the proof of membership in the first case. This proof is passed to `quotientPoint`, since $\pi_C(x)$ is defined only for $x \in C$. The two branches therefore give exactly the two values in the displayed formula. The values outside C serve only to represent the lift as a function on all of $\mathbb{R}^3$. The zero extension need not be continuous across the boundary of C . Regularity on Q is expressed through local extensions of $\tilde{f}|_C$ to open neighborhoods in

$\mathbb{R}^3$, with the required differentiability. Derivatives are taken within C and therefore do not depend on the assigned exterior values.

We encode the regularity of a configuration by a type `Regularity`, whose values are `finite k`, for $k \in \mathbb{N}$, and `smooth`. These represent C^k and C^∞ regularity, respectively. The keyword `inductive` declares the type by listing these alternatives and each line beginning with `|` introduces one of them.

```
inductive Regularity where
```

```
  | finite (k :  $\mathbb{N}$ )
  | smooth
```

We associate two operations with these regularity labels. The first assigns the differentiability order used by Mathlib. The second lowers finite regularity by one derivative, as needed for the magnetic component of a configuration, while preserving smooth regularity.

```
def Regularity.order : Regularity  $\rightarrow$   $\mathbb{N}\infty\omega$ 
```

```
  | .finite k  $\Rightarrow$  k
  | .smooth  $\Rightarrow$   $\infty$ 
```

```
def Regularity.predecessor : Regularity  $\rightarrow$  Regularity
```

```
  | .finite k  $\Rightarrow$  .finite (k - 1)
  | .smooth  $\Rightarrow$  .smooth
```

Both functions are defined by cases: each `|` introduces a possible input, and `$\Rightarrow$` specifies the corresponding output. The notation `.finite k` abbreviates `Regularity.finite k`, and `.smooth` abbreviates `Regularity.smooth`. The type $\mathbb{N}\infty\omega$ contains the natural differentiability orders, the smooth order ∞ , and an additional order ω for analyticity. Only the natural orders and ∞ are used here. Thus, `Regularity.order` sends `.finite k` to k and `.smooth` to ∞ . The operation `Regularity.predecessor` replaces C^k by C^{k-1} for $k \geq 1$ and leaves C^∞ unchanged. At $k = 0$ it returns C^0 , since subtraction in $\mathbb{N}$ is truncated at zero.

Let E, F be real normed vector spaces, let $S \subset E$, and let $f : E \rightarrow F$. We express regularity on S through local extensions: for every $x \in S$, there must be an open neighborhood U of x and a C^r map g on U such that $g(y) = f(y)$ for all $y \in U \cap S$. In our application, S is the closed covering cylinder C . This formulation includes regularity at its boundary. The corresponding Lean definition is

```
def HasLocalExtensions {E F : Type*
```

```
  [NormedAddCommGroup E] [NormedSpace  $\mathbb{R}$  E]
```

```
  [NormedAddCommGroup F] [NormedSpace  $\mathbb{R}$  F]
```

```
  (r : Regularity) (f : E  $\rightarrow$  F) (S : Set E) : Prop :=
```

```
     $\forall$  x  $\in$  S,  $\exists$  U : Set E, IsOpen U  $\wedge$  x  $\in$  U  $\wedge$   $\exists$  g : E  $\rightarrow$  F,
    ContDiffOn  $\mathbb{R}$  r.order g U  $\wedge$  EqOn g f (U  $\cap$  S)
```

The first lines specify the spaces and their structures. The declaration `NormedAddCommGroup E` provides addition, a zero element, additive inverses, and a norm on E ; addition is commutative, and the distance is given by $\|x - y\|$. The declaration `NormedSpace  $\mathbb{R}$  E` provides real scalar multiplication compatible with these operations and with the norm. Together, these declarations express that E is a real normed vector space. The corresponding declarations for F have the same meaning. The remaining arguments specify the regularity

label r , the function f , and the set S . The notation $S : \text{Set } E$ means that S is a subset of E , and $: \text{Prop}$ indicates that $\text{HasLocalExtensions } r \ f \ S$ is a mathematical assertion about these data. The body of the definition follows the stated extension condition. The symbols $\forall$, $\exists$, and $\wedge$ have the usual meaning. Thus, $\text{IsOpen } U \wedge x \in U$ says that U is an open neighborhood of x . The expression $\text{ContDiffOn } \mathbb{R} \ r.\text{order } g \ U$ states that g is C^r on U , with derivatives taken over $\mathbb{R}$, and $\text{EqOn } g \ f \ (U \cap S)$ states that $g = f$ on $U \cap S$. Although g is represented as a function on all of E , its regularity is required only on U .

For a function $f : Q \rightarrow E$, we apply this condition to its periodic lift $\tilde{f}$ on the cylinder C :

```
def HasRegularity {E : Type*}
  [NormedAddCommGroup E] [NormedSpace ℝ E]
  (r : Regularity) (f : ReferenceDomain → E) : Prop :=
  HasLocalExtensions r (periodicLift f) cylinder
```

Thus, $\text{HasRegularity } r \ f$ means that the lift of f admits local C^r extensions near every point of C , including its boundary. Since $\pi_C : C \rightarrow Q$ is a covering map, this gives C^r regularity on the reference solid torus in local coordinates.

We also require the position map $X : Q \rightarrow \mathbb{R}^3$ to be an embedding. At differentiability orders at least one, this means that X is C^r , is a homeomorphism onto its image, and has injective derivative at every point. In covering coordinates, these requirements are expressed by

```
def IsEmbeddingOfRegularity
  (r : Regularity) (X : ReferenceDomain → Vec) : Prop :=
  HasRegularity r X ∧ Topology.IsEmbedding X ∧ ∀ x ∈ cylinder,
  Function.Injective (fderivWithin ℝ (periodicLift X) cylinder x)
```

The three conditions have distinct roles. The first supplies the C^r regularity of X . The second, $\text{Topology.IsEmbedding } x$, says that X is a homeomorphism from Q onto its image, equipped with the topology inherited from $\mathbb{R}^3$. The third is the immersion condition: $\text{Function.Injective}$ requires the derivative to be injective. More precisely, $\text{fderivWithin } \mathbb{R} \ (\text{periodicLift } X) \ \text{cylinder } x$ is the Fréchet derivative of $\tilde{X}$ at x , computed using points of C . It is a continuous real-linear map $D_C \tilde{X}(x) : \mathbb{R}^3 \rightarrow \mathbb{R}^3$. At a boundary point, this is the derivative of any local C^1 extension. It is independent of the chosen extension: two such extensions agree on the interior of C , hence have the same derivatives there, and continuity gives equality of their derivatives at the boundary. Thus, the definition imposes the full immersion condition at boundary points as well as at interior points.

16.3. Configurations and their moduli space. The representative $(X, b, p) = (X, B \circ X, P \circ X)$ is stored as three functions on Q . The keyword **structure** collects these functions into one object.

```

structure Representative where
  position : ReferenceDomain → Vec
  magnetic : ReferenceDomain → Vec
  pressure : ReferenceDomain → ℝ
def IsConfiguration (r : Regularity) (c : Representative) : Prop :=
  IsEmbeddingOfRegularity r c.position ∧
  HasRegularity r.predecessor c.magnetic ∧ HasRegularity r c.pressure
abbrev Configuration (r : Regularity) := {c : Representative // IsConfiguration r c}

```

For a representative c , the fields $c.position$, $c.magnetic$, and $c.pressure$ are X, b, p , respectively. In particular, the last two are the *pulled-back* fields, not the ambient functions B, P . For finite $k \geq 3$, the configuration space is

$$\text{Emb}^k(Q, \mathbb{R}^3) \times C^{k-1}(Q, \mathbb{R}^3) \times C^k(Q, \mathbb{R}).$$

The subtype `Configuration r` attaches the required regularity conditions to the triple. It does not itself impose the MHS equations.

We equip this space with its usual product topology. The implementation uses the compact fundamental cylinder $C_0 = \overline{\mathbb{D}^2} \times [0, 2\pi]$, called `fundamentalCylinder`. The map `jets r f` records the derivatives $D^j \tilde{f}|_{C_0}$ for every permitted order j ; `Regularity.admits` expresses $j \leq k$ for finite regularity and allows every $j \in \mathbb{N}$ for smooth regularity. Each derivative space carries Mathlib's uniform convergence topology, `UniformFun`. Equivalently, for two maps f, g the topology is generated by

$$d_j(f, g) = \sup_{x \in C_0} \|D^j \tilde{f}(x) - D^j \tilde{g}(x)\|,$$

using $0 \leq j \leq k$ or all finite j , as appropriate. The instance `configurationTopology` induces the topology from the three jet maps, with orders $(k, k-1, k)$. Thus, the smooth case uses the full C^∞ topology.

Two configurations are equivalent if they differ by the action

$$(16.1) \quad (X, b, p) \mapsto (sOX \circ \Phi + c, \kappa Ob \circ \Phi, \kappa^2 p \circ \Phi + c_0),$$

where $s > 0$, $\kappa \neq 0$, $O \in O(3)$, $c \in \mathbb{R}^3$, $c_0 \in \mathbb{R}$, and Φ is a C^r diffeomorphism of Q . To express the last condition, `HasRegularLocalLifts r Φ` requires, near each $x \in C$, a C^r map F in Euclidean coordinates such that $F(y) \in C$ and $\pi_C(F(y)) = \Phi(\pi_C(y))$ for y in that neighborhood within C . The predicate `IsReparametrization` requires such lifts for both Φ and its inverse. With this convention, the full relation is

```

def ModuliEquivalent (r : Regularity) (u v : Configuration r) : Prop :=
  ∃ (s κ : ℝ) (0 : Vec ≃ℝ Vec) (c : Vec) (c0 : ℝ)
    (Φ : ReferenceDomain ≃ ReferenceDomain),
  0 < s ∧ κ ≠ 0 ∧ IsReparametrization r Φ ∧ ∀ q : ReferenceDomain,
  v.val.position q = s • 0 (u.val.position (Φ q)) + c ∧
  v.val.magnetic q = κ • 0 (u.val.magnetic (Φ q)) ∧
  v.val.pressure q = κ ^ 2 * u.val.pressure (Φ q) + c0

```

Here, `Vec ≃ℝ Vec` is the type of real linear isometric isomorphisms, hence orthogonal transformations. The notation `ReferenceDomain ≃ ReferenceDomain` specifies a bijection; its regularity and that of its inverse are imposed separately. The symbol $\bullet$ denotes scalar multiplication, and `u.val` is the representative underlying a

configuration. Notice that the scale s , the possibly negative magnetic factor κ , and the pressure shift c_0 are all included.

The moduli space is the quotient by this relation, with its quotient topology:

```
def ModuliSpace (r : Regularity) := Quot (ModuliEquivalent r)
def moduliClass (r : Regularity) : Configuration r → ModuliSpace r :=
  Quot.mk (ModuliEquivalent r)
instance moduliTopology (r : Regularity) : TopologicalSpace (ModuliSpace r) :=
  TopologicalSpace.coinduced (moduliClass r) inferInstance
```

Mathlib’s `TopologicalSpace.coinduced` constructs the quotient topology from the projection `moduliClass`. In mathematical terms, a subset of the quotient is open precisely when its inverse image under this projection is open in the configuration space. The following theorem ensures that equality in the quotient means exactly the action in (16.1):

```
theorem same_moduli_class_iff (r : Regularity) (u v : Configuration r) :
  moduliClass r u = moduliClass r v ↔ ModuliEquivalent r u v
```

The symbol $\leftrightarrow$ means “if and only if”. Lean’s general quotient `Quot` may be formed from a relation without first attaching a proof that the relation is an equivalence relation. The displayed theorem states that two configurations have the same moduli class if and only if they differ by the reference reparametrization and physical transformations specified in `ModuliEquivalent`.

16.4. The equations and the boundary condition. The physical fields are represented by functions $B : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ and $P : \mathbb{R}^3 \rightarrow \mathbb{R}$. Their values away from a neighborhood of the closed body play no role. The required smoothness is expressed by the following definition

```
def SmoothNear {E : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E]
  (Ω : Set Vec) (f : Vec → E) : Prop :=
  ∃ U : Set Vec, IsOpen U ∧ Ω ⊆ U ∧ ContDiffOn ℝ ∞ f U
```

This says that there is an open $U \supset \Omega$ on which the field is C^∞ . The showcase defines `basisVector i` as the i th coordinate vector and defines `gradient`, `curl`, and `divergence` by their usual Cartesian formulas, using Mathlib’s Fréchet derivative `fderiv`. Explicitly,

$$(\nabla P)_i = \partial_i P, \quad \text{curl } B = (\partial_2 B_3 - \partial_3 B_2, \partial_3 B_1 - \partial_1 B_3, \partial_1 B_2 - \partial_2 B_1), \quad \text{div } B = \sum_{i=1}^3 \partial_i B_i.$$

The helper vector forms a vector from its three coordinates, and `cross` is the usual cross product. Lean numbers these coordinates 0, 1, 2 rather than 1, 2, 3.

Boundary tangency is expressed through curves. For a smooth surface S , the condition $v \in T_x S$ means that $v = \gamma'(0)$ for a smooth curve in S with $\gamma(0) = x$. The definition used here is

```
def TangentTo (S : Set Vec) (x v : Vec) : Prop :=
  ∃ γ : ℝ → Vec, ContDiff ℝ ∞ γ ∧ γ 0 = x ∧ (∀ t, γ t ∈ S) ∧ fderiv ℝ γ 0 1 = v
```

The last expression evaluates the derivative of γ at 0 on the real number 1, and hence gives $\gamma'(0)$. For a smooth boundary, this is the usual tangency condition $B(x) \cdot n(x) = 0$. Consequently, the equations and their boundary condition are combined as follows:

```
def IsMHS (Ω : Set Vec) (B : Vec → Vec) (P : Vec → ℝ) : Prop :=
  (∀ x ∈ interior Ω, cross (B x) (curl B x) + gradient P x = 0 ∧ divergence B x = 0) ∧
  ∀ x ∈ frontier Ω, TangentTo (frontier Ω) x (B x)
```

Here, `interior Ω` and `frontier Ω` mean $\text{int } \Omega$ and $\partial\Omega$. The first line of the condition imposes the unforced MHS equations at every interior point; the second imposes tangency at every boundary point.

16.5. The axis and nested pressure surfaces. The round magnetic axis and pressure levels are defined directly by

$$\Gamma_R = \{(R \cos \theta, R \sin \theta, 0) : \theta \in \mathbb{R}\}, \quad \Sigma_p = \{x \in \Omega : P(x) = p\}.$$

In Lean these are `roundAxis R` and `pressureLevel Ω P p`:

```
def roundAxis (R : ℝ) : Set Vec :=
  Set.range fun θ : ℝ => vector (R * Real.cos θ) (R * Real.sin θ) 0
def pressureLevel (Ω : Set Vec) (P : Vec → ℝ) (p : ℝ) : Set Vec :=
  {x | x ∈ Ω ∧ P x = p}
def IsRegularLevel (Ω : Set Vec) (P : Vec → ℝ) (p : ℝ) : Prop :=
  ∀ x ∈ pressureLevel Ω P p, fderiv ℝ P x ≠ 0
```

The notation `Set.range` denotes the image of a map. A pressure level is regular when $dP_x \neq 0$ at each of its points. In Euclidean space this is equivalent to $\nabla P(x) \neq 0$. Nonemptiness is imposed separately when asserting that a regular level is a torus.

The predicate `IsEmbeddedTorus S` means that S is the image of a smooth embedding $Y : \mathbb{T}^2 \rightarrow \mathbb{R}^3$. More precisely, `torusLift Y` is the map $(\theta_1, \theta_2) \mapsto Y([\theta_1], [\theta_2])$ on $\mathbb{R}^2$. The definition requires this lift to be smooth with injective derivative, Y to be a topological embedding, and $Y(\mathbb{T}^2) = S$. These are the same embedding conditions as above, now for a two-dimensional torus.

To specify a foliation rather than merely a collection of toroidal levels, we require a smooth embedding

$$(16.2) \quad F : (0, 1] \times \mathbb{T}^2 \longrightarrow \mathbb{R}^3, \quad F((0, 1] \times \mathbb{T}^2) = \Omega \setminus \Gamma.$$

For every $r \in (0, 1]$, the image $F(\{r\} \times \mathbb{T}^2)$ must be an entire regular pressure level, and the slice $r = 1$ must be exactly $\partial\Omega$. The parameter domain of the foliation is $(0, 1] \times \mathbb{T}^2$. We represent it and its covering domain as follows:

```
abbrev FoliationDomain := Set.Ioc (0 : ℝ) 1 × Torus
def foliationCylinder : Set Vec :=
  {x | x 0 ∈ Set.Ioc (0 : ℝ) 1}
```

The notation `Set.Ioc (0 : ℝ) 1` denotes the interval $(0, 1]$: the left endpoint is excluded and the right endpoint is included. Since `Torus` is $\mathbb{T} \times \mathbb{T}$, the first line defines precisely $(0, 1] \times \mathbb{T}^2$. Its first coordinate labels the leaves, and the two circle coordinates parametrize each leaf.

The set `foliationCylinder` is

$$C_{\text{fol}} = \{x = (x_0, x_1, x_2) \in \mathbb{R}^3 : 0 < x_0 \leq 1\} = (0, 1] \times \mathbb{R}^2.$$

The corresponding covering projection is

$$\pi_{\text{fol}} : C_{\text{fol}} \longrightarrow (0, 1] \times \mathbb{T}^2, \quad \pi_{\text{fol}}(x_0, x_1, x_2) = (x_0, [x_1], [x_2]).$$

Thus, the first coordinate is unchanged, while the other two are taken modulo 2π .

For a parametrization $F : (0, 1] \times \mathbb{T}^2 \rightarrow \mathbb{R}^3$, we use the lift

$$\tilde{F}(x) = \begin{cases} F(x_0, [x_1], [x_2]), & x \in C_{\text{fol}}, \\ 0, & x \notin C_{\text{fol}}. \end{cases}$$

Its Lean definition is the following, with `X` denoting the map F :

```
def foliationLift (X : FoliationDomain → Vec)
  (x : Vec) : Vec := by
classical
exact if hx : x ∈ foliationCylinder then
  X (⟨x 0, hx⟩,
    (x 1 : PeriodicCircle),
    (x 2 : PeriodicCircle))
else
  0
```

In the first branch, `hx` is a proof that $0 < x_0 \leq 1$. The expression `⟨x 0, hx⟩` therefore specifies an element of $(0, 1]$. The other two entries give the classes of x_1, x_2 in $\mathbb{R}/(2\pi\mathbb{Z})$. As in `periodicLift`, the command `classical` permits the case distinction according to membership in the covering domain. The exterior value zero serves only to define the lift on all of $\mathbb{R}^3$. Smoothness is imposed through local extensions from C_{fol} , and derivatives are taken within that set. In particular, local extensions at $x_0 = 1$ express smoothness up to the outermost leaf. The endpoint $x_0 = 0$ does not belong to the parameter domain. The precise foliation condition is

```
def IsPressureFoliation (Ω Γ : Set Vec) (P : Vec → ℝ) : Prop :=
  ∃ X : FoliationDomain → Vec,
    HasLocalExtensions .smooth (foliationLift X) foliationCylinder ∧
    Topology.IsEmbedding X ∧ Set.range X = Ω \ Γ ∧
    (∀ x ∈ foliationCylinder,
      Function.Injective
        (fderivWithin ℝ (foliationLift X) foliationCylinder x)) ∧
    (∀ r : Set.Ioc (0 : ℝ) 1, ∃ p : ℝ, IsRegularLevel Ω P p ∧
      Set.range (fun θ : Torus => X (r, θ)) = pressureLevel Ω P p) ∧
    Set.range (fun θ : Torus => X (⟨1, zero_lt_one, le_rfl⟩, θ)) = frontier Ω
```

The local extension condition includes smoothness at the outer leaf $r = 1$. The equality with $\Omega \setminus \Gamma$ covers every point off the axis. The variable r here labels the leaves; it need not equal the pressure value p .

We combine the critical-set assertion, regularity off the axis, toroidal level sets, and the foliation into the predicate corresponding to clause (i):

```
def NestedPressureTori (Ω Γ : Set Vec) (P : Vec → ℝ) : Prop :=
  Γ ⊆ interior Ω ∧ {x | x ∈ Ω ∧ fderiv ℝ P x = 0} = Γ ∧
  (∀ p, (pressureLevel Ω P p).Nonempty → IsRegularLevel Ω P p →
    IsEmbeddedTorus (pressureLevel Ω P p)) ∧
  (∀ x ∈ Ω \ Γ, IsRegularLevel Ω P (P x)) ∧ IsPressureFoliation Ω Γ P
```

In this definition, the set equation says precisely $\{x \in \Omega : dP_x = 0\} = \Gamma$. In the main theorem we take $\Gamma = \Gamma_{NL}$. Clause (ii), namely $B(x) = 0 \iff x \in \Gamma_{NL}$ for $x \in \Omega$, will appear explicitly in the combined equilibrium condition below.

16.6. The extended stabilizer. For $g(x) = Ox + c$, with $O \in O(3)$ and $c \in \mathbb{R}^3$, membership in the extended stabilizer means

$$g\Omega = \Omega, \quad P(gx) = P(x), \quad B(gx) = \sigma OB(x) \quad (x \in \Omega), \quad \sigma \in \{1, -1\}.$$

There is one sign σ for the entire field. This is equivalent to the definition using $g_*B = \pm B$ and $P \circ g^{-1} = P$, since g maps Ω onto itself. In Lean it is written as follows:

```
def SignedStabilizes (Ω : Set Vec) (B : Vec → Vec) (P : Vec → ℝ)
  (0 : Vec ≅H [ℝ] Vec) (c : Vec) : Prop :=
  let g := fun x => 0 x + c
  g '' Ω = Ω ∧ (∀ x ∈ Ω, P (g x) = P x) ∧
  ∃ σ : ℝ, (σ = 1 ∨ σ = -1) ∧ ∀ x ∈ Ω, B (g x) = σ • 0 (B x)
```

Here $g '' \Omega$ denotes the image $g(\Omega)$. The orthogonal map O is unrestricted, so orientation-reversing isometries are included in the test.

The helper rotation θ is the map

$$R_\theta(x_1, x_2, x_3) = (\cos \theta x_1 - \sin \theta x_2, \sin \theta x_1 + \cos \theta x_2, x_3).$$

The exact stabilizer assertion in clause (iii) is

$$\text{Stab}_\pm(\Omega, B, P) = \{R_{2\pi j/N} : 0 \leq j < N\} \cong C_N.$$

It is formalized in Lean as follows:

```
def ExactCyclicSymmetry (N : ℕ) (Ω : Set Vec) (B : Vec → Vec) (P : Vec → ℝ) : Prop :=
  ∀ (0 : Vec ≅H [ℝ] Vec) (c : Vec), SignedStabilizes Ω B P 0 c ↔
  ∃ j : ℕ, j < N ∧ ∀ x : Vec, 0 x + c = rotation (2 * Real.pi * j / N) x
```

Thus every signed symmetry is one of the stated rotations, and every stated rotation is a signed symmetry. The equality in the last line is required on all of $\mathbb{R}^3$, so it identifies the Euclidean motion itself. This finite group contains neither a nontrivial continuous group of Euclidean symmetries nor an orientation-reversing symmetry.

16.7. The equilibrium and its family. For a representative $c = (X, b, p)$, put $\Omega = X(Q)$ and $\Gamma = \Gamma_{NL}$. The predicate `Equilibrium` $L \mathbb{N} c$ collects smooth embeddedness, the physical fields and their pullback identities, the MHS equations, and all of clauses (i)–(iii):

```
def Equilibrium (L : ℝ) (N : ℕ) (c : Representative) : Prop :=
  let Ω := Set.range c.position
  let Γ := roundAxis (N * L)
  IsConfiguration .smooth c ∧ ∃ (B : Vec → Vec) (P : Vec → ℝ),
    SmoothNear Ω B ∧ SmoothNear Ω P ∧
    (∀ q : ReferenceDomain,
      B (c.position q) = c.magnetic q ∧ P (c.position q) = c.pressure q) ∧
  IsMHS Ω B P ∧ NestedPressureTori Ω Γ P ∧
  (∀ x ∈ Ω, B x = 0 ↔ x ∈ Γ) ∧
  ExactCyclicSymmetry N Ω B P
```

The notation `let` introduces the local abbreviations Ω and Γ . The two pullback identities are

$$B(X(q)) = b(q), \quad P(X(q)) = p(q) \quad (q \in Q).$$

They ensure that the physical fields in the equations and stabilizer are those represented by c . In particular, the definition does not allow an unrelated choice of fields when forming the moduli curve. The equivalence $B x = 0 \leftrightarrow x \in \Gamma$ is the magnetic-axis assertion in clause (ii). Thus, `Equilibrium` has a stronger meaning here than merely being a solution of the MHS equations.

It remains to explain the smooth family and non-isolation in clause (iv). A family $c_\lambda = (X_\lambda, b_\lambda, p_\lambda)$ on $I = [a, b]$ is smooth when it extends to an open neighborhood $U \supset I$ and its three components are jointly smooth in the parameter and covering coordinates, including the boundary of Q . This is expressed by

```
def SmoothFamily (I : Set ℝ) (c : I → Representative) : Prop :=
  ∃ U : Set ℝ, IsOpen U ∧ I ⊆ U ∧ ∃ e : ℝ → Representative,
    (∀ lambda : I, e lambda.val = c lambda) ∧
  HasLocalExtensions .smooth
    (fun z : ℝ × Vec => periodicLift (e z.1).position z.2) (U ×s cylinder) ∧
  HasLocalExtensions .smooth
    (fun z : ℝ × Vec => periodicLift (e z.1).magnetic z.2) (U ×s cylinder) ∧
  HasLocalExtensions .smooth
    (fun z : ℝ × Vec => periodicLift (e z.1).pressure z.2) (U ×s cylinder)
```

The map e extends the representatives from I ; only its restriction to U is used in the smoothness conditions. The variable $z : \mathbb{R} \times \text{Vec}$ contains the parameter and spatial coordinates, so $z.1$ and $z.2$ select these two entries. The set $U \times^s \text{cylinder}$ is their Cartesian product. The local extensions are smooth in all these variables together. This condition therefore includes mixed derivatives and extension across both endpoints of I .

For each regularity r , the associated curve is

$$f : I \longrightarrow \mathcal{M}_{\text{emb}}^r(Q), \quad f(\lambda) = [X_\lambda, b_\lambda, p_\lambda].$$

The following predicate asserts that the configurations have regularity r , that f is continuous and injective, and that every member with $\lambda \in \text{int } I$ is non-isolated:

```
def ModuliCurve (r : Regularity) (I : Set ℝ) (c : I → Representative) : Prop :=
  ∃ h : ∀ lambda, IsConfiguration r (c lambda),
    let f := fun lambda => moduliClass r ⟨c lambda, h lambda⟩
    Continuous f ∧ Function.Injective f ∧
    ∀ lambda : I, lambda.val ∈ interior I →
      ¬ IsOpen ({f lambda} : Set (ModuliSpace r))
```

An element `lambda : I` is a real parameter together with its membership in I and `lambda.val` is the underlying real number. The witness h gives the regularity needed to form the moduli class. The last line says that the singleton $\{f(\lambda)\}$ is not open in the full moduli space, which is exactly non-isolation. Mathematically, this also follows from continuity and injectivity: if that singleton were open, its inverse image would be the open singleton $\{\lambda\}$ in I , impossible for an interior parameter. The Lean predicate includes this conclusion explicitly.

16.8. The formal theorem. With these definitions, the theorem recalled in Subsection 16.1 has the following Lean statement. The name `lambda` is the paper's parameter λ .

```
theorem equilibria_with_exact_cyclic_symmetry (L : ℝ) (hL : 0 < L) :
  ∃ a b : ℝ, 0 < a ∧ a < b ∧ b < 1 / 2 ∧
  ∃ N₀ : ℕ, 1 ≤ N₀ ∧ ∃ family : ℕ → Set.Icc a b → Representative,
    ∀ N : ℕ, N₀ ≤ N →
      SmoothFamily (Set.Icc a b) (family N) ∧
      (∀ lambda : Set.Icc a b, Equilibrium L N (family N lambda)) ∧
      (∀ k : ℕ, 3 ≤ k → ModuliCurve (.finite k) (Set.Icc a b) (family N)) ∧
      ModuliCurve .smooth (Set.Icc a b) (family N)
```

The arguments $(L : \mathbb{R})$ and $(hL : 0 < L)$ are the cell length and its positivity hypothesis. All other data are conclusions: there are real endpoints $0 < a < b < 1/2$, a threshold $N_0 \geq 1$, and one family of representatives. Mathlib's `Set.Icc a b` is the closed interval $[a, b]$. Thus the first line gives precisely a compact interval inside $(0, 1/2)$ with nonempty interior.

The family is indexed by natural numbers, and its stated properties are required whenever $N \geq N_0$. Since $N_0 \geq 1$, this expresses the paper's assertion for all sufficiently large integers. The line `Equilibrium L N (family N lambda)` supplies the equations and properties (i)–(iii) for every parameter in that same interval. Finally, the two `ModuliCurve` clauses give property (iv) for every finite $k \geq 3$ and for smooth regularity. Neither the interval nor the family is allowed to depend on k .

16.9. Source files and verification. All displayed declarations come from `Showcase.lean`. The two occurrences of `sorry` omit the proofs for exposition. The file `Showcase_WithProofs.lean` proves both statements. The proof files were checked with Lean 4.33.1. The proofs depend only on Lean's three standard axioms, `propext`, `Classical.choice`, and `Quot.sound`.

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DEPARTMENT OF MATHEMATICS, BROWN UNIVERSITY, 151 THAYER STREET, 02912 PROVIDENCE RI, USA.

Email address: javier_gomez_serrano@brown.edu

DEPARTMENT OF MATHEMATICS, BAR-ILAN UNIVERSITY, RAMAT-GAN 5290002, ISRAEL

Email address: lukas.liehr@biu.ac.il

MATHEMATICAL INSTITUTE, UNIVERSITY OF OXFORD, ANDREW WILES BUILDING, RADCLIFFE OBSERVATORY QUARTER, WOODSTOCK ROAD, OXFORD, OX2 6GG, UNITED KINGDOM.

Email address: mitchtaylor@shaw.ca